Kinovea Reference Manual

Video annotation & motion analysis

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Welcome to the manual for Kinovea.

Kinovea is a video annotation tool designed for motion analysis in sport. Its main features revolve around the study and capture of short sport video sequences.

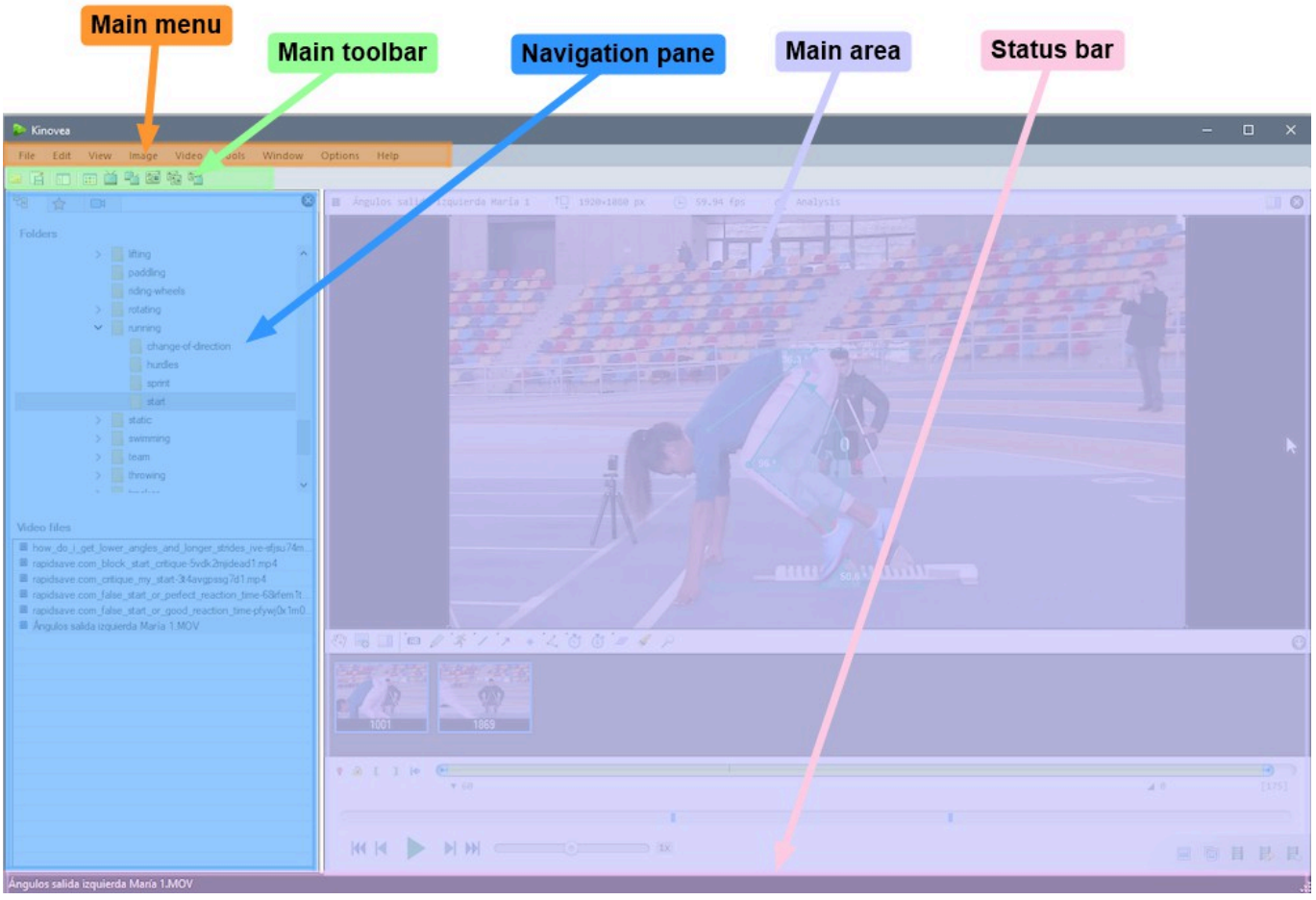
For a single-page overview of the features of Kinovea check the Features page on the website.

This manual is maintained by volunteers. If you find something that is confusing, wrong, or otherwise needs to be edited, let us know.

User interface

User interface

The following screenshot shows the main elements of the Kinovea user interface. These elements are then described in more detail in the sections below.



- The main menu
- The main toolbar
- The navigation pane
- The main area:
    - video player
    - video capture
    - dual mode
    - browser view
- The status bar

Main menu

The main menu is described in detail on the Main menu page.

Main toolbar

The main toolbar provides quick access to screen organization.

	Opens a video
	Saves annotations to the current KVA file
	Toggles the navigation pane visibility
	Closes screens and shows the browser view
	One playback screen
	Two playback screens
	One capture screen
	Two capture screens
	One playback screen and one capture screen

Navigation pane

The navigation pane is on the left hand side of the main Kinovea window. It provides access to the file system and connected cameras.

To show or hide the navigation pane use the menu View ▸ Navigation pane, or the shortcut F4.

It has the following tabs:

	File system
	Shortcuts
	Cameras

File system

The file system tab shows a tree view of the folders on your computer. The bottom part of the tab shows a list of supported files found in the selected folder.

Shortcuts

The shortcuts tab shows a tree view of bookmarked folders.

To add a shortcut, right click the folder and choose **Add to shortcuts**.

Cameras

The cameras tab shows a list of the cameras currently connected to the computer and a history of recently recorded videos.

The Manual connection button opens a dialog to manually add network cameras or camera simulators.

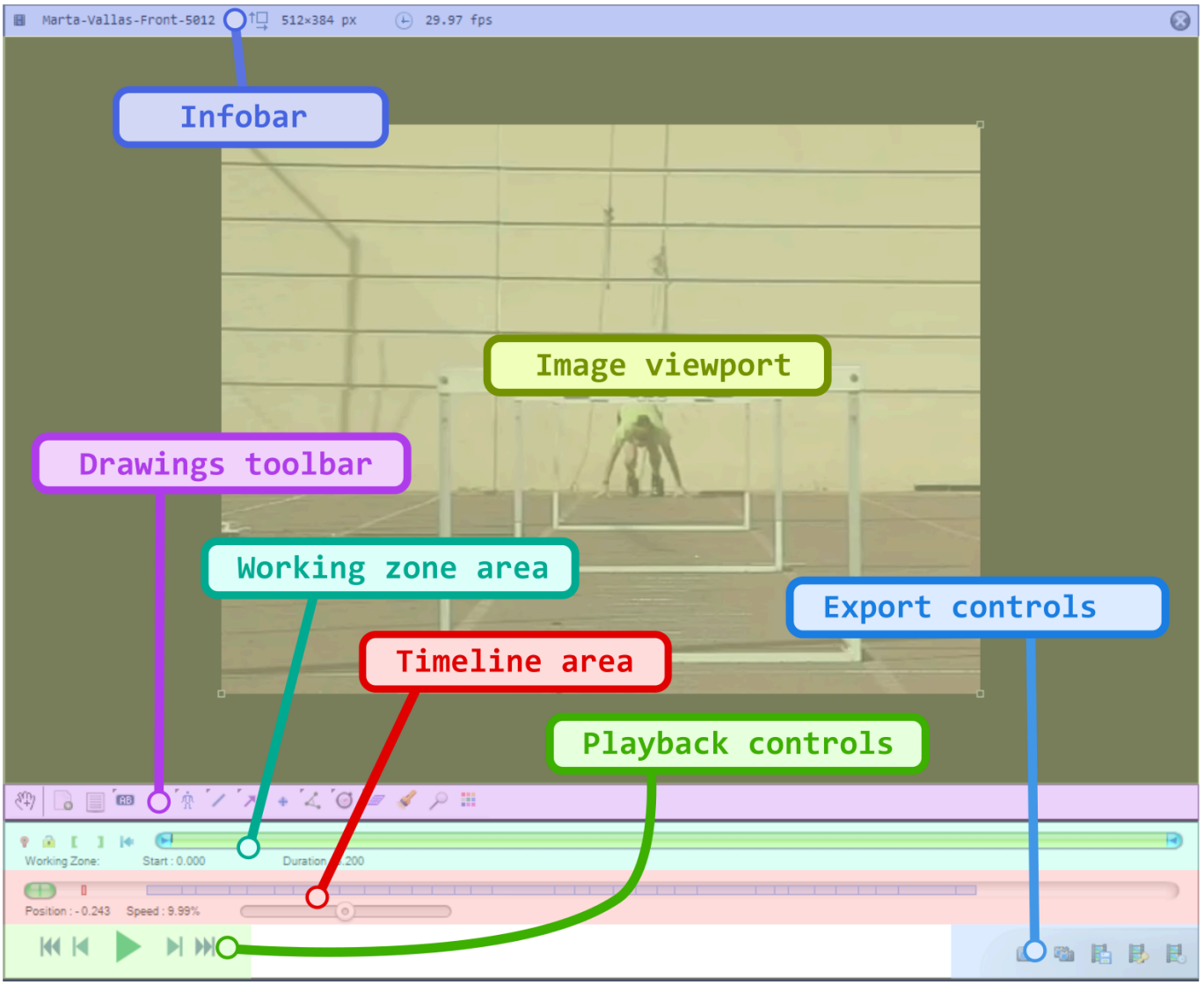
See Capture for more information about connecting cameras.

Video player

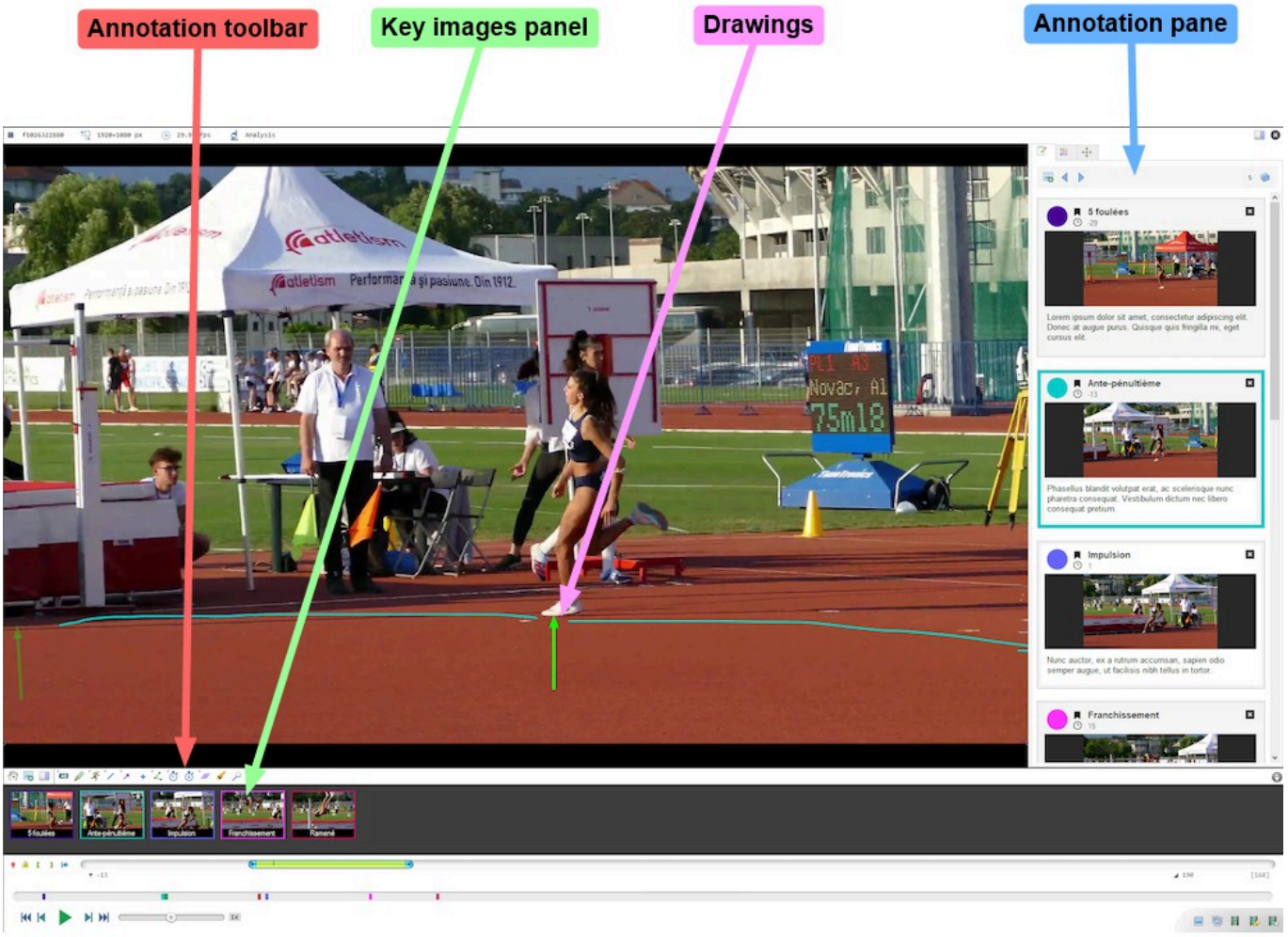
When opening a video file, Kinovea functions as a video player with annotation capabilities.

The following screenshots link directly to the corresponding documentation topic.

Playback, frame by frame and kinograms

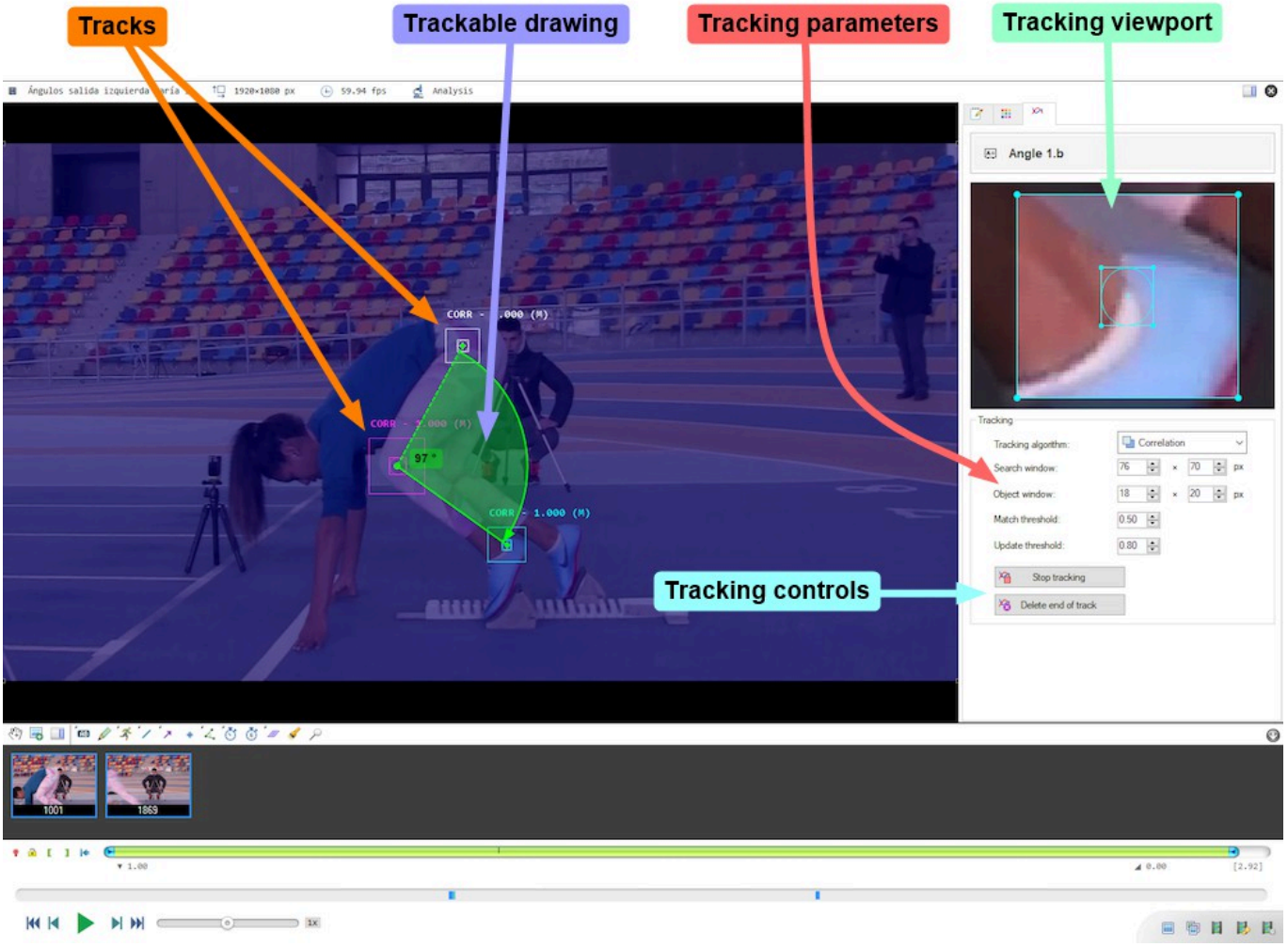


Annotations



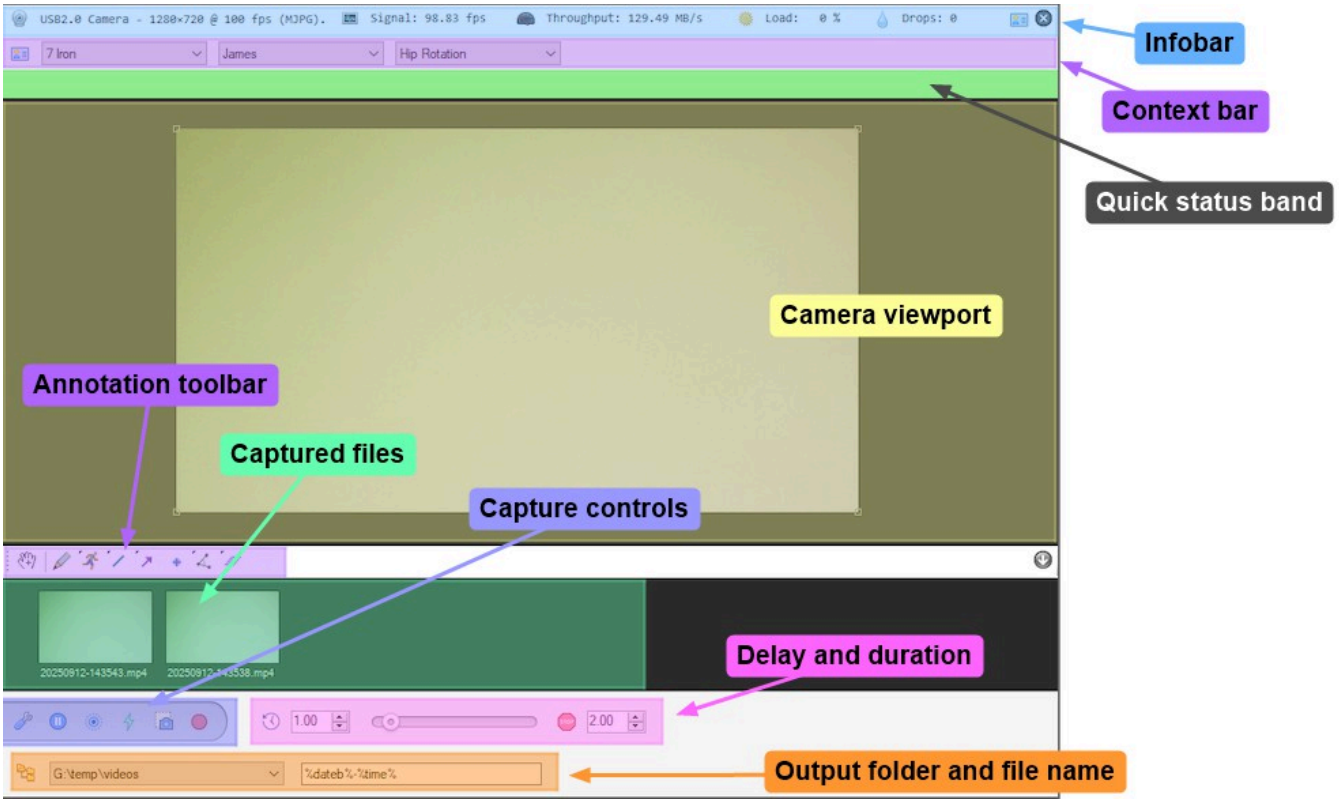
Measurement

Tracking



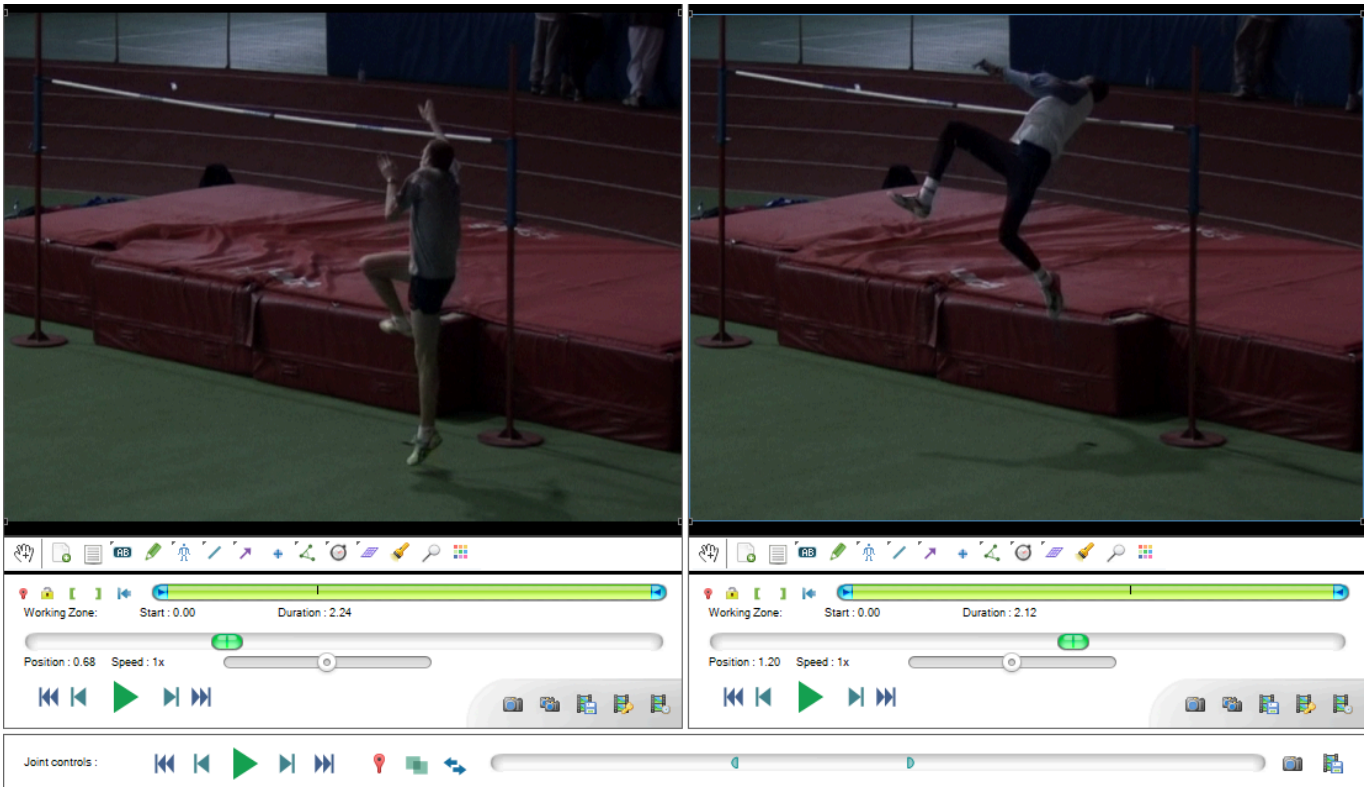
Video capture

When connecting to a camera or video capture device, Kinovea functions as a video capture tool.



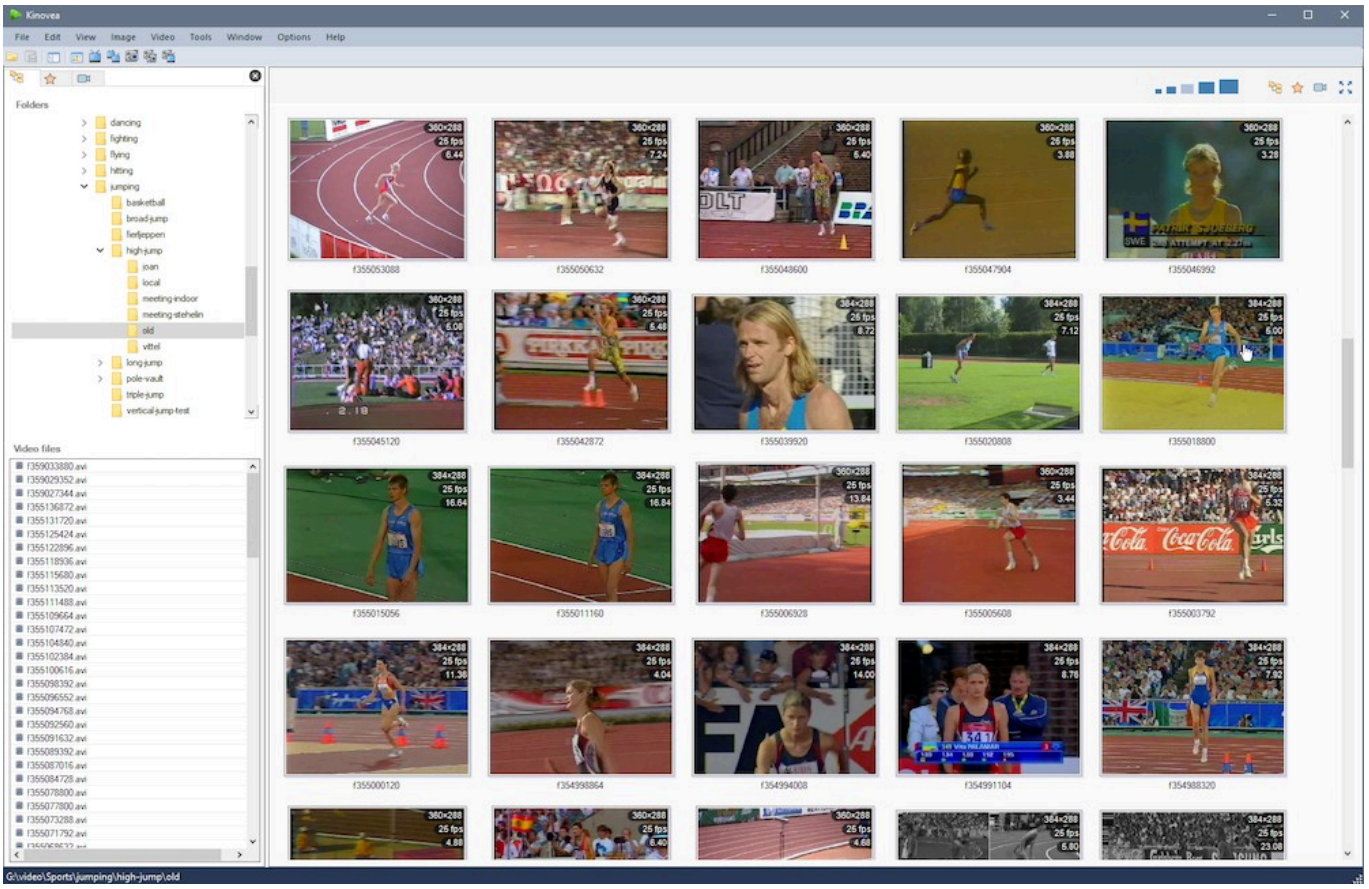
Dual mode

It is also possible to have two playback screens, two capture screens, or one of each side by side.



Browser view

When using Kinovea without any video opened or camera connected, it functions as a file browser.



Status bar

aiue

Main menu

File

Open video…	Opens a video file in a playback screen.
Open replay folder observer…	Selects a folder to monitor.
Load annotations…	Merges a KVA file into the current video.
Recent	Displays a list of recently opened videos.
Save annotations	Saves the current annotations file.
Save annotations as…	Saves the annotations to a specific file.
Export video	Saves the video with annotations painted on.
Export to spreadsheet	Saves annotations to one of several spreadsheet formats.
Close	Closes the video.
Quit	Closes Kinovea.

Edit

Undo	Moves back in the drawings operations history.
Redo	Moves forward in the drawings operations history.
Cut drawing	Places the active drawing in Kinovea's clipboard and deletes it.
Copy drawing	Places the active drawing in Kinovea's clipboard.
Paste drawing	Pastes the drawing from Kinovea's clipboard.

View

File explorer	Shows or hides the explorer panel.
Full screen	Places Kinovea in full screen mode.
One playback screen	Resets the workspace to one playback screen.
Two playback screens	Resets the workspace to two playback screens.
One capture screen	Resets the workspace to one playback screen.

Two capture screens	Resets the workspace to two playback screens.
Invert screens	Swaps screens.
Show/Hide common controls	Shows or hides the joint controls.

Image

Image format	Forces the aspect ratio of the images to a preset value.
Image rotation	Rotates the video or camera images.
Mirror	Flips the video or camera images horizontally.
Demosaicing	Performs color reconstruction on raw videos.
Deinterlace	Removes interlacing artifacts.

Video

Configure video timing...	Opens a dialog to change the capture framerate or the nominal framerate.
Overview...	Displays a mosaic of the images in the video.
Reverse...	Inverts the order of the video frames.

Tools

Observational reference	Displays preset drawings that can be addded to the active video.
Camera test grid	Displays a test pattern to the active capture screen.
Coordinate system	Shows or hides the coordinate system on the active video.
Camera calibration	Displays the lens distortion calibration dialog.
Scatter diagram...	Displays the scatter diagram dialog.
Linear kinematics...	Displays the linear kinematics plot.
Angualr kinematics...	Displays the angular kinematics plot.
Angle-angle diagram...	Displays the angle-angle plot.

Options

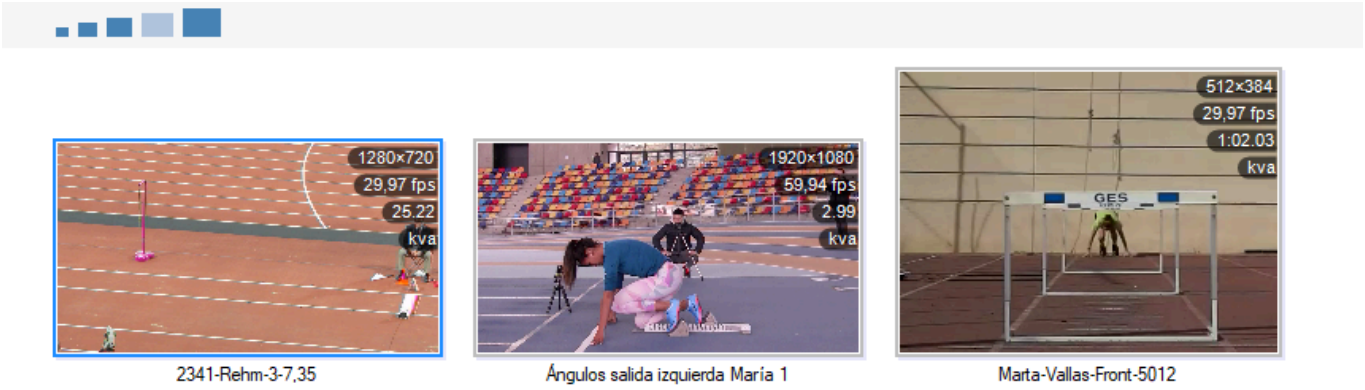
Language	Changes the language of the user interface.
Time	Changes the preferred time representation.
Workspace	Saves or exports the current state of the screens.
Preferences	Opens the preference pages.

Help

Check for updates…	Checks if a new version of Kinovea is available.
Open log folder…	Opens Kinovea's application data folder.
www.kinovea.org	Opens Kinovea's website in the default web browser.
About Kinovea…	Opens the About dialog.

Browser view

The browser view is displayed when there are no playback or capture screens loaded. It shows thumbnails corresponding to the active tab of the explorer panel.



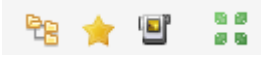
In the case of the file explorer and shortcut tabs, it displays thumbnails of the files in the selected folder. In the case of the camera explorer it displays thumbnails of all the cameras connected to the computer.

Size buttons



The size buttons change the size of the thumbnails.

Navigation pane buttons



These buttons change the active tab in the navigation pane. The rightmost button enable or disable full screen mode.

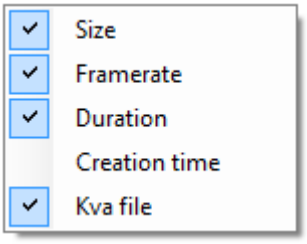
Content preview

Thumbnails of video files contain multiple frames of the video. The extra frames can be viewed using the sections at the bottom of the thumbnail.



Properties overlay

Some metadata can be overlaid on the thumbnails of videos or images. To configure which metadata is overlaid right click the background of the explorer to bring up the context menu.



Enabling the metadata `Kva file` will make it display “kva” on the thumbnail when annotations exist as a sidecar file to the main video.

Preference pages

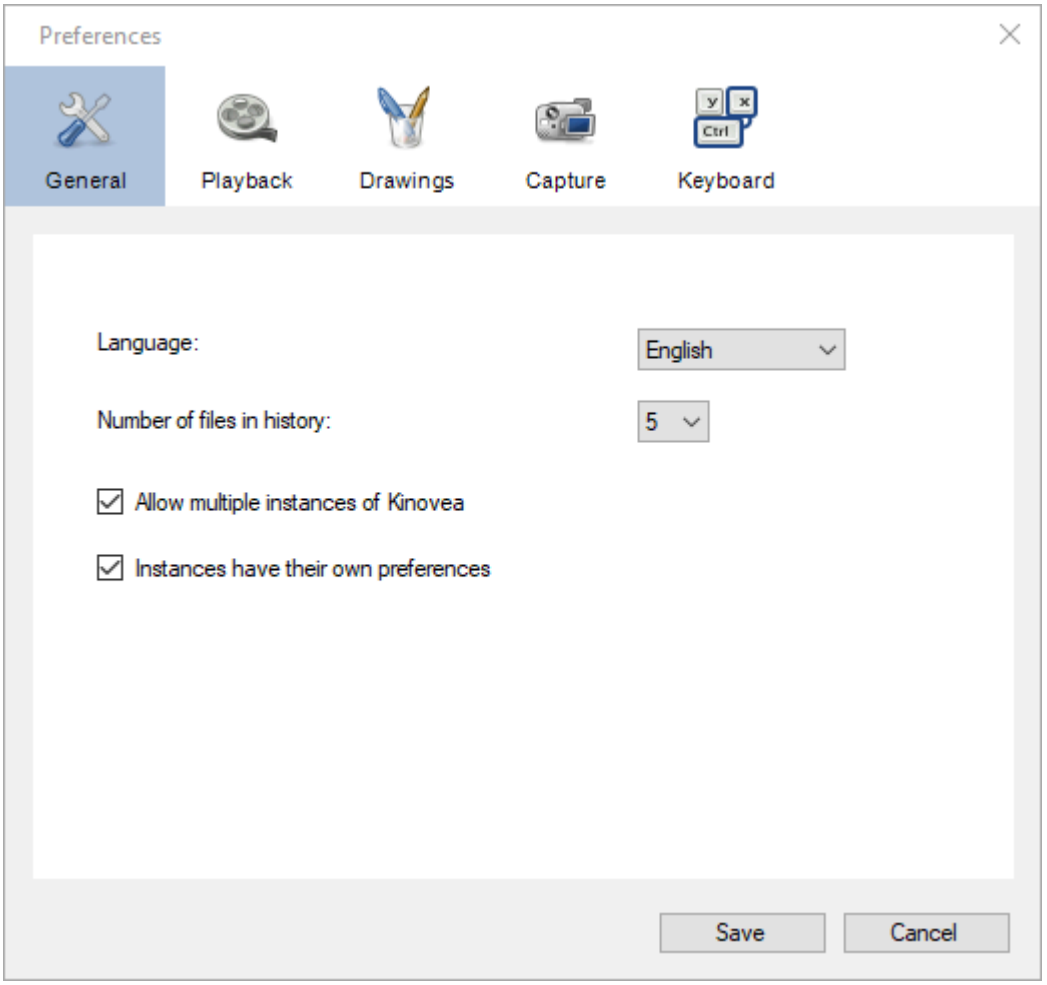
The Kinovea preferences contain settings to control how Kinovea behaves.

The various pages are reached by clicking on the buttons at the top of the dialog. Each page may contain multiple tabs.

The following preference pages are available:

General

General



Language

This option lets you change the language of Kinovea's user interface.

The translations are contributed by the community. If you would like to contribute please check the translation project at Weblate.

Number of files in history

This options controls how many recently opened files and replay folders are remembered and listed under the menu File ▸ Recent.

Allow multiple instances of Kinovea

When this checkbox is checked it is possible to run multiple Kinovea at the same time. This can be used to record more than two cameras or play more than two videos at the same time. Each instance is identified by a name or a number.

See also: ../instances.

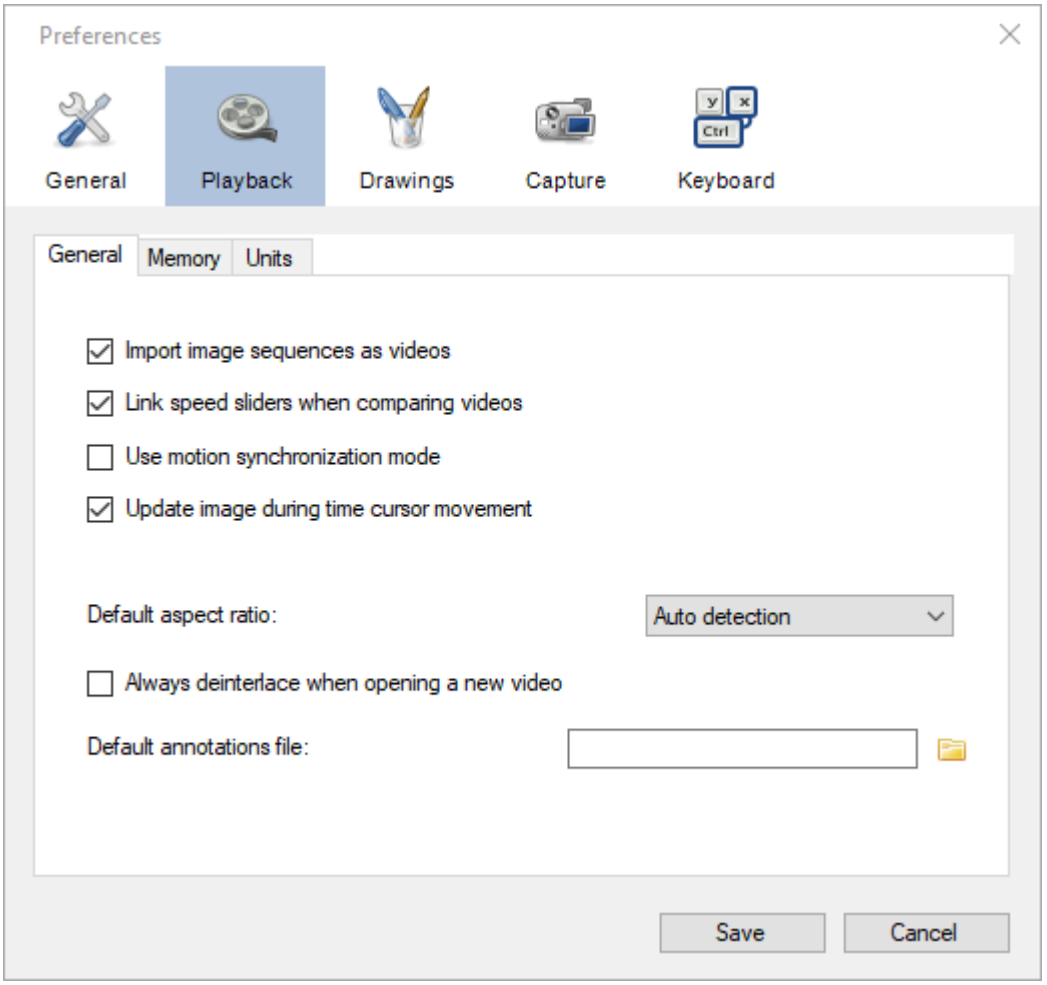
> **Note**
>
> Video synchronization does not work across different instances.

Instances have their own preferences

When this checkbox is checked each instance of Kinovea will maintain its own preferences, independently of all other instances. This option is only available in the first instance started.

Playback

General



Import image sequences as videos

When this option is enabled and an image file is opened, Kinovea will try to detect an **image sequence** by looking for consecutive numbers in the name of other files of the same folder. If an image sequence is detected, all the image files will automatically be loaded and the whole collection interpreted as a video.

Example of file names interpreted as a sequence:

- image_001.png, image_002.png, image_003.png, etc.
- test01.jpg, test02.jpg, test03.jpg, etc.

The numbers can be anywhere in the file name. Aside from the numbers the rest of the name should be identical for all files. The amount of leading zeroes in the numbering scheme should be consistent between all files of the sequence.

A gap or a change in the numbering system will be interpreted as the end of the sequence. The detection is bidirectional: any file can be used to load the sequence, the resulting video will always start at the file with the lower number found.

Link speed sliders when comparing videos

When this option is checked and two videos are compared, changing the playback speed in one player will automatically change the playback speed in the other player to match it.

> **Note**
>
> The playback speed is matched according to the real time speed, taking into account differences in video file framerate and capture framerate. If one video is in slow motion and the other is at normal speed, this option should still work as expected as long as the capture framerate of the slow motion video is correctly configured.

Update image during time cursor movement

When this option is not checked the video display will be paused when manually moving the timeline cursor.

Default aspect ratio

This option defines the default image aspect ratio configured any time a video file is opened. It is the same as manually configuring the menu **Image ▸ Image format**.

The following options are available:

- `Auto detection`
- `Force 4:3`
- `Force 16:9`

The `Auto detection` option uses the image size and the pixel aspect ratio found in the video file metadata to calculate the image height. The other options will change the height of the video to match a 4:3 or 16:9 aspect ratio.

Always deinterlace when opening a new video

This option forces the deinterlacing mechanism to be enabled for all opened files. It is the same as manually configuring the menu **Image ▸ Deinterlace**.

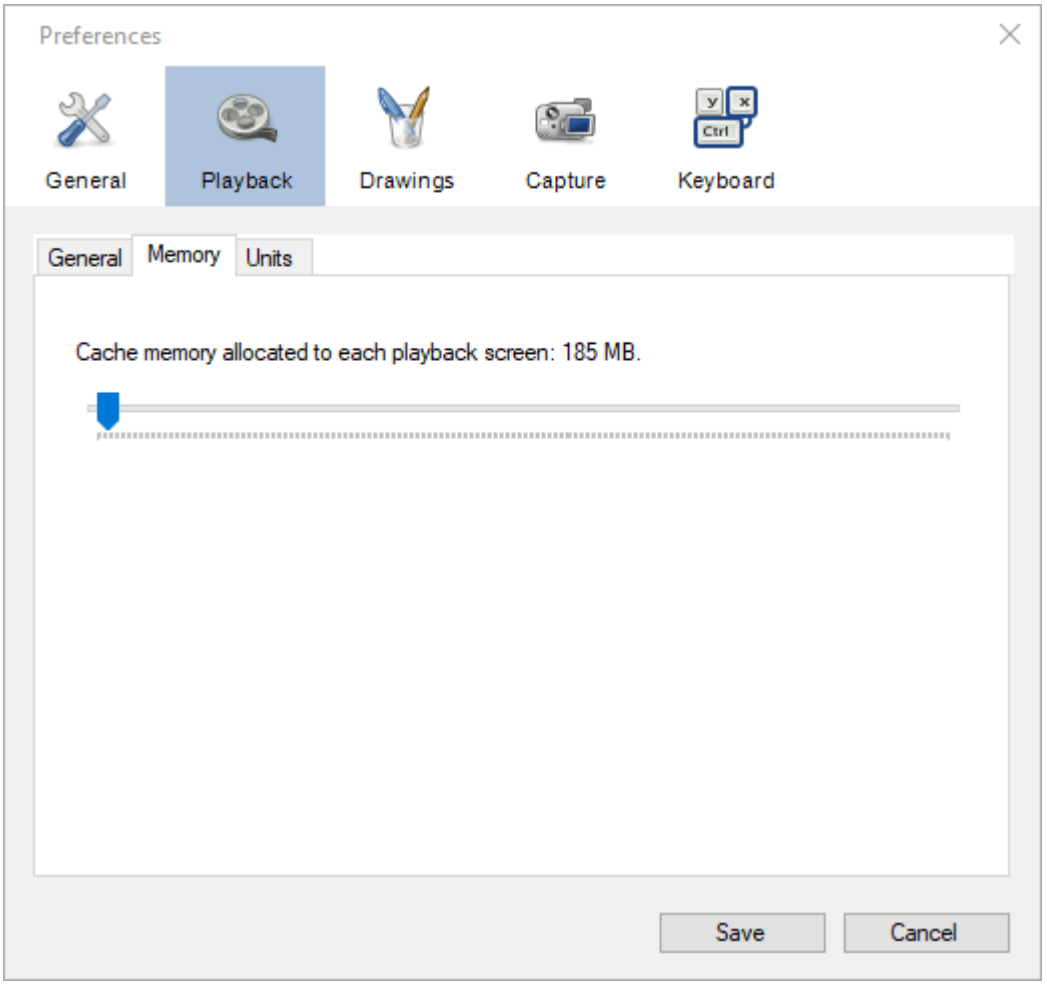
Default annotations file

This option lets you point to a .KVA file containing video annotations that will be automatically loaded when any video is opened.

Other annotations files can still be loaded on top of the video by using the sidecar file method or through the menu File ▸ Load annotations. They will be merged with each others.

See also: Saving and loading annotation files.

Memory



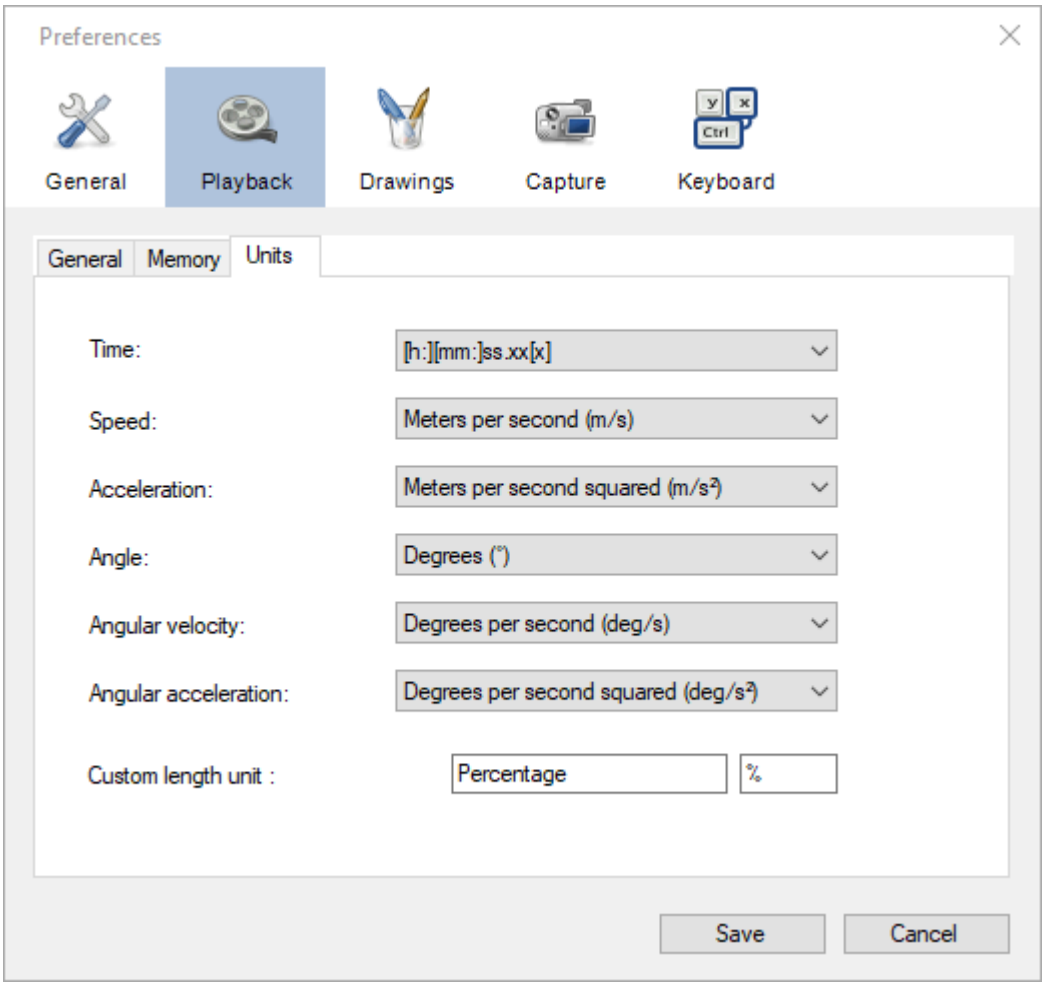
Cache memory allocated to each playback screen

The cache memory is used to load the video content in system memory and speed up navigation. When the active video section (working zone) fits in the cache memory it will be automatically loaded into this cache. If the video section does not fit in the cache the memory will not be consumed.

When using side by side comparison each playback screen can use at most half the memory amount configured.

In the case of multiple instances of Kinovea, each instance has its own cache memory.

Units



Time

This option controls the format of all time-related information displayed in the program [1]. It is used in the timeline position and duration, in chronometers and clocks, and in exported files.

The following options are available:

Format	Example	Description
[h:][mm:]ss.xx[x]	1:10.48	Textual timecode.
Frame number	1762	Rank of the current frame.
Total milliseconds	70480	Integer number of milliseconds.
Total microseconds	1284	Integer number of microseconds.
Ten thousandth of an hour	904	Ten thousandths of an hour
Hundredth of a minute	542	Hundredths of a minute

Format	Example	Description
[h:][mm:]ss.xx[x] + Frame number	1:10.48 (1762)	

When using textual timecode if the real time framerate is higher than 100 fps, thousandths of seconds are displayed. Hours and minutes are only displayed when necessary.

Note

The time starts at the **time origin**. The time origin can be configured to be anywhere in the video. Video locations that are before the time origin are displayed as negative numbers. If the time origin is not manually defined, the time origin is automatically set to the start of the current video section.

Speed

The unit for speed is used in the trajectory tool and in the `Linear kinematics` window when setting the measurement display option to `Speed`, `Horizontal velocity` or `Vertical velocity`. It is also used in the Angular kinematics window when using Tangential velocity.

The following options are available:

Unit	Symbol
Meters per second	m/s
Kilometers per hour	km/h
Feet per second	ft/s
Miles per hour	mph

Note

If no spatial calibration has been performed the speed unit will automatically be **Pixels per second (px/s)**.

Acceleration

The unit for acceleration is used in the trajectory tool and in the `Linear kinematics` window when setting the measurement display option to `Acceleration`, `Horizontal acceleration` or `Vertical acceleration`. It is also used in the `Angular kinematics` window when using `Tangential acceleration`, `Centripetal acceleration` or `Resultant acceleration`.

The following options are available:

Unit	Symbol
Meters per second squared	m/s^2
Feet per second squared	ft/s^2

Note

If no spatial calibration has been performed the acceleration unit will automatically be **Pixels per second squared (px/s^2)**.

Angle

The unit for angle is used in tools measuring angles and in the `Angular kinematics` window when setting the data source option to `Angle` or `Total displacement`.

The following options are available:

Unit	Symbol
Degrees	°
Radians	rad

Angular velocity

The unit for angular velocity is used in the `Angular kinematics` window when setting the data source option to `Angular velocity`.

The following options are available:

Unit	Symbol
Degrees per second	deg/s
Radians per second	rad/s
Revolutions per minute	rpm

Angular acceleration

The unit for angular acceleration is used in the `Angular kinematics` window when setting the data source to `Angular acceleration`.

The following options are available:

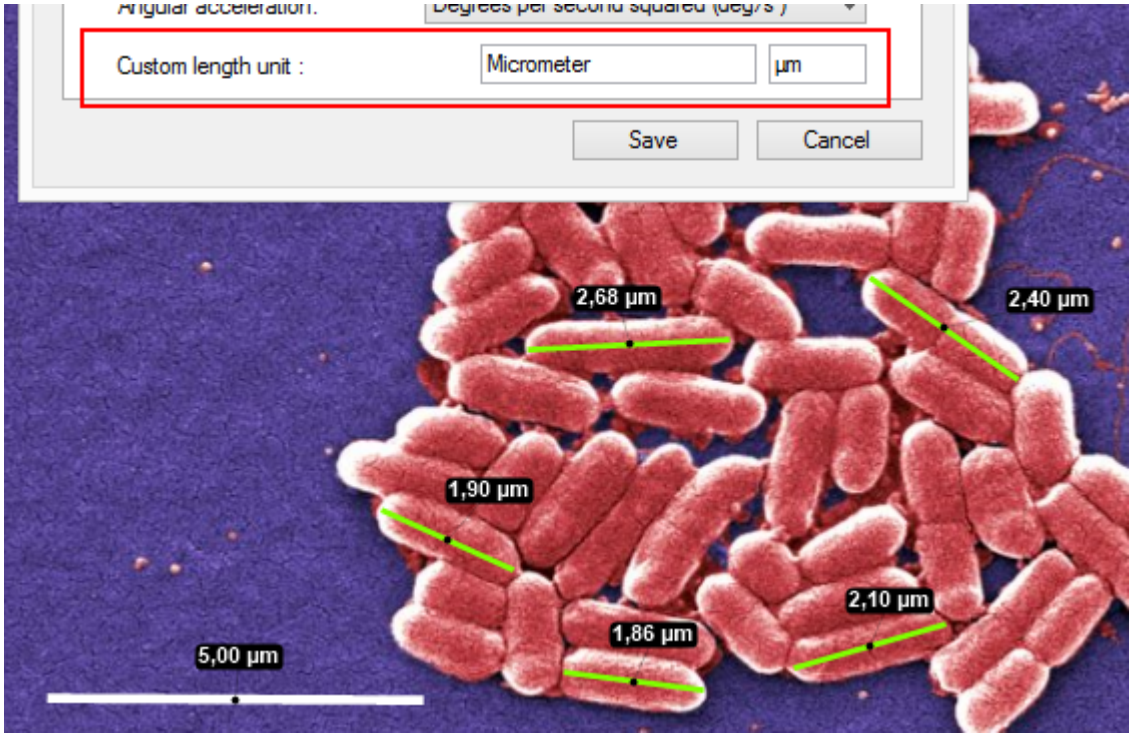
Unit	Symbol
Degrees per second squared	deg/s^2
Radians per second squared	rad/s^2

Custom length unit

This option defines the name and symbol for an additional length unit. The built-in length units are: millimeters, centimeters, meters, inches, feet and yards.

This custom length unit will appear at the bottom of the length unit drop down in the spatial calibration dialogs.

The scale factor between pixels and this unit is defined during the calibration process in the same manner as for other length units.



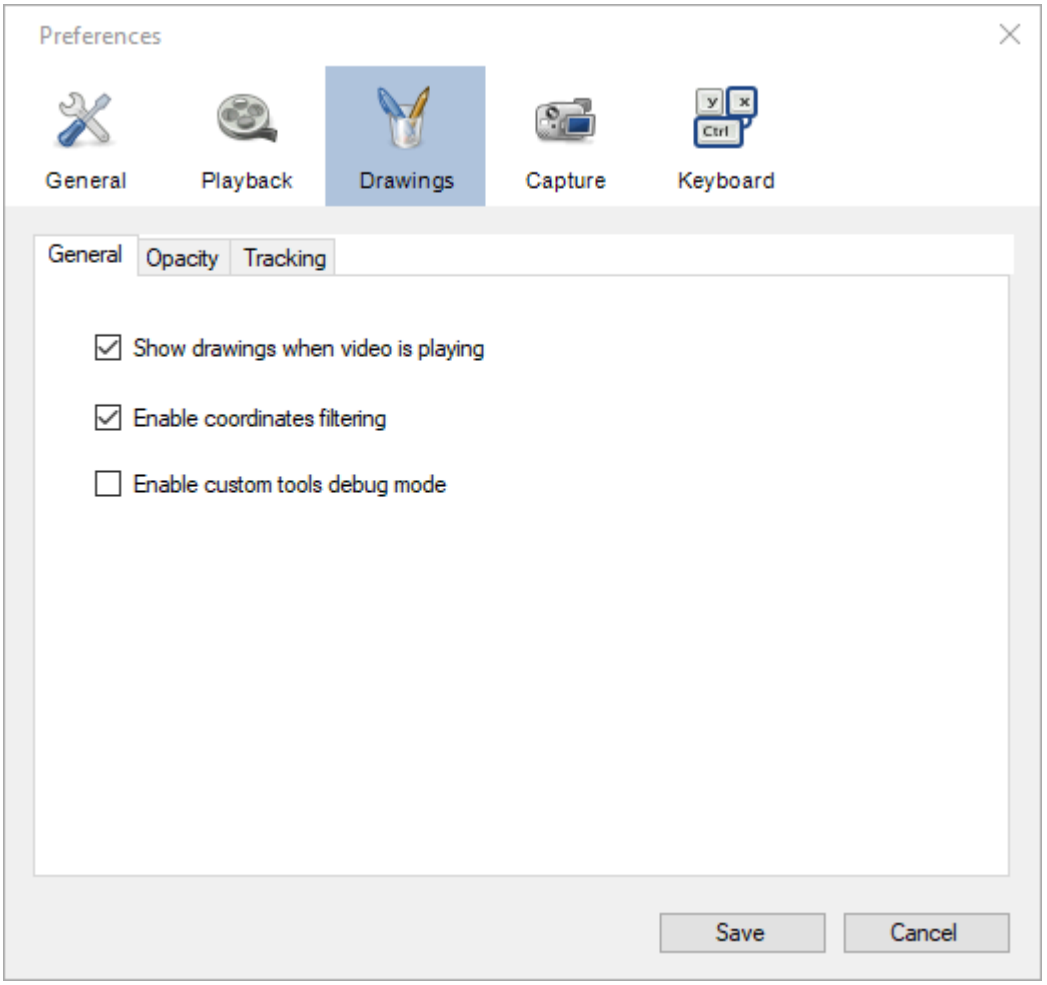
Using the custom length unit to add micrometers to the list of built-in length units.

Footnotes

[1] With the exception of the time axis in the kinematic analysis dialogs. In these dialogs the time is always displayed numerically, either in milliseconds or normalized.

Drawings

General



Show drawings when video is playing

This option controls whether drawings are visible or not when the video is in continuous playback.

Enable coordinates filtering

This option controls the filtering of spatial coordinates for trajectories and tracked drawings. Due to the digitization process the raw coordinates are noisy and the resulting quantities, especially derivatives like speed and acceleration, are less accurate than they could be. Carefully filtering the coordinates remove a lot of this noise and provide more accurate measurements.

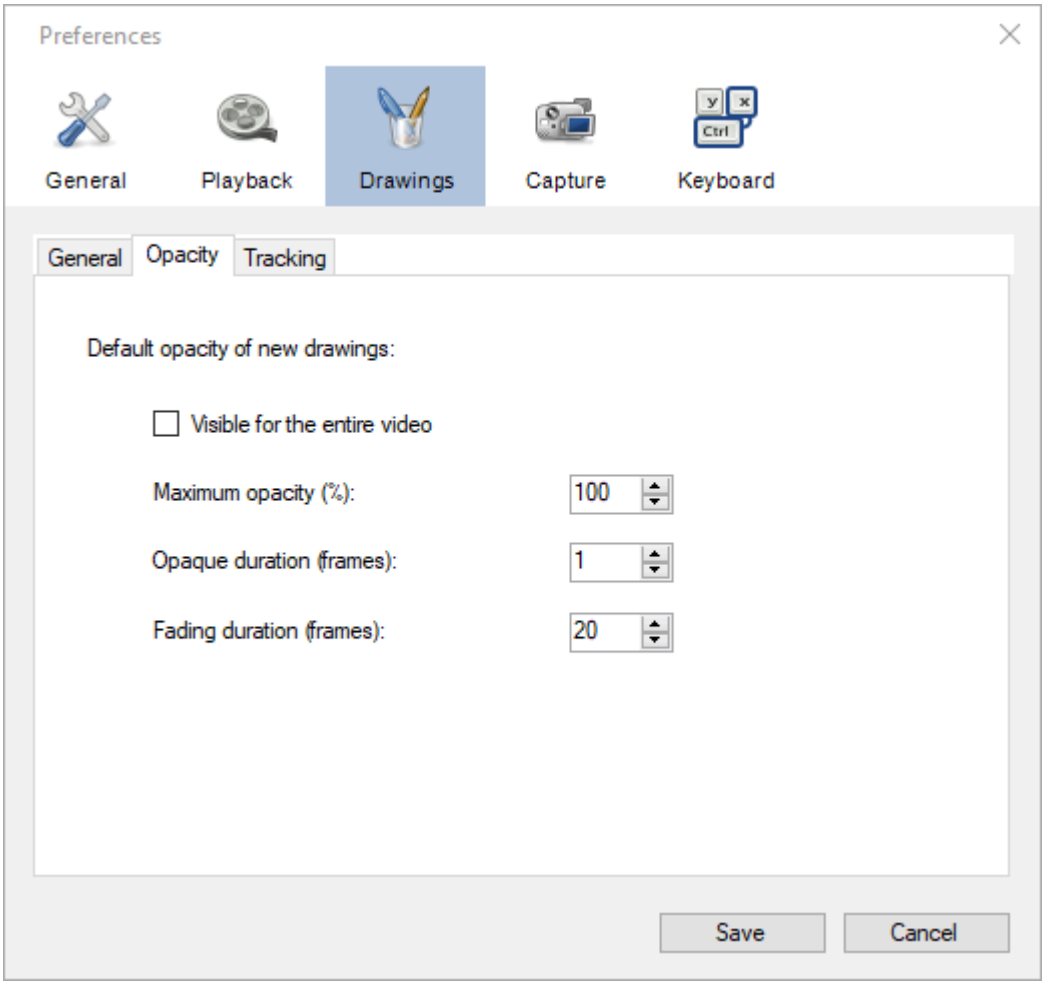
You can uncheck this option if you would rather export the raw coordinates and perform the filtering yourself.

Enable custom tools debug mode

When this checkbox is ticked custom tools will display extra information about the name and relations of points, handles and segments. This option can be used by tool authors to facilitate design.

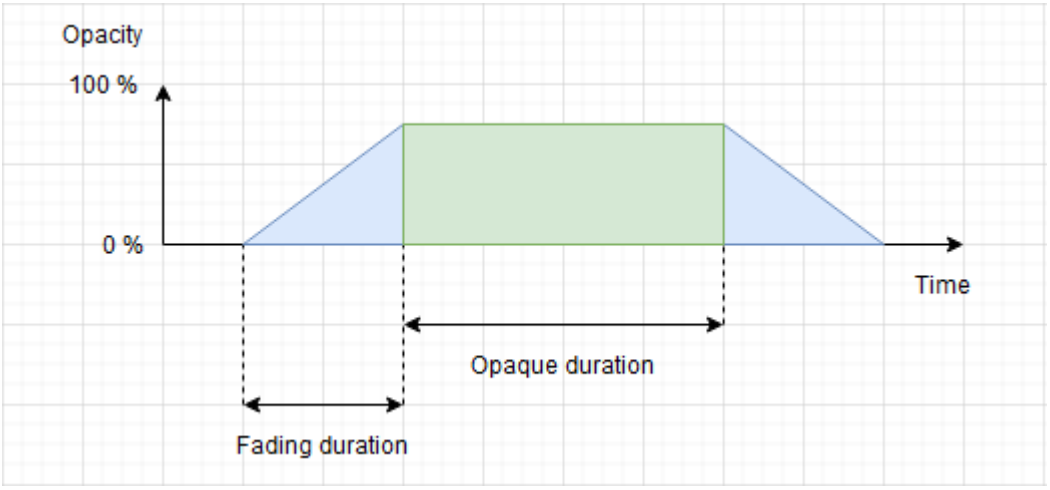
See also: /annotation/tools.

Opacity



The options on this tab control the default visibility of drawings. Drawings can be customized independently by bringing their context menu and selecting options in the Visibility sub menus.

In the most general case a drawing's visibility over time follows the following pattern:



Visible for the entire video

When this option is checked new drawings are visible for the entirety of the video. The opaque duration and fading duration options are ignored.

Maximum opacity

This option controls the opacity used during the opaque section. A value of 100% means the drawing will not let the background show through. A value less than 100% means the drawing will be somewhat transparent.

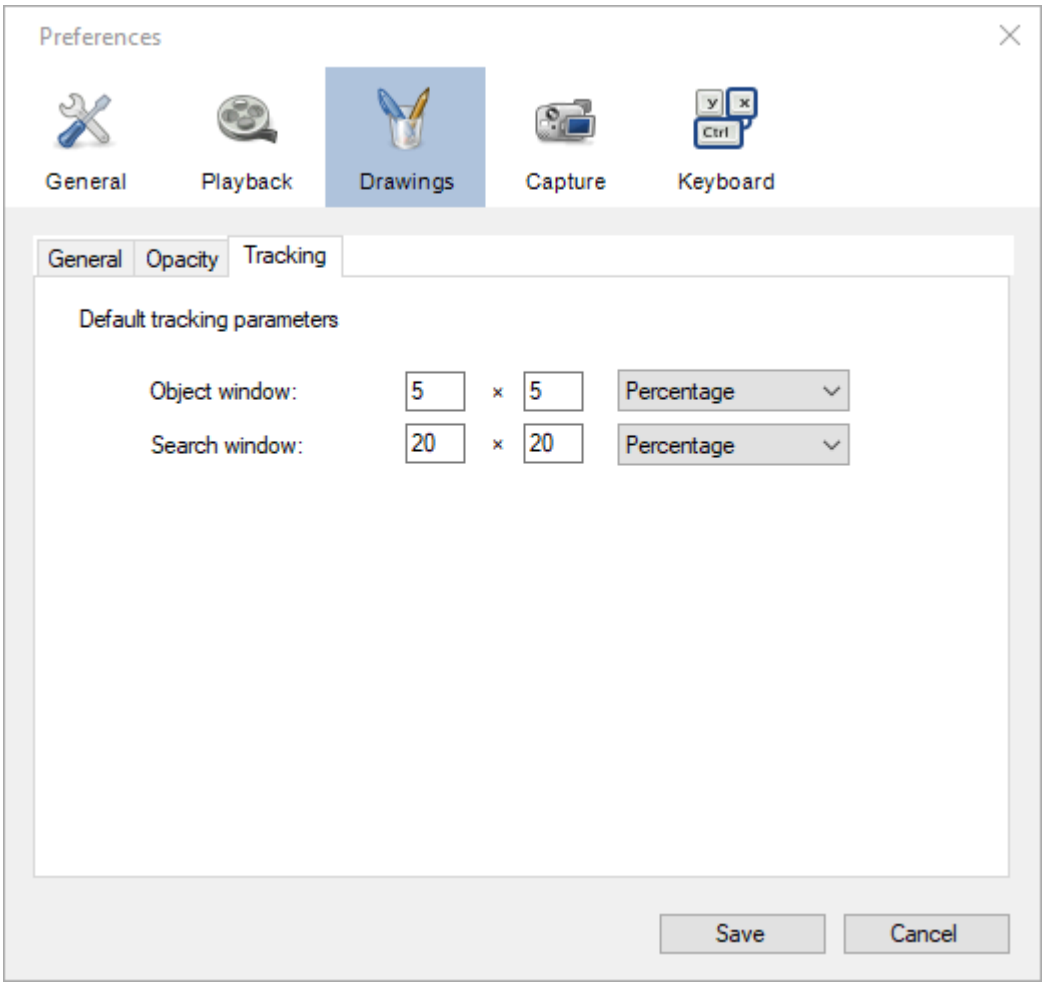
Opaque duration

This option controls how long the drawing stays at its maximum opacity level before fading out. This section starts at the keyframe onto which the drawing was added.

Fading duration

This option controls the duration of the ramps before and after the maximum opacity, until the drawing becomes completely invisible.

Tracking



The options on this page control the default parameters for trajectories and drawing tracking. Trajectories can also be configured independently by bringing their context menu and going in their configuration window. The tracking parameters of other drawings cannot be modified after creation.

Object window

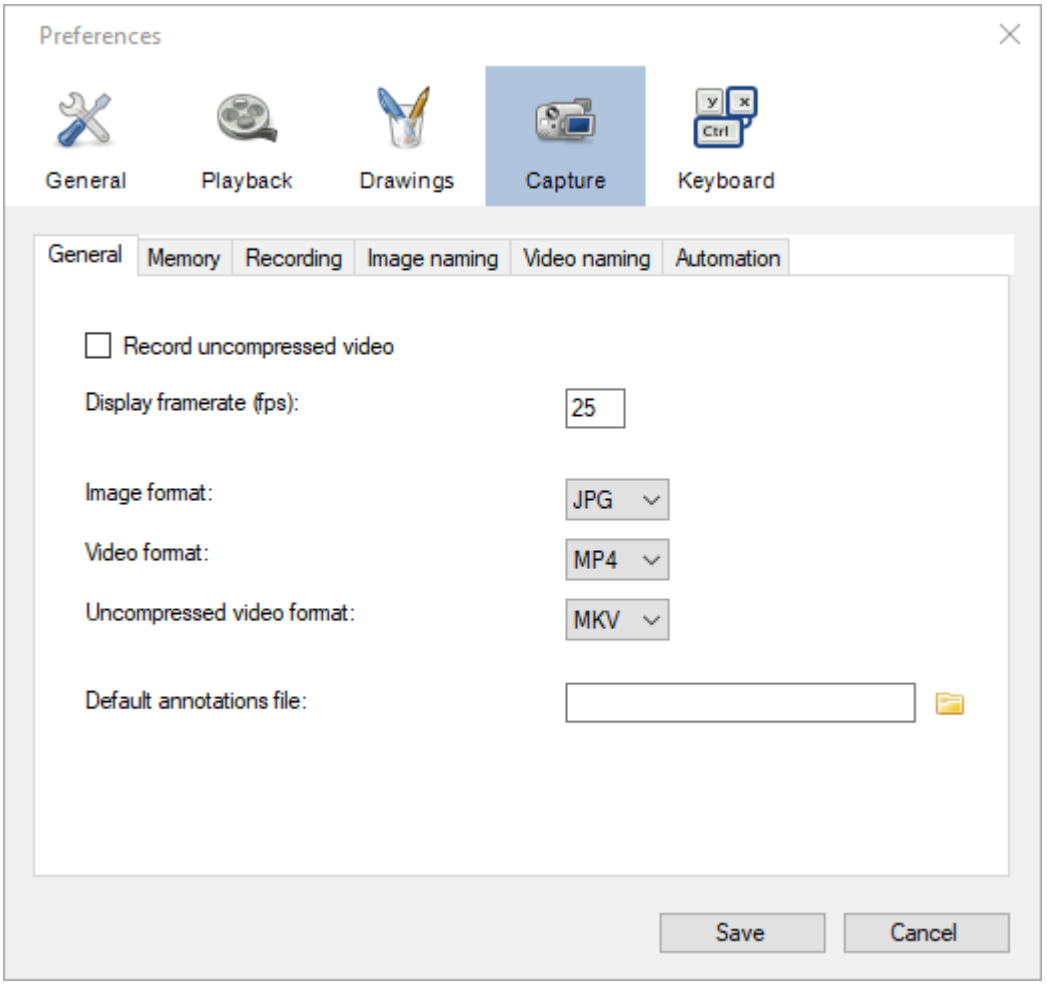
The object window size defines the size of the patch of image around the tracked point that is being looked for in other frames. This should be set to be as small as possible in order to avoid including the background in the tracked patch.

Search window

The search window size defines the area in which the point is looked for. This should be large enough to compensate for the object change of position from one frame to the next. However search window too large can lead to the tracking algorithm picking a different object in another part of the image, if this object looks similar to the tracked object.

Capture

General



Record uncompressed video

When this option is checked the video are recorded without compressing the video frames to JPEG first. When this option is unchecked the video frames are compressed to JPEG using high quality settings to retain fidelity.

This option can be used to minimize dropped frames, by trading off space for speed (due to the JPEG settings used, image quality shouldn't really be much impacted by this option). When using a fast storage medium it can be faster to store the large uncompressed images compared to the time it takes to compress them.

For image snapshots the compression depends on the file format.

Note

When configuring cameras to use MJPEG streams this option is ignored. The MJPEG streams are always saved as-is in their container format.

Note

When configuring cameras to use raw Bayer pattern images you must enable this option to be able to reconstruct the color information at playback time.

Warning

Not all video players can play back uncompressed files.

Display framerate

This option defines the frequency at which the camera images are updated in the capture screen.

While recording the computer resources are shared between displaying the camera stream and recording it to the storage medium. The highest priority is always given to recording but lowering this value can help reduce the overall load on the computer and improve recording performance.

Image format

This option defines the default image format used when saving snapshots.

The following image formats are available:

- JPG
- PNG
- BMP

Video format

This option defines the default video container format used when recording compressed videos. All videos are compressed using an MJPEG codec.

The following container formats are available:

- MP4
- AVI

- MKV

Uncompressed video format

This option defines the default video container format used when recording uncompressed videos.

The following container formats are available:

- MKV
- AVI

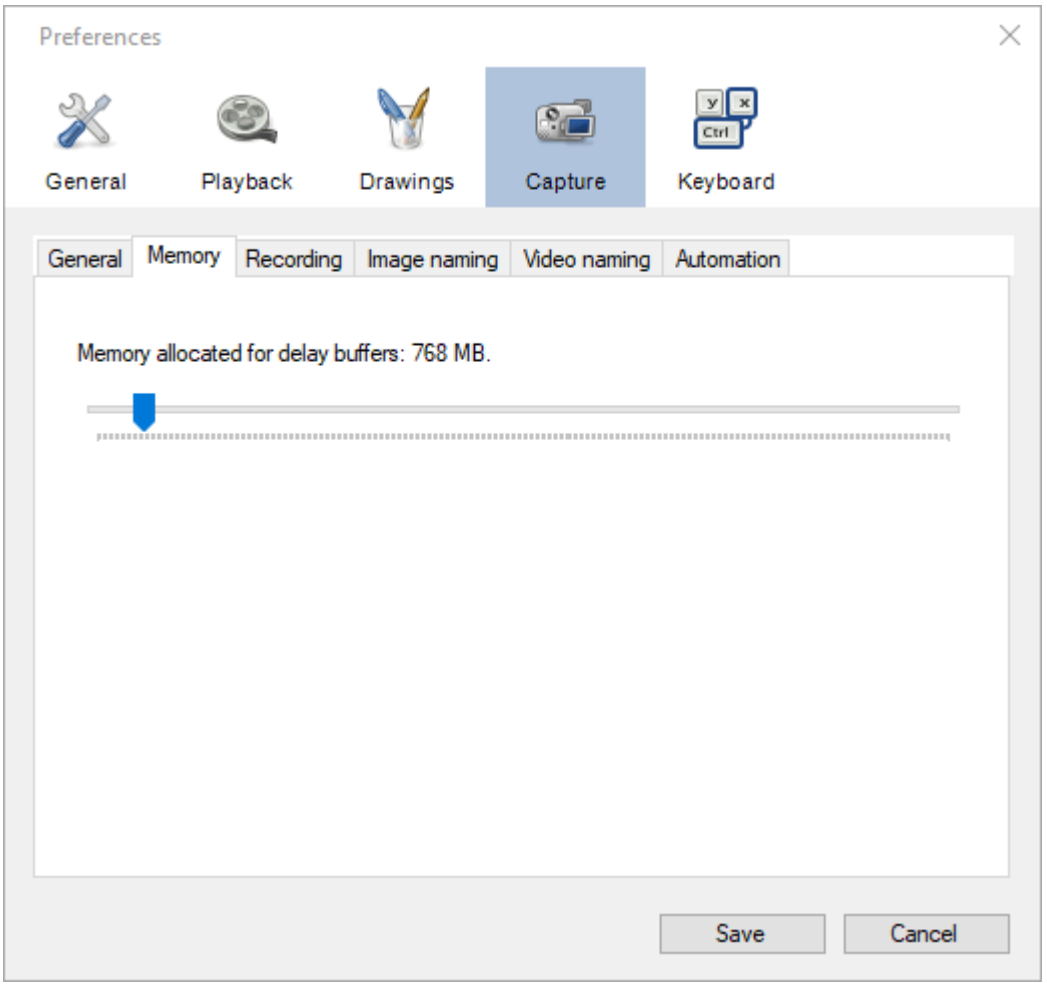
Default annotations file

This option lets you point to a .KVA file containing annotations that will be automatically loaded when any camera stream is opened.

Other annotations files can still be loaded on top of the camera stream by using the menu **File ▸ Load annotations**. They will be merged with each others.

See also: Saving and loading annotation files.

Memory



Memory allocated for delay buffers

This option controls the amount of memory used to remember old frames, in the context of the delayed view of the camera feed. By extension, this option defines the maximum delay configurable in the capture screen. The maximum delay is based on image size, image format and capture framerate.

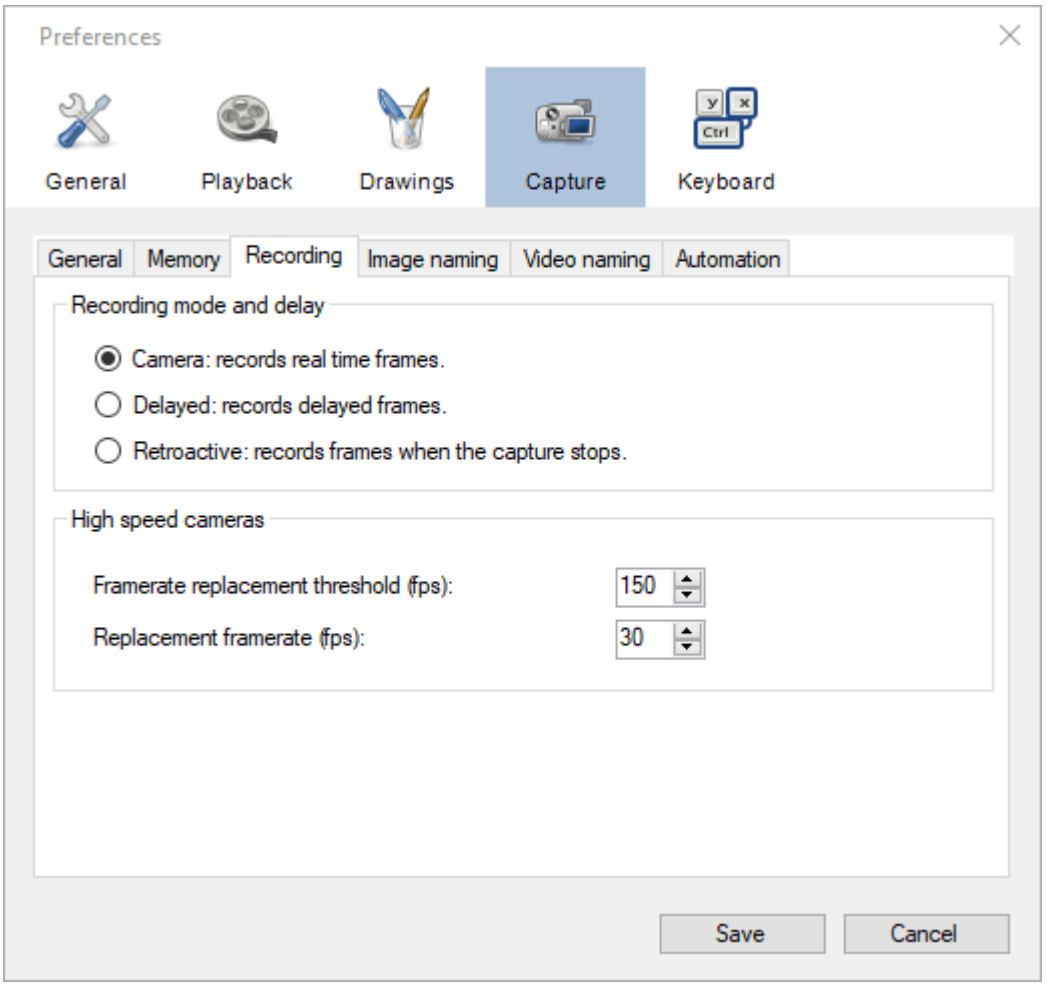
When using two capture screens at the same time each screen uses half the memory amount configured.

In the case of multiple instances of Kinovea, each instance has its own delay buffer memory.

Warning

Unlike the cache memory in the playback screen, this amount of memory is always allocated and used as soon as a capture screen is opened.

Recording



Recording mode and delay

The recording mode option defines how the recording sub-system interacts with the delay buffer to create the output video.

When the camera transmits a new frame it is always put in the delay buffer and the recording sub-system always takes frames from the delay buffer to create the output video.

Camera

When using this recording mode the delay value set in the capture screen is ignored.

Recording is performed on the fly, the saved frame is always the most recent frame sent by the camera.

Delayed

When using this recording mode the delay value set in the capture screen is taken into account.

Recording is performed on the fly, the saved frame is taken from the delay buffer based on the delay value.

This can be used to record actions happening before the moment the record button is hit or triggered.

Retroactive

When using this recording mode the recording is not performed on the fly. Instead, at the end of the recording process, when clicking the stop recording button or when the maximum recording duration is reached, the camera feed is paused, the delay buffer is frozen, and the video file is created all at once.

The delay value is taken into account for creating the recording.

This mode offers the best recording performances and minimizes dropped frames, at the cost of a reduced maximum length for created videos and a temporary freezing of the camera feed.

High speed cameras

The options in this group let you alter the framerate written in the metadata of the output file. This influences the amount of resources required to replay the file and the apparent speed of the action.

A camera might be capable of producing and transmitting 1000 frames per second but the computer will not be able to play the file back at that speed and the monitor won't be able to refresh itself fast enough either. To work around this problem it is usual to reduce the framerate of the output file to a more typical one. Recording devices normally apply this transformation automatically. This results in a video that appears to be in slow motion.

Framerate replacement threshold

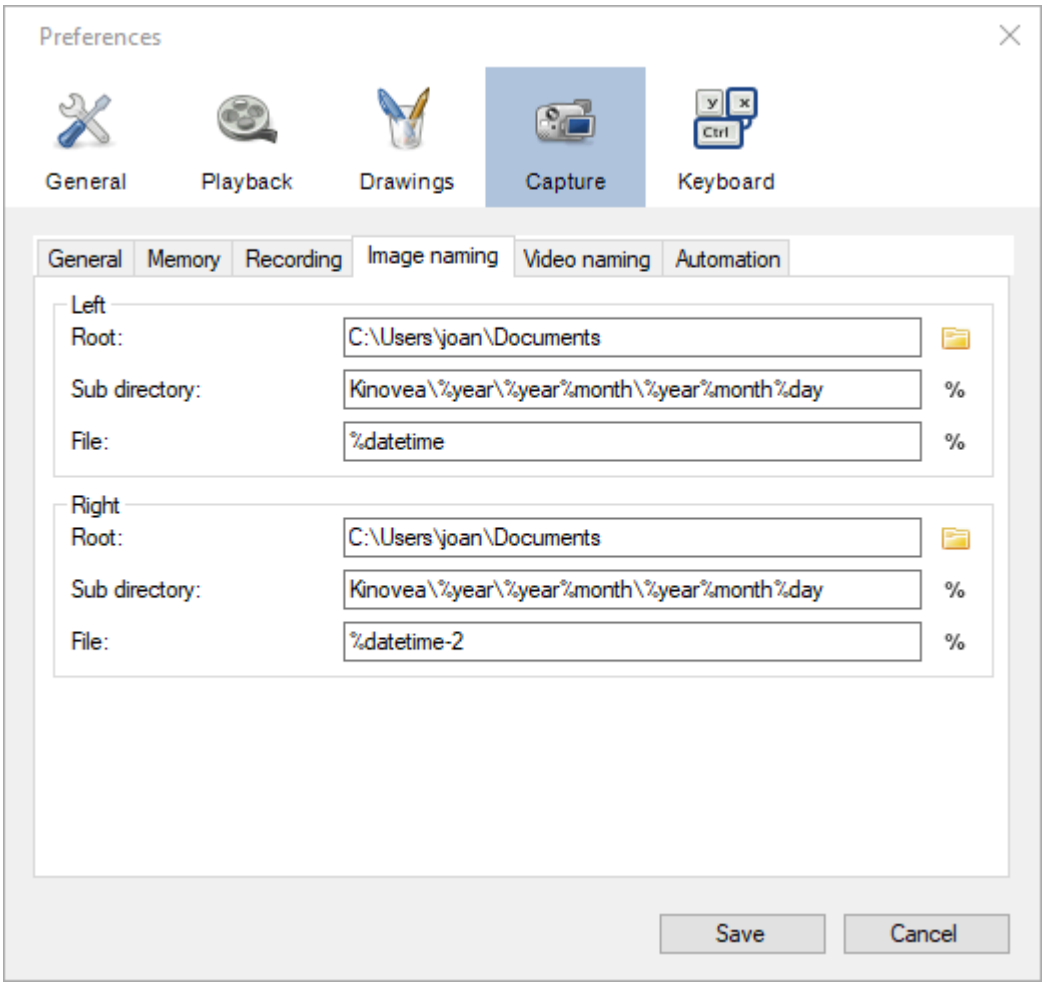
This option controls the framerate from which the output file is modified to use a lower one.

If the camera is configured to send images at a higher framerate than this value, the actual framerate stored in the file metadata will be the replacement framerate. If the camera is configured to send images at a lower framerate than this value, no change will happen.

Replacement framerate

This value defines the final framerate written in the file metadata when the framerate configured in the camera is above the threshold.

Image naming



The options on this page let you configure the automated naming system for image snapshots of the camera stream.

The final path and file name is created by concatenating the `Root`, `Sub directory` and `File` values. Each field can contain special macros referring to context variables that are automatically inserted in the final path.

If no context variable are used at all, the file naming system will prepare the next recording by automatically incrementing a counter and appending a number to the file name.

If the computed value result in the same name as an existing file the capture screen will prompt for overwrite confirmation.

To view the list of available context variables click the `%` button next to the `Sub directory` or `File` fields.

The following context variables are available:

Macro	Description
%year	The current year
%month	The current month as a number from 01 to 12.
%day	The current day of the month from 01 to 31.
%hour	The current hour from 00 to 23.
%minute	The current minute from 00 to 59.
%second	The current second from 00 to 59.
%date	The current date in the format “YYYYMMDD”.
%time	The current time in the format “HHMMSS”.
%datetime	The current date and time as “YYYYMMDD-HHMMSS”.
%camalias	The camera alias.
%camfps	The configured framerate for the camera.
%recvfps	The framerate really received from the camera.
%%	This is replaced by an empty string.

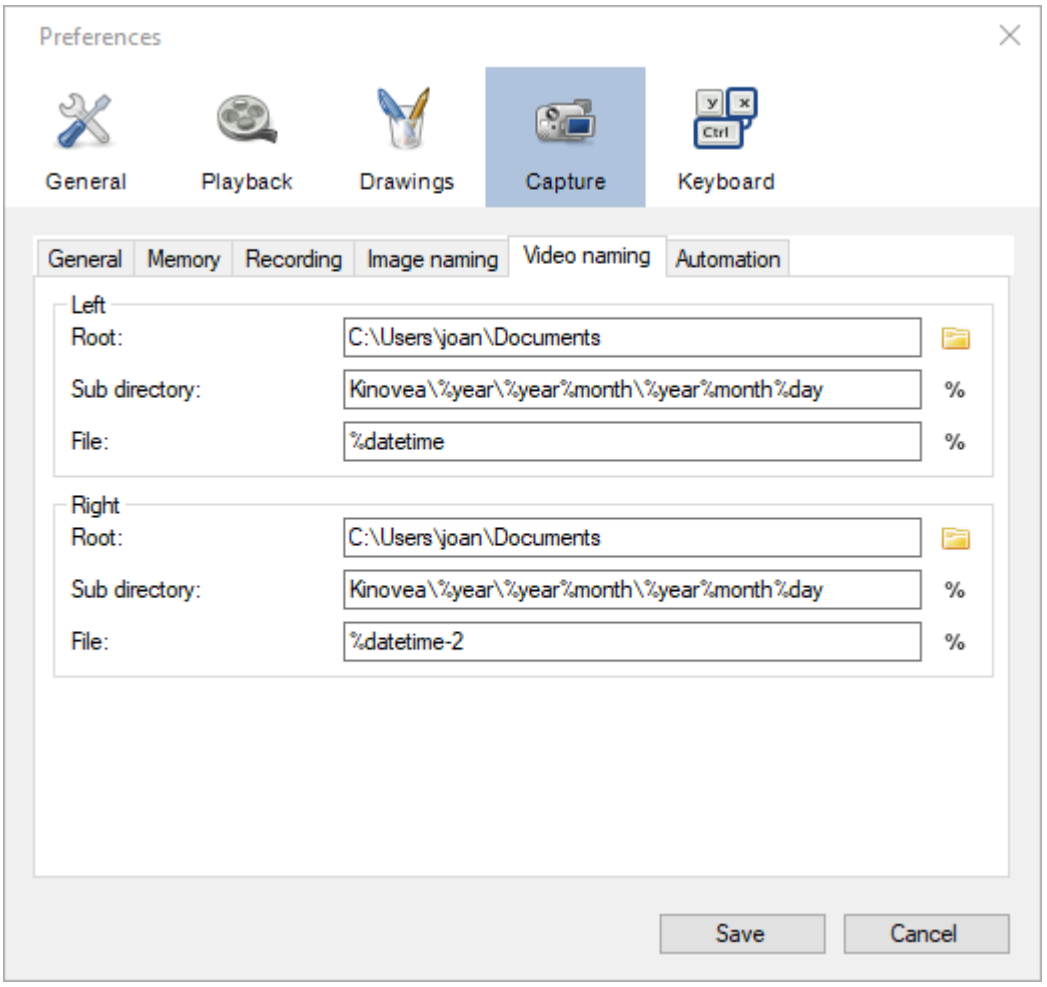
Anything that is not exactly part of a macro is copied verbatim to the output. Some examples assuming the current date and time is October 20th, 1968 at 16:00:00 (4 PM):

```
%year-%month-%day: 1968-10-20.
%hour-%minute-%second: 16-00-00.
%datetime: 19681020-160000
%date_text: 19681020_text
%date-%camalias: 19681020-mycamcorder
```

Note

If you want to use a completely static file name and bypass the automated counter increment for consecutive recordings, use the `%%` macro variable. Be aware that this will require you to either enter the filename manually for every recording or overwrite an existing file.

Video naming



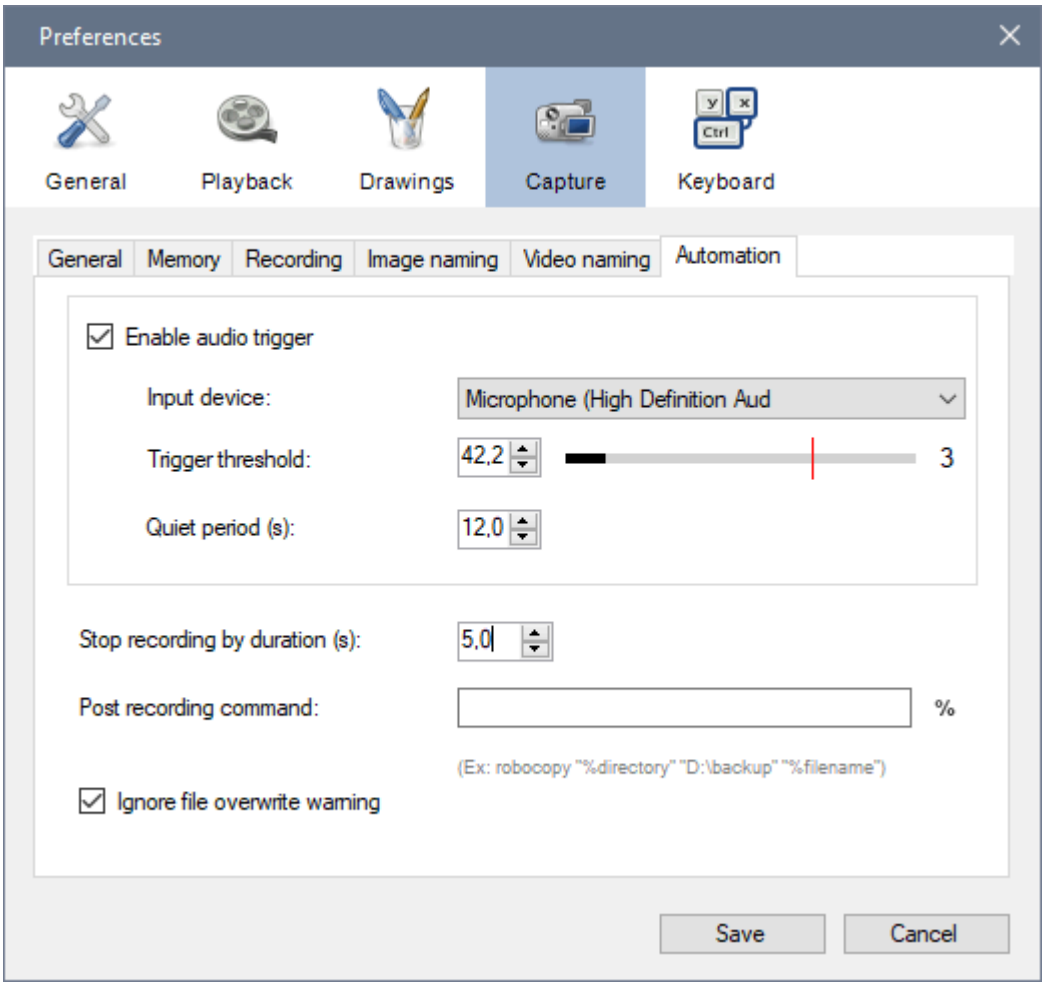
The options on this page let you configure the automated naming system for video recordings of the cameras streams.

The options are similar to the ones in Image naming.

Warning

Avoid using the Windows system drive as a target for camera recording to minimize concurrent access and resource sharing.

Automation



Audio trigger

Enable audio trigger

When this option is checked Kinovea measures the volume level on the microphone and triggers the start of the recording when this volume goes above the configured threshold.

Note

The audio trigger mechanism can be disarmed for individual cameras from the capture screen controls.

Input device

This option lets you select which microphone is used to trigger recordings.

Trigger threshold

The trigger threshold defines the volume level required to trigger recordings. You should see the black line moving laterally as the microphone picks up sounds. The vertical red line represents the trigger level.

The counter on the right is incremented each time the trigger is reached and reset when the threshold value is changed. You can use this to get immediate feedback while figuring out the appropriate configuration.

Idle time

The idle time defines the amount of time after each recording during which the audio trigger is automatically disarmed.

Stop recording by duration

This option defines the maximum duration for recordings. Recordings started manually or by audio trigger will be stopped right after they reach this duration. Setting the value to 0 disables the option and requires manually stopping the recording process.

This option is orthogonal to delayed recording. For example if the camera is configured with a 2-second delay and the maximum duration is set to 5 seconds, the created video will last 5 seconds as configured: the first 2 seconds are actions that happened before the recording trigger and the last 3 seconds are actions that happened after the recording trigger.

In combination with the audio trigger this option lets you record multiple sequences without manually interacting with the computer.

> **Note**
>
> This value is a lower bound, the final video might be slightly longer than configured due to internal processing and alignment with frame boundaries.

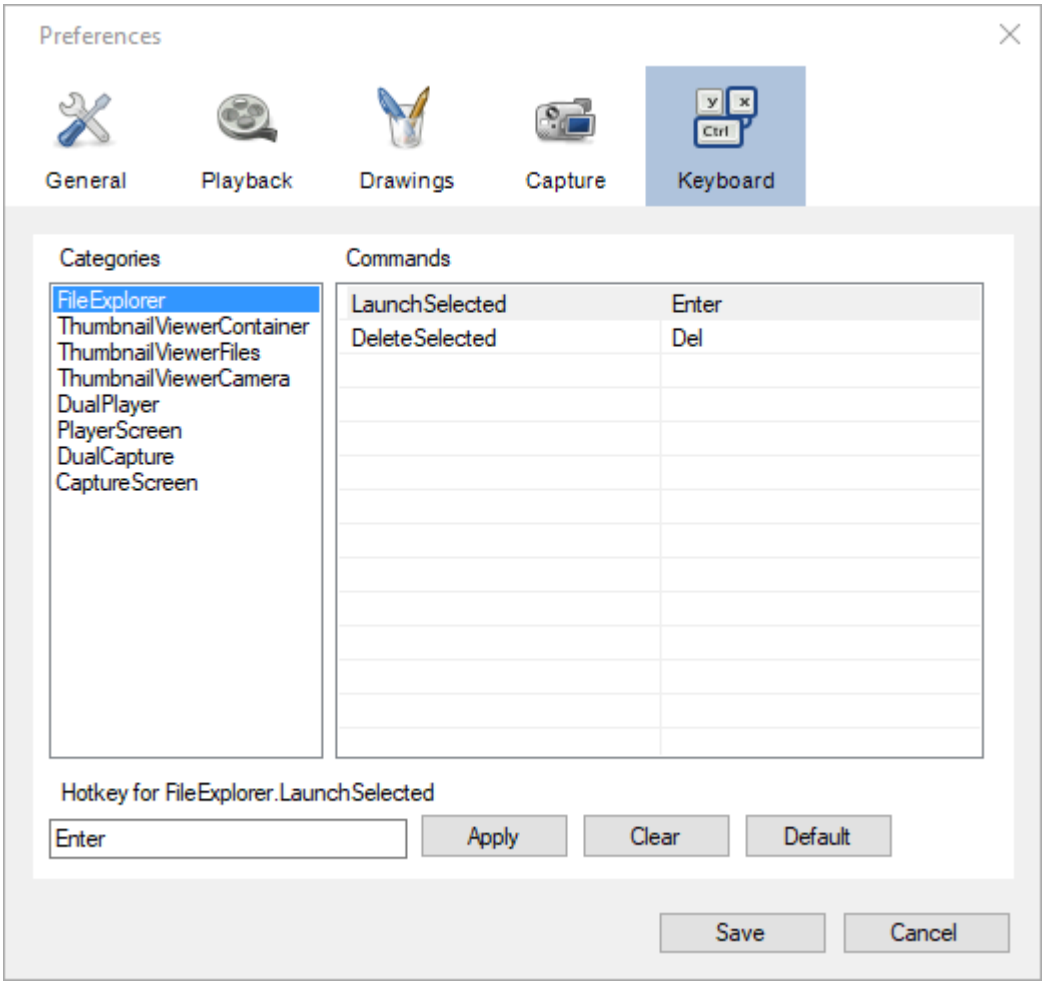
Ignore file overwrite warning

This option bypasses the overwrite confirmation dialog when the recording about to start points to an existing file. If the option is checked the existing file is irremediably deleted and overwritten by the new one.

This option can be used if you are limited in space and do not need to save all sequences. In this scenario you can continuously record to a single file and manually copy it to a different location only when you really want to keep it.

Keyboard

General



This page lets you view and change the keyboard shortcuts for Kinovea internal commands. The shortcuts are grouped in categories based on which part of the user interface the shortcut is active on. The Commands list view displays each command and the corresponding keyboard shortcut.

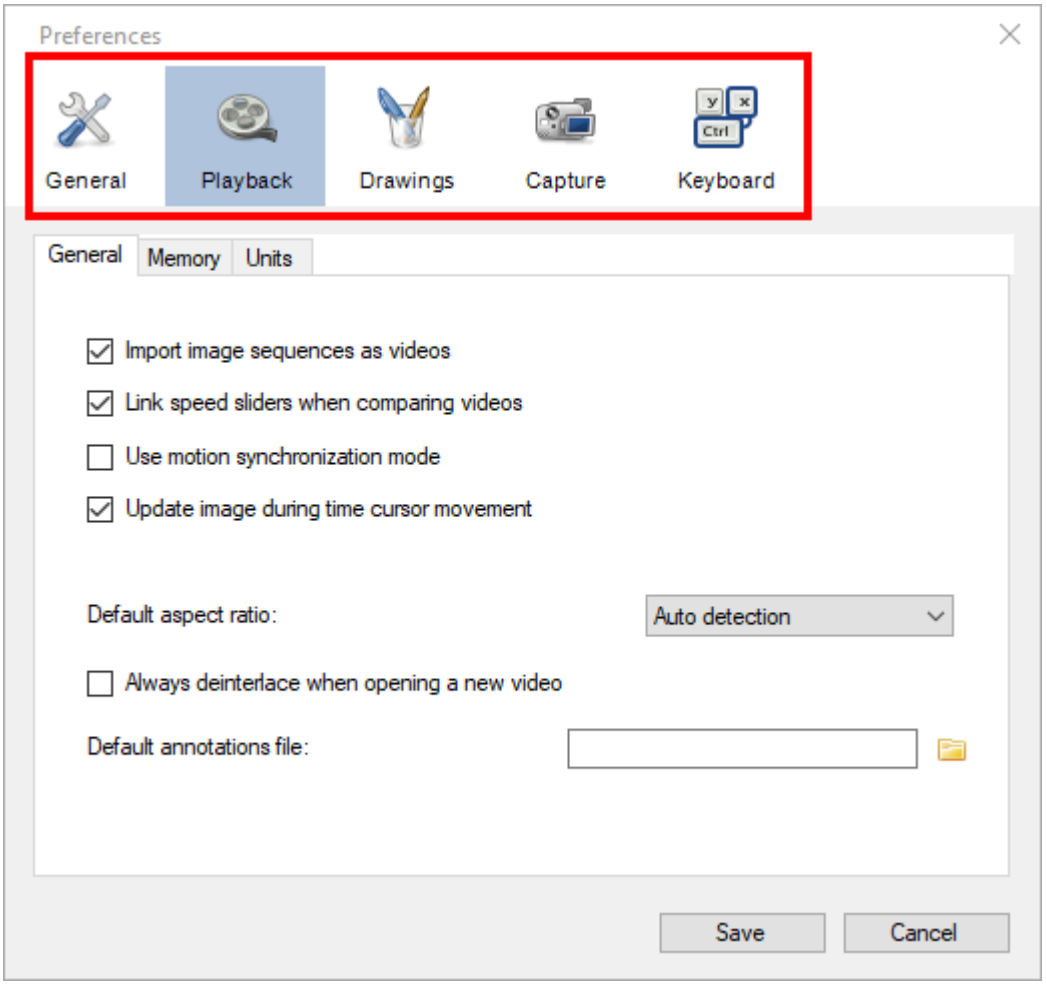
Modifying a keyboard shortcut

To modify a keyboard shortcut select the corresponding category and command in the command list. The existing keyboard shortcut will be displayed in the text box at the bottom left of the page.

Click in the text box and perform the sequence of keys that you wish to use as a replacement shortcut. Click the `Apply` button to commit the new shortcut to the preferences.

Click the `Clear` button to remove the existing keyboard shortcut for the active command.

Click the Default button to restore this particular command to its default keyboard shortcut.



Glossary

A list of important concepts used in Kinovea and in this documentation.

Annotations

Annotations in Kinovea combine all the pieces of information you add to the video. This includes drawings and comments but also image transforms like rotation or mirror, calibration data, tracking data, kinogram data, etc. They are saved in KVA files.

Calibration

Calibration is required to make measurements in real world units such as meters or seconds. Kinovea has a number of calibration sub-systems:

- Spatial calibration by line for a perfectly frontal view.
- Spatial calibration by a rectangle in perspective for a non-frontal view.
- Lens distortion calibration.
- Time calibration (origin and “scale” from capture frame rate).

Camera motion estimation, Camera tracking

Camera motion estimation calculates the rotation and zoom of the camera between frames. Drawings then take this motion into account when they are drawn so they appear fixed in the world even if the camera moves.

Capture folders

Context, Context variables

Delay, Delay buffer

Demosaicing, Demosaicking, Debayering

Drawing, Drawing tools

Kinovea drawings are vector based and their points can be moved after they are created. Some drawings can be tracked over time.

Frame rate

Kinovea deals with multiple frame rates:

- camera frame rate: frame rate *configured* in the camera (ex: 200 fps)
- capture frame rate: frame rate actually *captured* by the camera (ex: 195 fps)
- capture screen “display frame rate”: refresh rate of the capture screen (ex: 10 fps)
- video frame rate: nominal playback rate stored as a metadata in the video file (ex: 30 fps)
- playback frame rate: frame rate at which the video is being played back.

When capturing from a high speed camera (capture frame rate > 150 fps) it is customary to set the video frame rate to a lower value (ex: 30 fps) to allow for smooth playback.

Knowing the capture frame rate is important for time-based measurements such as speed and acceleration.

Intra-only, Intra-frame, All-i, All-Intra

Intra-only is an encoding strategy for video files where each frame is encoded independently from the others. This is good for frame by frame navigation and Kinovea uses this by default when capturing and exporting videos.

It contrasts with Long-GOP encoding. Smartphones, action cameras and web videos typically use Long-GOP encoding to save space while high-end cameras and camcorders often have the option to use intra-only encoding.

Key frame, Key image

Kinematics

Quantities such as position, velocity or acceleration. These can be computed from the pixel coordinates of tracked points and are supported in Kinovea. Acceleration values should not be used directly though as they are too sensitive to noise.

Kinetics, Dynamics

Quantities such as force, torque, momentum or energy. These require mass and inertia properties of the objects to be computed and are not supported in Kinovea.

Kinogram

A composite image combining multiple frames from the video sequence. The source frames can be taken at regular intervals or at specific events such as maximum extension of a joint.

KVA

KVA is the main file format used by Kinovea to save annotations. This is an XML based format.

Long-GOP

Long-GOP is an encoding strategy for video files where most frames depend on other frames to be decoded. This is good for compression but makes it more difficult to navigate frame by frame. See Intra-only.

Markerless tracking, Human pose estimation

When the subject does not wear any markers tracking often fails to follow the joint positions accurately. In such cases we can use manual digitization by going over the video frame by frame, or we cane use markerless tracking using machine learning models.

Markerless tracking using machine learning models is not directly supported in Kinovea but you can use Sports2D to process videos and export a .TRC file that can be imported in Kinovea.

Plane of motion

Kinovea is a 2D program and all measurements are done on a single plane. This plane can either be parallel to the camera sensor (frontal view) or in perspective.

Screen, Player screen, Capture screen

The primary user interface elements where videos and camera streams are displayed. Screens are loaded in windows. A window can have zero, one or two screens.

Stabilization, Image stabilization

2D image stabilization based on a previously tracked point, to remove camera shake. If the camera undergoes rotation or zoom use camera tracking instead.

Stream format, Pixel format

The stream format describes the content of the video frames as delivered by the camera. Camera vendors tend to use different names for the same thing but in the end there are only a few formats we are interested in:

- MJPEG, JPEG, etc. Good for frame by frame navigation, low bandwidth and low CPU usage.
- RGB, RGB8, RGB24, RGB32. Avoid this if possible. High bandwidth and high CPU.
- BayerRG8, BayerBG8, etc. Use this if available and check “Enable software demosaicing”.
- Mono8, Y800, for monochrome cameras.

Synchronization

When two videos are loaded side by side in the same Kinovea window they can be synchronized by aligning their time origin. Synchronization works on videos with different length, frame rate, and even different capture frame rate (one video in slow motion not the other). It supports video playback and frame by frame navigation using common controls.

Template matching

The main algorithm used in object tracking. It looks for a patch of the image in the consecutive images. This algorithm will not recognize the template if it rotates. Thus it works best when using round markers of a single color contrasting with the background.

Time origin

This is the “zero” time reference for time-based measurements. It can be placed on any frame and is also used for synchronization between two videos. When using capture by trigger the time origin is set to the trigger frame.

Tracking, Object tracking

Also known as “digitisation”. Tracking consists in following the position of a point or an object over time. This can be done manually or automatically using various algorithms. It is supported for single points or for multi-point drawings such as lines, poly-lines or angles. For full human models consider using markerless tracking. This is what enable kinematics measurements.

There is also camera tracking which follows the motion of the camera itself.

Window

A window is an instance of Kinovea that can be opened and closed independently. Windows have names and remember their state between sessions. Windows contain zero, one or two screens and can be referenced by workspaces.

Working zone

A segment of the video defined by an in-point and an out-point. The playback and all operations are limited to this zone. If the memory allows it, the working zone is cached in memory for faster frame by frame navigation.

Workspace

A workspace in Kinovea is a collection of related windows that can be started together. The workspace itself doesn't contain any data apart from the list of windows.

Observation

Introduction

Supported file types

Kinovea comes equipped with video decoding libraries to supports many file types and does not need nor use any third party codecs. It supports the following:

- Most video file formats, including MP4, MPEG, MOV, AVI, MKV, M4V, FLV, WMV, among others.
- Animated GIFs.
- Raster image files PNG, JPEG and BMP.
- SVG vector images.
- Image sequences.

Image sequences must have the same size and be named following a numbering pattern. They are loaded as a single video.

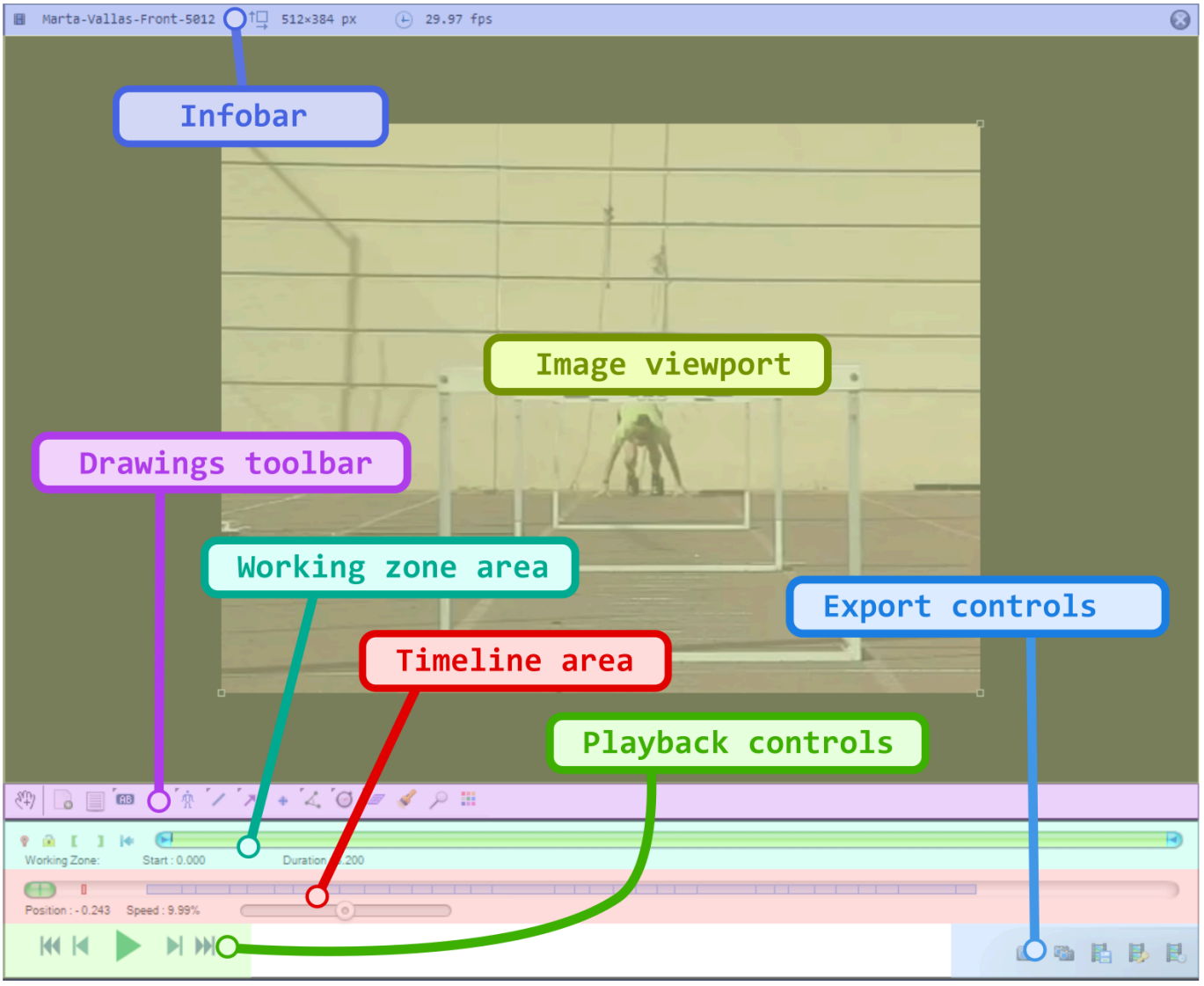
Opening files

To open a video in Kinovea do any of the following:

- Drag and drop the file from Windows explorer onto the Kinovea window.
- Use the menu `File ▸ Open` and select the file.
- Use the menu `File ▸ Recent` and select a recently opened file.
- Use the explorer panel to locate and double click on a file in the file explorer, the shortcut explorer, or the recently captured files, or drag and drop it into an opened playback screen.
- Double click on a thumbnail in the thumbnail explorer.

Playback screen user interface

The playback screen is divided in the following areas:



- Video viewport
- Infobar
- Annotation toolbar
- Working zone area
- Timeline area
- Playback controls
- Export controls

Some functions are not directly visible in the screenshot

- Keyboard navigation
- Mouse navigation

- Context menu

Video viewport

The image viewport is the main area where the video is played back.

Double clicking the image maximizes it in the viewport area. The handles at the four corners of the image can be used to change its size.

Infobar

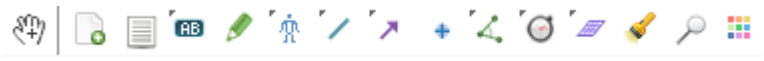
The infobar contains the name of the file, the image size and the frame rate.

The file name can be clicked to access a context menu to switch between normal video mode and replay folder observer mode.

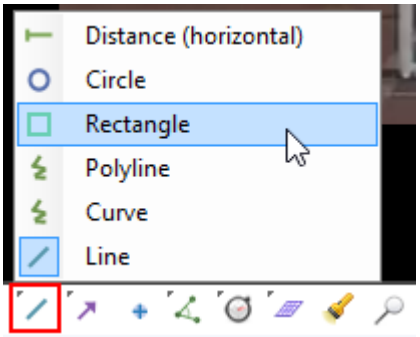


Annotation toolbar

The annotation toolbar contains buttons to create new key images, select the active tool and open the color profile.



The toolbar contains more tools than those immediately visible. Buttons that host extra tools have a little black triangle in the top-left corner. The extra tools can be accessed by right-clicking or long-clicking the primary button.



Working zone area

The working zone defines the segment of the video that the player is working with. The play head loops within the working zone.



Marks the current time as the time origin. This makes time values relative to this moment.

Locks the working zone start and end point to avoid changing them by mistake.

Sets the starting point of the working zone within the video.

Sets the ending point of the working zone within the video.

Resets the working zone to the whole video.

You can also update the working zone boundaries by directly manipulating the blue end points.

Timeline area

The timeline area displays the current position within the video, time markers and the speed control.



Time markers

Time markers are the colored rectangles inside the timeline gutter, they provide information about annotations.

The red marker is the time origin.

The other markers are key images, chronometers or trajectories. Their color corresponds to the color of the drawing.

Playback controls



From left to right the buttons provides the following functions:

- Returns to the start of the video or working zone.
- Goes back one frame.
- Starts playback.
- Goes forward one frame.
- Goes to the end of the video or working zone.

The playback loops to the start when it reaches the end of the video or working zone.

Speed slider

The speed slider goes from 0 to twice the nominal speed of the video.

The displayed speed value takes into account the slow motion factor configured such that the speed is shown as a percentage of the real world action speed. For example if a video is filmed at 240 fps and saved into a file as 24 fps, the video will normally play back at 10% of the real world speed. In this case the speed control will go from 0 to 20% with a mid-point at 10%.

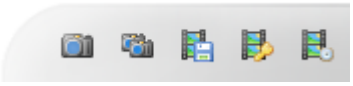
Warning

If the video cannot be played back at its nominal speed for performance reasons the playback speed value will automatically be lowered down.

Performance for playing back depends on the displayed image size, the frame rate and the file format.

Export controls

The export controls provide ways to export videos and images of the current file.



See also: Exporting videos.

Keyboard navigation

The following keyboard shortcuts are defined by default:

← Left arrow	Step backward one frame

→ Right arrow	Step forward one frame
↑ Page up	Big jump backward
↓ Page down	Big jump forward
SHIFT + ↑ Page up	Small jump backward
SHIFT + ↓ Page down	Small jump forward
↖ Home	Jump to the start
↘ End	Jump to the end
SHIFT + ← Left arrow	At the start, will force jump back to the end

- The big jump is 10% of the video length
- The small jump is 1% of the video length

Mouse navigation

There are two modalities to use the mouse to navigate in time:

- Use the **mouse wheel** to move forward and backward.
- Hold the ALT key and drag the image left or right to move backward or forward.

Context menu

The context menu provides quick access to more functions.

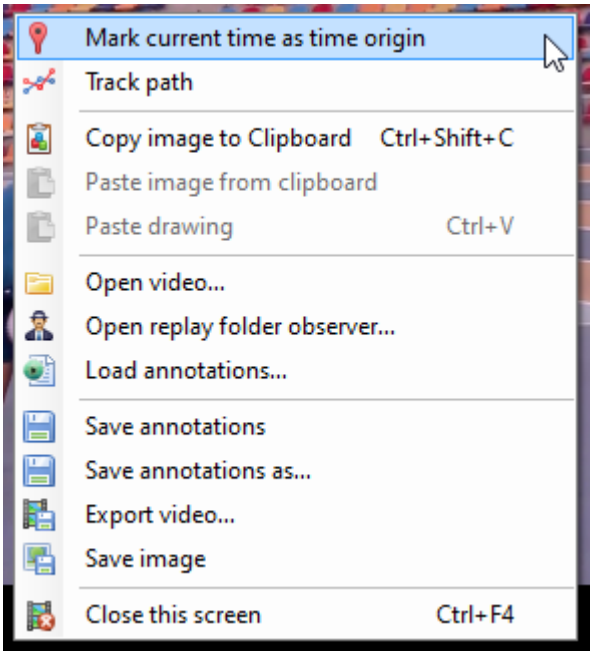


Image transformation

The following image transformations can be applied:

- Aspect ratio
- Rotation
- Mirroring
- Deinterlacing
- Demosaicing
- Stabilization
- Zoom & magnification

Aspect ratio

To change the aspect ratio of the image use the menus under **Image ▸ Image format**.

Some devices use non-rectangular pixels and don't fill the corresponding pixel aspect ratio value in the file metadata. In these case it might be necessary to force the aspect ratio to a known value.

The following options are available:

- Auto detection
- Force 4:3
- Force 16:9

Rotation

To change the orientation of the image use the menus under **Image ▸ Image rotation**.

The following options are available:

- No rotation
- 90° clockwise
- 90° counter-clockwise
- 180°

Videos filmed in portrait mode on a smartphone typically already have a flag embedded in the video file and don't need to be manually rotated.

Mirroring

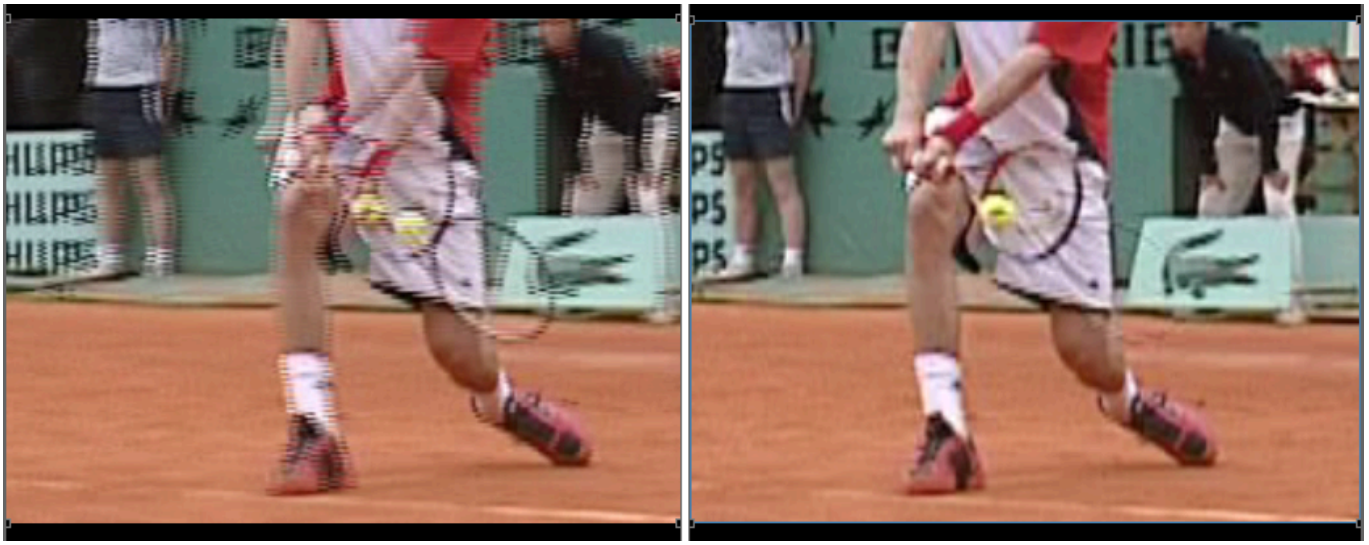
To flip the image horizontally use the menu Image ▸ Mirror.

Deinterlacing

To deinterlace the video use the menu Image ▸ Deinterlace.

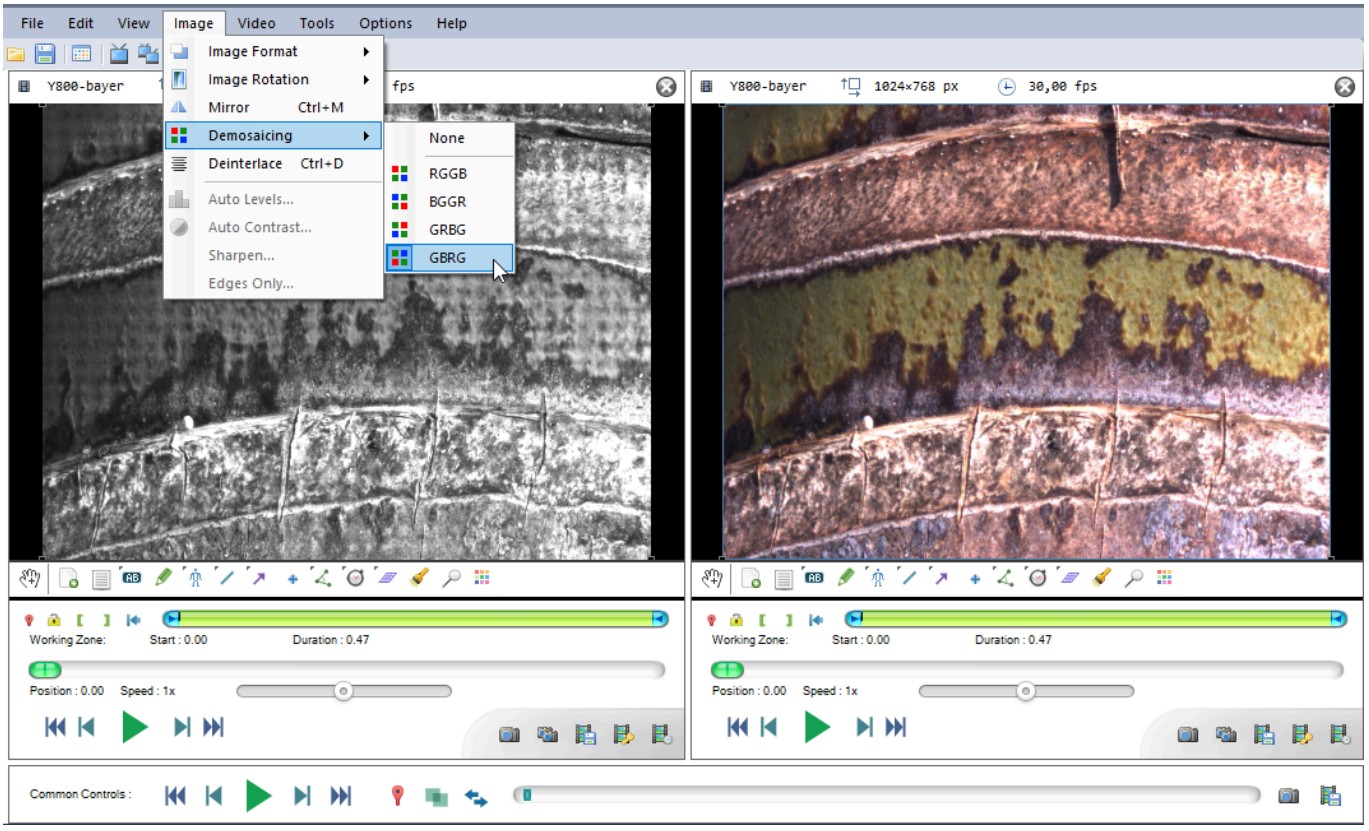
Some capture devices store video using an interlaced format. Interlaced videos store half images at twice the frame rate, alternating odd and even rows. This causes a combing artifact when the filmed motion is fast as objects or subjects move during the half frame interval.

The deinterlacing algorithm reconstructs full images by combining rows from adjacent frames.



Demosaicing

Videos saved in Bayer mode contain the raw sensor data before color reconstruction. The color can be rebuilt using the menu under Image ▸ Demosaicing.



The following options are available:

- RGGB
- BGGR
- GRBG
- GBRG

The appropriate option to select depends on the device and mode used during recording.

Stabilization

To stabilize the image by a previously tracked object, use the menu **Image ▸ Stabilization** and select a track.

The tracked object will be kept at the same position in the image, stabilizing the video around it.

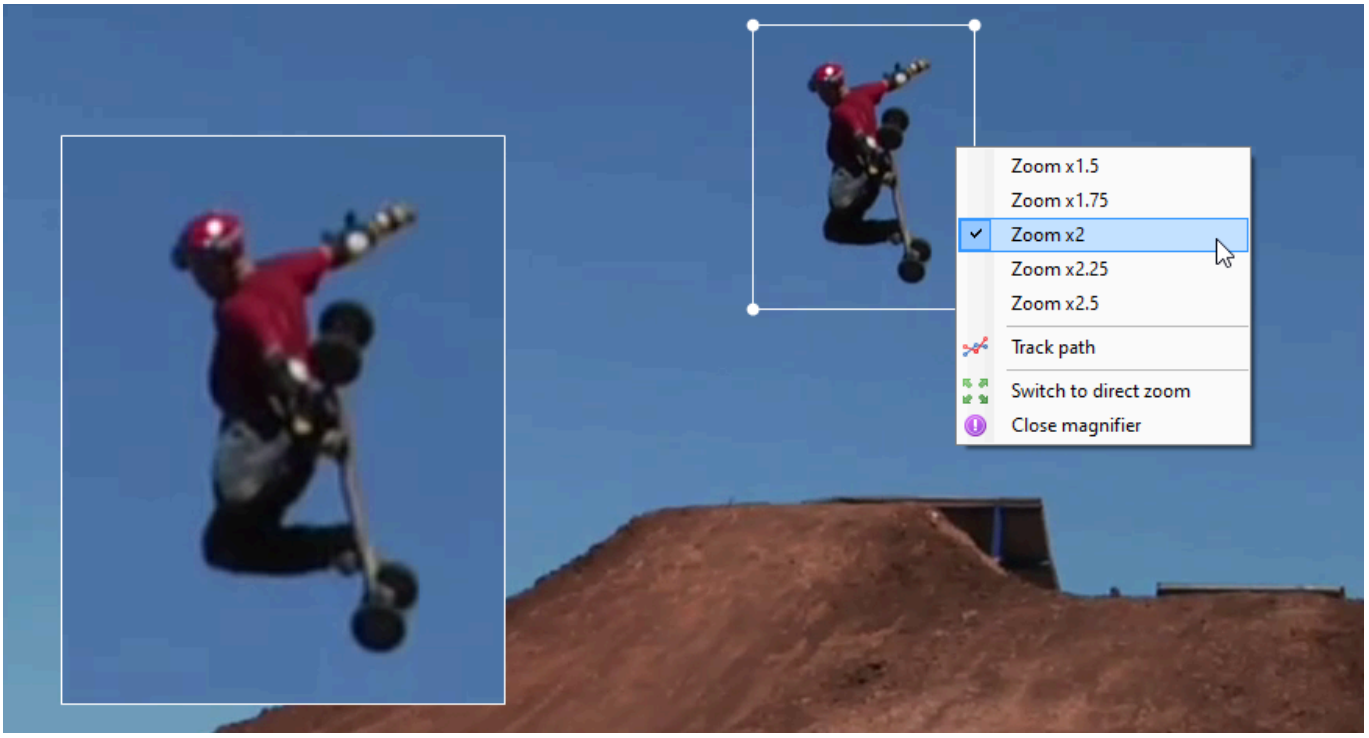
This is a simple 2D stabilization suitable for small camera shake or to “lock in” on an object or body part. If the camera is undergoing a rotation or zoom, consider using the Camera motion estimation function.

Zoom & magnification

To zoom into the image use `CTRL` + **mouse wheel** or `CTRL` + `Minus` and `CTRL` + `Plus`.

`CTRL` + `Numpad0` resets the zoom level.

The magnifier tool can also be used to create a picture in picture magnified area.



Kinograms

Introduction

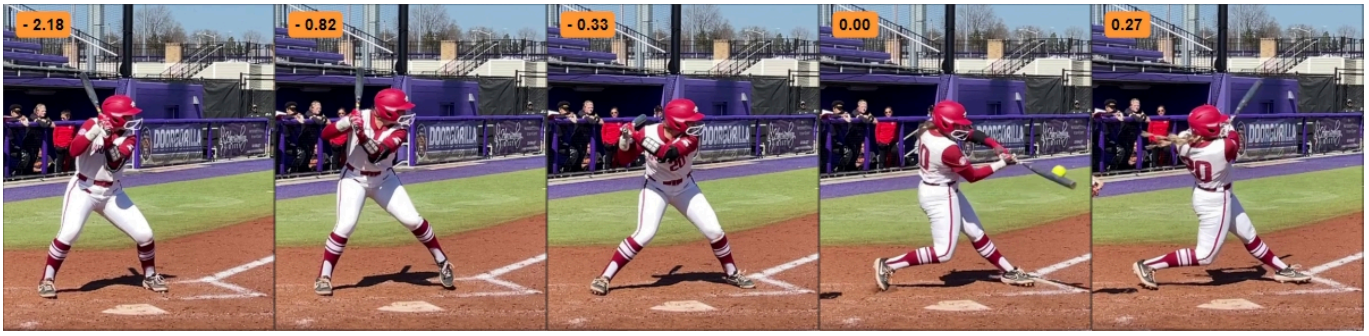
Kinograms show multiple frames of the video at the same time.

The two main use-cases for kinograms are 1. the full decomposition of the motion, typically using a large grid and regular intervals, and 2. the creation of reports highlighting the posture at standard key moments of the motion, using fewer images at non regular intervals.

Full decomposition:



Key points:



To enter the Kinogram mode, use the menu Video ▸ Kinogram.

To exit the Kinogram mode right click in the image background and use the menu **Exit Kinogram**. Alternatevely, use the exit button at the top right of the playback screen or select another video mode.

General

When the Kinogram mode is activated the main viewport displays a collection of frames taken from the video, by default spaced at regular intervals.

The playback controls are still active and the current frame is highlighted in blue. The drawing tools can be used to add annotations on top of the composite.



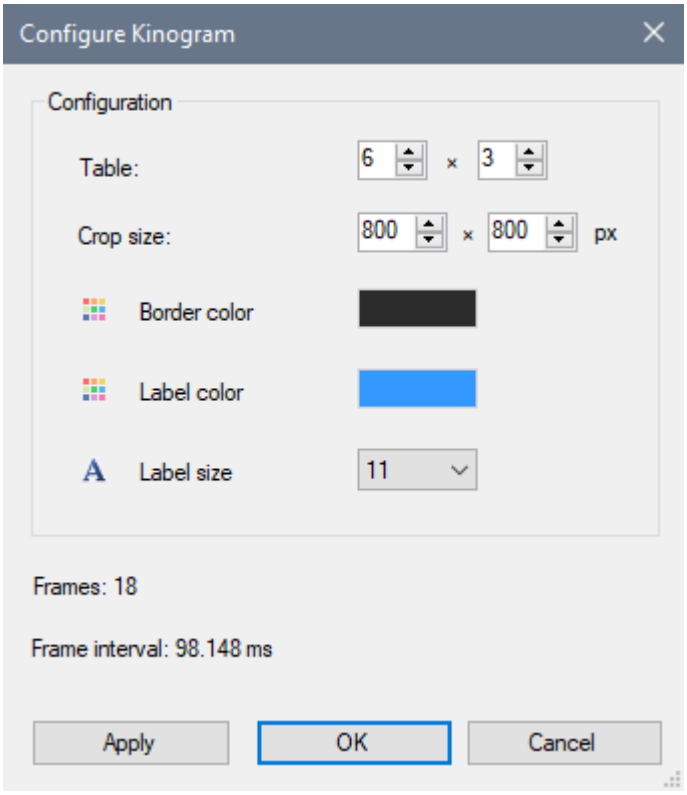
Adjusting the content of the cells

Using the left mouse button, drag the images around to select which area of the original video frame should be displayed inside a particular cell.

To adjust the *time* reference of a cell, use the ALT key and scroll backward or forward with the mouse wheel while hovering above the cell.

Parameters

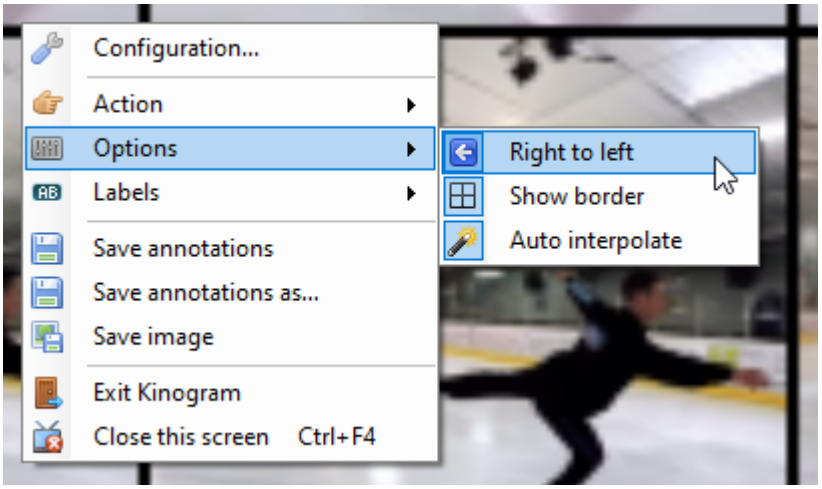
The parameters of the kinogram can be controlled from the context menu **Configuration...**.



- Table : The number of columns and rows of the kinogram.
- Crop size : the size of the area of the original images extracted into the cells of the kinogram.
- Border color : Color of the border between the cells.
- Label color: color used for the time or frame number labels.
- Label size: font size used for the time or frame number labels.

Toggles

Other options are available via the Options menu.

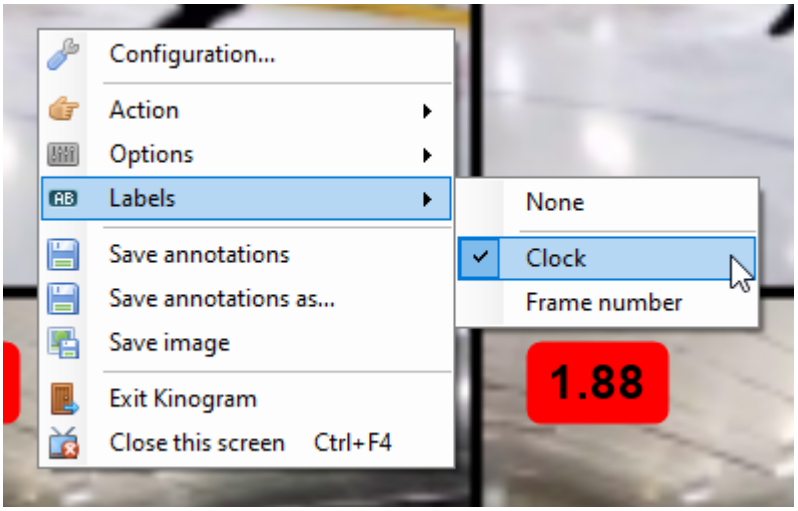


- Right to left : Enable this if the athlete is moving right-to-left. This configures the kinogram to use the top-right cell as the first cell and progress right to left for each row.
- Show border: whether to show the outline of the cells.

- Auto-interpolate: when this option is enabled dragging the cell content will automatically interpolate the neighboring cells to match the motion. If this is disabled you can still trigger it manually by using the menu **Action ▸ Interpolate tiles**.

Information labels

Small labels giving information about each cell can be shown by using the Labels menu.



- Clock: shows the time of the image in the cell, taking into account the time origin and scale if configured.
- Frame number: shows the cell index within the Kinogram.

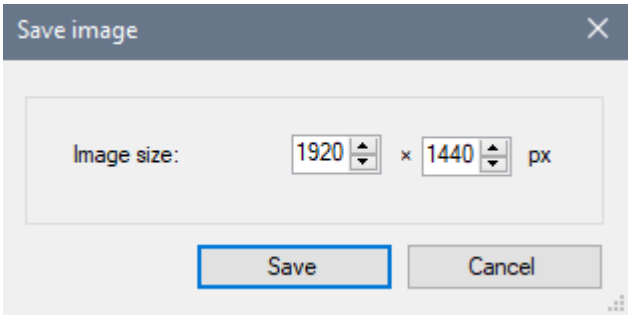
The labels can be moved around by dragging. To move all the labels at once, use the `SHIFT` key while dragging them. This is typically used to move all the labels to a particular corner of their cell.

To reset the labels to their default location use the menu **Action ▸ Reset label positions**.

Saving

The kinogram configuration is saved as part of the KVA file (menu **Save annotations...**) and will be restored when the video is opened again.

To save the kinogram image itself use the menu **Save image** in the context menu of the kinogram mode. This will ask for the output image size. Since the kinogram is a composite of images of the original video you may save it at a larger resolution without loosing quality.


Save image
Image size:
1920
×
1440
px
Save
Cancel

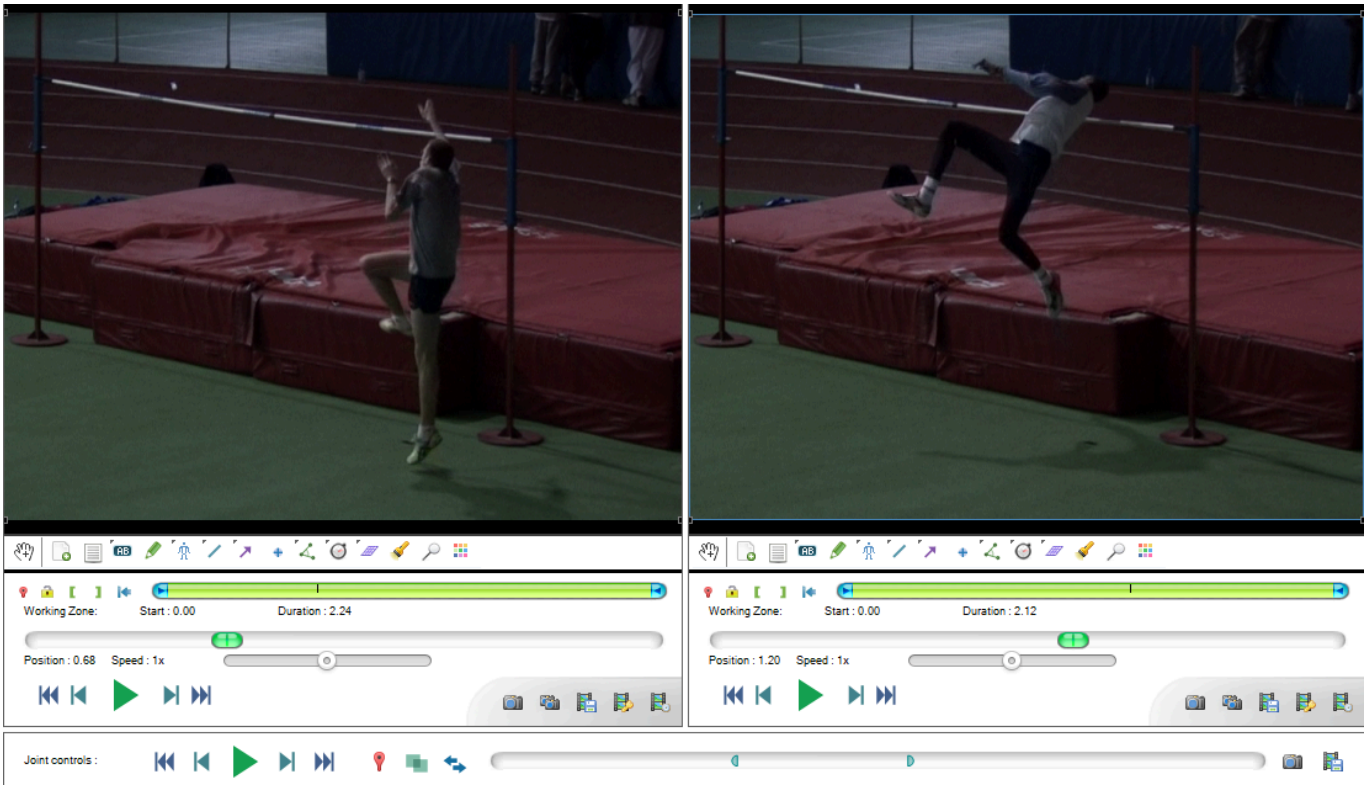
Comparison

Comparison

Comparison

Two videos can be opened side by side using the `Two playback screens` button or the menu **View ▸ Two playback screens** .

The timeline cursor in each playback screen can be moved independently.



Superposition

Superposition paints each video on top of the other at 50% opacity. This is a basic mechanism to compare motion in-situ if the videos were filmed in the same environment with a static camera.



Synchronization

The synchronization system supports both synchronized playback and synchronized frame by frame navigation of two videos.

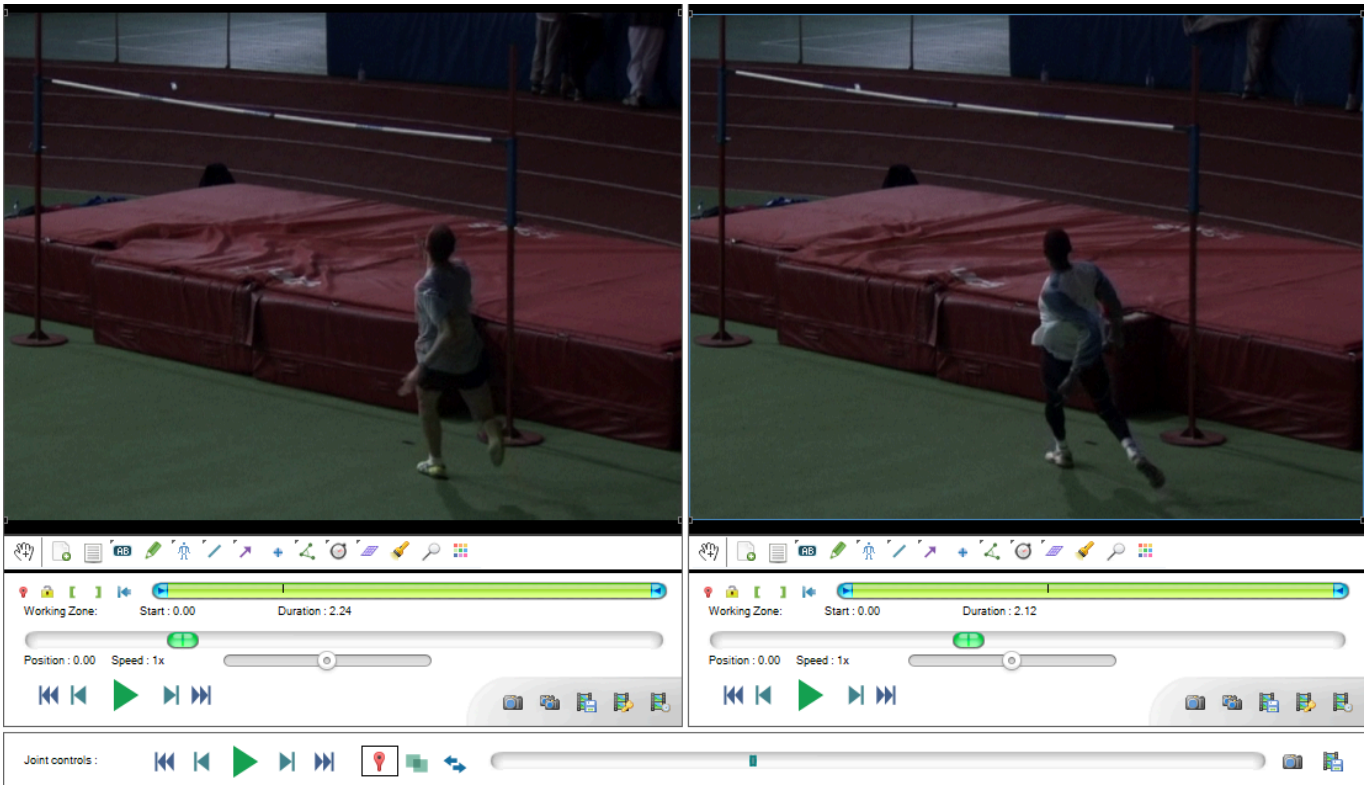
The videos can be heterogenous:

- different frame rates
- different durations
- different image sizes
- different capture frame rates (one video is slow motion and not the other)

Synchronization point

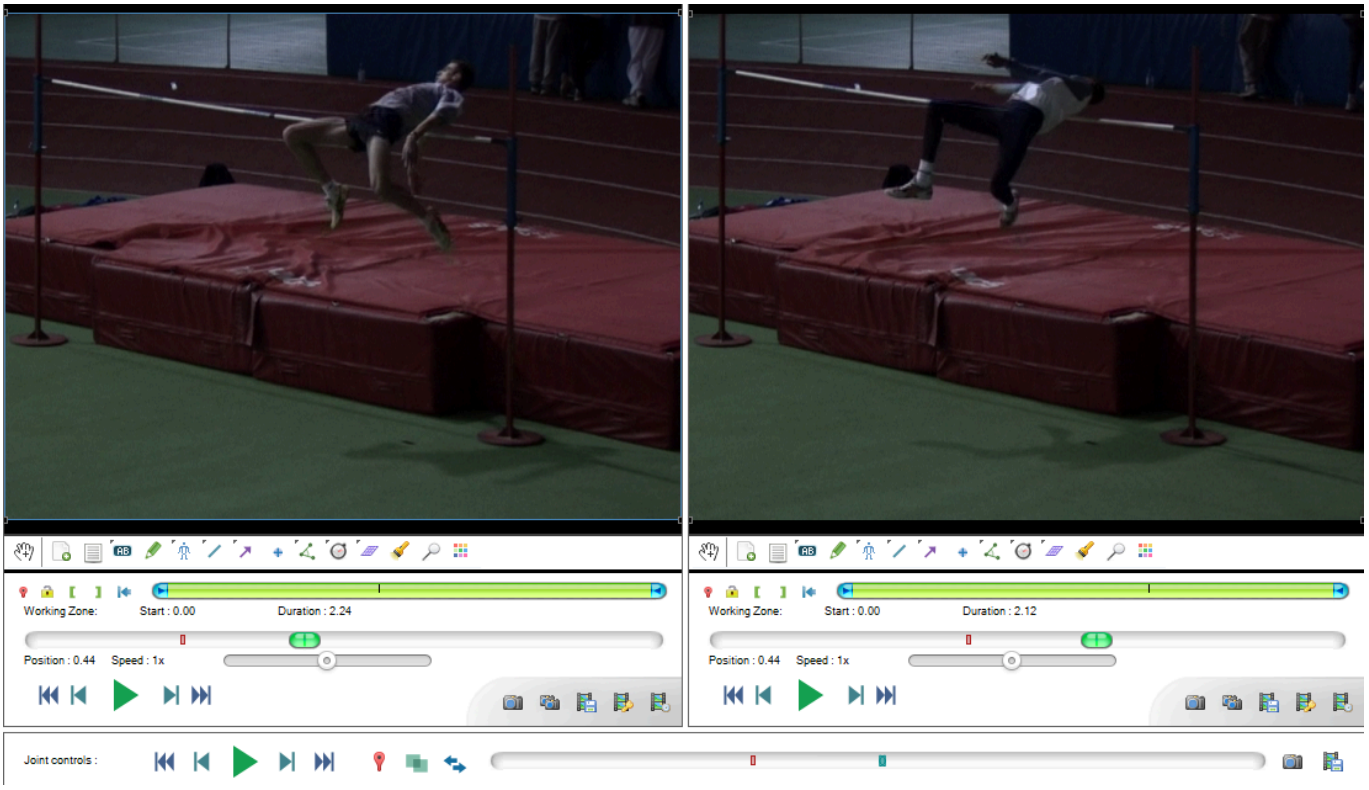
Two videos can be synchronized by setting their time origin to a common event visible on both videos. When the videos are synchronized they will pass through their time origin at the same time.

To set the time origin in a video move to that point in the video and click the `Mark current time as time origin` button or right click the background of the video and choose the **`Mark current time as time origin`** menu. Alternatively you can move each video to the correct point independently and use the `Synchronize videos on the current frames` button in the joint controls area.



During joint-playback, one video may start and/or end playing before the other. They will then join up at the end of their respective loop and restart the cycle.

To perform joint frame-by-frame navigation, move the cursor in the joint timeline or use the joint controls buttons.



Joint controls



The playback controls are the same as for individual playback screens but act on both videos at the same time.

The other buttons have the following functions:

- Set the time origin in both videos to their respective current time.
- Enable or disable superposition of the videos into one another.
- Swap the playback screens.

Linked speed controls

By default the speed controls in each playback screen are linked with each other: lowering the playback speed in one video lowers it in the other.

The speed controls are independent of the frame rate of the video files so this should apply a similar slow motion factor to both videos and keep the comparison meaningful.

If you are confident that you do not want the speed sliders to be linked together you may change the option in **Options ▸ Preferences ▸ Playback ▸ General ▸ Link speed sliders when comparing videos**.

Dual replay

Normally in the capture-replay workflow the captured video is replayed as soon as possible by the replay screen after it is created.

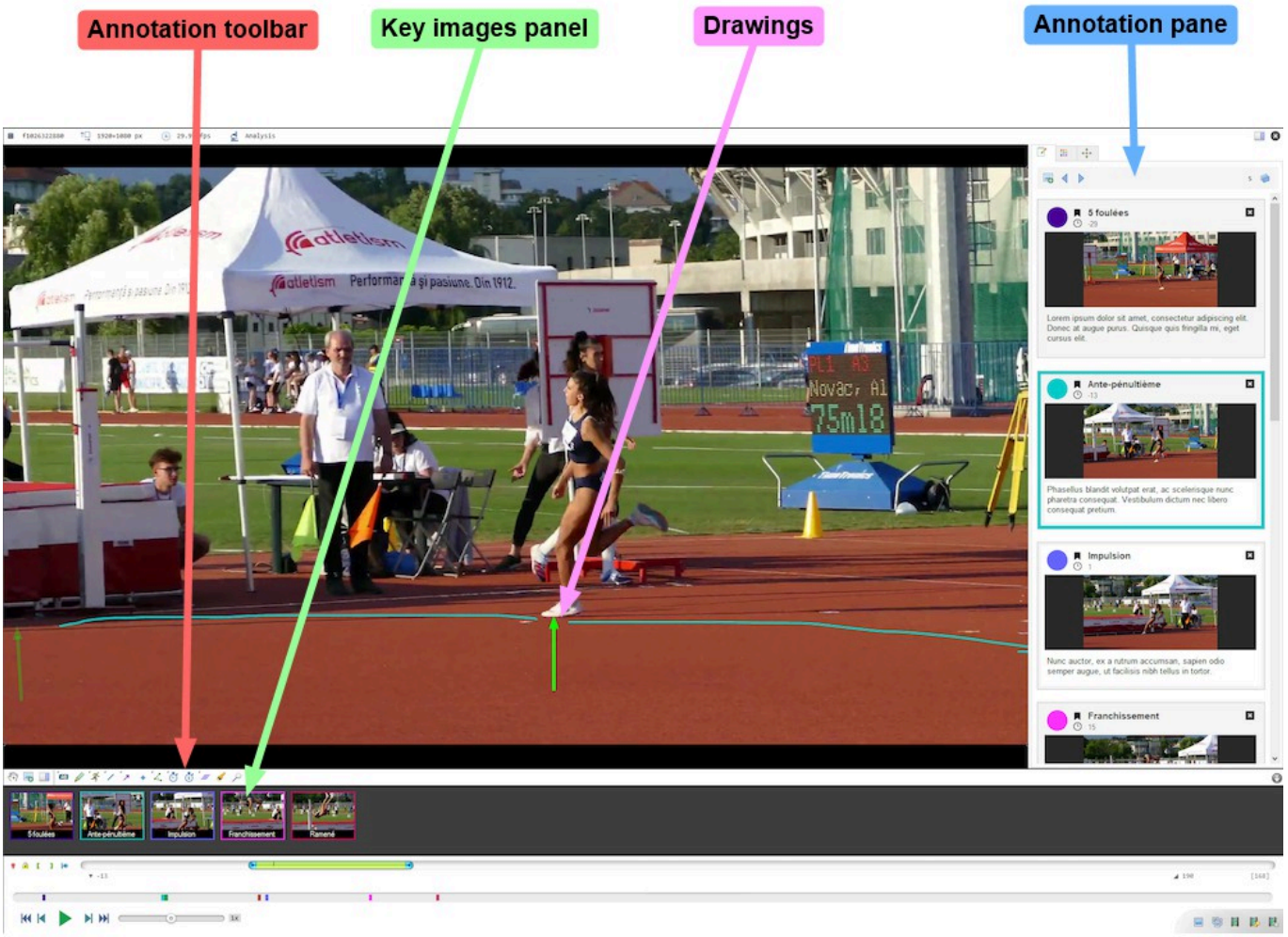
In the case of a dual replay scenario where both player screens are replay folder observers monitoring the capture folder of their respective capture screen, the synchronization engine waits for both videos to be available before starting the playback.

Annotation

Annotation user interface

Annotations are used to document key points, add commentary, drawings or insert reference images into the video. Certain tools can also be used to measure times, repetitions, angles or distances.

The following screenshot describes the parts of the interface used for annotating videos.



- The annotation toolbar has tools for adding drawings and key images.
- The list of key images shows thumbnails of the key images added to the video.
- The drawings are added on top of the video.
- The annotation pane has tabs for editing key images and drawings.

Some annotation features are not immediately visible in the screenshot

- Other tools can be found in different places of the interface.
- Big pointers can be used for presentation purposes.
- The background layer sits between the video and the drawings.

Annotations can be saved and loaded using annotation files.

Certain external formats can be imported, see importing data.

Annotation toolbar



The left part of the toolbar has the following general function buttons:

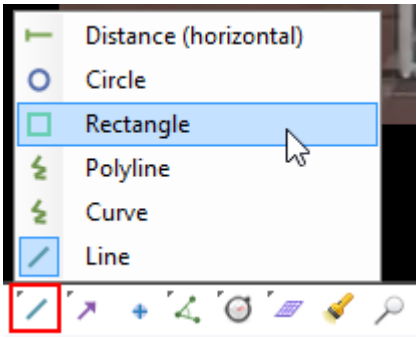
	Manipulates drawings or pans the image.
	Adds a key image at the current time.
	Toggles the annotation pane.

The rest of the toolbar has the drawing tools.

There are more tools than those immediately visible. Buttons with a small arrowhead in the top-left corner contain other tools that can be accessed by doing a right click or a long click (click and hold) on the button.



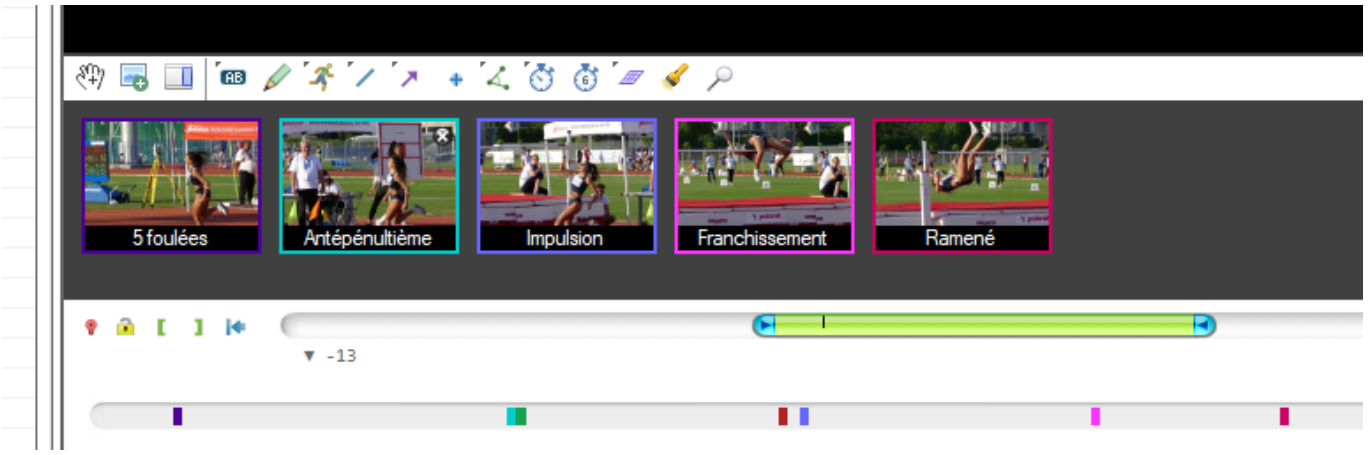
A flying menu opens with the extra tools available.



For a description of the individual drawing tools see General annotation tools, Posture annotation tools, Measuring time, Counting events and measuring cadence, Measuring angles, Measuring positions and distances.

To stop using a particular tool and come back to the hand tool you can also use the Escape key.

List of key images



The list of key images presents an overview of all added key images. It can be used to delete or move them in time using the context menu or dragging them to the timeline.

More about key images can be found in Key images.

Drawings

Drawings are vector-based, they can be manipulated and resized after they have been added to the video.

They have a context menu that can be used to access tool-specific functions, copy and paste, and deletion.

For a description of the individual drawing tools see General annotation tools, Posture annotation tools, Measuring time, Counting events and measuring cadence, Measuring angles, Measuring positions and distances.

For editing style and opacity options see Drawings properties.

The drawings can be attached to filmed objects even when the camera is moving, see Camera compensated annotations.

Warning

Drawings are created in key images. Deleting the key image deletes all the drawings in it.

Annotation pane

The annotation pane has tabs to edit key images and drawings.



	Key images
	Drawings properties
	/tracking/object_tracking

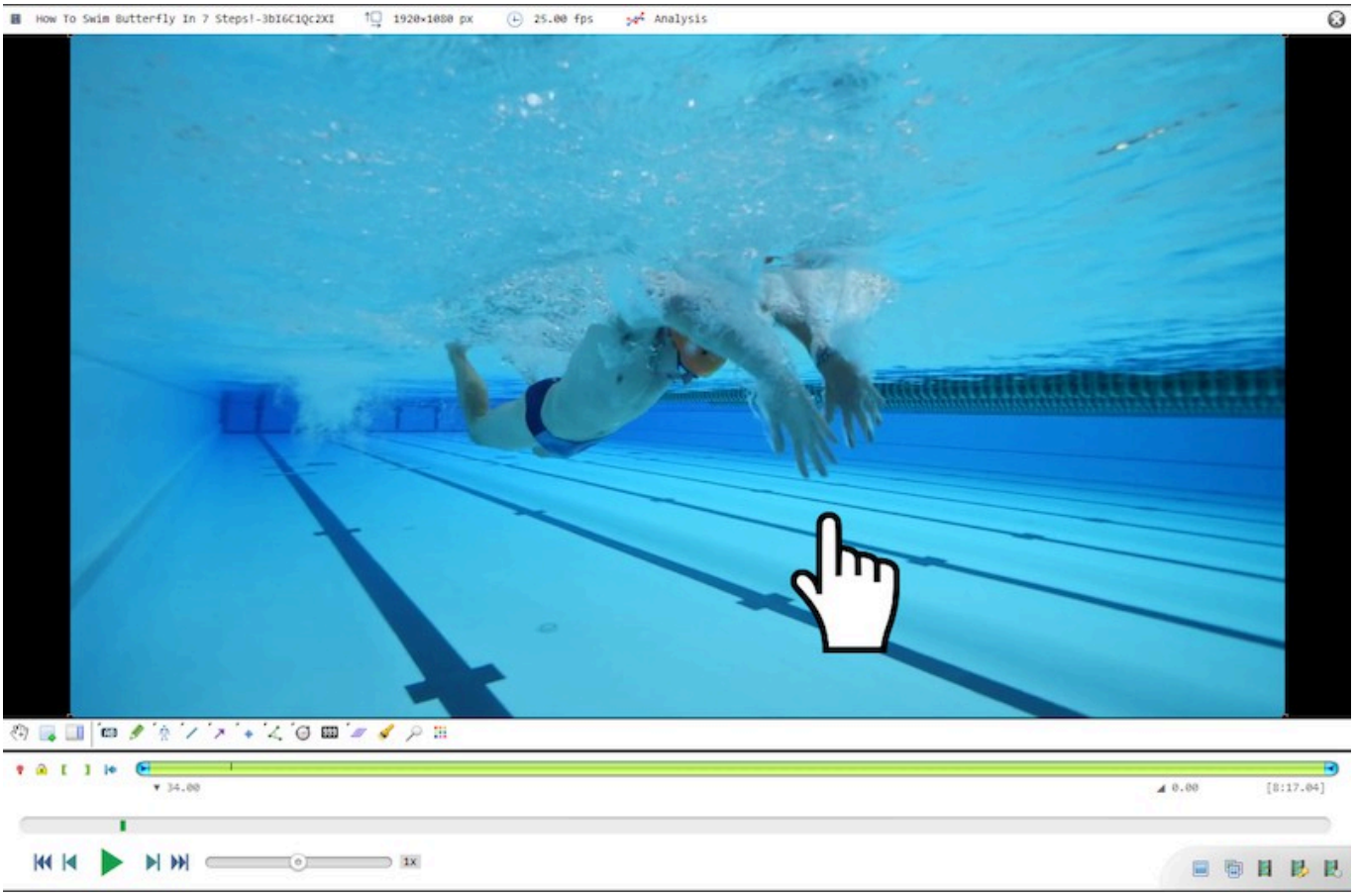
Other tools

Some tools are not part of the annotation toolbar and can be found in other places in the interface.

- Trajectory objects are created by right clicking the video viewport and menu **Track path**
- The coordinate system is available in the main menu **Tools ▸ Coordinate System**
- The camera test grid is available in the main menu **Tools ▸ Camera Test Grid**
- Inserting an image or SVG as a reference is done using the main menu **Tools ▸ Import Image**

Big pointers

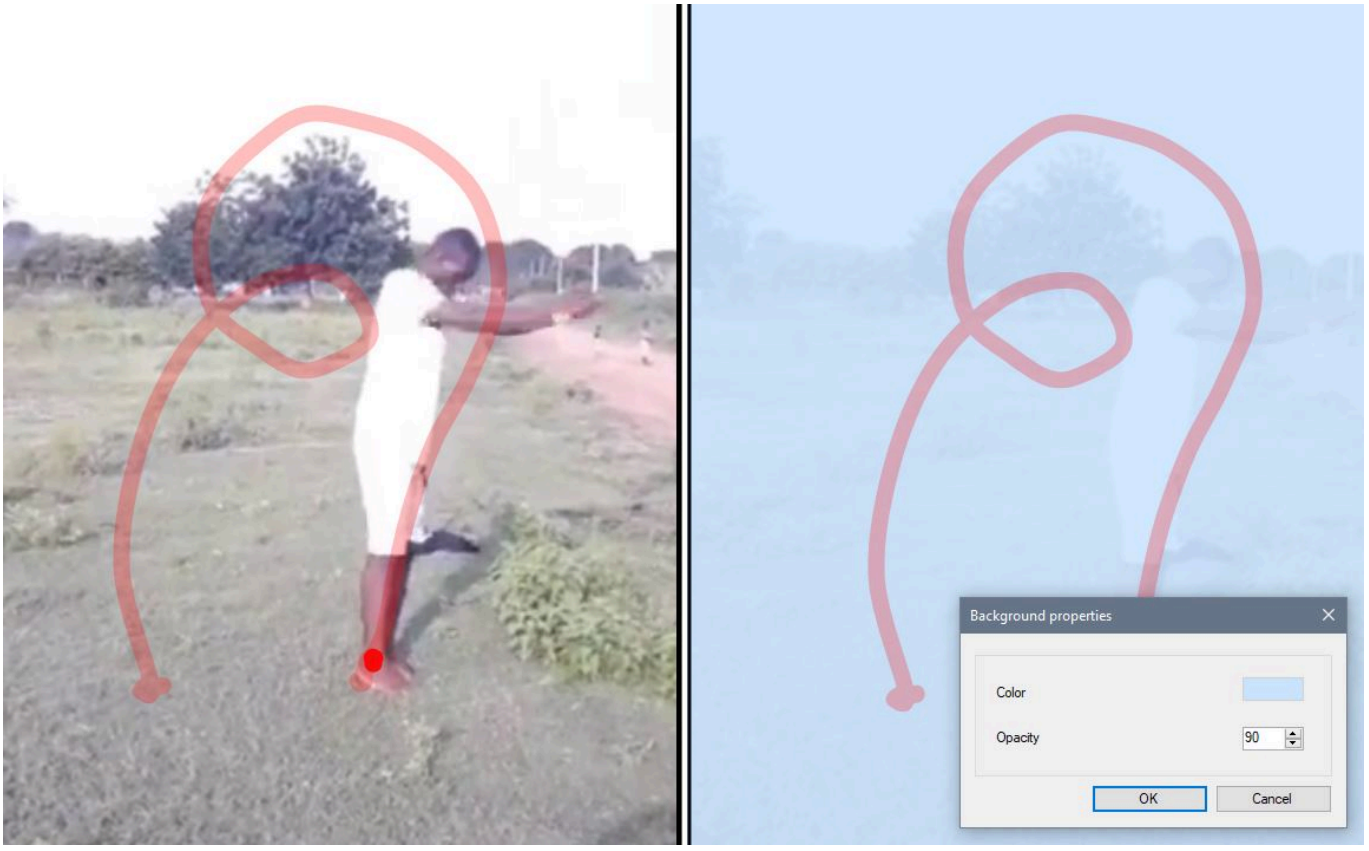
For presentation purposes the hand tool can use a large image instead of the normal cursor. To enable this mode go to the main menu **Options ▸ Pointers** and select a pointer from there. This list of image-based pointers can be customized by adding new images to the “Pointers” directory under application data.



Background layer

The background layer sits between the video image and the drawings. It can be used to make the drawings more visible against the video or, at full opacity, show only the annotations and trajectories on a plain background.

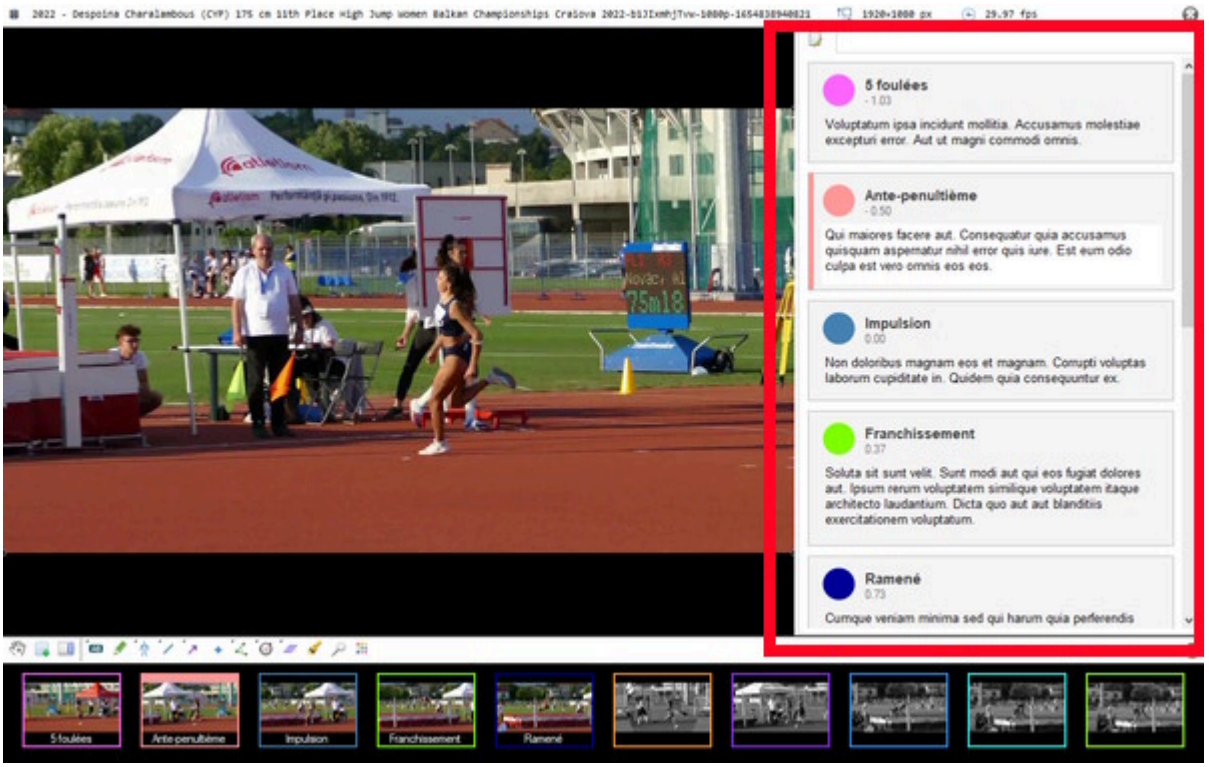
To change the background layer color and opacity right click in the background of the video and choose **Background...**. This option is also available in the main menu **Tools ▸ Background...**


Background properties
Color
Opacity
90
OK
Cancel

Key images

Video frames marked as key images show up in the key image panel below the viewport as well as in the annotation pane.

Key images can be used to quickly navigate to and annotate important moments of the video. They have a title, color and comments associated with them.



Key image panel under the video viewport

The key image panel under the video viewport has small thumbnails of all the key images.

images/annotation/keyimage-thumbnails.jpg

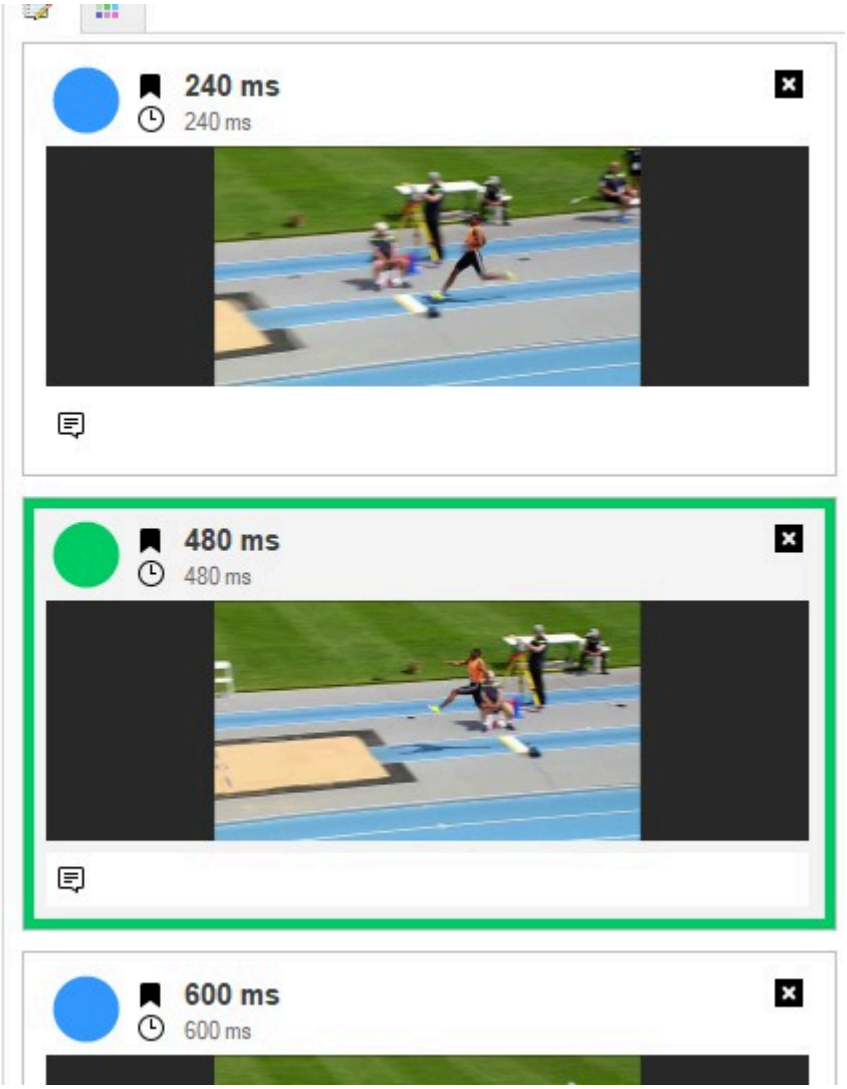
Clicking a thumbnail will move the video to the time of the key image.

Key images that are outside the working zone are grayed out.

Annotation pane

The annotation pane can be opened by clicking the Annotation pane button in the tool bar or by double clicking a key image thumbnail in the bottom panel.

The key image tab in the annotation pane shows a card for each key image and is where you can change the title, color, and add comments to the selected key image.



The toolbar at the top of the key image panel contains the following buttons:

|Add|	Adds a new key image at the current time.
|Previous|	Moves to the previous key image in the list.
|Next|	Moves to the next key image in the list.
Counter	Number of key images.
|Show|	Toggle button to show all key images or only those within the working zone.

The key image cards supports the following functions:

- Color box: Click to open a color picker to change the color of the key image.
- Title: Click to edit the title of the key image.

- Time: Time of the key image. This field is not editable.
- Comment field: Click to add comments to the key image.
- Delete: Deletes the key image.

Changing the time of a key image

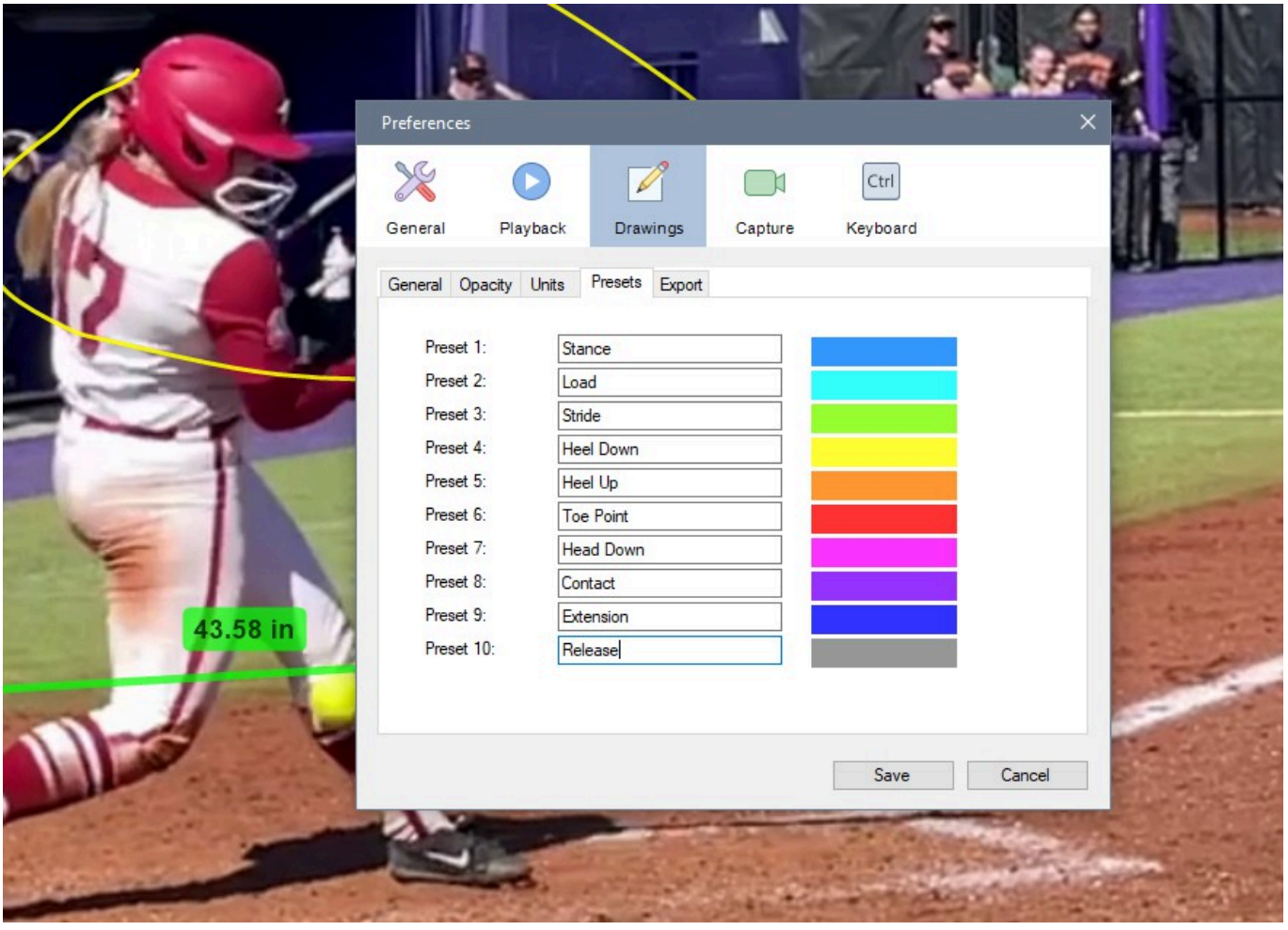
To change the time of a key image you can drag its thumbnail from the bottom panel and drop it to a new position in the timeline.

Alternatively you can move the play head to the desired time, then right click the thumbnail and select `Move to current time` to set the time of the key image to the current frame in the viewport.

Key image presets

It is possible to create key image presets to quickly add key images with a predefined title and color, using a keyboard shortcut.

To define the presets, go to the preferences via menu `Options ▸ Preferences ▸ Drawings ▸ Presets`.



General annotation tools

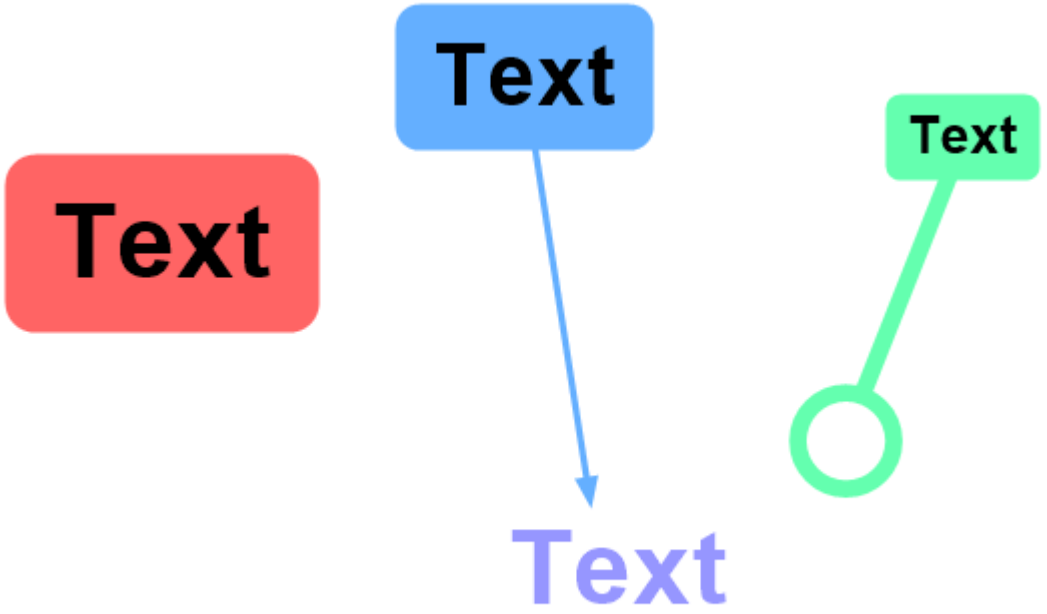
Text

- Text

The text tool can be used to create small labels. It has an optional “leg” with an arrow or circle at the end to point at things or circle them.

The width of the leg can be configured independently of the text size. The background can be transparent or filled.

To edit the text of an existing label, double click it and start typing. To change the font size visually you can drag the bottom right corner of the label.

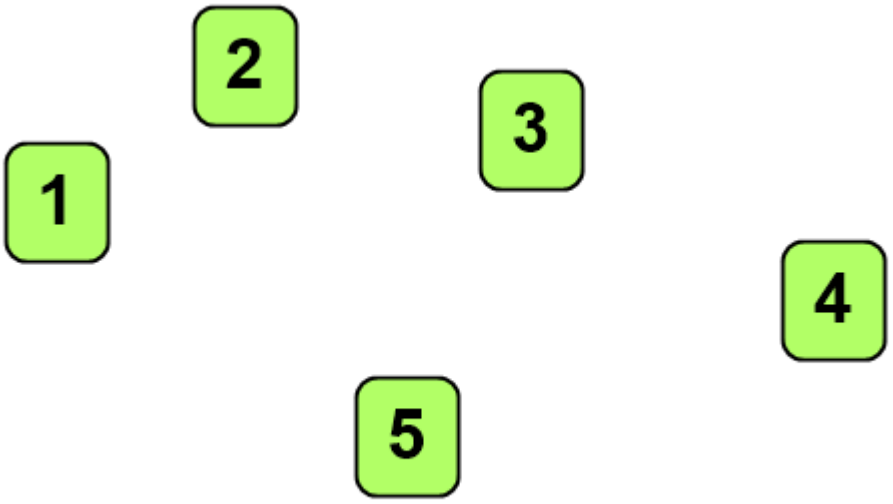


Numbers

- Numbers

The number tool can be used to add consecutive numbers.

There is only one number sequence per video, all the numbers are using the same style and size and are part of the same sequence. Using the delete menu only removes that particular number.



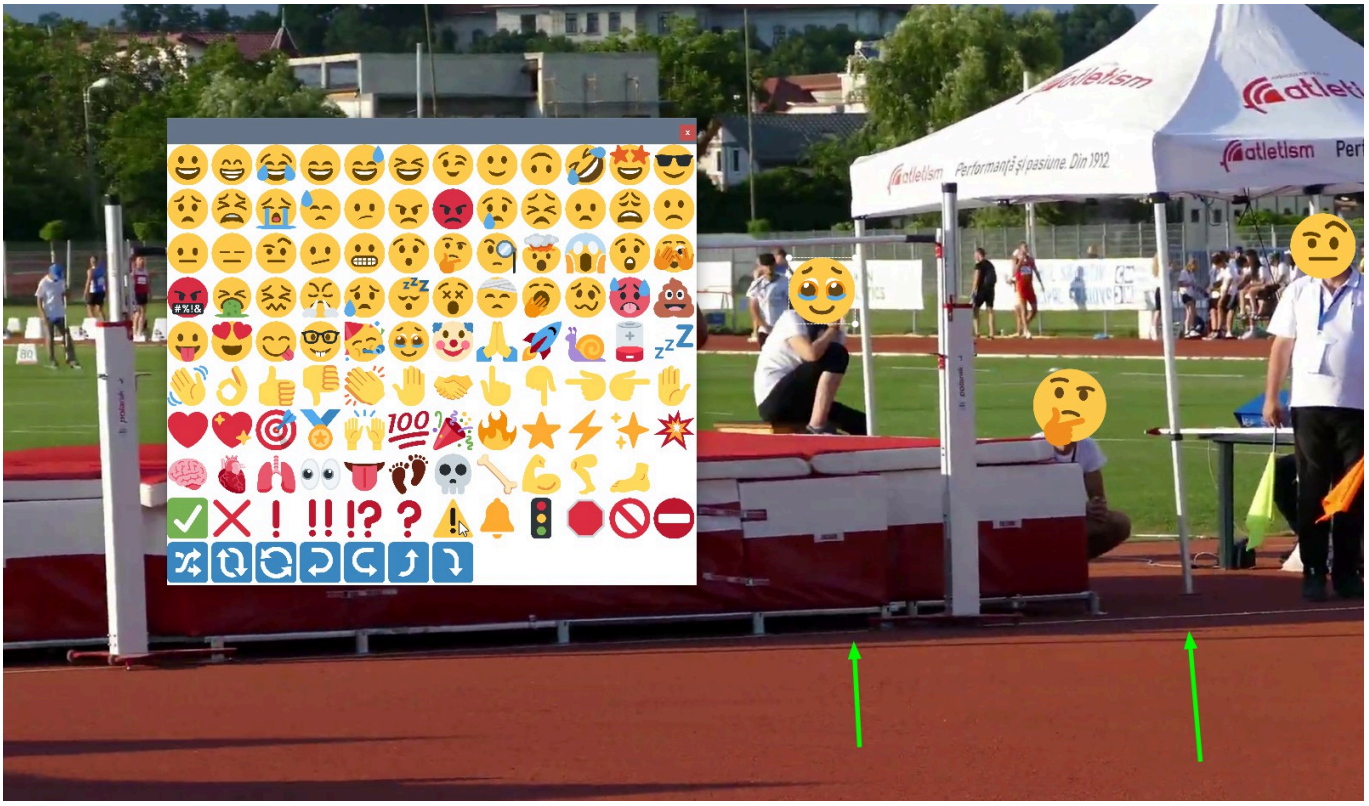
Stickers

Stickers are pre-defined emoji images that can be added to the video to quickly illustrate a point without the need to create a text object or attach a comment.

To change the emoji used by a sticker double click it to open the selection dialog.



Stickers can also be used for anonymization purposes on static images.



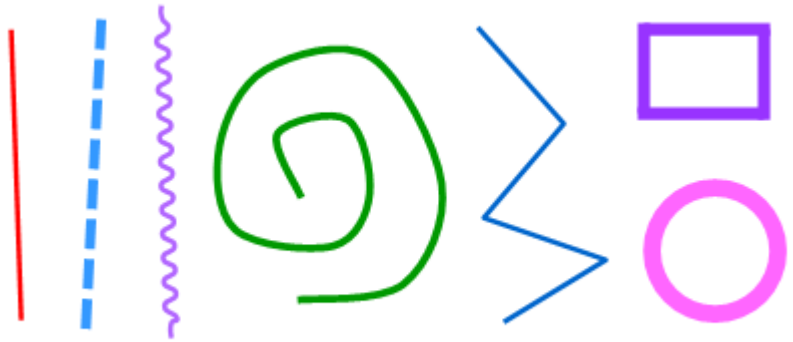
Pencil

Pencil tool

The pencil tool is used to create free-hand drawings. The width of the stroke can be modified afterwards but the path is fixed.

Line family

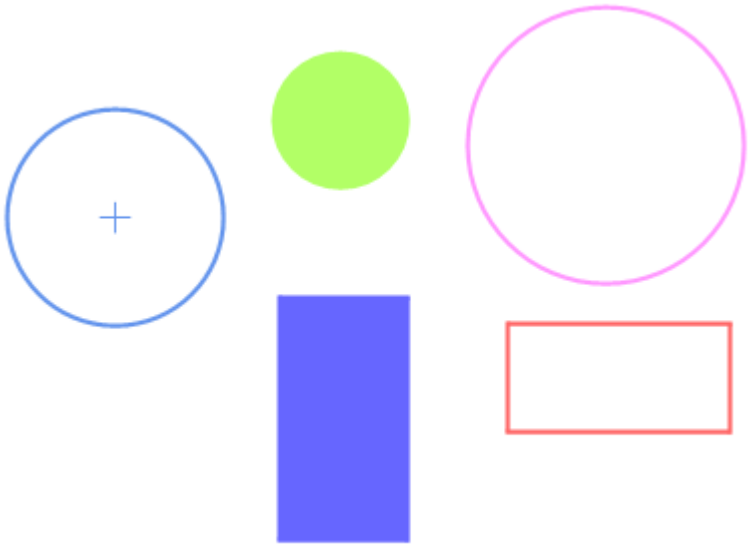
- Line
- Curve
- Polyline
- Rectangle
- Circle
- Distance horizontal



The line tool and the circle tool can also be used to measure segment lengths and the radius or circumference of circles, see Measuring positions and distances for more information.

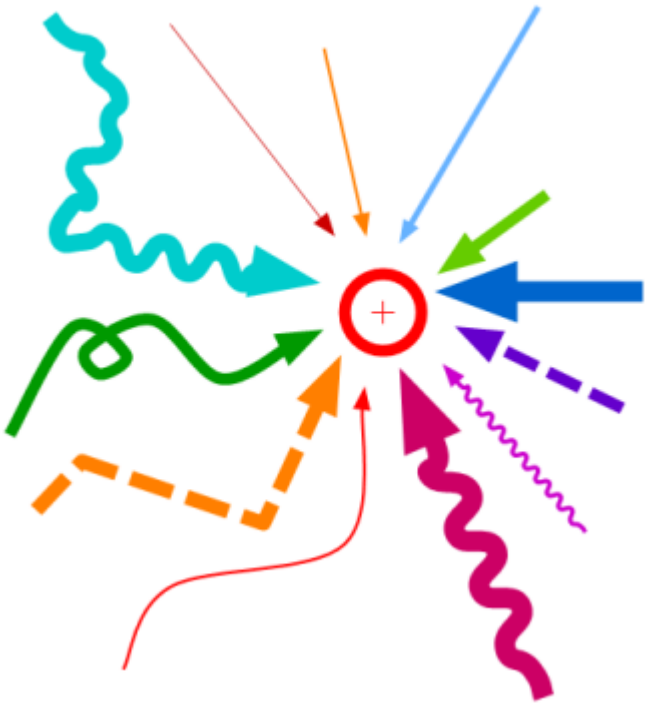
Furthermore the line tool can be used as a calibration stick. See Spatial calibration for more information.

The circle and rectangle tools have an option toggle to be drawn as filled shapes. The circle tool also has an option toggle to draw its center.



Arrow family

- Arrow
- Arrow - Dash
- Arrow - Squiggly
- Arrow - Curve
- Arrow - Polyline
- Arrow - Polyline dash
- Arrow - Polyline squiggly

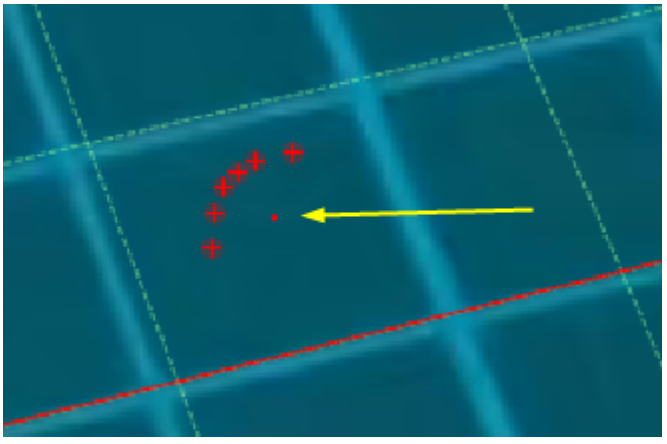


Marker

- Marker tool

The marker tool identifies the location of an object or joint. See also: Measuring positions and distances.

It has an option toggle to be displayed as a single-pixel dot.



Angle family

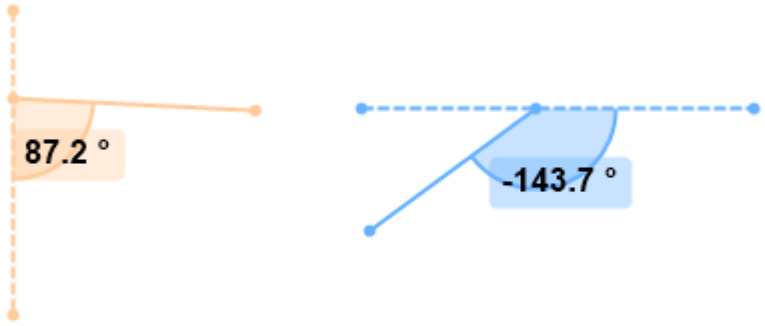
- Angle tool
- Goniometer

- **Angle to horizontal**
- **Angle to vertical**

Tools in the angle family measure angles on a particular plane.

The angle tool has options to choose the angle orientation. It also has an option to display the full circle instead of just the angle arc, in order to make it clearer that the measurement is sitting in a particular plane.

See also: Measuring angles.



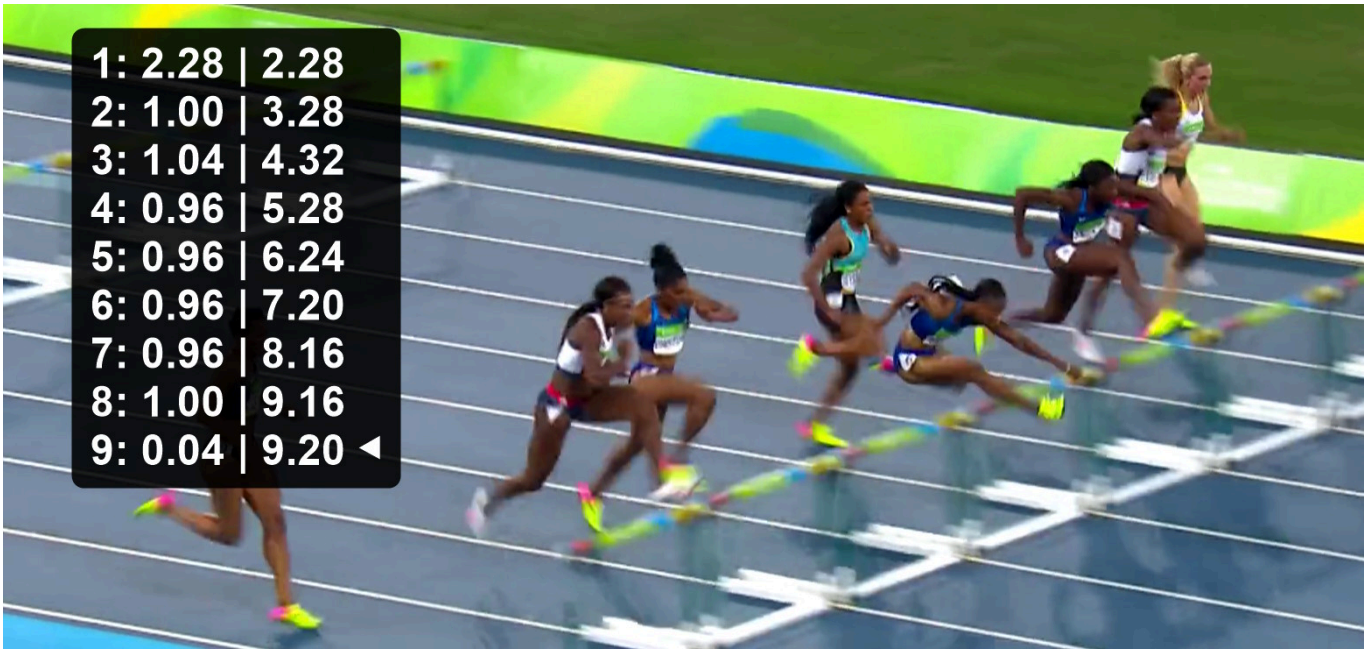
Stopwatch family

- **Stopwatch**
- **Clock**

The stopwatch is used to measure one or more time intervals, can associate a label and a tag with each time segment.

The clock tool shows the video time relatively to a specific frame and can be used as a simple count down.

See also: Measuring time.

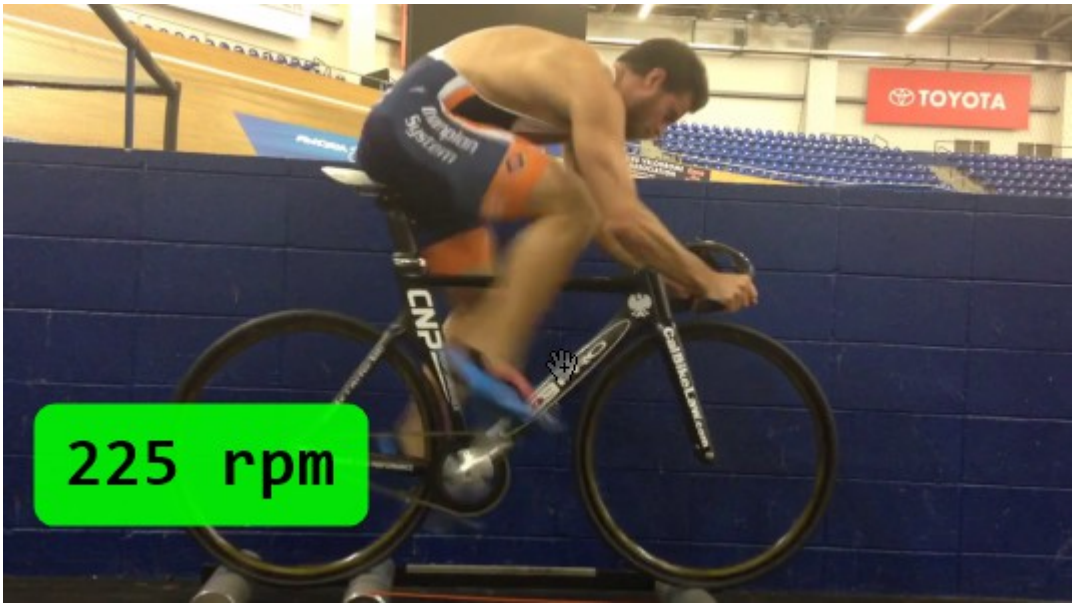


Counter

- Counter

The counter tool is used to count occurrences of an event or cycle. It can be used to measure cadence or repetitions.

See also: Counting events and measuring cadence.

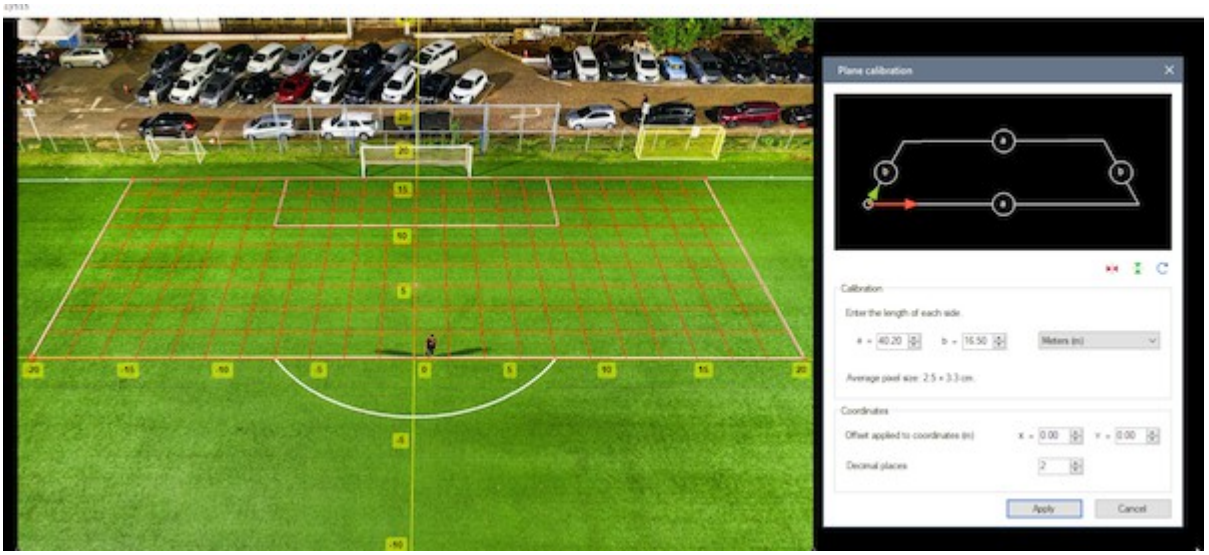


Grid tools

- Perspective grid
- Grid

The grid tools are used to materialize a plane of motion or are used for calibration purposes.

See also: Spatial calibration.



Spotlight tool

- 🖌️ **Spotlight tool**

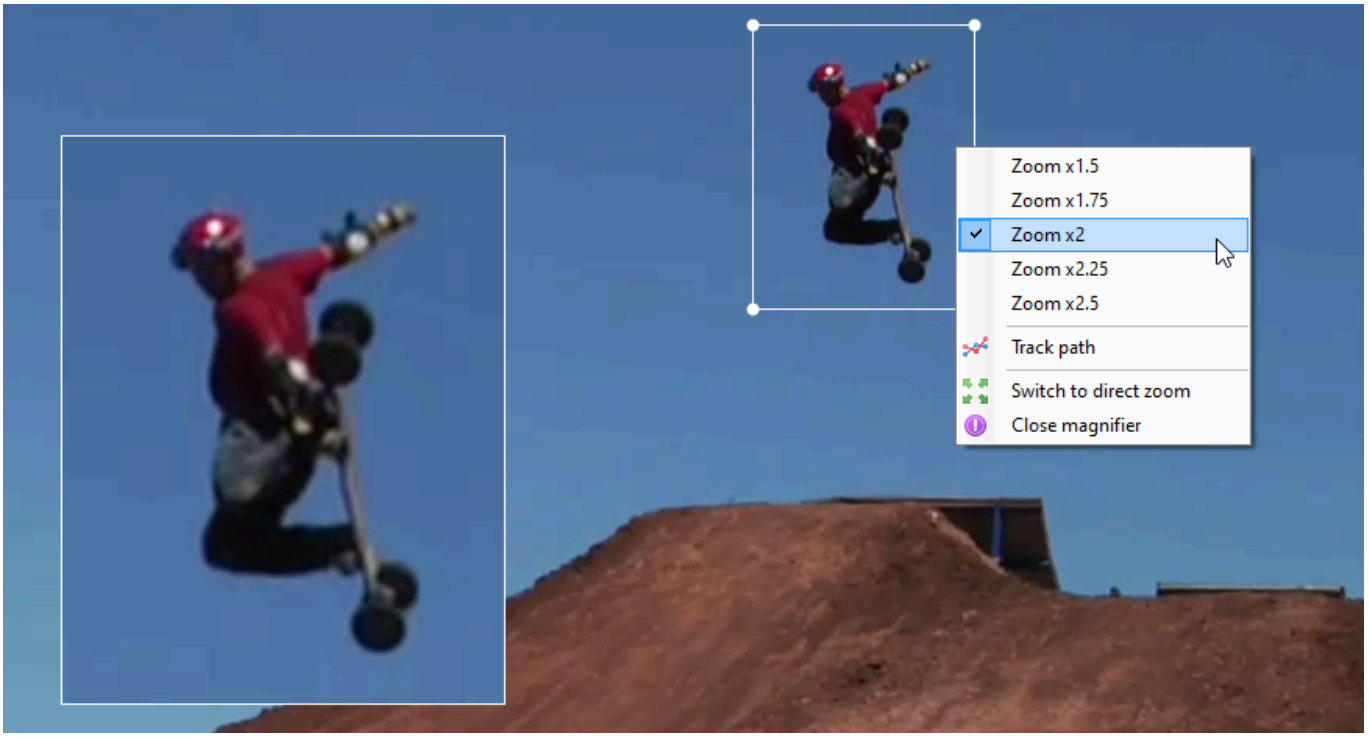


The spotlight tool is used to highlight a particular area of the image by dimming the rest. Multiple spotlights can be added throughout the video.

Magnifier

- 🔎 **Magnifier**

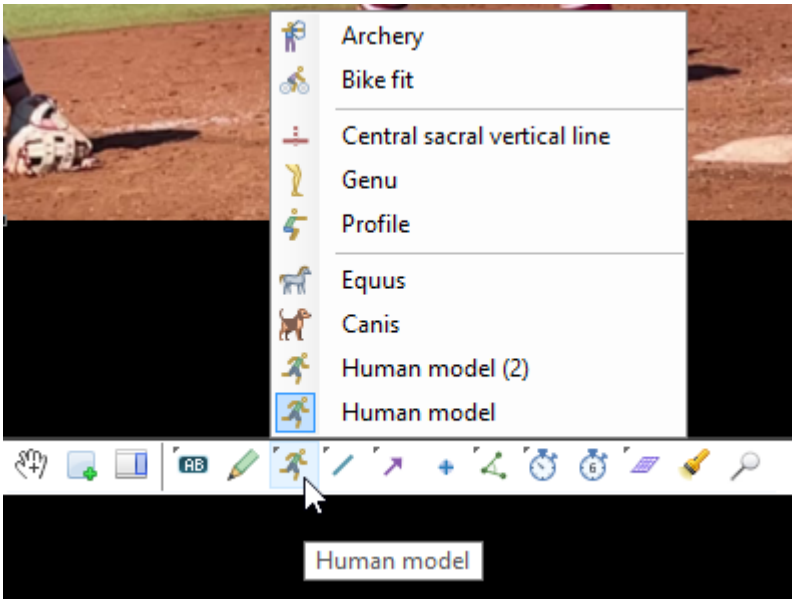
The magnifier creates a picture-in-picture effect with an enlarged version of the current image displayed within the original image. This is a display mode rather than a normal drawing tool, it is not saved in the KVA file.


Zoom x1.5
Zoom x1.75
Zoom x2
Zoom x2.25
Zoom x2.5
Track path
Switch to direct zoom
Close magnifier

Posture annotation tools

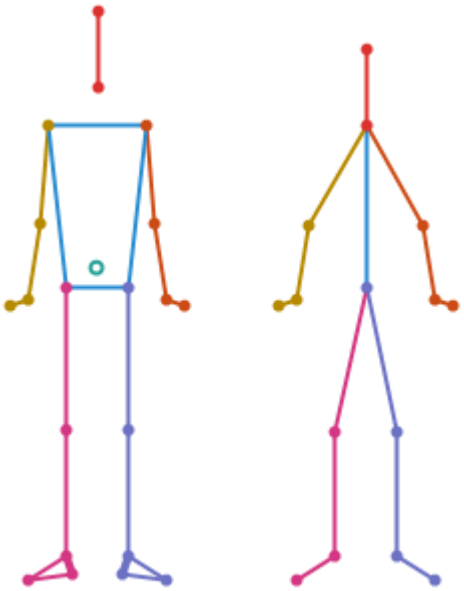
Posture tools are typically activity-specific, they are used to annotate alignments or relative orientations of the segments.

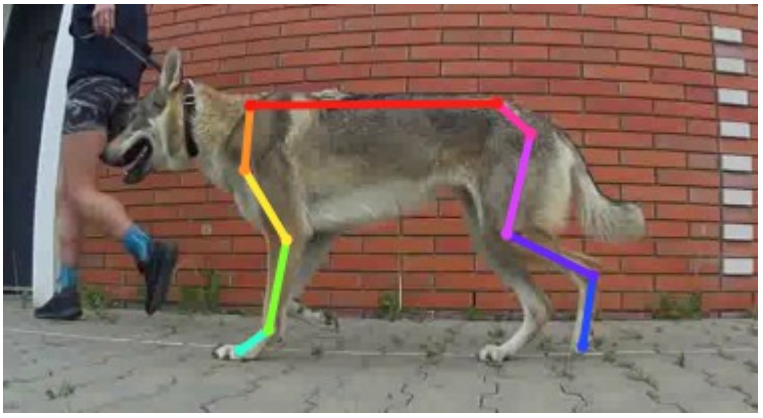
They are found under the Human model button in the toolbar.



Body models

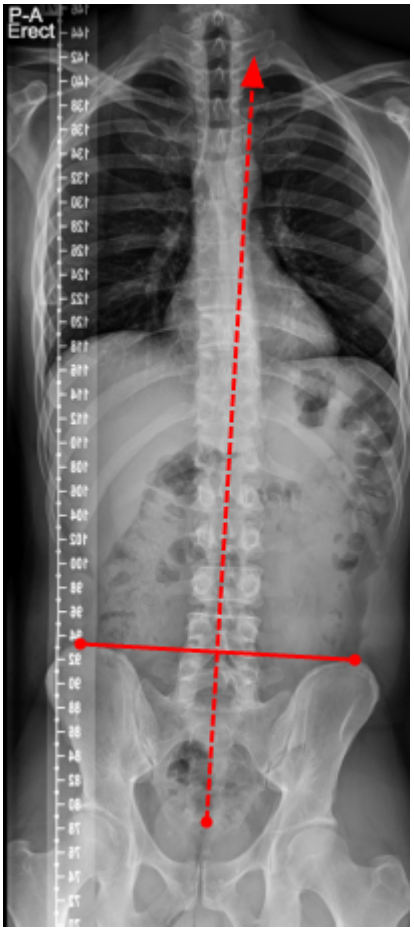
- Human model
- Human model (2)
- Canis
- Equus





Anatomy

- Profile
- Genu
- Central sacral vertical line



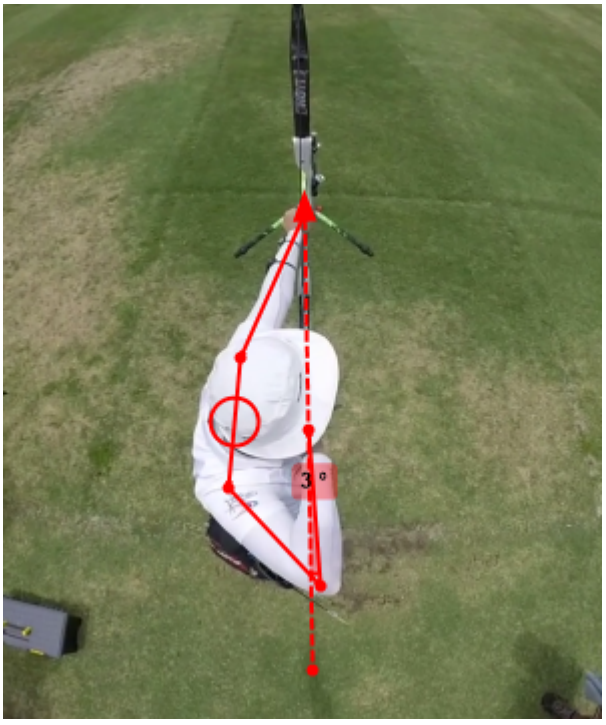
Cycling

- 🚴 Bike fit



Archery

- 🏹 Archery



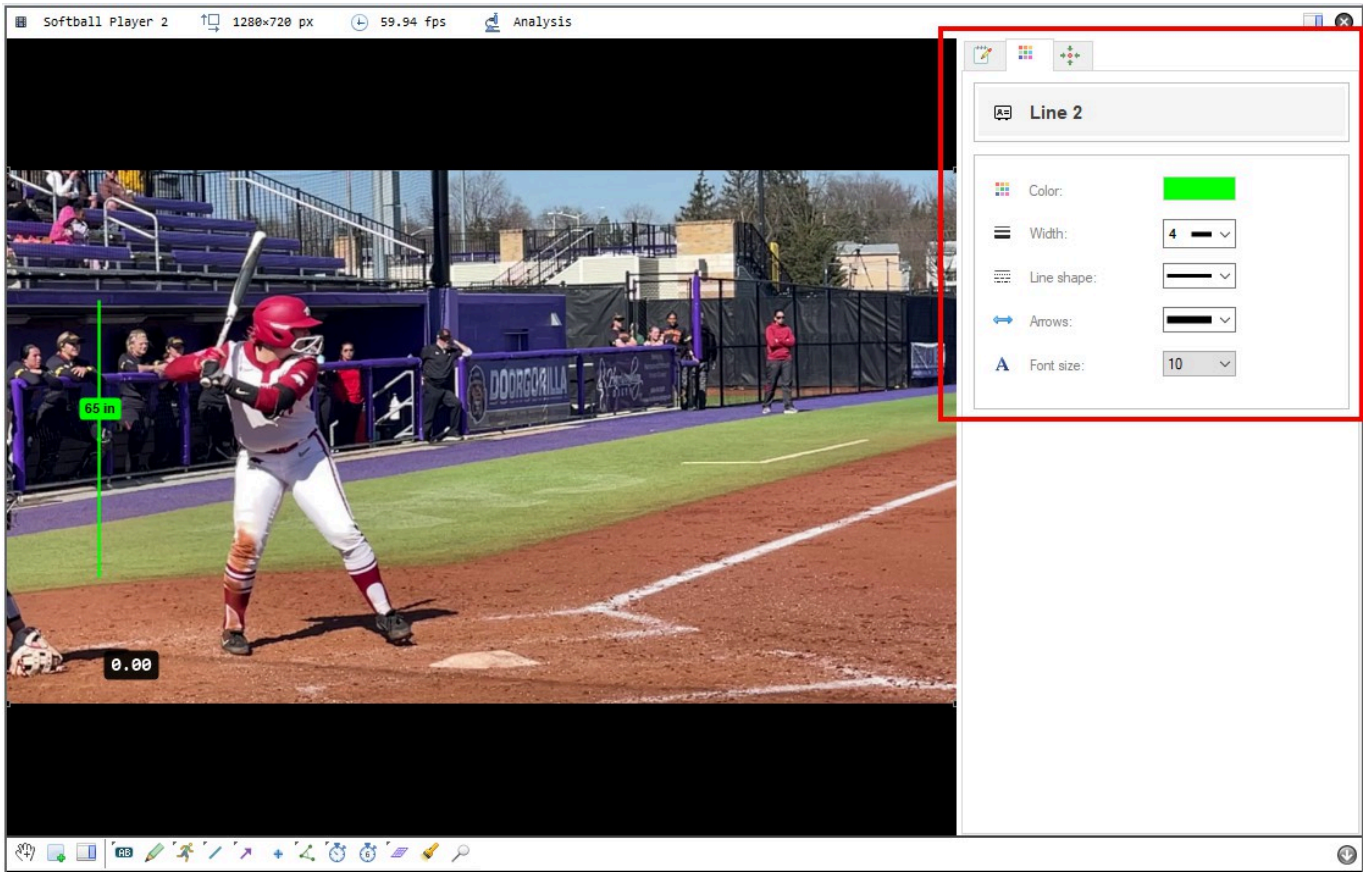
Drawings properties

Drawings typically have the following properties

- Name
- Visual style (color, size, line type, etc.)
- Options (tool-specific toggles)
- Measure label (for tools that are used to measure times, distances or angles)
- Visibility settings (opacity and fading profile)

Name and style

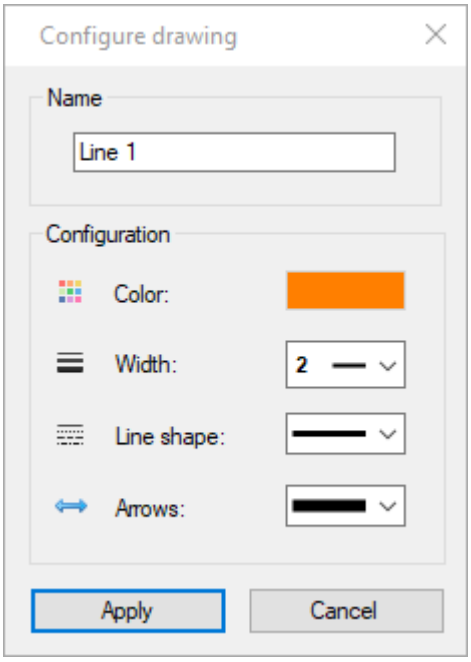
To change the name and style of a drawing use the properties tab in the annotation pane. Click on the drawing to load it in the property editor.



Note

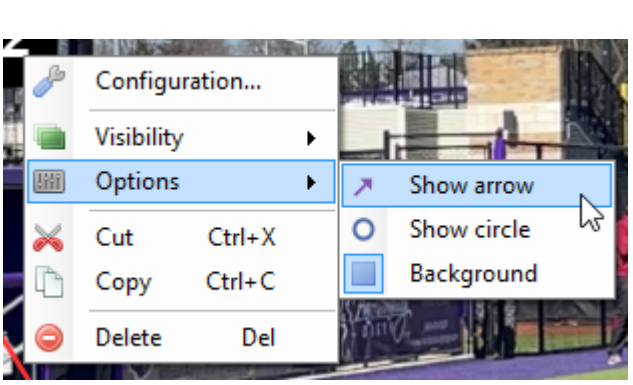
When the style of a drawing is changed it becomes the default style for new drawings of the same type.

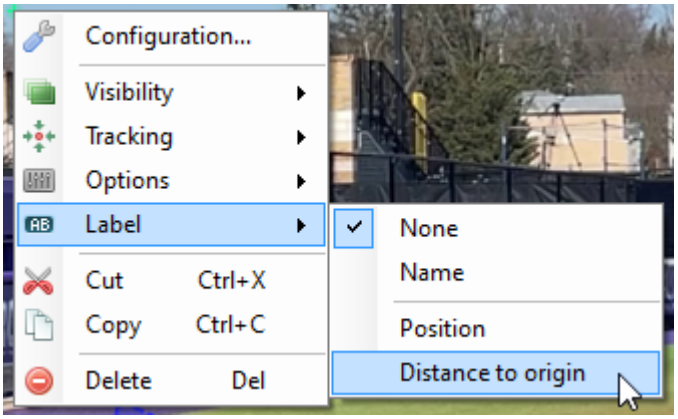
You can also right click the drawing and select Configuration... to open the configuration dialog.



Options and measure labels

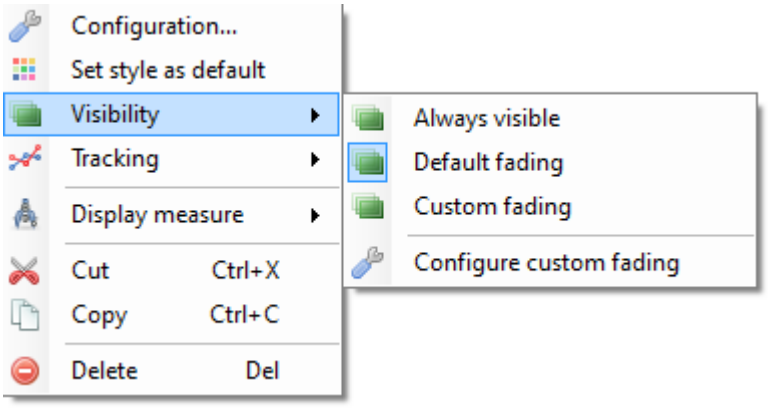
Currently the options and measure labels are not avaiable in the annotation pane and are accessed through the context menu.





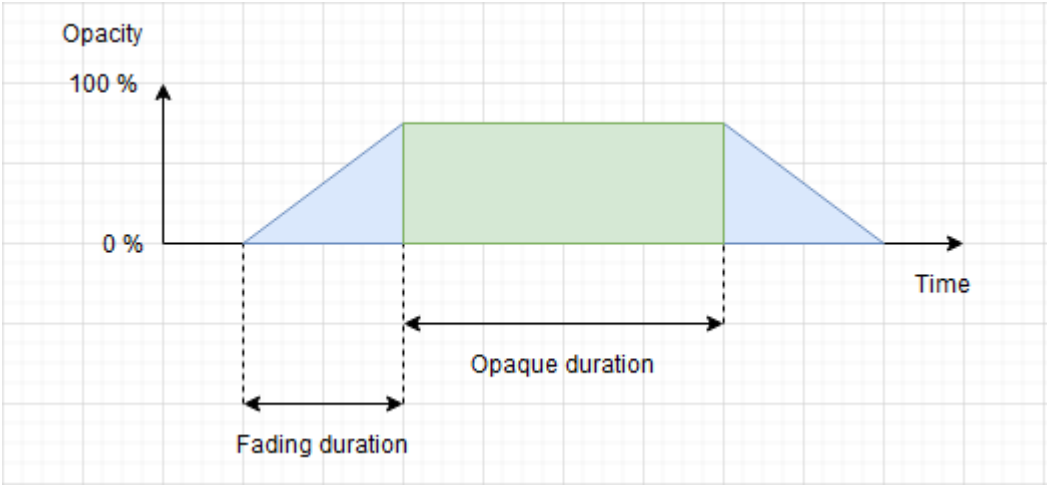
Visibility profile

Drawings have a visibility profile that controls their opacity throughout the video.



New drawings start with the default opacity options set in Options ▸ Preferences ▸ Drawings ▸ Opacity. Each drawing can then be configured to use different visibility options.

In general terms drawings have a fade-in ramp, an opaque section and a fade-out ramp. The default options make drawings fully visible on their key image and fade in and out of the neighboring frames. When drawings are tracked they stay opaque during the section of video where they are tracked.



Always visible

With this option the drawing is visible for the entirety of the video. The opaque duration and fading duration options are ignored.

Warning

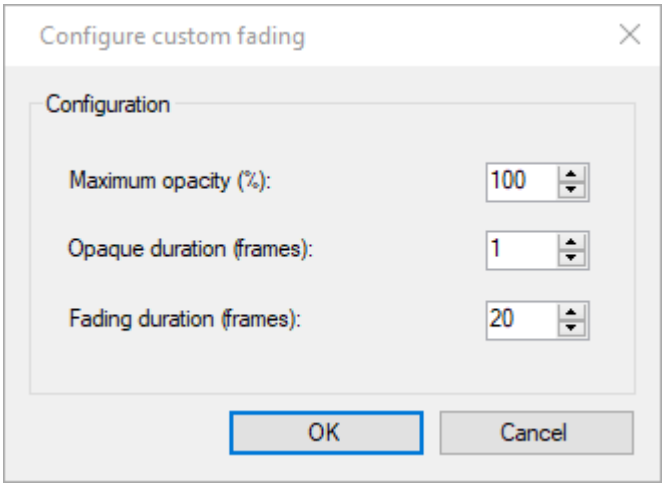
The drawing is still attached to the key image it was added to. It will be deleted if the parent key image is deleted.

Default fading

With this option the drawing uses the default opacity configuration set in **Options ▸ Preferences ▸ Drawings ▸ Opacity**.

Custom fading

With this option the drawing uses a custom configuration defined through the `Configure custom fading` dialog.



Maximum opacity (%)

This option controls the opacity used during the opaque section. A value of 100 % means the drawing will not let the background show through. A value less than 100 % means the drawing will be somewhat transparent.

Opaque duration (frames)

This option controls how long the drawing stays at its maximum opacity level before fading out. This section starts at the keyframe onto which the drawing was added.

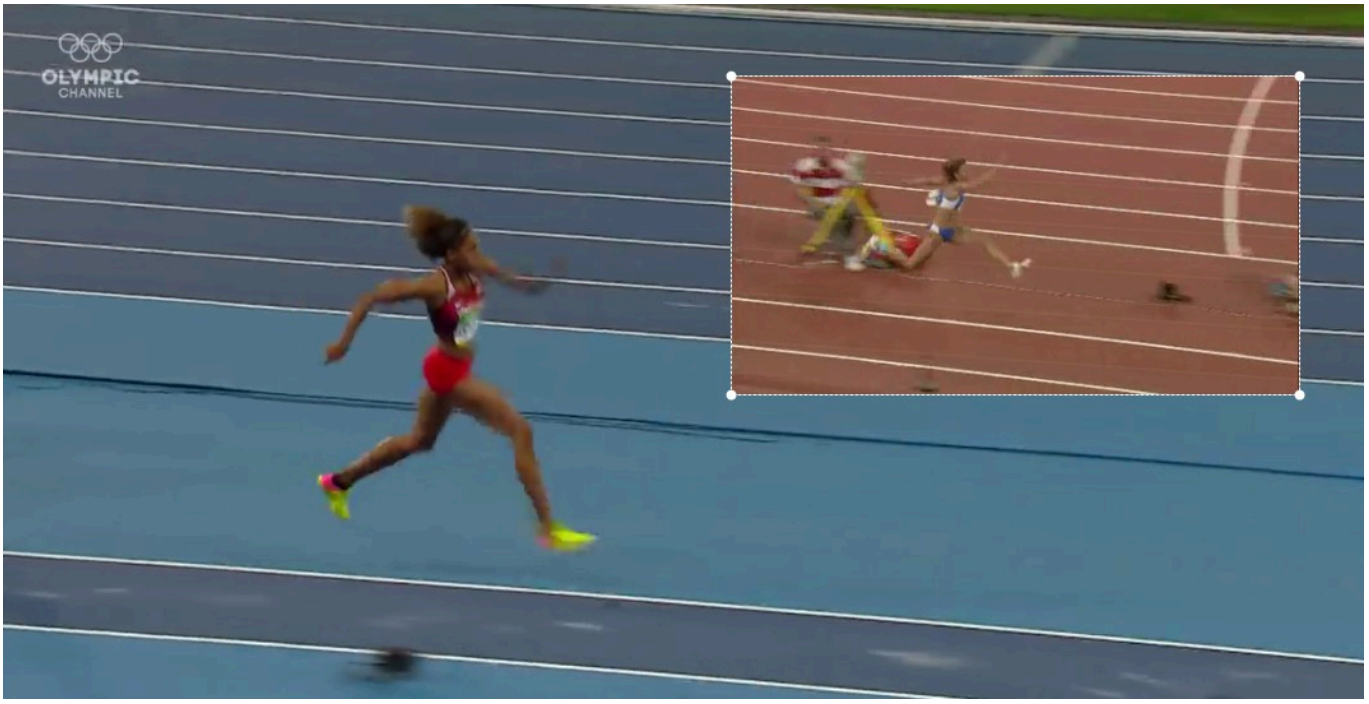
Fading duration (frames)

This option controls the duration of the ramps before and after the maximum opacity until the drawing becomes completely invisible.

Inserting images and vector drawings

Inserting external images

You can insert images and SVG vector drawings into videos to create a picture in picture effect.



To import an image or SVG file into the current video use the menu **Tools ▸ Import image...**.

To import an image that you have copied to the clipboard in another application, right click the video background and use the menu **Paste image from clipboard**.

To transfer an image from one screen to the other, use the menu **Copy image to clipboard** and then paste it in the other video.

Saving and loading annotation files

Annotations are saved in KVA files. These are XML files with a .kva extension.

The term annotation in Kinovea has a broad meaning and encompasses anything that is not directly available in the video file itself. The KVA file stores the following data:

- key images timeline positions, names, color and comments
- drawings attached to key images and their properties
- trajectories, stopwatches, clocks and counters.
- timelines of tracked objects
- image-level overrides such as aspect ratio, rotation or mirroring
- stabilization settings
- spatial calibration
- custom time origin
- capture frame rate
- user override frame rate
- working zone start and end
- the coordinate system object parameters
- background layer opacity and color
- kinogram configuration
- camera motion data
- lens distortion configuration

Saving annotations

To save the current annotations use the menu **File ▸ Save** or the shortcut CTRL + S.

The preferred workflow is to save a KVA file alongside the video as a "sidecar" file. This way the annotations are automatically loaded the next time you open the video. For example if the video file is `video.mp4` save as `video.kva`. This is the default option when saving.

Kinovea lets you choose a different name or location but you will have to manually load the annotations each time you open the video.

Annotations are typically only relevant to the video they were created on, but it is possible to import them into other videos as a way to share key image templates, calibration data or other configuration options. This can be interesting for videos filmed during the same session with the same camera setup.

Loading annotations

To load an external annotation file into an opened video use the menu **File ▸ Load annotations**.

The imported annotations are merged together with the existing annotations you might have already added to the video.

If the imported annotations were created on a different video, a conversion step may take place to adapt the drawings dimensions and time codes to the new image size and frame rate.

Loading external annotations can be used to import calibration settings between videos filmed during the same session without having to perform the calibration for every video.

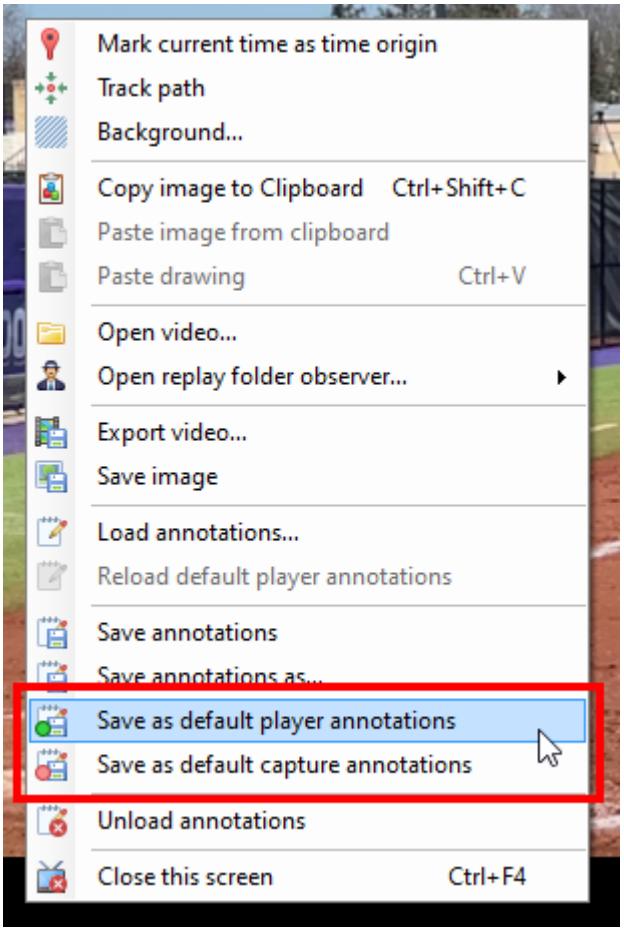
Unloading annotations

To completely get rid of all annotations on the current video, including calibration data and other metadata, use the menu **Unload annotations**.

If instead you wish to delete all drawings but not the other metadata like image rotation, the simplest way is to delete all key images.

Default annotation file

To automatically import a specific annotation file into *every* video, you can define a default annotation file. On the video with the annotations you want to save as default, use the context menu **Save as default player annotations** or **Save as default capture annotations**.



This will create a copy of the file, save it in the application data directory, and add a reference to this location in the preferences under `Options ▸ Preferences ▸ Playback ▸ Default annotation file` or `Options ▸ Preferences ▸ Capture ▸ Default annotation file`.

You can also directly edit this path and point it to a particular file of your choosing.

When opening a new video, if the file exists it will be automatically loaded before any other annotations. The sidecar KVA file will then be merged into it.

Capture annotations

When recording video from Kinovea, annotations are automatically created and saved in a sidecar KVA file next to the recorded video file. This file contains the capture frame rate, the camera rotation and mirror options, and the time origin is set to the start of the recording (ex: trigger time).

If drawings were added in the capture screen, they are also saved in this KVA file. It is possible to disable this behavior in `Options ▸ Preferences ▸ Capture ▸ Recording ▸ Exported annotations`.

In a capture-replay workflow, if these drawings are modified on the player side and saved, it is possible to send them back to the capture screen using the following menu on the capture screen `Reload linked annotations`. Only drawings on the first key image are reloaded, as the capture

screen only has one key image.

This workflow can be useful in a self-training context: posture guides can be adjusted on the recorded video much easier compared to the live camera feed.

Importing external data

In addition to loading annotations created in Kinovea you may import annotations created in other software.

Two external formats are supported: OpenPose body poses and SRT subtitles.

OpenPose

OpenPose is a software that detects human body poses in images.

To learn more about OpenPose check the project page: https://github.com/CMU-Perceptual-Computing-Lab/openpose

The result of running the OpenPose program on a video is a set of JSON files containing data for one or more human postures. OpenPose uses a 25-point body model.

This is not meant to be used for measurements but for general posture assessment.

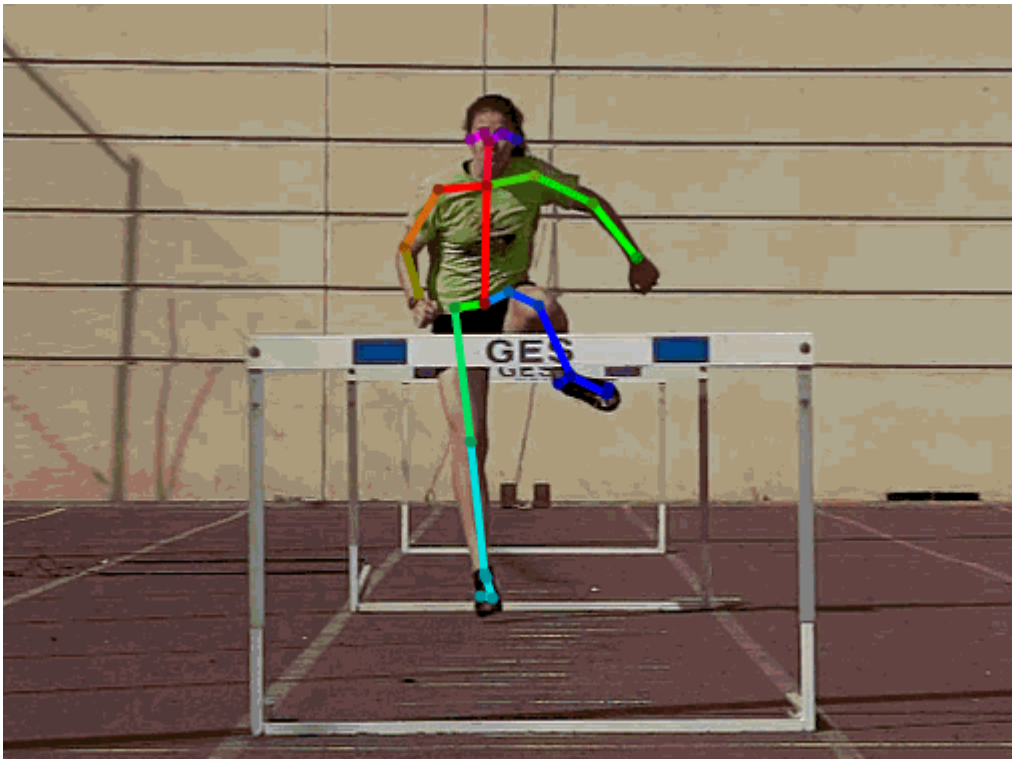
The workflow to import OpenPose data into Kinovea is the following:

Run the OpenPose software on the video, using the *write_json* option. This creates a set of .json files in the output directory. Each file contains descriptors for the detected poses.

```
> bin\OpenPoseDemo.exe --video examples\media\video.avi --write_json output/
```

In Kinovea, use the menu **File ▸ Load annotations...** and point to the first JSON file of the series.

Each detected human pose will be converted to a human model drawing using a dedicated model.



SRT SubRip Text subtitles

SRT is a simple subtitle file format containing a series of subtitles and the corresponding start and end timestamps.

To load an SRT subtitle on top of a video in Kinovea, use the menu `File ▸ Load annotations...` and point to the SRT file.

A label object will be created for each subtitle block with its visibility set up according to the timestamps for the subtitle.

Measurement

Measurement

Introduction

UI and Measurable quantities

Calibration sub-systems

- Calibration of space
- Calibration of lens distortion
- Calibration of time

Measurement accuracy

- Calibration validation
- Guidelines

Making measurements

- Coordinate system
- Measuring time
- Measuring repetitions and cadence
- Measuring positions, distances and velocity
- Measuring angles and angular speed

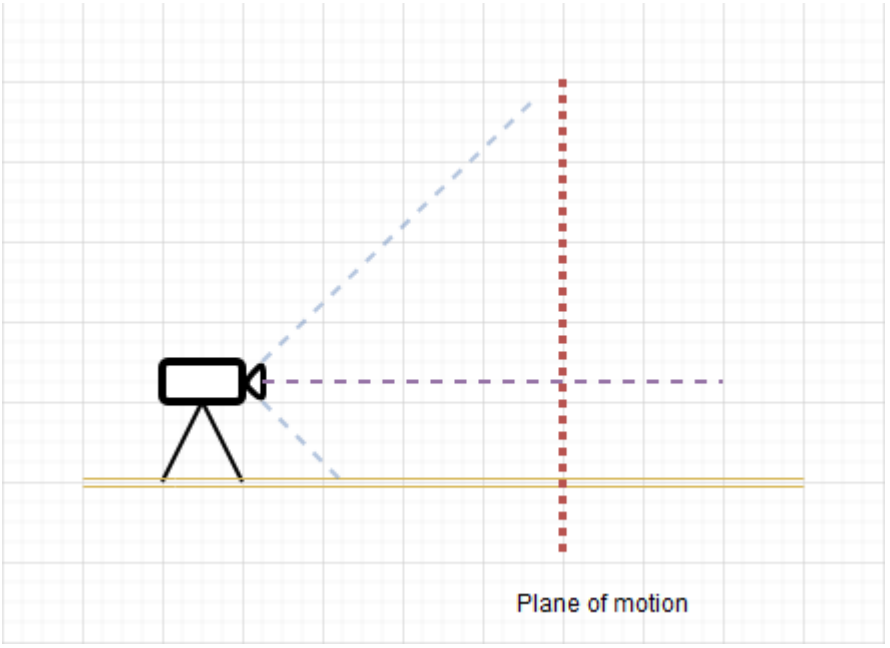
Spatial calibration

In order to use Kinovea to make measurements on the video, it is necessary to calibrate the transformation of pixels in the image into real world units. Kinovea supports two calibration mechanisms: calibration by line and calibration by plane.

All measurements in Kinovea must sit on a 2D plane. If the motion you want to study is on a plane parallel to the image plane (orthogonal to the camera optical axis), you may use calibration by line. Otherwise, if you are measuring points on the ground, for example, you should use the calibration by plane. If the motion is happening in arbitrary 3D space you cannot measure it in Kinovea.

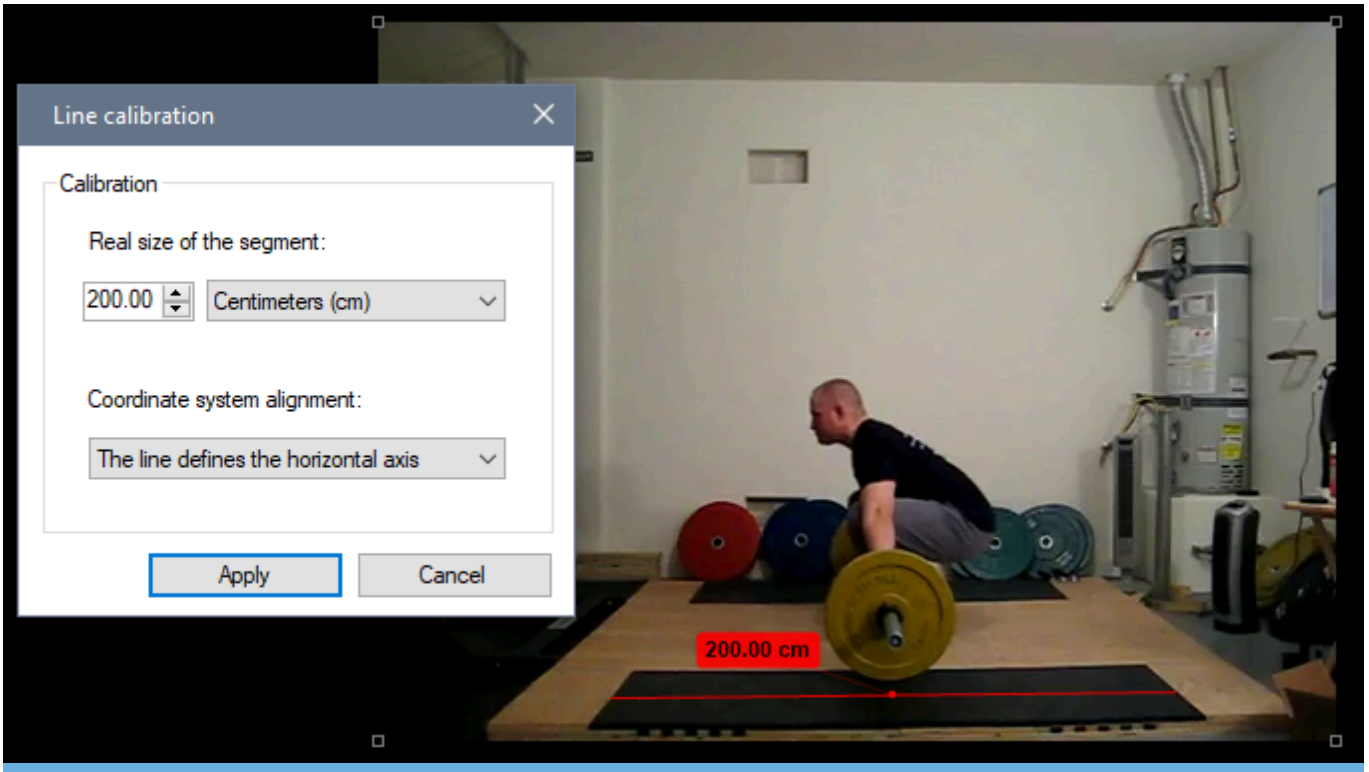
Line calibration

Line calibration is possible when the motion is sitting on a 2D plane parallel to the camera plane.



To perform line calibration follow these steps:

- Have an object of known length visible in the video.
- Add a line object and place it on top of the object of known length.
- Right click the line and select the `Calibrate` menu.
- Enter the real-world length of the object.



Note

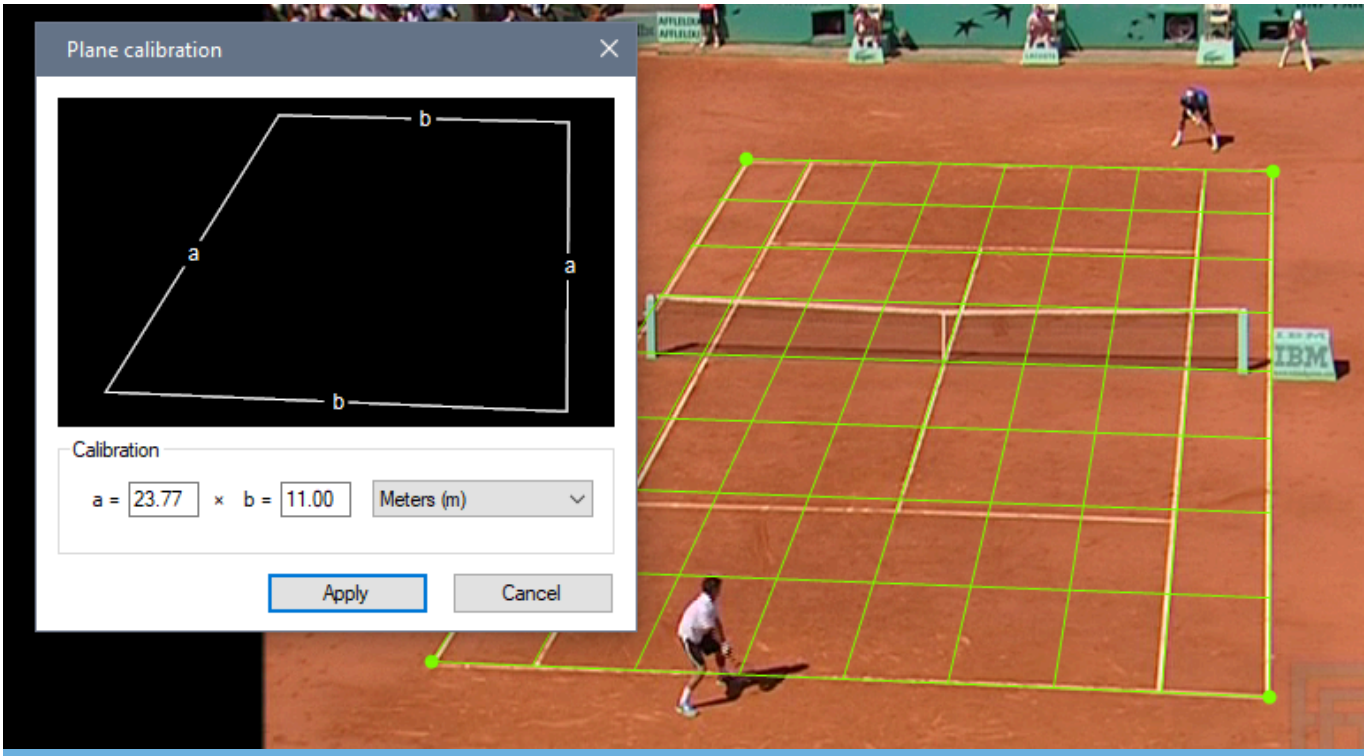
Modifying the calibration line after calibration already took place will update the calibration to use the new pixel length of the line and keep the real-world length as configured.

Plane calibration

Plane calibration is possible when the motion is sitting on an arbitrary 2D plane visible in the video.

To perform plane calibration follow these steps:

- Have a rectangle of known dimensions visible in the video.
- Add a perspective grid object and move its corners to match the rectangle.
- Right click a corner and select the **Calibrate** menu.
- Enter the real-world width and height of the rectangle.



Note

Modifying the calibration plane after calibration already took place will update the calibration to use the new pixel dimensions of the plane while keeping the real-world dimension as configured.

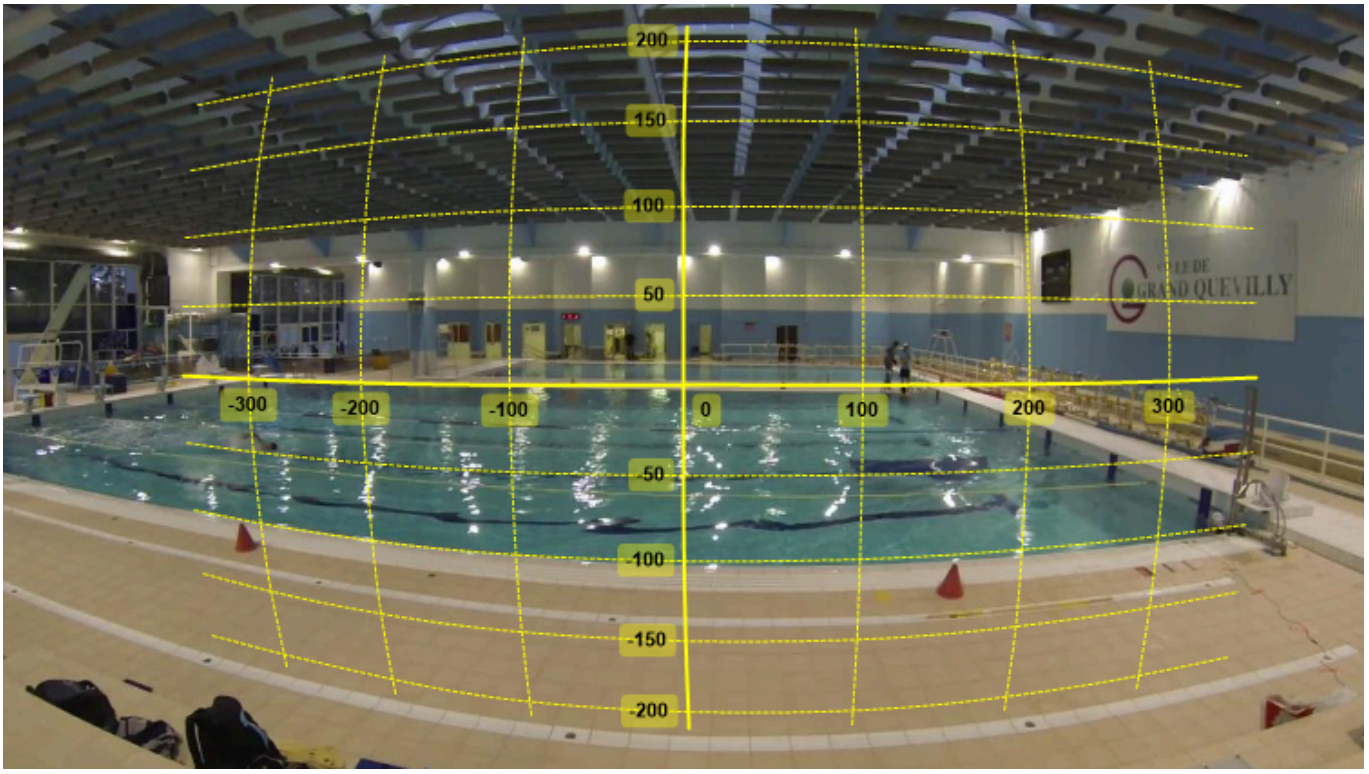
Lens distortion calibration

Revamped:

https://www.kinovea.org/en/forum/viewtopic.php?id=1977

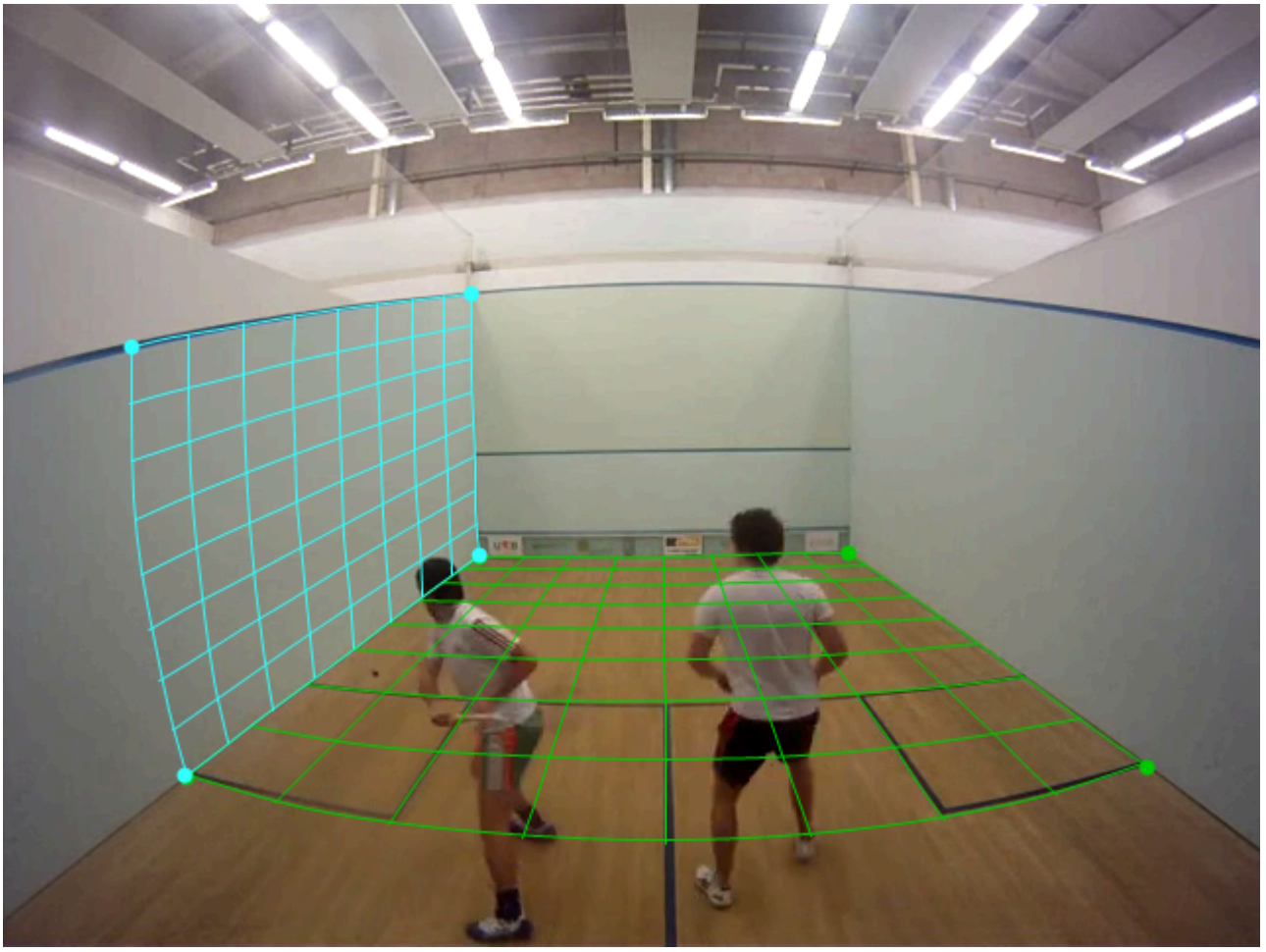
Introduction

When using cameras with significant distortion is it important to calibrate it in Kinovea before making measurements. The lens distortion calibration rectifies the coordinates before they are passed to the spatial calibration.



Barrel distortion exposed by a GoPro action camera and the default coordinate system after lens distortion calibration.

Lens calibration is compatible with both line calibration and plane calibration. The coordinate system, lines and grid objects are drawn distorted to follow the distortion. The images themselves are not rectified.



Distortion coefficients

The description of the lens distortion is done in the lens calibration dialog in the menu **Tools** ▸ **Camera calibration**.

There are multiple approaches to finding coefficients that best match a particular camera lens and camera mode, they are described below.

All the approaches involve filming a highly structured pattern in order to facilitate the distortion estimation, whether it is done visually or programmatically. This pattern can be a chess board, a brick wall, or any structured image displayed on a flat screen.

The following parameters are used during calibration:

- `fx`, `fy`: focal length in pixels.
- `cx`, `cy`: pixel coordinates of the optical center.
- `k1`, `k2`, `k3`: radial distortion coefficients.
- `p1`, `p2`: tangential distortion coefficients.

Visual estimation

In this approach the distortion is estimated visually by directly changing the distortion coefficients and trying to align the grid on the image. It is the least accurate approach but can still provide decent results.

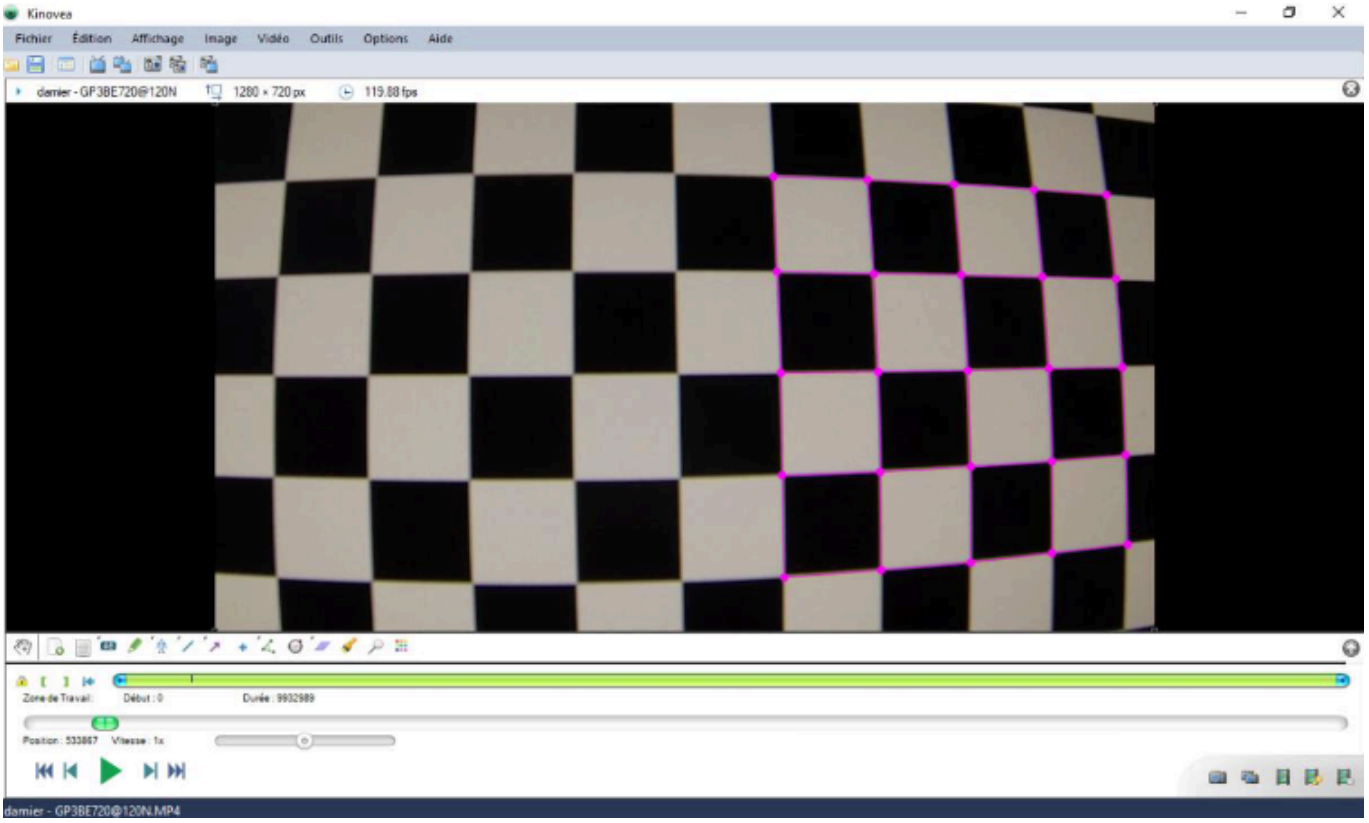


Follow these steps:

- Film the pattern straight on.
- Load the video and open the lens calibration dialog.
- Tweak k1 and k2 distortion coefficients to match the distorted grid image.
    - k1 is the distortion of the first order and should be used as a starting point to get the grid to align as best as possible in the central region of the image.
    - k2 can be used to counteract the main distortion at the periphery of the image.
- You can play with the cx, cy, p1, p2 parameters as well to adjust the grid to the image.

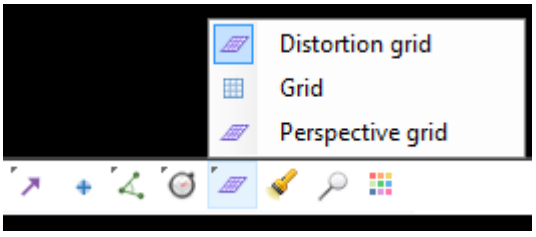
Distortion grid

In this approach the distortion is estimated programmatically by Kinovea from a set of distortion grid objects manually placed over images of the video.



Follow these steps:

- Film a chess board or structured pattern from various angles.
- Select some images (for example five) and add a Distortion grid object.



- Map each point of each grid onto corners on the filmed pattern.

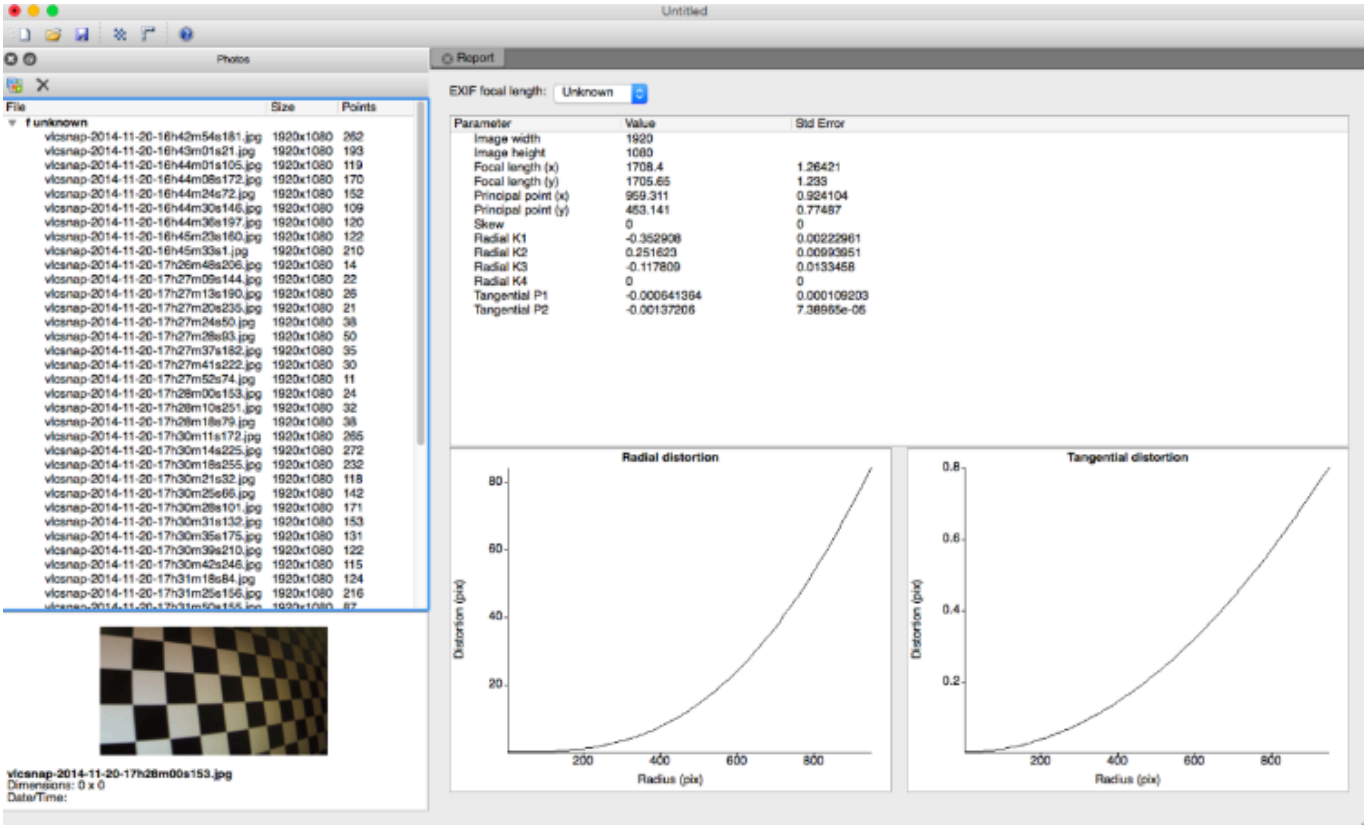


- Open the lens calibration dialog and click the `Calibrate camera` button in the lower left.

This will compute and fill the distortion parameters.

Agisoft Lens

Agisoft Lens is an automatic lens calibration software.



Note

Agisoft Lens as a free, standalone desktop software is no longer published by Agisoft.

To use Agisoft Lens to calibrate the lens distortion follow these steps:

- Open Agisoft Lens and display the chessboard using the `Show chessboard` button.
- Film the chessboard displayed from up close in such a way that only parts of the chessboard are visible, not the background of the monitor.
- Open the video in Kinovea and export at least five images to separate files.
- In Agisoft Lens import the images using the `Add photos` button.
- Start calibration by using the `Calibrate` button, selecting the following options:
    - Fit cx, cy;
    - Fit k1, k2, k3;
    - Fit aspect;
    - Fit p1, p2.
- Use Save calibration to save an XML file of the solved parameters.
- In Kinovea open the lens calibration dialog and do **File ▸ Import ▸ Agisoft Lens**.

This will convert and fill the distortion parameters.

Exporting and importing distortion parameters

The distortion parameters are saved in the KVA file but if you want to re-use the same parameters on a different video you can export them to a separate file. Use the menus **File ▸ Save** and **File ▸ Open** in the lens calibration dialog.

Note

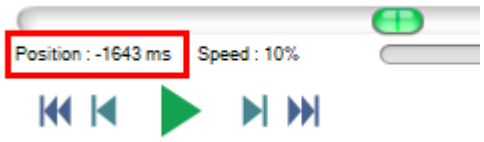
Any change of camera model, lens, or configuration options involving image resolution or zoom requires a new calibration procedure.

Time calibration

Time origin

Time information in Kinovea is relative to a specific point in the video. By default, this is the start of the video or working zone.

To set the time origin to an arbitrary point in the video navigate to this point and use the `Mark current time as time origin` button or context menu. All displayed times will be relative to this origin, using negative time before the event and positive time after it.



Setting the time origin to another point of the video can be useful to quickly get timing information in relation to a specific event, for example, a golf ball impact, a pitcher's release point, a long jumper's take-off, a starting gun trigger.

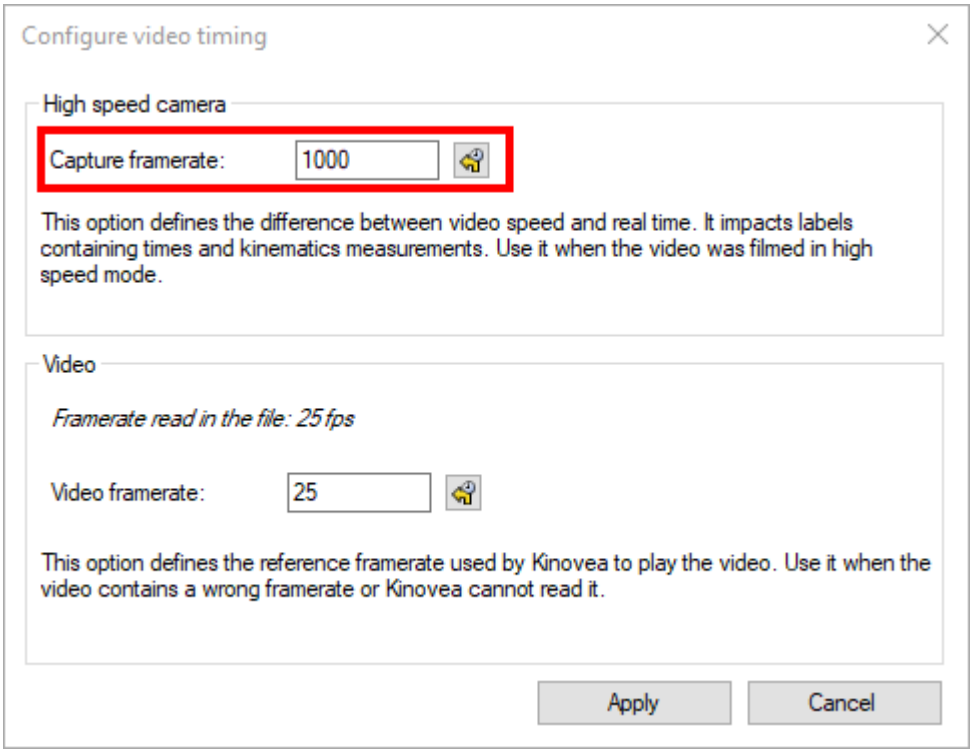


Capture frame rate

When a video is captured with a high speed camera (or using the high speed or slow motion function of a smartphone), the generated video file often has a frame rate different from the capture frame rate. For example a video filmed at 1000 fps may be saved to a file with a more typical playback frame rate of 25 fps.

In this case the video will play back slower than real time, which is expected, but the time-related information and calculations would be erroneous if they were based on the playback framerate.

To work with this type of video it is important to configure (or “calibrate”) the time scale. This is done by going to `Video ▸ Configure video timing` and filling the capture frame rate.

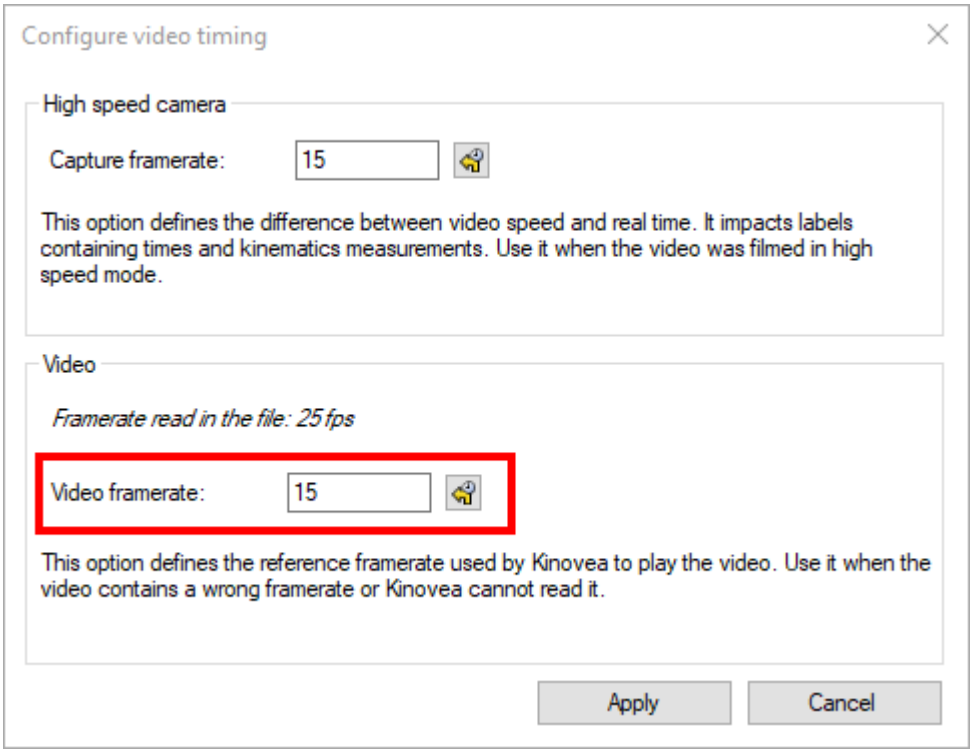


Changing this option does not change the nominal speed at which the video is played back. In other words setting the speed control at its mid-point will still play back the video at the same slow motion rate as before. Instead, this option changes the time coordinates of the images.

Broken playback frame rate

In some cases a video is saved with a frame rate which is just plain wrong, or Kinovea cannot read it. For example a USB camera might claim that it is capturing video at 25 fps but the video stream is actually transfered at 15 fps.

In this case the video will play back at the wrong speed and the time calibration will be wrong. If you know the real playback frame rate at which the video is supposed to be played back, you may enter it in `Video ▸ Configure video timing`.



Changing this option does change the nominal speed at which the video is played back.

Calibration validation

Guidelines to minimize errors

The following guidelines are designed to minimize the systematic and random errors inherent to measuring spatial motion on 2D images.

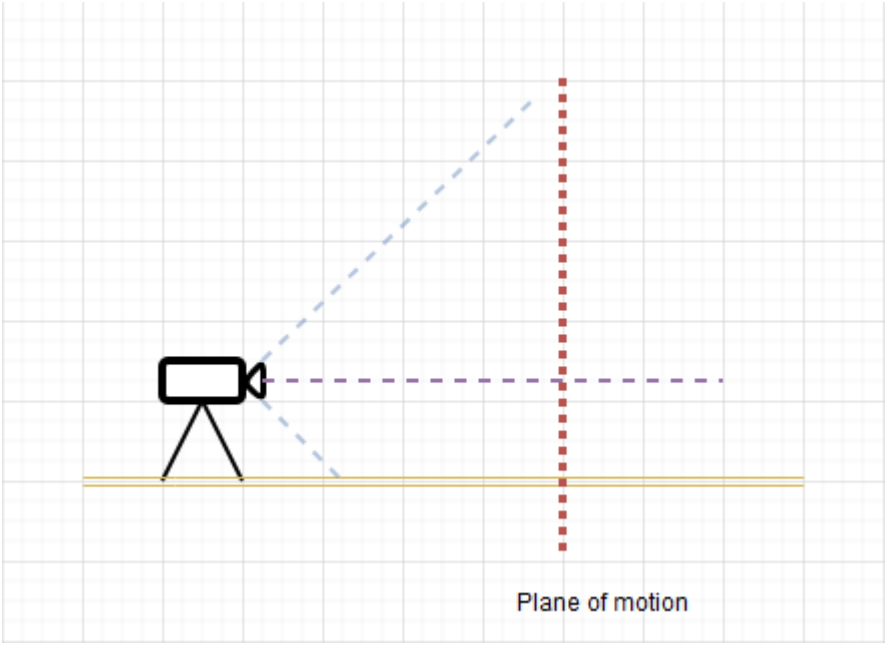
Camera setup

Mount the camera on a tripod and avoid camera motion

The camera must remain stationary for the images to provide a stable frame of reference. If the camera is moving relative to the scene, the plane of motion will change over time and the calibration from one video frame cannot be used on other frames.

Align the camera plane with the plane of motion

When using calibration by line the camera must be aligned with the plane of motion, the camera optical axis must be orthogonal to the plane of motion.



Maximize the camera distance to the scene

To minimize perspective distortion. place the camera as far as possible from the scene and zoom in. This will reduce errors due to points moving in and out of the plane of motion.

Camera and digitization

Minimize or correct lens distortion

Use a lens with as little distortion as possible to minimize the curvature of the plane of motion in the images.

This distortion can be taken into account in Kinovea using the lens calibration dialog.

See also: lensdistortion.

Use an appropriate exposure time

To minimize motion blur, decrease the exposure time as much as possible (increase shutter speed).

Maximize image size and image quality

Use the largest possible image resolution to improve the precision of the digitization process when locating points on the images.

Use the zoom function

Use the zoom function in Kinovea to position the points more precisely. Kinovea uses fractional pixel coordinates for calibration and measurements.

Scene setup

Use a large reference object

The scaling object used for calibration should occupy as much of the image as possible.

Use a vertical or horizontal reference

To align the axes of the coordinate system with the real world use a plumb line or other object of known direction as the calibration object. If the real world vertical line is not parallel to the image side you can set the calibration line to define the vertical axis.

Avoid measuring objects outside the plane of motion

Everything that is measured must sit on the calibrated plane of motion. Coordinates and measurements using points physically outside the plane of motion will be inaccurate.

Use small round markers

Use small markers on the object or joints. The markers shape should not vary when rotated, for example a disk or concentric disks should be suitable.

Coordinate system

All measurements are based on the coordinate system defined by the current calibration.

To show or hide the coordinate system use the menu `Tools ▸ Coordinate system`.

Line calibration

The first placed point of the line becomes the default origin of the coordinate system. The line direction becomes an axis of the coordinate system based on the option selected in Coordinate system alignment.

The following alignment options are available:

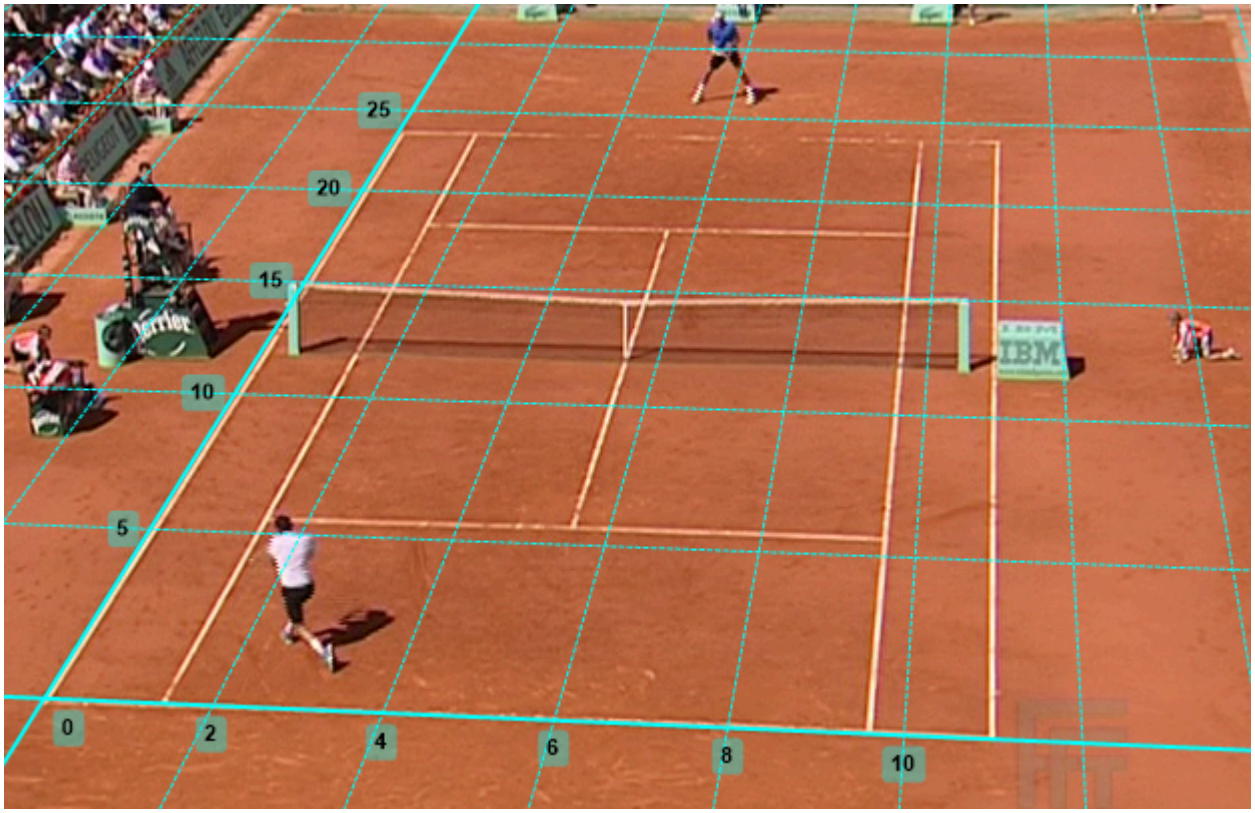
- `The line defines the horizontal axis`
- `The line defines the vertical axis`
- `Aligned to image axes`

If `Align to image axes` is selected the orientation of the line is ignored.



Plane calibration

The bottom-left corner of the grid becomes the origin of the coordinate system. The coordinate system axes are aligned with the calibration grid object.



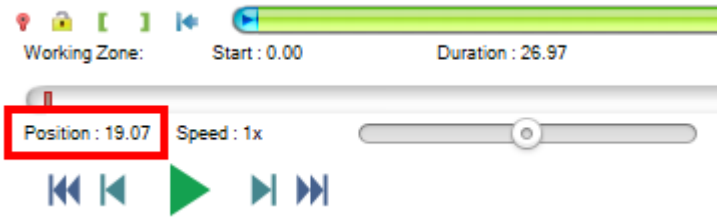
Origin of the coordinate system

The coordinate system's origin is independent from the calibration object. It can be moved around by dragging the axes.

Measuring time

Time position

The current time position is visible under the main timeline.



Time unit

The time unit can be changed from the menu Options ▸ Time or from Options ▸ Preferences ▸ Playback ▸ Units ▸ Time. The following options are available:

Format	Example	Description
[h:][mm:]ss.xx[x]	1:10.48	Textual timecode.
Frame number	1762	Rank of the current frame.
Total milliseconds	70480 ms	Integer number of milliseconds.
Total microseconds	1284 µs	Integer number of microseconds.
Ten thousandth of an hour	904	Ten thousandths of an hour
Hundredth of a minute	542	Hundredths of a minute
[h:][mm:]ss.xx[x] + Frame number	1:10.48 (1762)	

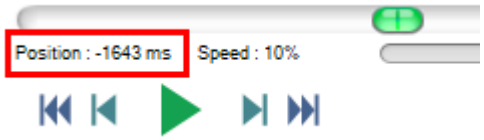
Note

The configured time unit is used for the time position and duration, the clock and stopwatch tools, and when exporting to spreadsheets. The kinematics dialog always uses total milliseconds as the time unit.

Time origin

Times displayed are relative to the time origin defined for the video. By default the time origin is at the begining of the video but you can change it in order to display times relative to a particular event of the video; such as a ball impact, a jump take-off, a release point, a race start, etc.

Video positions before the time origin have negative times.



The time origin can be changed manually by using the time origin button or by right clicking the image and selecting **Mark current time as time origin**. The current video frame becomes the time origin.

Relative clock

To annotate the video with the current video time use the clock tool.



By default a clock object simply displays the current video time using the global time origin. Each clock tool may also have its own custom time origin independent from the global one. To define a custom time origin Right click the clock tool and choose **Mark current time as time origin for this clock**.



The clock object can be identified by name by using the **Show label** menu and changing the object name in its configuration dialog.

Time intervals

To measure a time interval use the Stopwatch tool .



To measure the time elapsed between two points of the video follow these steps:

- Add a stopwatch object to the video.
- Move to the start of the interval, right click the stopwatch object and choose **Start the stopwatch**.
- Move to the end of the interval, right click the stopwatch object and choose **Stop the stopwatch**.



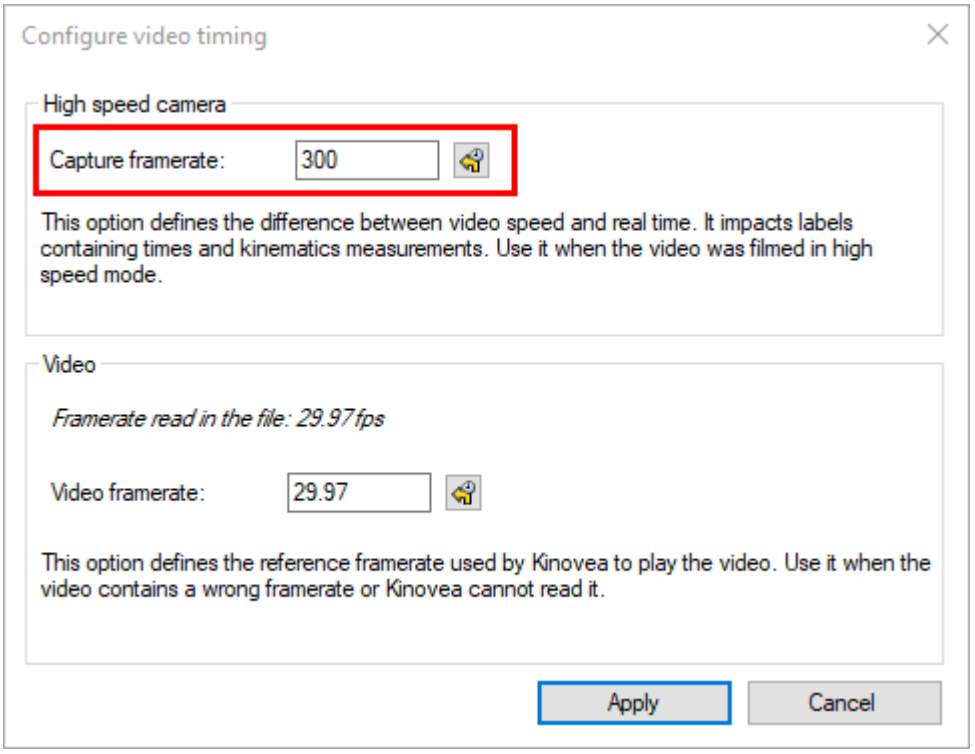
In order to measure multiple time intervals you must add several stopwatch objects.

Time scale

Videos filmed with high speed cameras or in high speed mode can have a video frame rate different from the capture frame rate. For example a video filmed at 1000 fps is typically saved with a “standard” playback rate of 30 fps. This makes the final video appear in slow motion even when the speed slider is set to 1x.

In this case it is important to tell Kinovea about the original capture frame rate for times to be correct. This impacts time positions and time intervals.

Open the video timing dialog from menu **Video ▸ Configure video timing...** and in the top part of the dialog, enter the capture frame rate.



Note

When capturing videos with Kinovea this option is automatically set based on information found in the KVA file saved next to the recording.

Counting events and measuring cadence

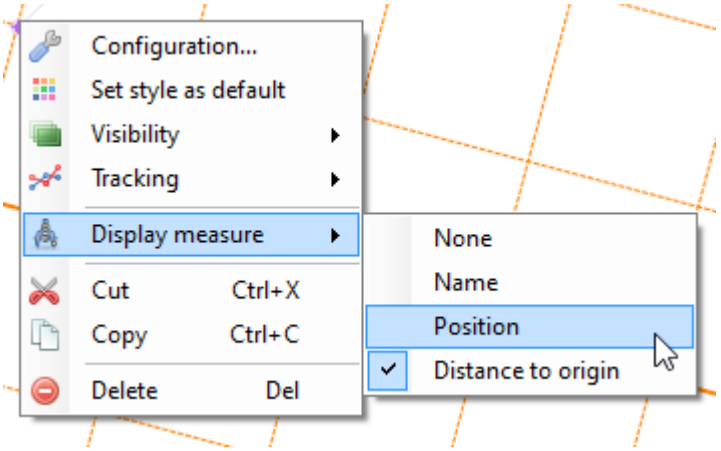
https://www.kinovea.org/en/forum/viewtopic.php?id=1979

Measuring positions and distances

Introduction

To measure positions and distances on video images, it is necessary to first perform calibration.

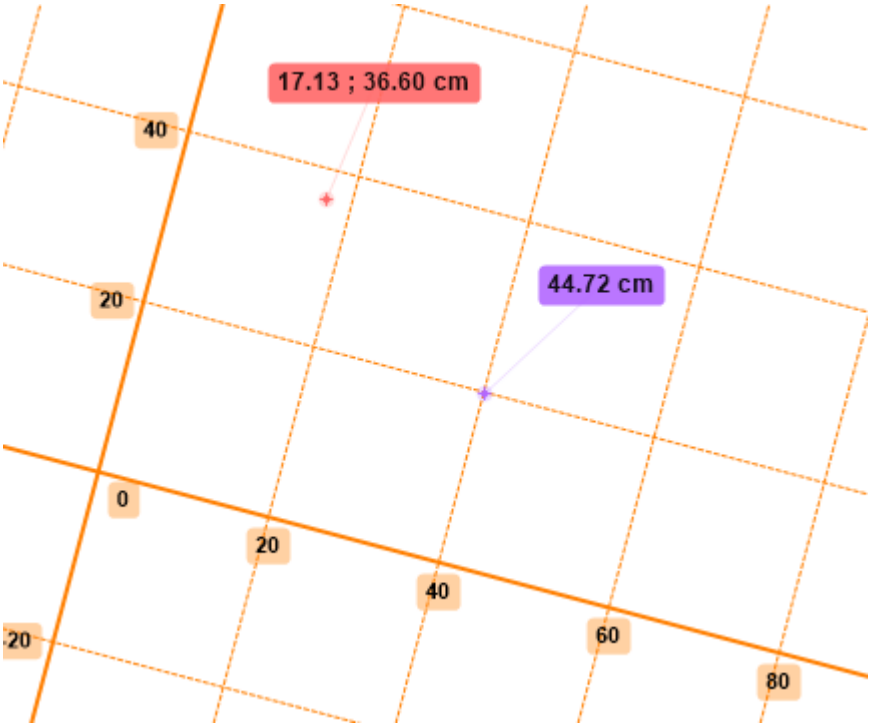
To display a measurement right click the object and choose an option under **Display measure**.



The unit of distance is the unit used during calibration.

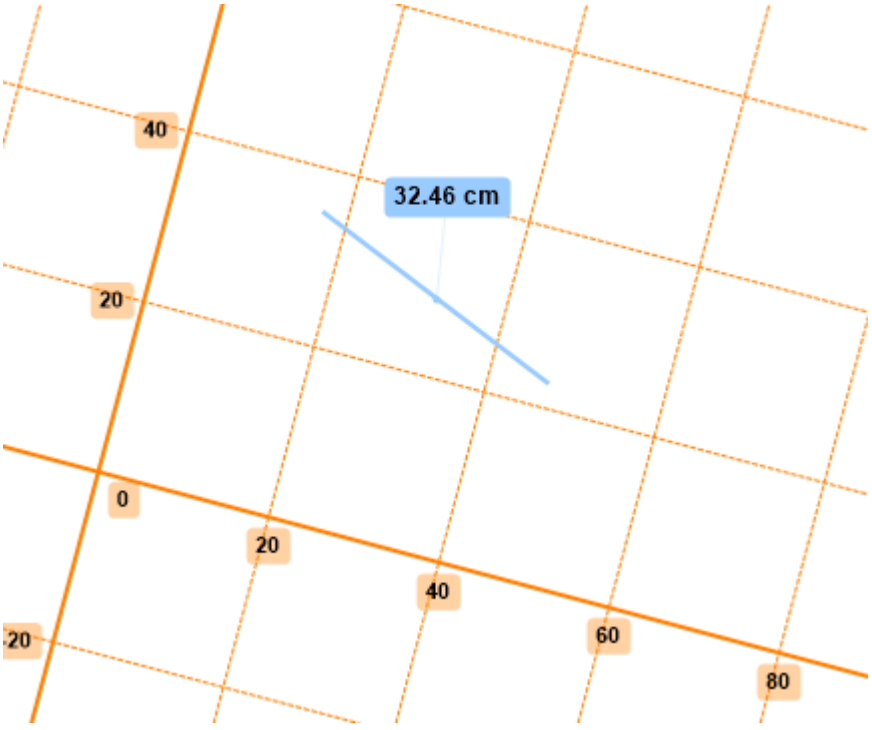
Points

The marker tool can be used to measure the position of a point in the current coordinate system or its distance to the origin.



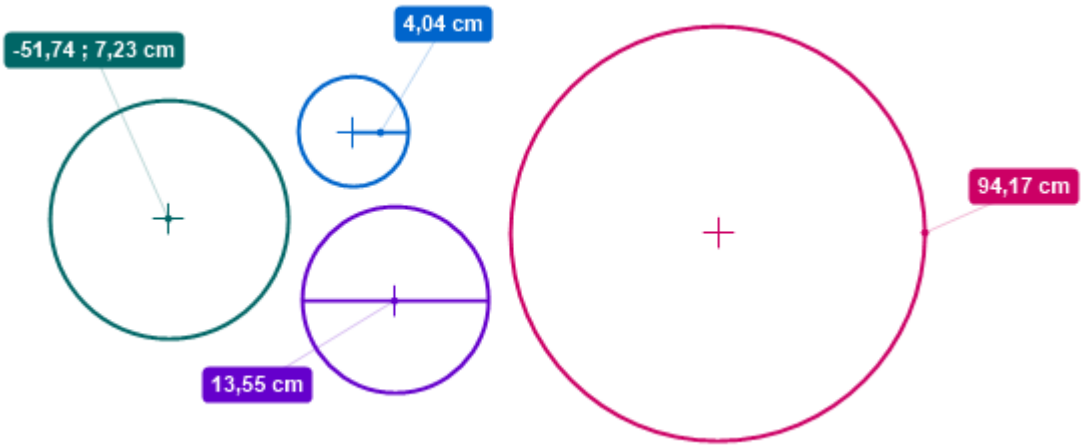
Lines

The line tool can be used to measure the distance between two points.



Circles

The circle tool has display options to show the center position, radius, diameter, and circumference of the circle.


-51,74 ; 7,23 cm
4,04 cm
94,17 cm
13,55 cm

Measuring angles

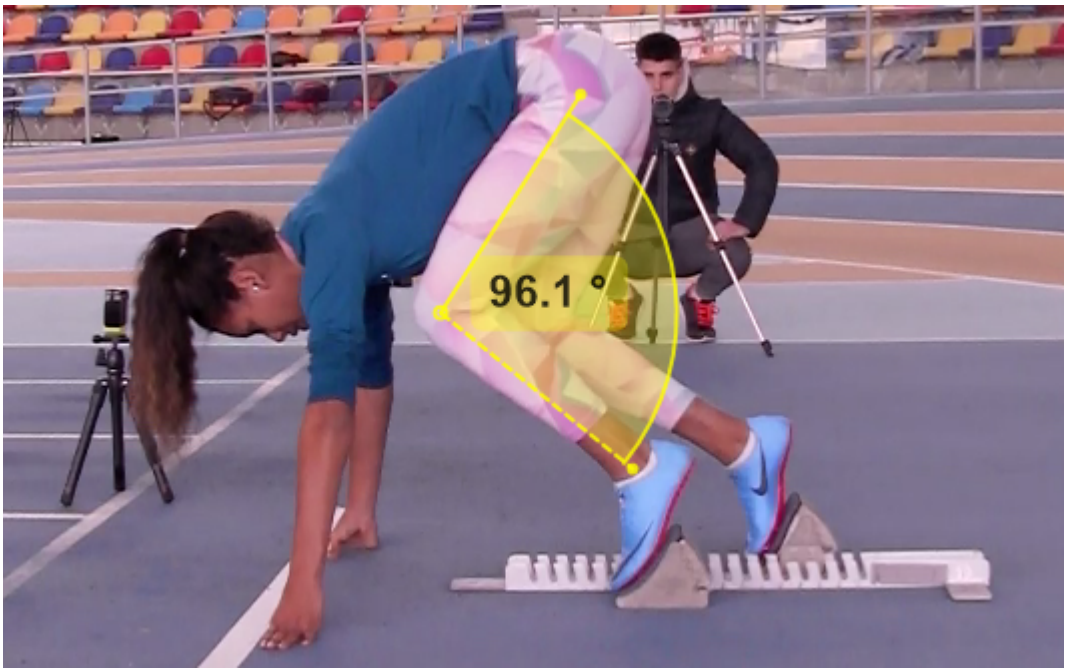
Introduction

To measure angles on planes that are not orthogonal to the camera optical axis it is necessary to first perform plane calibration.

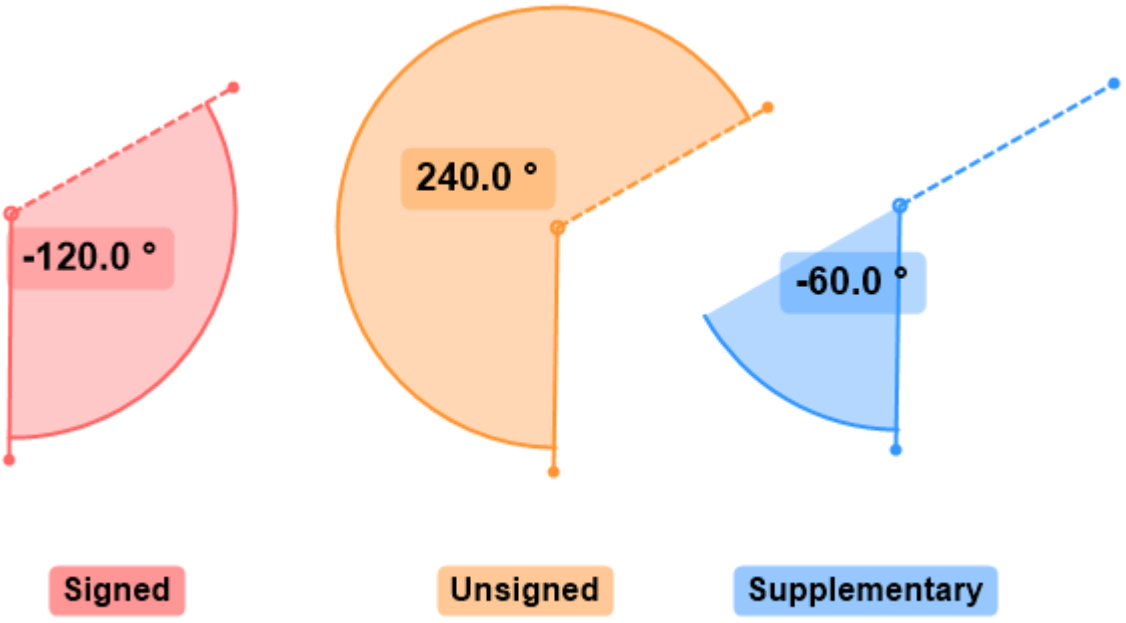
The unit for angles can be configured in **Preferences ▸ Playback ▸ Units**. The options are `Degrees (°)` and `Radians (rad)`.

Angle

To measure an angle, add an angle object and position its end points.



By default the angle runs counter-clockwise, starting from the dashed leg. The context menu options let you switch between signed or unsigned angle, change from counter-clockwise to clockwise, and switch to supplementary angle.



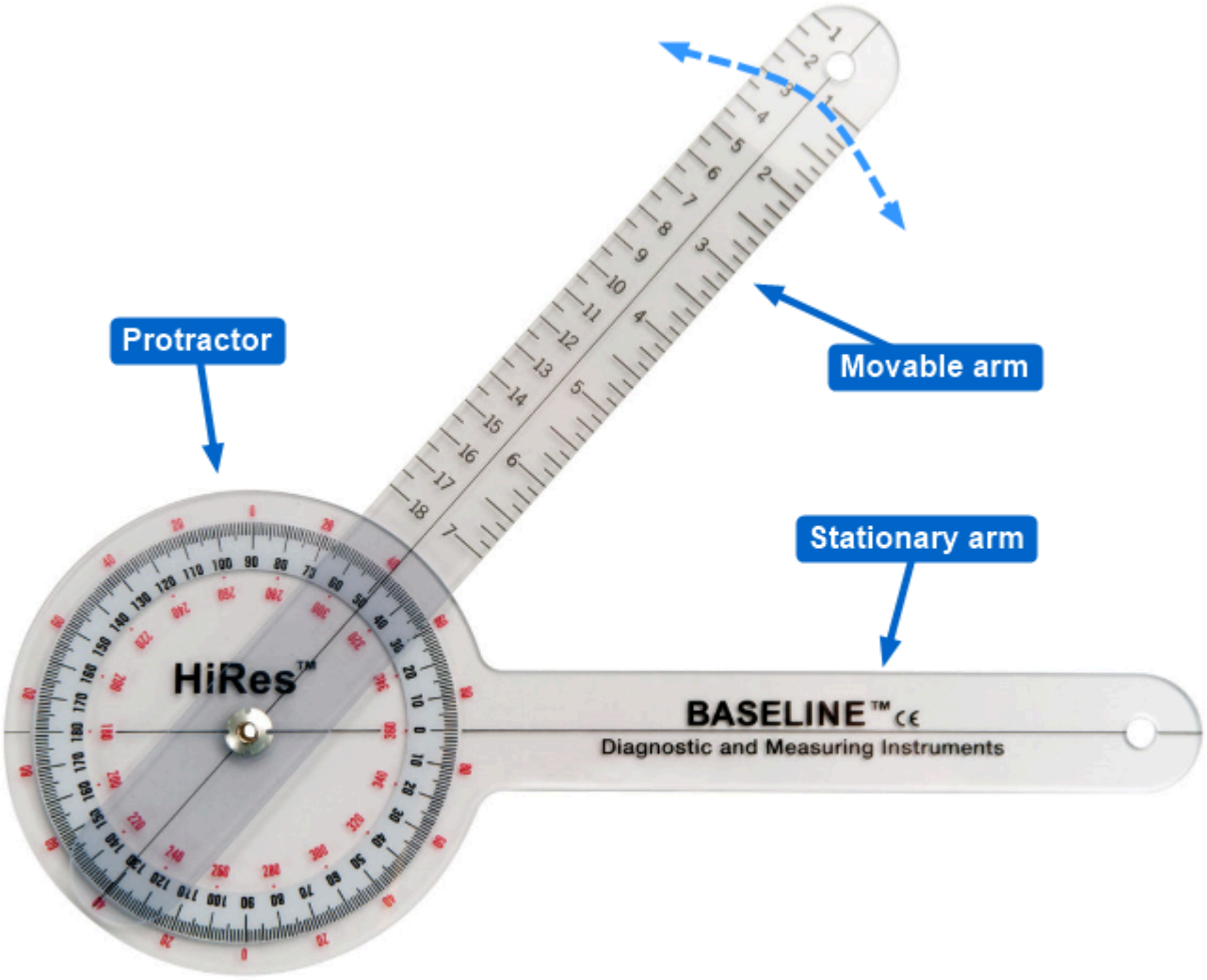
Note

Always keep in mind that it is not possible to measure angles in arbitrary space from a 2D image. Angles can only be measured when the three points lie on a known 2D plane. This must be either the image plane or the plane calibrated using plane calibration.

Goniometer

The goniometer tool lets you measure the extension or flexion of a body segment relative to a referenced anatomical angle or neutral position.

A physical goniometer combines two arms and a protractor. One arm is called the stationary arm and the other is called the movable arm. The protractor part contains multiple graduated rings that allow the physician to pick a reference axis when reading the angle.

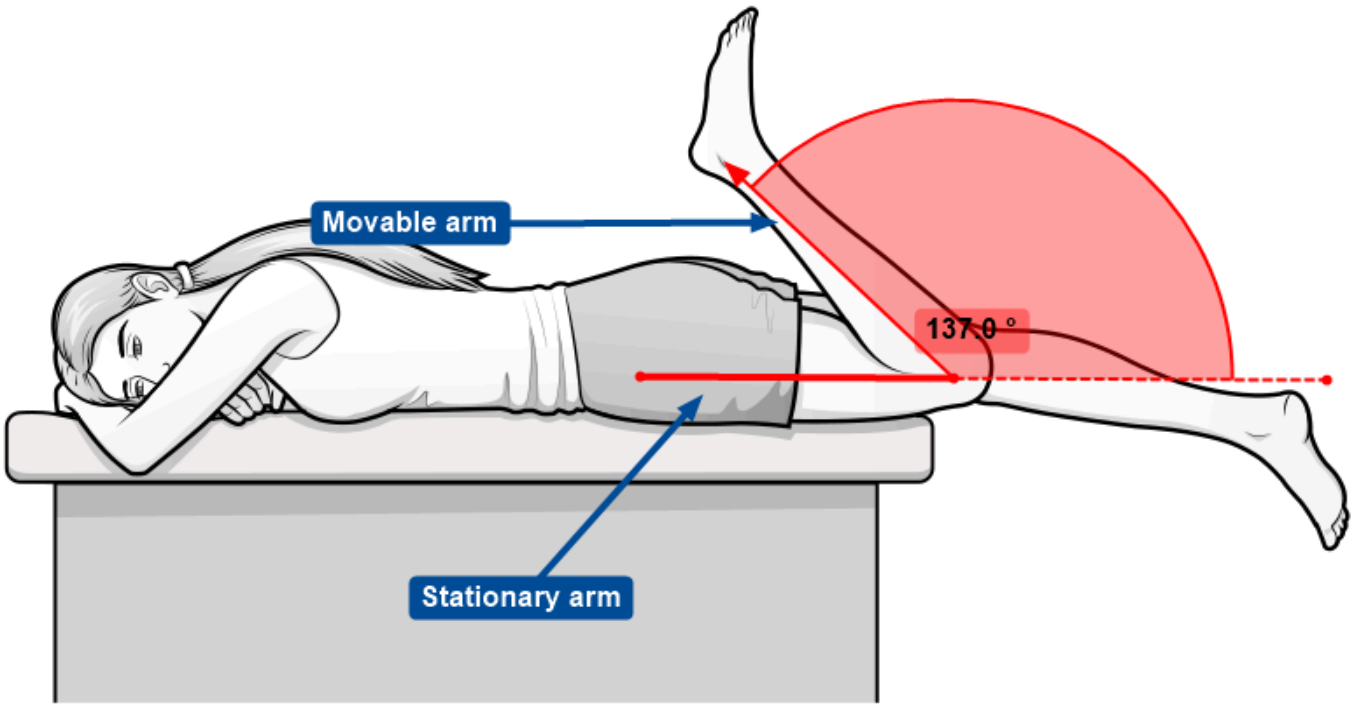


Typical plastic goniometer

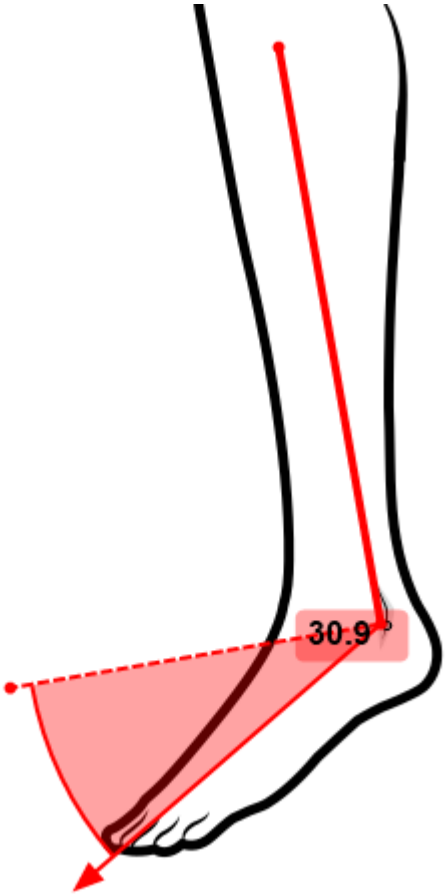
The stationary arm is aligned with the reference segment to materialize the neutral position and the movable arm is aligned with the segment for which we are measuring the range of motion.

Using a goniometer instead of a simple protractor makes it easier to align the protractor with the body segments and helps standardize the way the measurements are made for the range of motion of specific joints.

In Kinovea, the goniometer tool has three branches: The stationary arm is the thick plain arm. The movable arm is the one with the arrow at the end. The dashed line is used to define the protractor reference axis in relation to the stationary arm. This branch rotates by 45° increments relatively to the stationary arm.



Knee flexion.

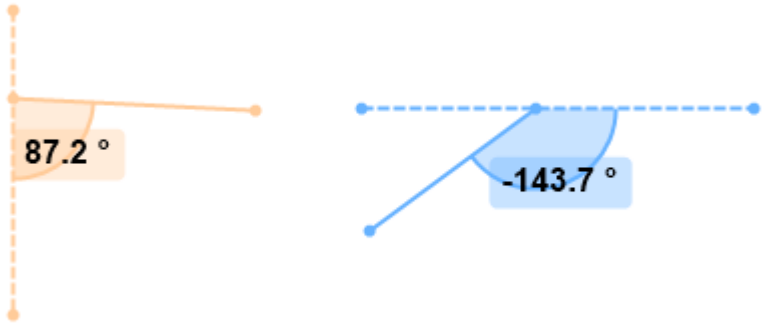


Plantar flexion. The reference axis is set perpendicular to the leg.

This tool is conceptually similar to a real goniometer with 8 protractor rings but the reading is simplified by showing only one measurement at a time.

Angles relative to image axes

The angle-to-horizontal and angle-to-vertical tools let you measure angles relatively to the horizontal or vertical axes of the image. The dashed line represent the reference axis.



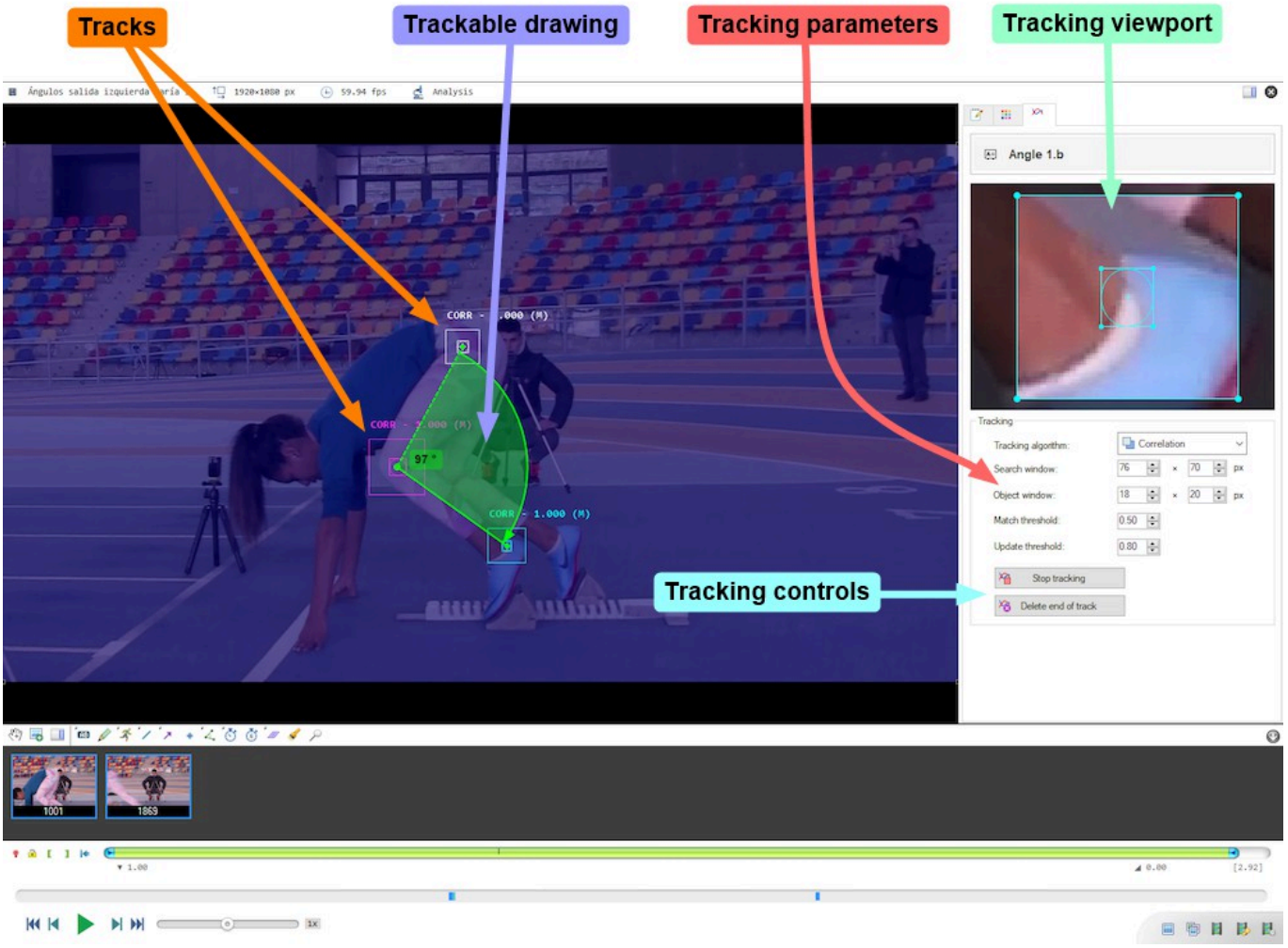
Tracking

Tracking

This section covers the features related to tracking points, objects, angles, polylines and other drawings, establishing moving coordinate systems and the estimation of the camera motion.

By itself tracking does not require calibration or a perfect frontal view. It is possible to get qualitative insights from the visual path of points or objects over time without calculating real world kinematics. To calculate kinematics quantities such as speed or acceleration, calibration is necessary.

The following screenshot illustrates the main user interface for tracking points and drawings:



Tracking of points, objects and people

- The fundamental tool of tracking is the “track”, visually represented by trajectory objects.
- Kinovea implements several tracking algorithms with their own parameters.
- Trackable drawings like angles, polylines or markers can have tracks attached to them.

- For full body tracking, when manual digitization is too time consuming and template-based tracking fails, we can turn to markerless tracking using external machine learning frameworks.

Moving camera or moving coordinates

- The coordinate system or the calibration objects themselves can be tracked.
- 2D image stabilization uses a track to compensate for camera or object translation.
- To estimate more complex camera motion there is a dedicated mode with its own interface.
- Camera compensated annotations stick to the real world despite camera motion.

Trajectory object

Tracking algorithms

Trackable drawings

Markerless tracking

Tracked coordinates

Camera motion estimation

Camera compensated annotations

Kinematics

Kinematics

Introduction

To measure linear or angular speed and other kinematics quantities follow these steps:

- Establish a line or plane calibration.
- Optionally compensate for lens distortion.
- For high speed videos make sure time is calibrated correctly as well.
- Track a point trajectory or track an object containing an angle.
- The measured data can be displayed in two ways:
    - As a small label on the object in the viewport.
    - In a dedicated kinematics diagram.
- To export the data, use the export options in the kinematics diagrams.

Measurement labels

To display the value as a label attached to the object, right click the object and choose an option under the **Display measure** menu. Each measurable object has specific options based on the quantities it can measure.

Kinematics diagrams

The following kinematics diagrams are available:

- Scatter diagram
- Linear kinematics
- Angular kinematics
- Angle-angle diagram

Data filtering

Rationale

Due to the digitization process the raw coordinates are noisy and the resulting quantities, especially derivatives like speed and acceleration, are less accurate than they could be. Carefully filtering the coordinates remove a lot of this noise and provide more accurate measurements.

Data shown in the kinematics diagrams is computed using filtered coordinates. This filtering can be disabled under Preferences ▸ Drawings ▸ General ▸ Enable coordinates filtering.

Filtering mechanism

The coordinates are passed through a low pass filter to remove noise. The filter does two passes of a second-order Butterworth filter. The two passes (one forward, one backward) are used to reset the phase shift [1].

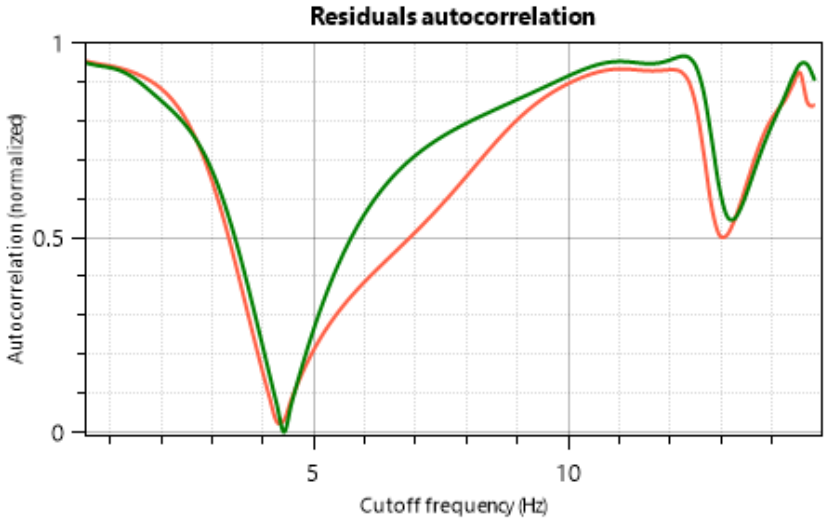
To initialize the filter, the trajectory is extrapolated for 10 data points on each side using reflected values around the end points. The extrapolated points are then removed from the filtered results [2].

The filter is tested on the data at various cutoff frequencies between 0.5Hz and the Nyquist frequency. The best cutoff frequency is computed by estimating the autocorrelation of residuals and finding the frequency yielding the residuals that are the least autocorrelated. The filtered data set corresponding to this cutoff frequency is kept as the final result [3].

The autocorrelation of residuals is estimated using the Durbin-Watson statistic.

Cutoff frequency visualization

For trajectories, the cutoff frequency can be visualized in the About tab of the diagram dialog. The calculated cutoff frequency depends on the data and is different for each trajectory object.



Cutoff frequencies (Hz)

Source	X	Y
Trajectory 1	4.266	4.411

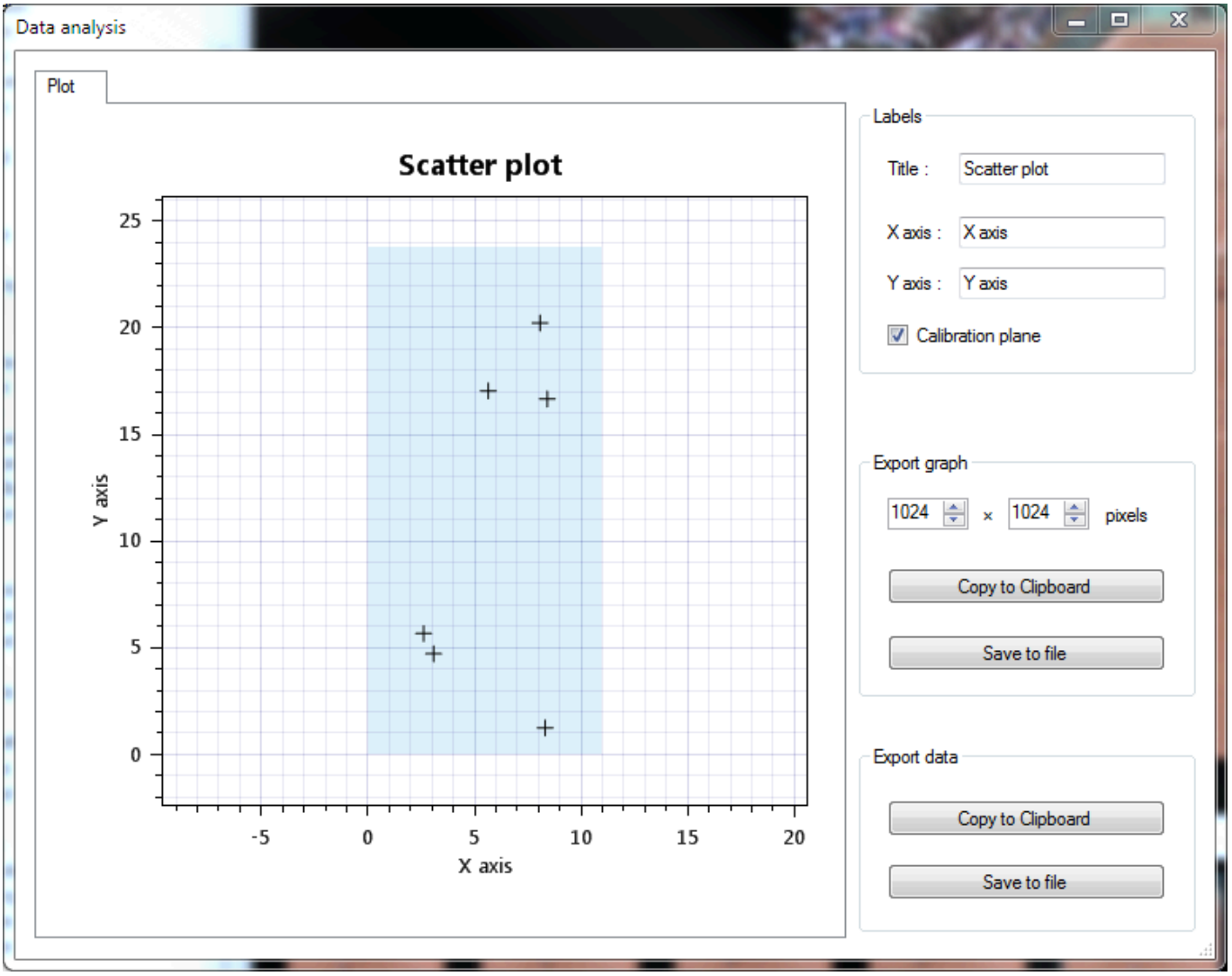
Footnotes

[1] Winter, D. A. (2009). Biomechanics and motor control of human movements (4th ed.). Hoboken, New Jersey: John Wiley & Sons, Inc.

[2] Smith G. (1989). Padding point extrapolation techniques for the butterworth digital filter. J. Biomech. Vol. 22, No. s/9, pp. 967-971.

[3] Challis J. (1999). A procedure for the automatic determination of filter cutoff frequency for the processing of biomechanical data., Journal of Applied Biomechanics, Volume 15, Issue 3.

Scatter diagram



Description

The scatter diagram displays the positions of the marker objects in the video. The axes are based on the calibrated coordinate system.

This can be used to get a bird's eye view of the contact points of a ball on the ground for example.

Data source

The data source is the whole set of marker objects.

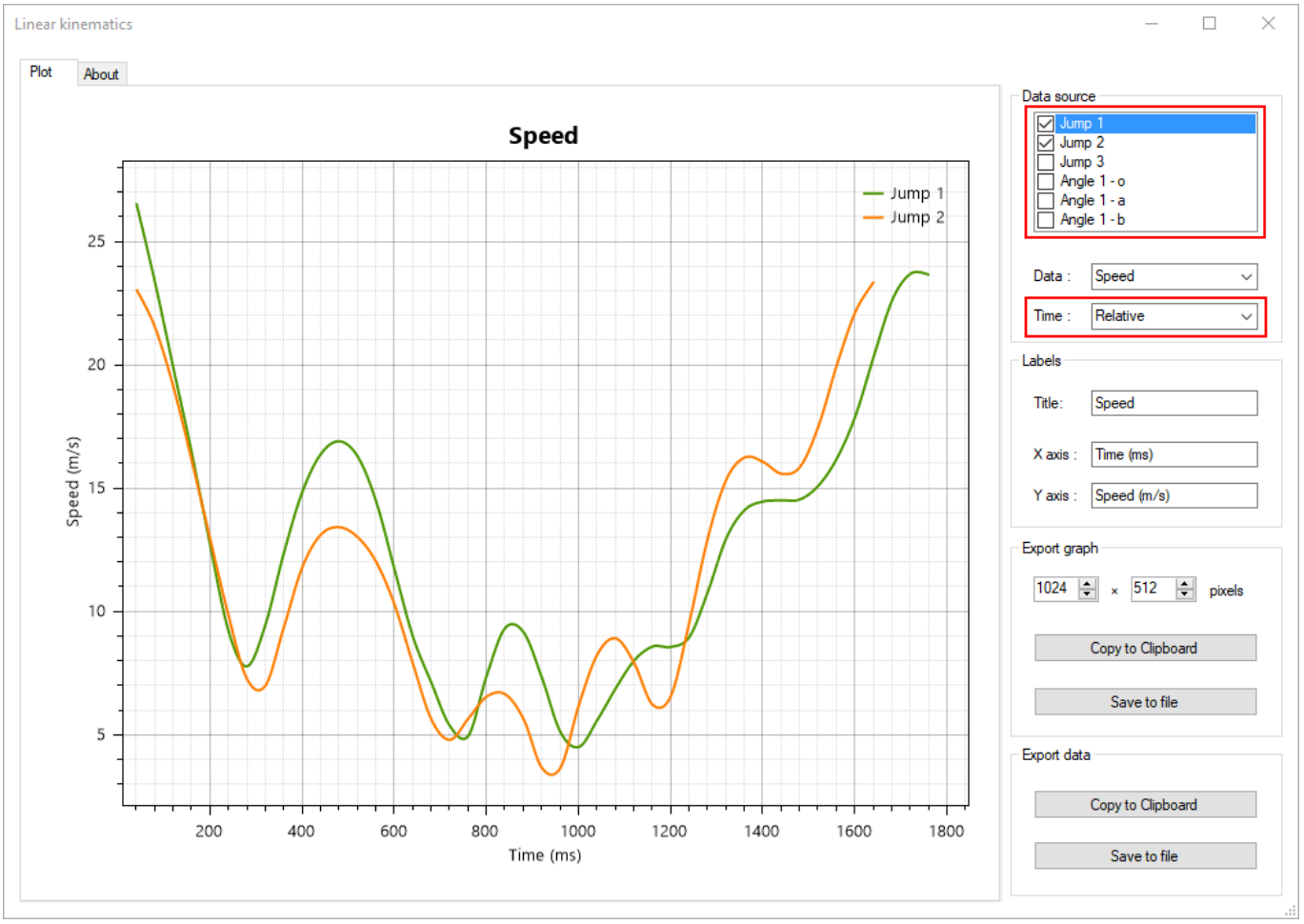
Calibration plane

This option highlights the area covered by the perspective grid used to calibrate the coordinate space.

Export options

The data can be exported to an image or to tabular data. For tabular data the points are sorted by time; the first column is the time in milliseconds, the second and third columns are the X and Y positions.

Linear kinematics



Description

The linear kinematics diagram displays line charts of points trajectories and tracked objects points.

This can be used to study and compare the evolution of position, speed, or acceleration over time.

Data source

The data source list contains a list of all tracked points. Data from individual points can be hidden from the line chart by unchecking the checkbox in front of the point name.

Points in drawings with multiple points are listed by the name of their drawing and a point identifier.

Data type

The Data type drop down sets the kinematic quantity used in the line chart. The following options are available:

- Horizontal position
- Vertical position
- Total distance
- Total horizontal displacement
- Total vertical displacement
- Speed
- Horizontal velocity
- Vertical velocity
- Acceleration
- Horizontal acceleration
- Vertical acceleration

Total horizontal displacement and total vertical displacement computes the resultant displacement between the first point and the current point of the trajectory. This is similar to the horizontal and vertical position but makes it relative instead of absolute.

> **Note**
>
> Be mindful that acceleration values are very sensitive to noise as they are the second derivative of the digitized position.

Time type

The Time type drop down sets the time scale used in the line chart. The following options are available:

- Absolute : times are in normal video time.
- Relative : times are expressed relative to the start of each trajectory.
- Normalized : times are expressed as a fraction of each trajectory duration.

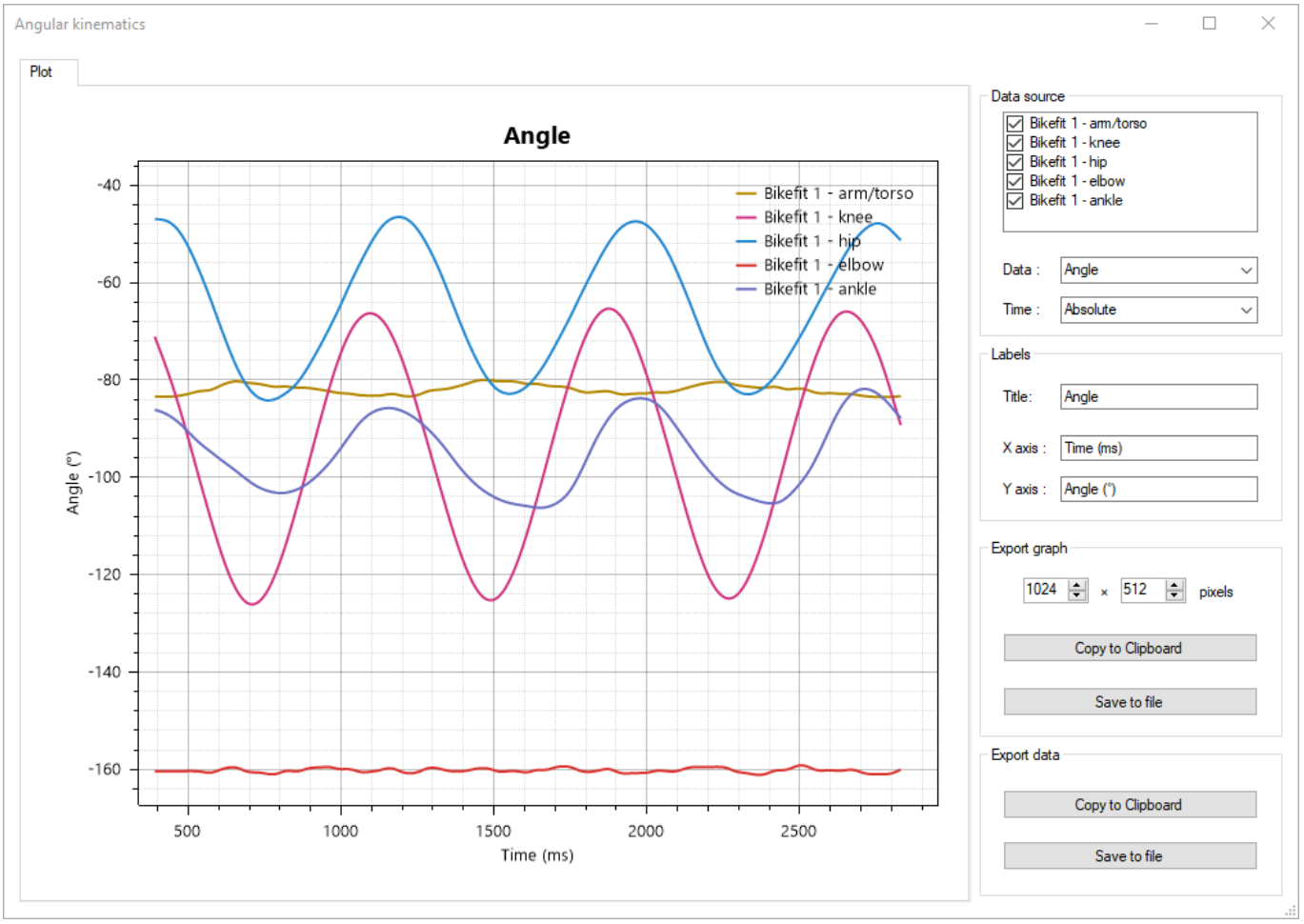
Relative time aligns events to their start, this can be used to compare trajectories that did not occur at the same time in the video.

Normalized time can be used to compare the shape of time series that did not last the same duration.

Export options

The data can be exported to an image or to tabular data. For tabular data the points are sorted by time; the first column is the time in milliseconds, the other columns are values from the time series.

Angular kinematics



Description

The angular kinematics diagram displays line charts of tracked angles.

This can be used to study and compare the evolution of angle, angular speed or angular acceleration over time.

Data source

The data source list contains a list of all tracked angles. Data from individual angles can be hidden from the line chart by unchecking the checkbox in front of the point name.

Data type

The Data type drop down sets the kinematic quantity used in the line chart. The following options are available:

- Angle
- Total displacement
- Angular velocity
- Tangential velocity
- Angular acceleration
- Tangential acceleration
- Centripetal acceleration
- Resultant acceleration

Note

Be mindful that acceleration values are very sensitive to noise as they are the second derivative of the digitized position.

Time type

The Time type drop down sets the time scale used in the line chart. The following options are available:

- Absolute : times are in normal video time.
- Relative : times are expressed relatively to the start of each trajectory.
- Normalized : times are expressed as a fraction of each trajectory duration.

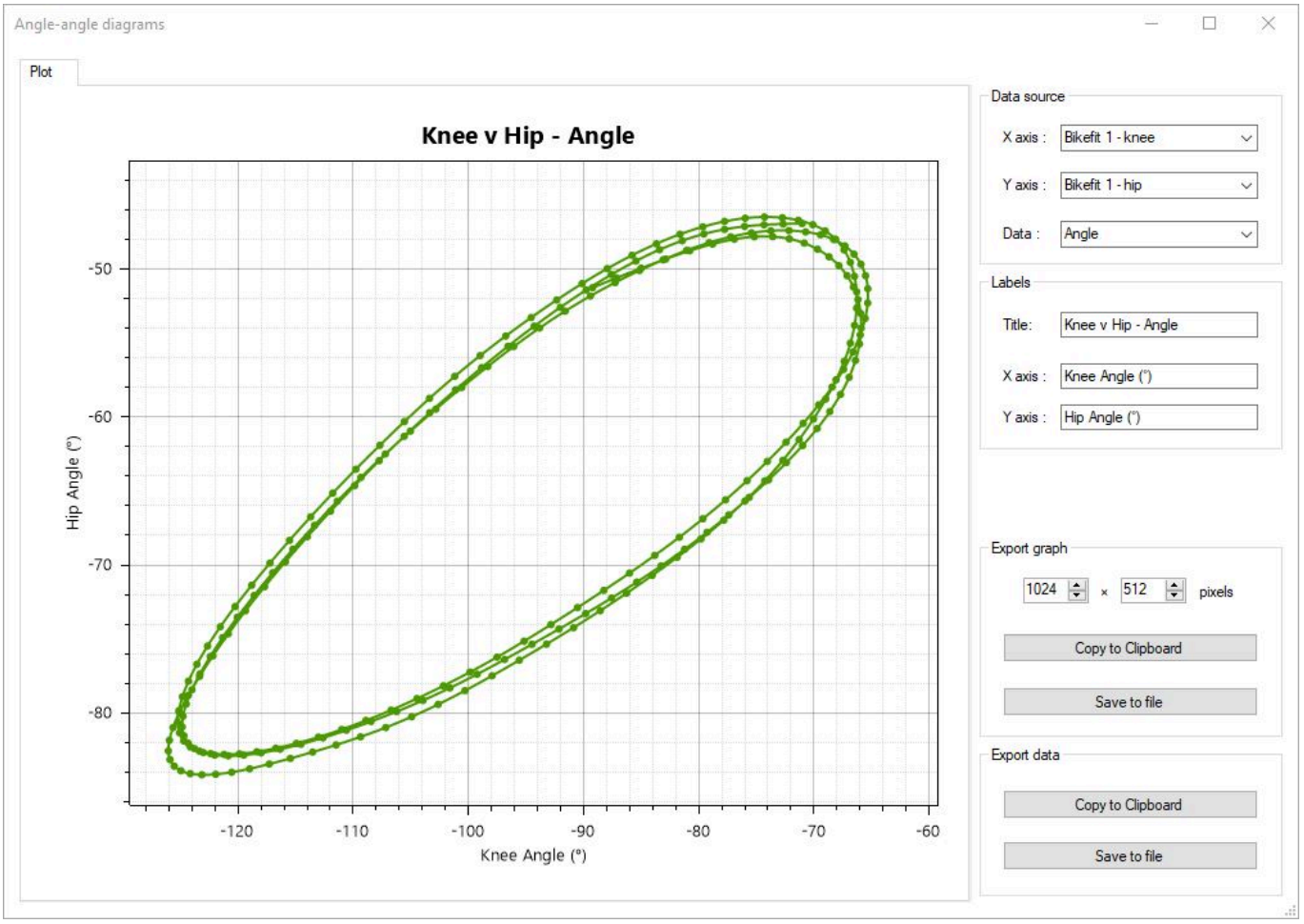
Relative time aligns events to their start, this can be used to compare trajectories that did not occur at the same time in the video.

Normalized time can be used to compare the shape of time series that did not last the same duration.

Export options

The data can be exported to an image or to tabular data. For tabular data the points are sorted by time, the first column is the time in milliseconds, the other columns are values from the time series.

Angle-angle diagram



Description

The angle-angle diagram is a qualitative plot showing the dynamics of a movement pattern over multiple trials. This type of diagram correlates two angular measurements over time to highlight their interaction.

For example, this can be used to study the coordination of the knee and hip during flexion or extension.

Data source

The data source lists the angle objects that are plotted on each axis.

Data type

The data type drop down sets the kinematic quantity used for the variable-variable chart. The following options are available:

- Angle
- Angular velocity
- Tangential velocity

Export options

The data can be exported to an image or to tabular data. For tabular data, the output has only two columns, correlating the first angle value with the second value at the same point in time.

Export

Export

Exporting videos

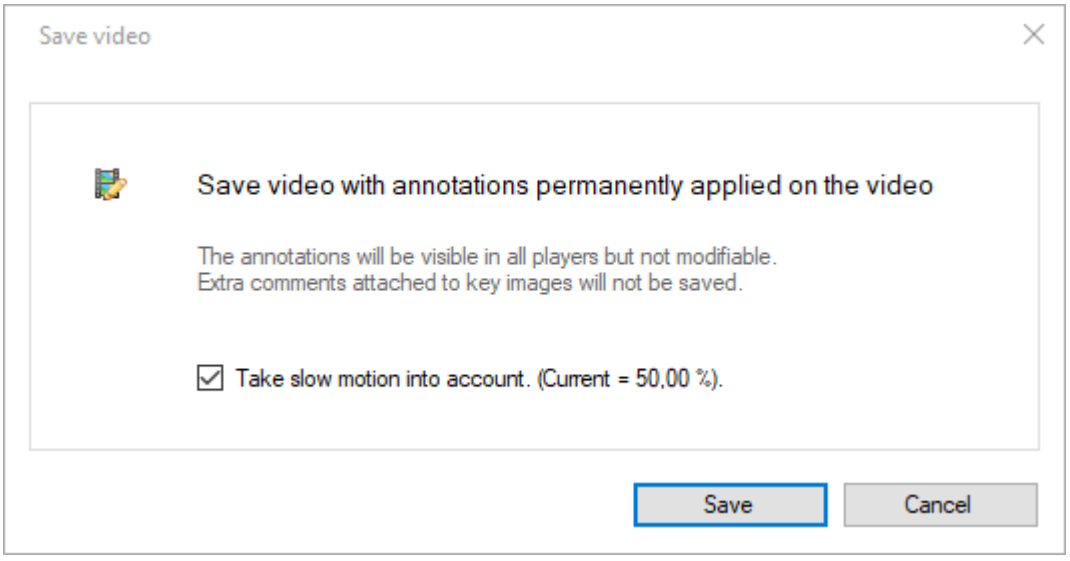
The video and image export options can be accessed from the export toolbar in the playback screen.



Export a video with annotations painted on

To export the current working zone as a new video use the Video export button, the menu File ▸ Export video, or the context menu Export video.

This will open an export dialog with an option to take the slow motion into account. If checked the resulting video will use the current speed slider to determine the output framerate.



The output video format can be selected in the file name selection dialog between MKV (Matroska), MP4, or AVI. The video codec is always MJPEG.

Note

Drawings added to the video are painted on the output images and will no longer be modifiable. Comments added at the key image level and other non directly visible annotations are not saved in the output.

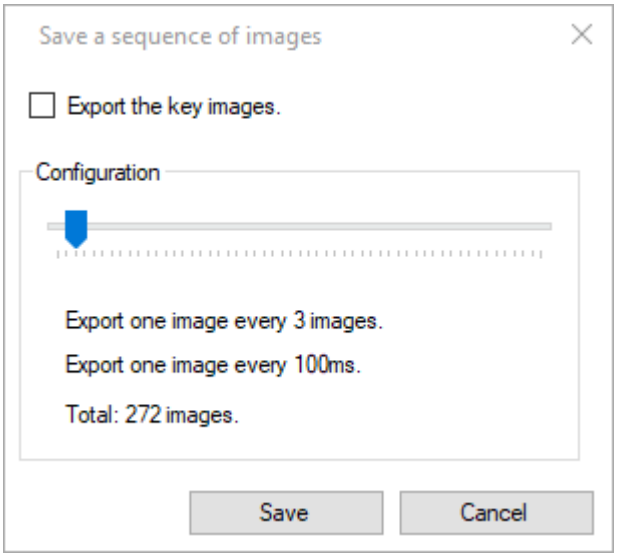
Export an image

To export the current image use the Export image button or the Export image entry of the context menu.

The output image format can be selected in the file name selection dialog between JPEG, PNG, or BMP.

Export an image sequence

To export a series of images from the video, use the Export sequence button in the playback screen. This brings up a dialog to configure the frequency at which the images are taken from the video.



If the checkbox Export the key images is checked, the frequency slider is ignored and only the key images are exported.

Export a dual video or dual image

When using two playback screens, use the Export video or Export image buttons in the joint controls to create a single output containing both input side by side.

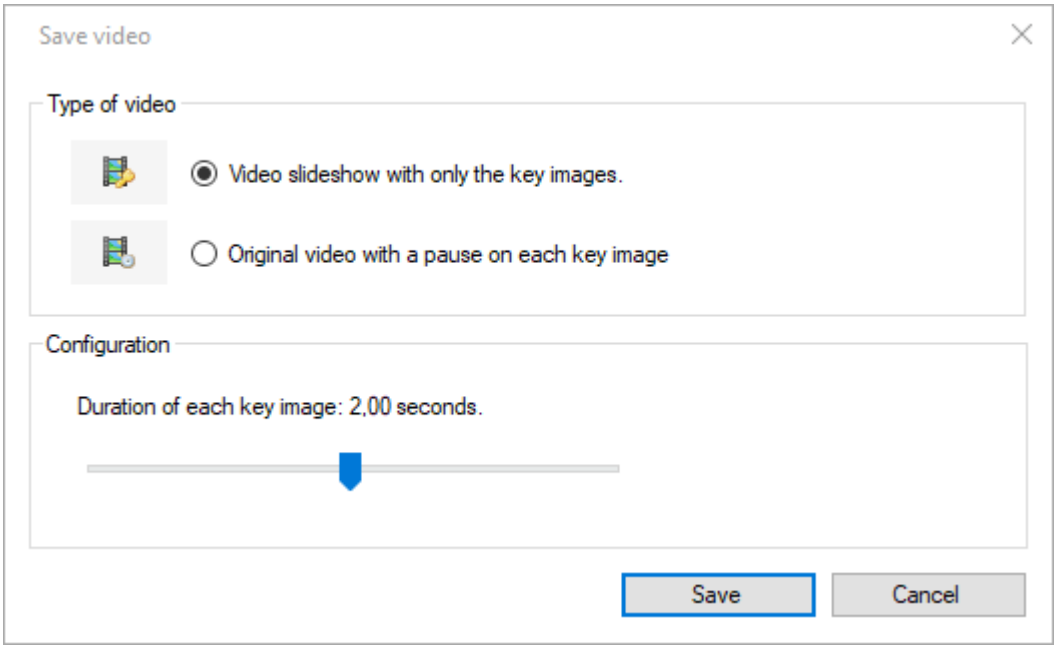
The input videos will be combined frame by frame using the configured synchronization point.

Export a slideshow or a video with pauses

To export a slideshow video where only the key images are shown, use the Export slideshow button.

To export a video where the video is paused on key images but otherwise plays the action normally, use the Export paused button.

Both options bring up the same dialog to configure the duration of each key image in the output video.



Exporting images

Exporting text documents

Exporting spreadsheet documents

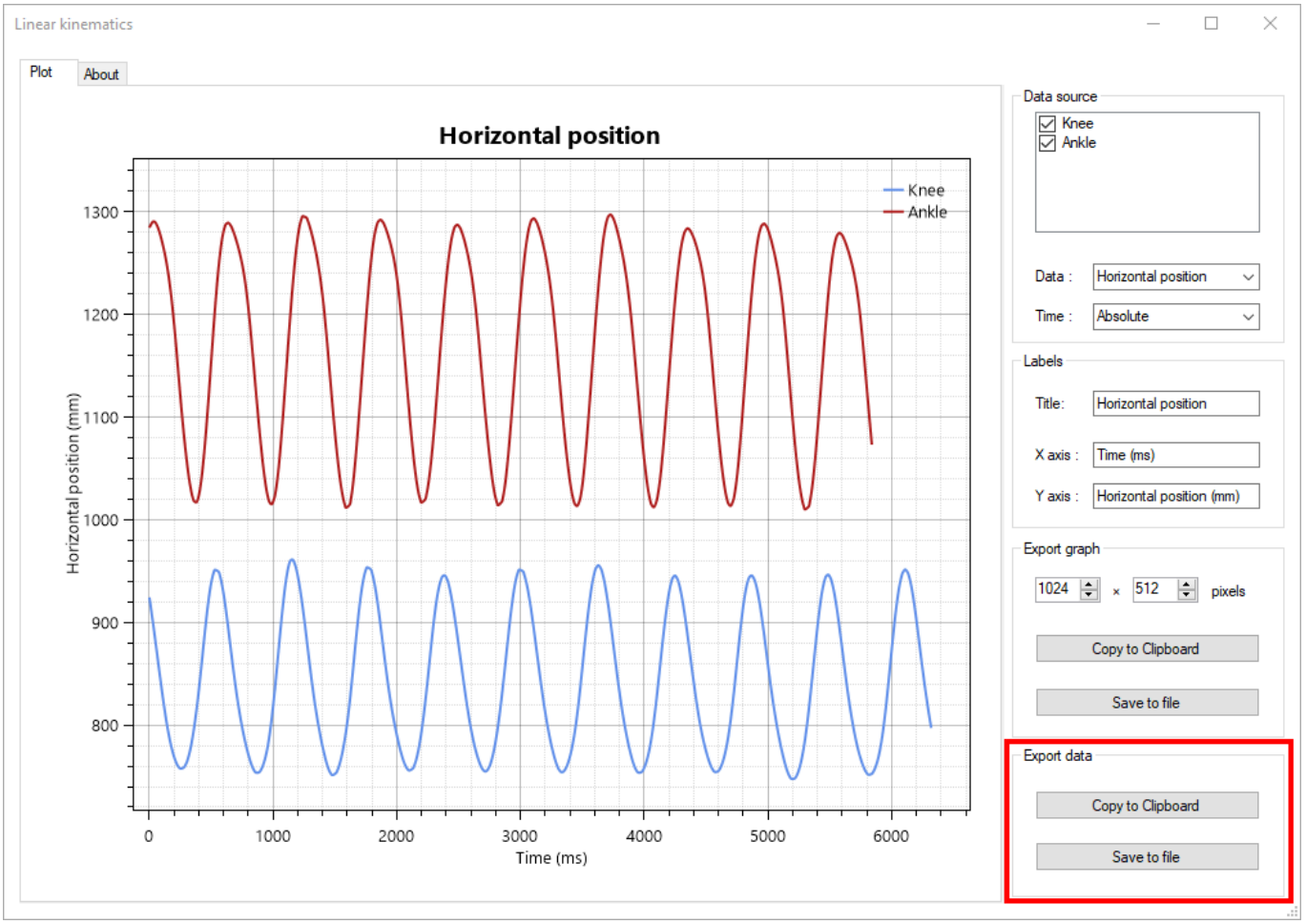
Save annotations

To save annotations use CTRL + S or the menus **Save** or **Save as...**. This saves annotations to a KVA file, the native annotation format of Kinovea.

Export measurements from kinematics dialogs

To export measurements such as point positions, linear velocities, or angular accelerations, use the kinematics dialogs. The kinematics dialogs are found under the **Tools** menu: **Scatter diagram**, **Linear kinematics**, **Angular kinematics**, **Angle-angle diagram**.

To export data use the export options at the bottom right of the dialogs.



This will export the data in CSV format, either to the clipboard or to a file. The first column is the time, either in milliseconds or normalized, and the other columns are the data sources.

The measurements displayed and exported from the kinematics dialog use the filtered coordinates. The filtering process is described in the About box of the `Linear kinematics` dialog. You can control filtering from the preferences at `Options ▸ Preferences ▸ Drawings ▸ General ▸ Enable coordinates filtering`.

Export data to spreadsheet

Another option to export data is to use the converters menus under `File ▸ Export to spreadsheet`.

The following options are available:

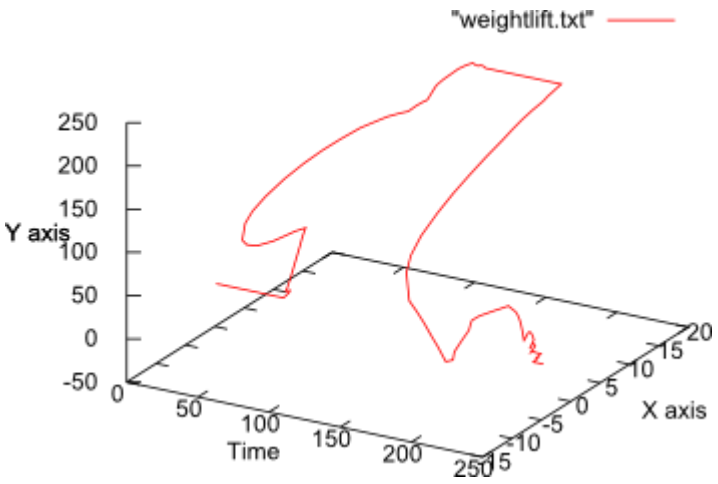
- `LibreOffice (.odf)`
- `Microsoft Excel (.xml)`
- `Web (.html)`
- `Gnuplot (.txt)`

The underlying mechanism for these menus is to **convert** the annotation data into the output format: it does not perform any higher level computations. This approach has the following differences with the export from the kinematics dialogs:

- The coordinates do not use filtering.
- Only the coordinates are exported, not any higher level measurements like speed or acceleration.
- Each object is exported independently in its own table.
- Key images times are exported.
- Stopwatches are exported.
- The time column uses the configured timecode format and may not be numerical in nature.

To use Gnuplot to plot the trajectory data on a 3D graph with time as the third dimensions, you can use the following commands:

```
gnuplot> set xlabel "Time"
gnuplot> set ylabel "X-AXIS"
gnuplot> set zlabel "Y-AXIS"
gnuplot> set ticslevel 0
gnuplot> splot "weightlift.txt" using 1:2:3 with lines
```



Citation examples

If you used Kinovea in your research we would very much appreciate it if you included it in your bibliography.

You can find examples of formatted citations in the About dialog under the Citation tab.

Note

Kinovea is an open source project and is not published by a company, thus there is no meaningful “city” or “country” of origin as is sometimes requested by journals for software references.

Capture

Introduction

Supported devices

Kinovea supports the following broad categories of cameras and capture devices:

- DirectShow compatible devices.

  Webcams and other USB 2.0 cameras, HDMI capture cards and IEEE1394 camcorders.

- Machine vision cameras.

  High quality and/or high speed capture devices. USB 3.0 Vision or GigE Vision devices compatible with the GenICam standard.

- Network cameras and smartphones.

  Anything that can serve MJPEG streams (RTSP streams are not supported).

- Camera simulator.

  A virtual device for testing purposes.

These sub-systems work alongside each other and you may connect multiple devices of different types at the same time.

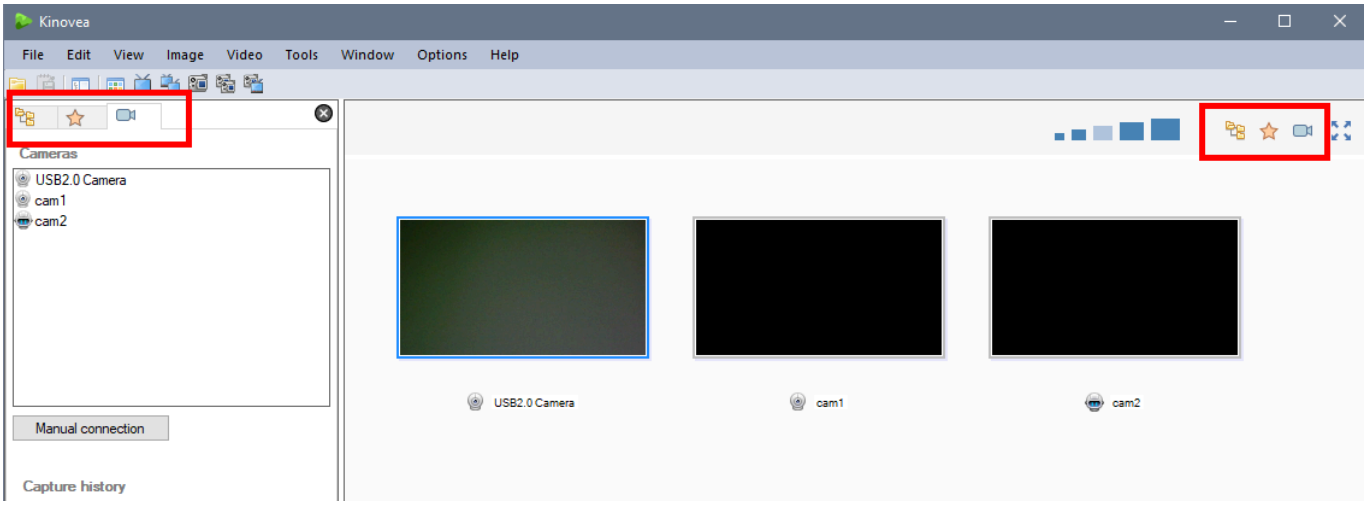
You can actively stream up to two cameras in a single instance of Kinovea. If you need more than two cameras or if you prefer to dedicate each camera to its own window, you can run multiple instances of Kinovea at the same time and connect one in each instance.

Listing connected cameras

While no capture or player screen is loaded, the main area of Kinovea displays the browser view.

To view the list of cameras currently connected to the computer select the camera tab in the navigation pane. Kinovea will start monitoring the system to detect connected cameras and the browser view will display thumbnails of the discovered cameras.

When this tab is selected and no capture screen is yet open, Kinovea will constantly monitor the system to detect new cameras. When a capture or player screen is open, Kinovea stops monitoring for new cameras.



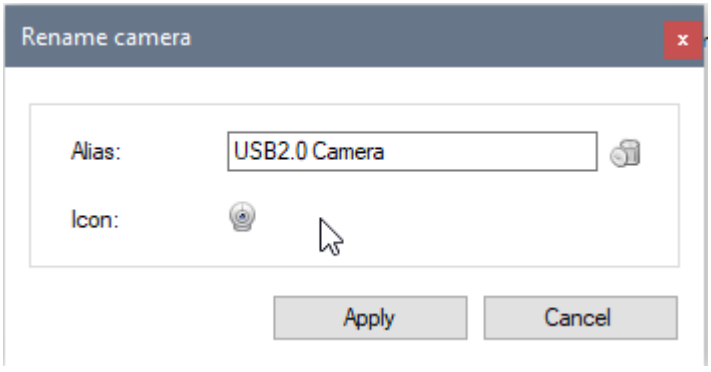
Opening a camera

You can open a camera by doing any of the following:

- Double click on the camera thumbnail in the browser view,
- Double click on the camera name in the camera list view,
- Drag and drop the camera from the camera list view to the browser view,
- Drag and drop the camera from the camera list view into an already opened capture screen,
- Right click the camera thumbnail and choose **Open**.

Camera name and icon

You can rename or change the icon for the camera by right-clicking the thumbnail and choosing **Rename**.



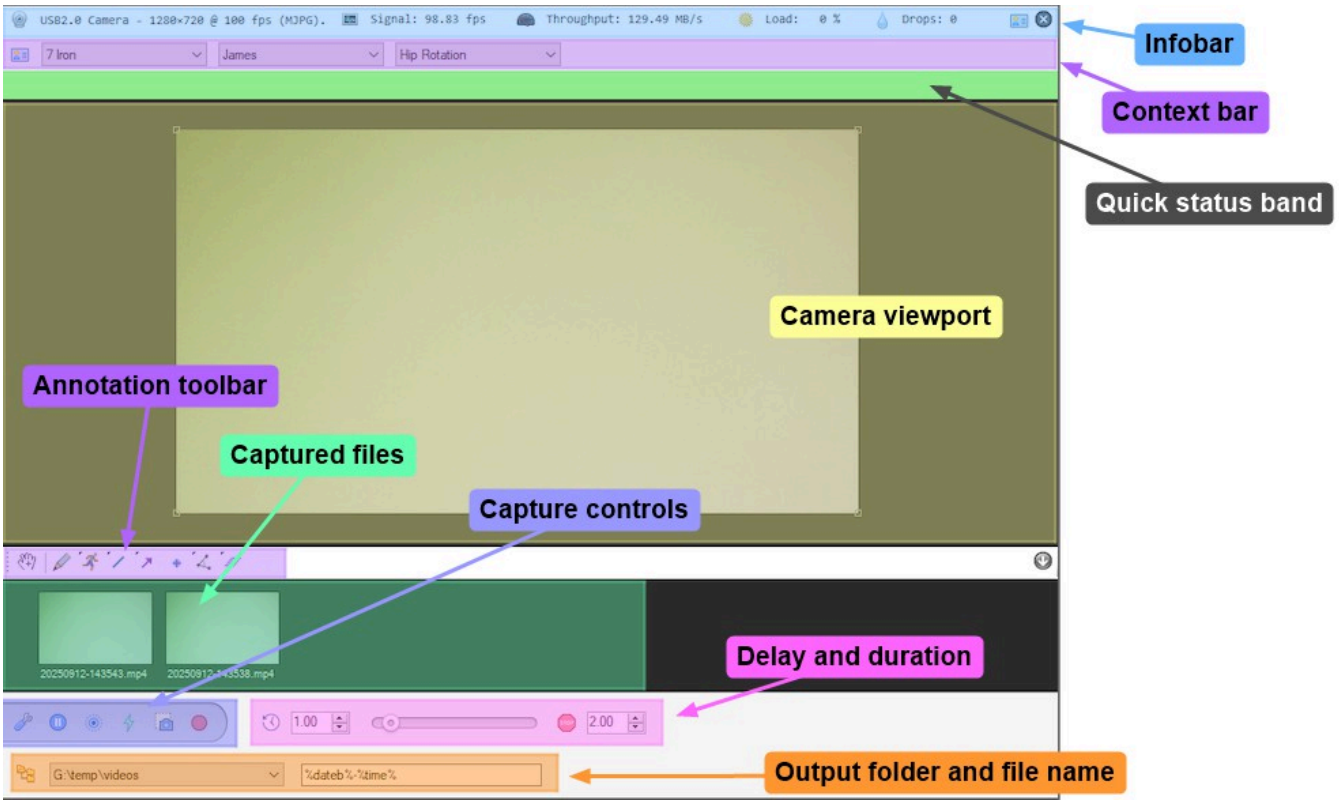
Camera settings

When you modify the settings of a camera, these settings are saved and will be applied automatically the next time you open that camera.

You can reset these settings by right-clicking the thumbnail and choosing **Forget custom settings**.

Capture screen user interface

The capture screen is divided in the following areas:



At the top:

- The Infobar contains information about the connected camera and streaming performances.
- The Context bar has drop downs of context variables to tag the captured files.
- The Quick status band gives a visual indication of the current status for recording.

In the middle:

- The Camera viewport is where the camera image and drawings are visible.

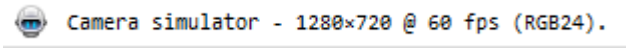
At the bottom:

- The Annotation toolbar contains drawing tools usable on the capture screen.
- The List of recently captured files displays thumbnails of the files recorded in the session.
- The Capture controls area has buttons to configure the camera, pause or restart the stream, switch between live and delayed view, arm and disarm the capture trigger, save an image, and start/stop recording.
- The Delay and duration area controls the delay of the stream and the duration of recordings.

- The Output folder and file name area lets you pick the output folder and define the file naming pattern of recorded files.

Infobar

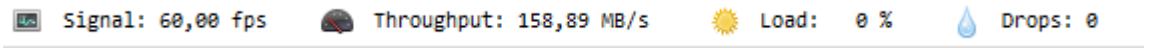
The infobar contains information about the connected camera and streaming performances.



The first part of the infobar displays the alias of the camera and the current configuration for image size, frame rate and image format. Clicking in this area of the infobar will bring up the camera configuration dialog.

The frame rate indicated is the one configured, the actual frame rate sent by the camera might be different for various reasons like low light levels or hardware limitations.

The second part of the infobar displays the following live statistics:



Signal (fps)

This is the frequency at which Kinovea is receiving images from the camera. The value is in frames per second.

Many cameras will reduce their frame rate based on various external factors. For example, when a camera uses auto-exposure and the exposure duration computed by the device is incompatible with the selected frame rate.

Throughput (MB/s)

This is the amount of data that passes through Kinovea as it processes the stream. The value is in megabytes per second, with the convention of one megabyte as 1024 kilobytes.

This value is related to the the image size, frame rate and format, possible on-camera image compression, link bandwidth, and possible post-processing done at the driver level.

You can use this value to estimate the necessary speed for your storage medium to write the uncompressed stream.

In the case of a non-compressed stream in RGB24 format, the value is calculated as follows:

```
Throughput = (width × height × 3 × frame rate) / (1024*1024)
```

Load (%)

This value describes how much Kinovea is struggling to keep up with the camera framerate. It is computed as the time taken to process one frame divided by the interval between frames.

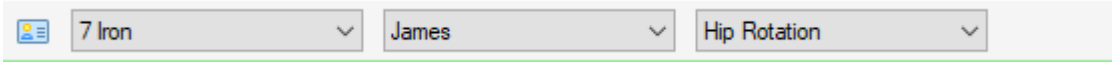
When this value is near 100% it means it takes Kinovea the same amount of time to process one frame as the time budget it has for that frame, if it goes over 100% dropped frames may occur.

Drops

This is the number of frames that could not be processed by Kinovea during the current or last recording session.

If this value is non-zero, some frames are missing from the output video and any measurements involving time will not be perfectly accurate.

Context bar



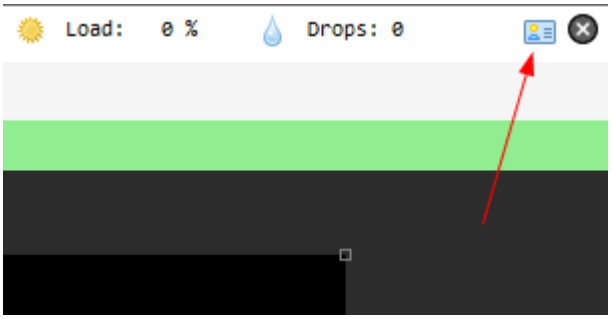
The context bar contains drop down lists of context variables. It is only visible if any variables have been configured.

Context variables are used to tag the captured files with information like the name of the person being filmed, the type of exercise being performed, the equipment being used or any other relevant context.

The variables are configured from Options ▸ Context.

These variables can be used in the naming of the captured files and in the path of the capture folder to create a self-structuring file hierarchy. See Context variables.

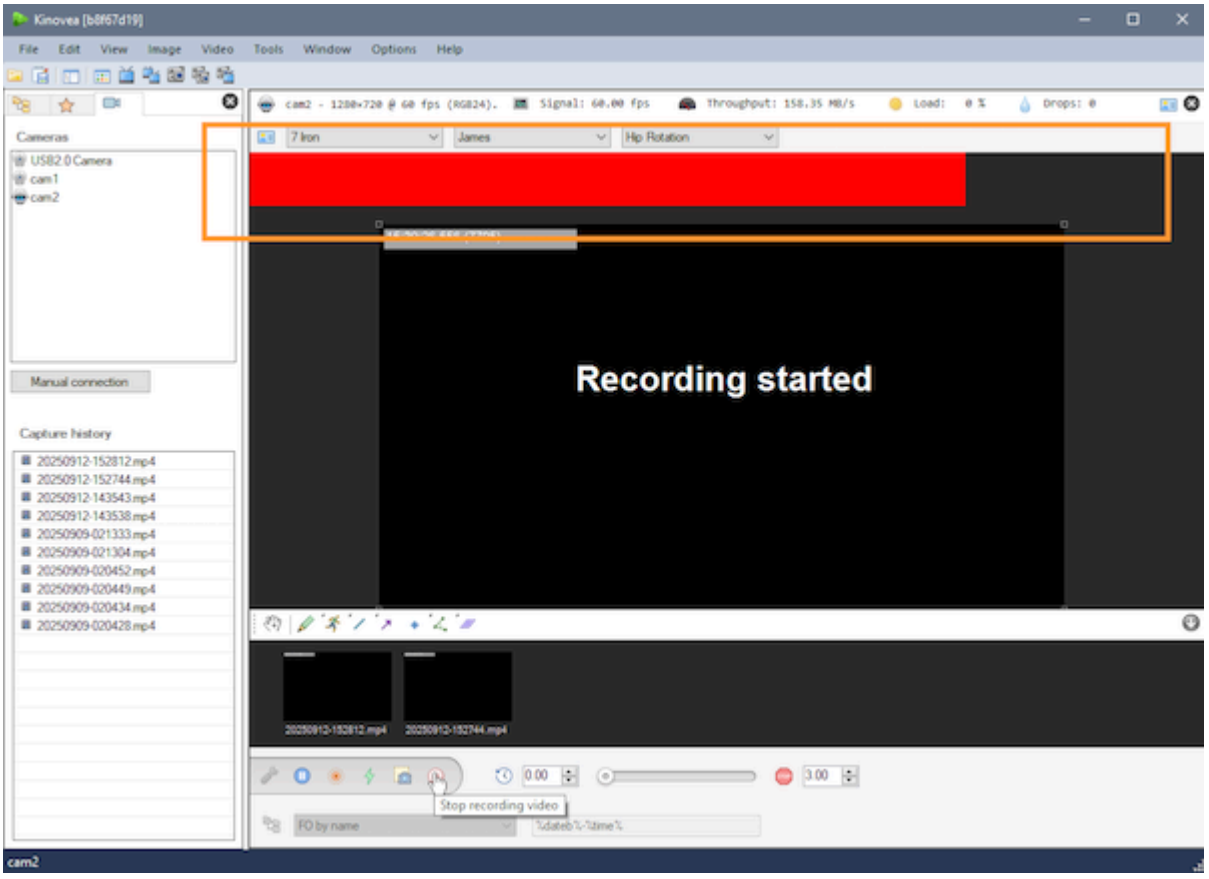
To disable the context bar and ignore the context variables, click the identity card icon in the top right corner of the capture screen.



If the context bar is disabled but the path or name of the captured files contains context variables, the recording will be cancelled and a warning message will be displayed.

Quick status band

The quick status band is a colored band at the top of the viewport that gives a quick visual indication of the current status of the capture screen.



The color codes are as follows:

Color	Status
Purple	The camera is not connected or there is no camera selected
Yellow	The camera is connected but the stream is paused

Color	Status
Blue	The camera is streaming, but the capture trigger is disarmed or in the idle period
Green	The camera is streaming, and the capture trigger is armed
Red	Recording is in progress

If the recording has a maximum duration the quick status bar will progressively reduce in size as the recording approaches the maximum duration.

Camera viewport

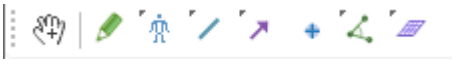
The camera viewport is the main area where the camera image is visible.

The image itself can be moved around by dragging with the mouse and resized using the manipulators at the corners of the image or by using `CTRL` + **mouse scroll**. Drawings on the capture screen can go outside the image area.

If the image stays black, there might be a problem with the available USB bandwidth or power, or the exposure duration might be too short.

If nothing is visible at all, not even the black image rectangle, the camera did not connect correctly; for example, the camera might be in use in another application at the same time.

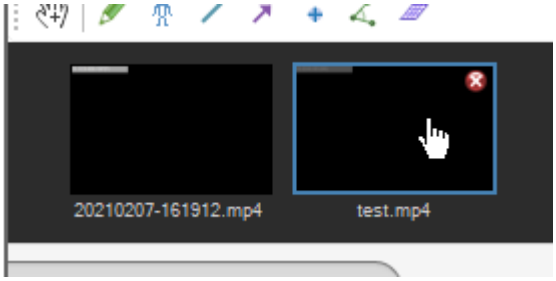
Annotation toolbar



The annotation toolbar contains drawing tools usable on the capture screen. Some tools available in the playback screen are not available in the capture screen.

Some buttons may give access to multiple tools. To access the other tools, right click the button or perform a long press on the button.

List of recently captured files



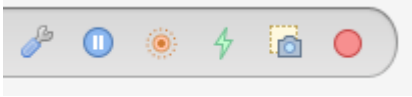
This area displays thumbnails of the recently captured files.

Double clicking on a thumbnail opens the file in a playback screen. The close button removes the thumbnail from the list without deleting the file.

Right-clicking on a thumbnails brings a context menu with the following options:

Open	Opens the video in a playback screen.
Locate in Windows Explorer	Opens the Windows Explorer on the parent folder of the video.
Rename	Make the file name label editable and save the new file name.
Hide	Removes this thumbnail from the list without deleting the file.
Delete	Sends the file to the Windows trash.

Capture controls



Configure camera

Displays the camera configuration dialog to change options like the image size or frame rate.

Pause camera / *Start camera*

Pauses or restarts the camera stream.

When the camera stream is paused it is possible to review the last few seconds of action by adjusting the delay slider.

The view is live / *The view is delayed*

Sets the camera view to live or delayed. This is independent from the recording it only affects the view.

The capture trigger is armed / *The capture trigger is disarmed*

Arms or disarms the capture trigger.

If capture by trigger is disabled globally this will always show as disarmed. If capture by trigger is enabled globally but it is disarmed here, trigger events will be ignored.

The trigger options can be configured in Options ▸ Preferences ▸ Capture ▸ Trigger.

Save image

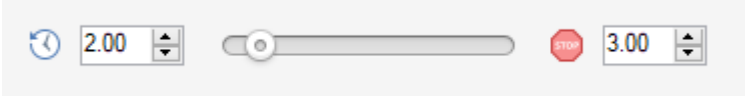
Saves the image currently displayed to an image file in the selected capture folder and with the defined file name.

Start recording video / *Stop recording video*

Starts or stops recording the video. The video is recorded based on the global format and recording mode options. It is saved in the selected capture folder with the defined file name.

See userinterface/preferences/capture for more information.

Delay and duration



The delay numerical field and slider let you adjust the amount of delay, in seconds, of the camera stream.

The delay setting can be ignored for the viewport image by using the Live view button in the capture controls. In this case the delay is still taken into account when recording if the recording mode is set to Delayed or Retroactive.

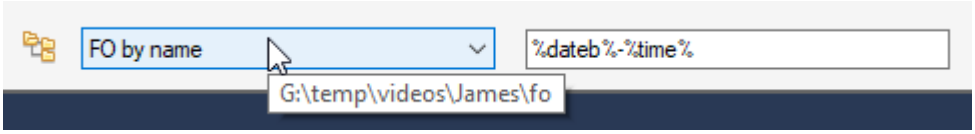
The maximum amount of delay depends on the camera configuration — hardware compression, image format, image size, frame rate — and the memory allocated in the delay cache under Options ▸ Preferences ▸ Capture ▸ Memory.

The duration numerical field let you set a maximum duration for the recording. This is the total duration of the resulting file.

Note

If delay is set to 2 seconds and maximum duration is set to 3 seconds, the resulting file will be 3 seconds long and contain 2 seconds of action *before* the recording button was pressed or triggered, and 1 second of action *after* the recording button was pressed or triggered.

Output folder and file name



Capture folder

The capture folder drop down list let you select the folder where captured images and videos will be saved.

This list of folders can be configured from Options ▸ Preferences ▸ Capture ▸ Folders. This preference page can be accessed by clicking on the folder button to the left of the drop down list.

The file name and capture folder path can use variables that are replaced by their current value when creating the file. This can be used to create self-structuring folder hierarchies for example by automatically saving in sub-folders by date and name.

Hover the mouse over the capture folder drop down to see a tooltip with the actual path target.

See Capture folders, Context variables and Capture preferences for more information.

File name

The file name field lets you define either a plain file name or a file name pattern using variables.

When using a plain file name a number will be automatically appended before the next recording to avoid overwriting existing files.

When using a file name pattern with variables, the recorded files will be named according to the pattern and the variables will be replaced by their current value.

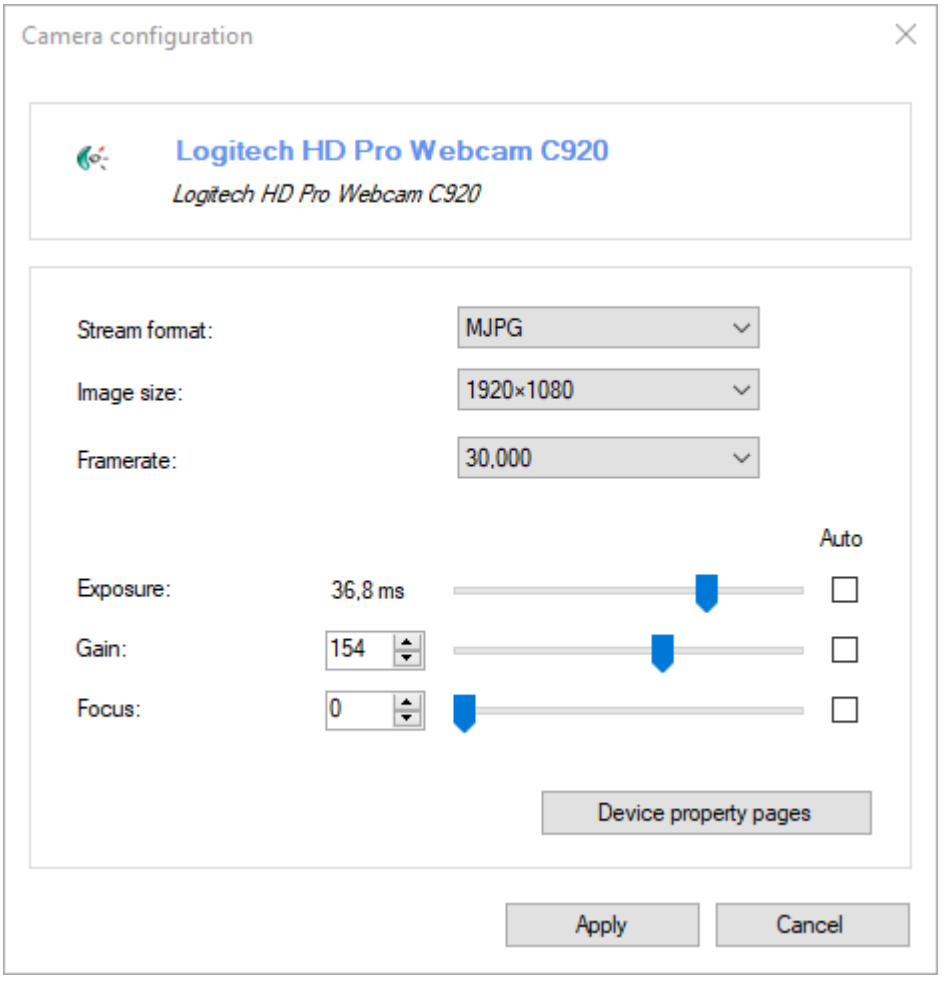
You can configure the global default file name and insert variables in a graphical way in the preference pages **Options ▸ Preferences ▸ Capture ▸ Files**.

DirectShow compatible devices

General

Kinovea supports devices with a DirectShow driver. These devices include USB video class (UVC) webcams, HDMI to USB converters, Mini DV camcorders, and analog video converters.

Configuration



The exact list of available options depends on the camera.

Stream format

The avalaible stream formats depend on the brand and model of the camera.

The typical stream formats include:

- `RGB` , `RGB24` , `RGB32` : the images are not compressed.
- `YUV` , `YCbCr` , `YUY2` , `I420` : the images are not compressed.

- `MJPEG`: the images are compressed on the camera.

Using the `MJPEG` stream format can lower the bandwidth requirements and improve framerate.

Image size

This defines the image size of the stream and impacts the available options in the framerate dropdown. Devices in this category only support a predefined set of image sizes.

Framerate

This is the list of framerates available for the selected image size.

Exposure

This value is related to the amount of time the sensor is exposed.

Changing the exposure duration lets you find a tradeoff between motion blur and light requirements. Lowering the exposure duration reduces motion blur and increases the amount of light required to capture the scene.

For most cameras brands, the unit of this value is not known and it is exposed as an arbitrary number. For some camera brands the value is shown in milliseconds if a special property is exposed by the driver or the values where derived manually.

This value is a limiting factor for the framerate. If this value is too high the framerate is lowered automatically by the camera.

Gain

This is the amount of amplification of the signal captured on the sensor. Increasing this value increases the apparent brightness but can introduce noise in the image.

This value has no impact on the framerate.

Focus

This drives motorized lenses to adjust focus.

Other options

Options that are not supported in Kinovea can be modified using the `Device property pages` button.

Machine vision cameras

Installation

Machine vision cameras are supported via a plugin distributed separately from the main Kinovea download.

- Download `Kinovea.GenICam-<version>.zip`.
- Unzip the archive somewhere and copy the `GenICam` folder into Kinovea application data folder under the `Plugins\Camera` sub-folder. You can find the application data folder via Kinovea menu **Help ▸ Open application data folder...**.
- You must still install the drivers and software from your camera vendor separately. If the installer for the vendor runtime or SDK has various options make sure to check the options mentioning “GenICam” or “GenTL”.
- Restart Kinovea.

Supported devices

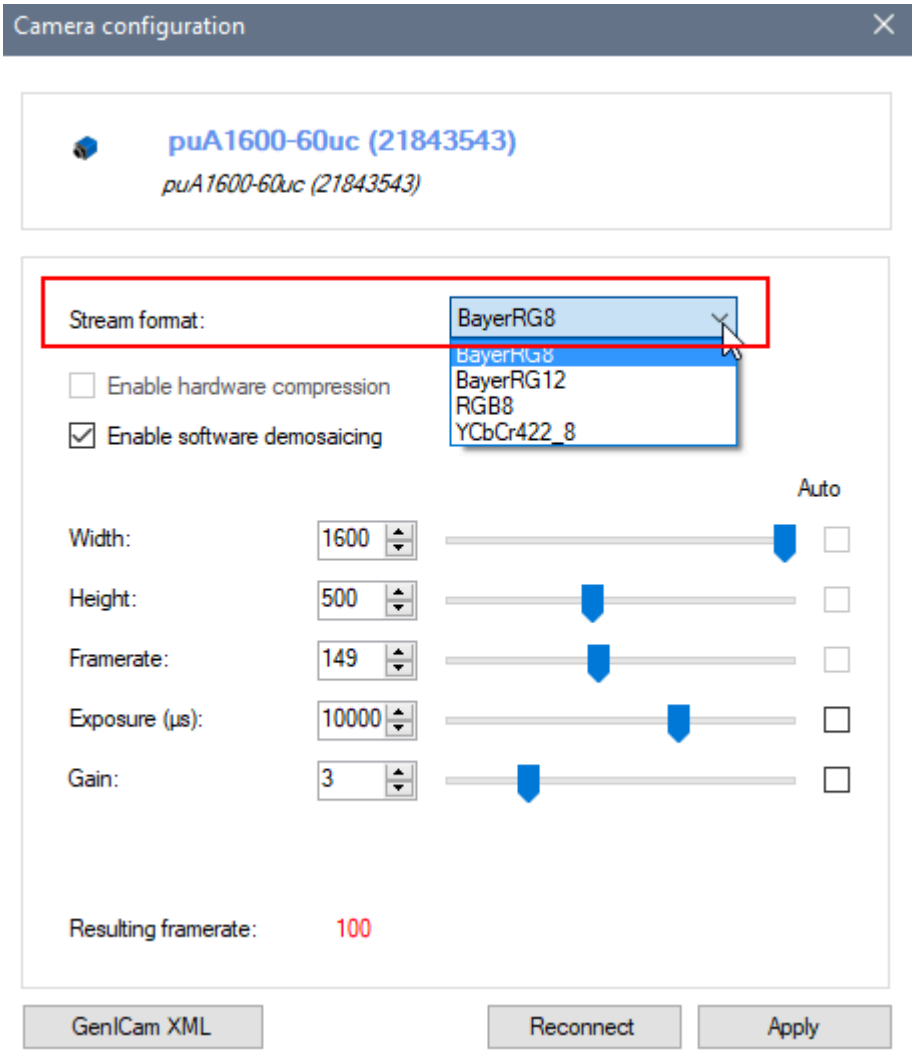
Cameras from the following vendors are reported to work:

- Allied Vision
- Basler
- Baumer
- Daheng imaging
- IDS
- Teledyne FLIR
- Vision Datum

Known issues:

- With IDS cameras, reducing the image resolution may result in a black image.
- Daheng cameras using GigE interface may not work (USB 3 cameras do work).
- When using two Teledyne FLIR cameras of the same model at the same time it can crash. One camera should work.

Configuration



Stream format

The available stream formats depend on the brand and model of the camera.

The suggested workflow for color cameras is to use a raw stream format (for example `BayerRG8`) and enable software demosaicing to reconstruct color images on the fly.

Enable hardware compression

This option is available if the camera supports hardware compression.

Enable software demosaicing

This option can be used when the stream format is a raw format (for example `BayerRG8`) to reconstruct a color image on the fly.

Note

When using a non raw stream format the camera is performing the demosaicing itself on the camera hardware. This is not necessarily faster than doing in on the computer and it requires 3 times the bandwidth to transfer the color images. The top frame rate advertised by the vendor is usually for a raw stream format.

It is also possible to not enable this option, save the raw stream format to disk, and only reconstruct the color information later when loading the video in the player.

To enable this workflow go to **Options ▸ Preferences... ▸ Capture ▸ General**, and check `Record uncompressed video`. When reading the video back choose the appropriate option under the menu **Image ▸ Demosaicing**. This option creates enormous files.

Width

The width of the images. This usually has no effect on the maximum framerate possible as the cameras always read entire sensor rows.

Height

The height of the images. This usually does impact the maximum framerate possible as only the required rows are read from the sensor.

Frame rate

The target frame rate. Whether this frame rate is actually reached or not depends on the stream format, image size, exposure time and the camera hardware. If the frame rate cannot be sustained, the `Resulting frame rate` value will be displayed in red.

If the `Auto` checkbox is checked, the camera will ignore the value and always send the maximum frame rate possible based on the rest of the configuration and the camera hardware. If the `Auto` checkbox is not checked, the camera will use at most the configured value, if it is possible for the hardware to do so. The manual configuration can be interesting if you want to use a specific frame rate that is less than the maximum possible.

Note

After changing the image size or stream format you must click on `Reconnect` for the maximum frame rate information to be updated.

Exposure (µs)

This is the amount of time the sensor is exposed, in microseconds. Changing the exposure duration lets you find a tradeoff between motion blur and light requirements. Lowering this value reduces motion blur but increase the amount of light required to capture the scene.

This value is a limiting factor for the framerate. For example a value of 20 milliseconds implies that there cannot be more than 50 images per second captured.

Gain

This is the amount of amplification of the signal captured on the sensor. Increasing this value increases the apparent brightness but can introduce noise in the image.

This value has no impact on the framerate.

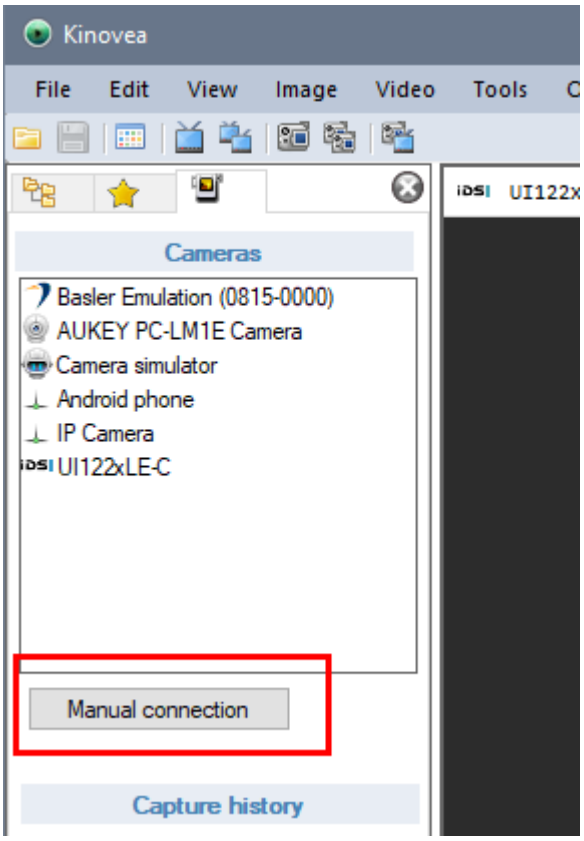
Resulting frame rate

This value is the actual frame rate at which the camera will send images, based on the other values and the camera hardware capabilities.

Network cameras and smartphones

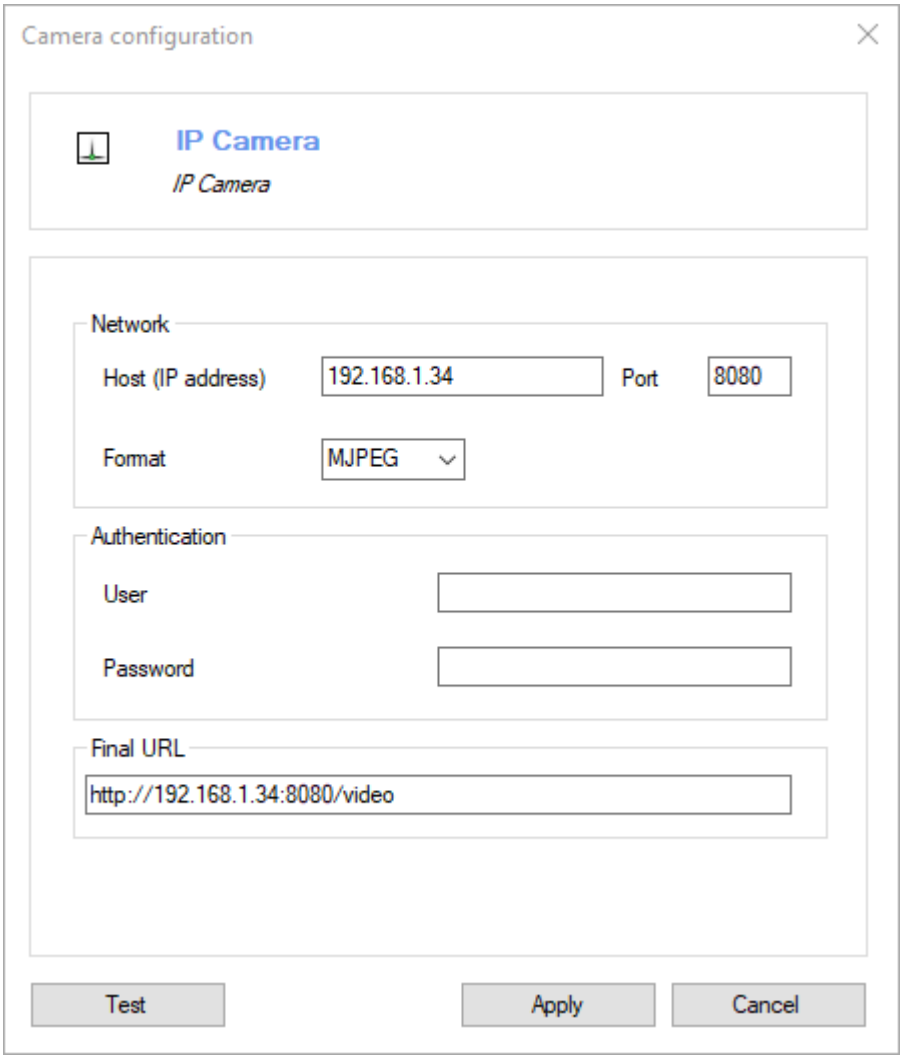
General

In order to add a network or IP camera in Kinovea use the `Manual connection` button on the camera tab of the explorer panel.



This dialog brings the configuration options described below. The same options are available later by clicking on the `Configure camera` button in the capture screen.

Configuration



The values for the options depend on the particular brand of network camera and your own network configuration. The most important setting is the `Final URL`, this is the URL used by Kinovea to connect to the camera stream.

Clicking the `Test` button will try to connect to the camera and report success or failure.

Smartphones

Smartphones can be used as network cameras over the WiFi network by using a dedicated application, provided it supports streaming in MJPEG. For example “IP Webcam” by Pavel Khlebovich can be used on Android devices.

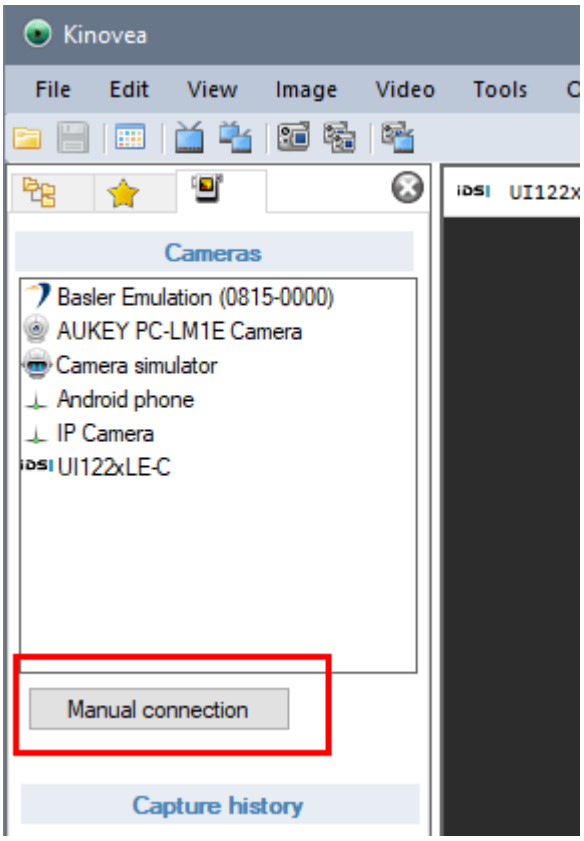
Consult the application to find the URL of the server and the IP address of the smartphone.

Camera simulator

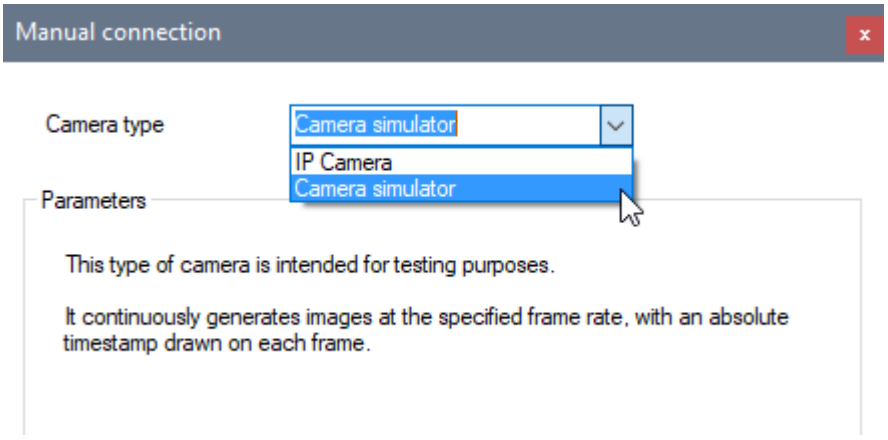
General

The camera simulator is a virtual camera in Kinovea. This camera can be used to evaluate the expected performances of real hardware on a particular computer.

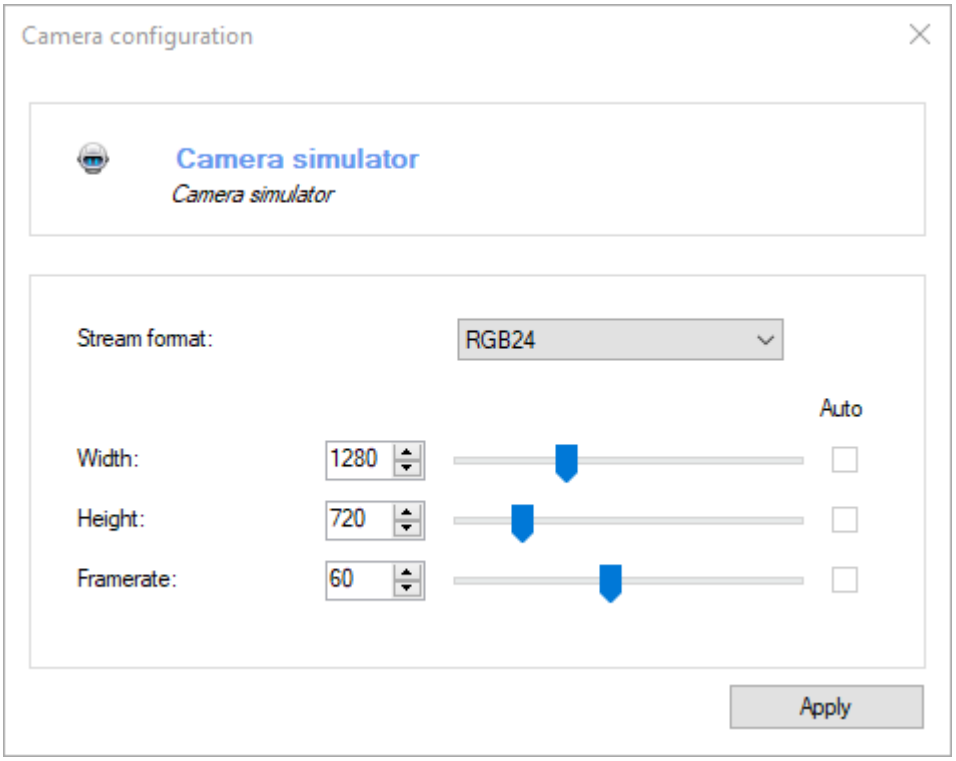
In order to add a camera simulator use the `Manual connection` button on the camera tab of the explorer panel.



In the manual connection dialog, use the `Camera type` drop down and select `Camera simulator`.



Configuration



Stream format

The following stream formats are available:

- `RGB24` : the images are not compressed.
- `JPEG` : the images are compressed.

When using the `JPEG` format, a set of images is compressed in advance and images are rotated to the output. This is to avoid any computer slow downs due to compression when using the simulator to evaluate if a particular camera configuration would be sustainable; as the JPEG compression would be done in the camera.

When using the `RGB24` format, the current time and image sequence number is stamped on the image.

Width

The width of the images.

Height

The height of the images.

Frame rate

The target frame rate of the stream.

Recording

Recording options

Starting and stopping recording

Recording can be started manually by using the record button Record in the capture controls, or automatically by configuring an audio trigger.



To configure the audio trigger, see Options > Preferences > Capture > Automation.

Replaying the captured video

The videos recorded during the session are added to the list of recently captured files.

A recording can be played back by double clicking the thumbnail of the recently captured file or by right clicking and choosing Open.

New recordings can also be played back automatically by using a replay folder observer on the target folder of the capture screen.

Output options

The path to save the recording and the file name of the output video are based on the options under Options ▸ Preferences ▸ Capture ▸ Video naming.

The duration of the recording is either set manually when using the stop recording button, or automatically when using the Stop recording by duration option in Options ▸ Preferences ▸ Capture ▸ Automation.

Recordings can be compressed or uncompressed based on the option set in Options ▸ Preferences ▸ Capture ▸ General.

Recordings can ignore or take into account the live delay based on the recording mode set in Options ▸ Preferences ▸ Capture ▸ Recording.

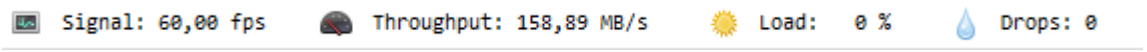
For high speed cameras the framerate set in the metadata of the output video can be customized by configuring a replacement framerate in Options ▸ Preferences ▸ Capture ▸ Recording.

Performances

When the recording process is not fast enough to sustain the camera framerate, images are skipped (dropped) and not added to the output video. This can corrupt time measurements made on the output video, as this assumes stable frame rate.

In order to make the most out of the camera without any frame drops, it is important to identify the bottlenecks and configure the camera and Kinovea according to your requirements and the trade offs you are interested in.

Use the infobar to get feedback about performances.



See also: Capture screen user interface.

Recording mode

The recording mode `Retroactive` should not yield any frame drops but the duration of the output videos is limited by the amount of memory allocated for the delay buffer. Furthermore, this mode renders the camera unavailable for a small duration after the recording, while Kinovea performs the actual export.

The other two modes record on-the-fly and can record for arbitrary long periods (based on storage space) and without post-recording pause. However they require more configuration if the camera does not already produce compressed images.

Compression

For cameras that do not produce an already compressed stream, Kinovea may compresses the images on the fly on the CPU. This process is usually slow compared to the frame rate that the camera can sustain.

You can disable compression on recordings under **Options ▸ Preferences ▸ Capture ▸ General**.

When compression is disabled, the amount of data to store on file is between 5 to 10 times larger. In this configuration, the speed of the storage medium might become the new bottleneck. Solid state drives (SSD) are very much recommended instead of hard drives (HDD). NVMe SSD offer even better performances. Ultimately it is also possible to configure a RAM Drive to further increase the storage speed.

Automation scenario

It is possible to setup Kinovea to record and replay videos multiple times in a row without manual interaction.

To do this set the recordings to start from the audio trigger and stop from the recording duration preset. Add a replay folder observer monitoring the capture folder, this will automatically open and play the last recorded video.

Capture folders

Recording triggers

Context variables

Delay and replay

Live delay

Feature description

The live delay function lets you delay the display of the live stream.

This function is very useful for self-coaching: set the delay to be approximately the total time of the exercise plus the time necessary to come back to the computer. The camera and Kinovea can then be left unattended. By the time you complete your exercise and come back to the computer, you should see a replay of the action.

The same approach can be used with a group of students or athletes, forming an uninterrupted queue of intertwined feedback loops.

Applying delay

To delay the display of the camera stream use the delay slider or the delay input box. The delay amount is in seconds.



The maximum amount of delay depends on the camera configuration — hardware compression, image format, image size, frame rate — and the memory allocated in the delay cache under `Options ▸ Preferences ▸ Capture ▸ Memory`.

If you wish to use a delay that is larger than the maximum that can be set using the slider, increase the size of the memory buffer under `Options ▸ Preferences ▸ Capture ▸ Memory`. Note however that this buffer is always allocated, even if you do not use the delay function.

Note

When using two capture screens at the same time the memory buffer is shared between the screens.

Recording modes

It is possible to record video while delay is active. The option selected for the recording mode impacts whether the delay is taken into account for the recording or not.

Recording mode	Delay
Camera	No
Delayed	Yes
Retroactive	Yes

Combining delay and recording can be used to record actions happening before the moment the record button is hit or triggered.

Time origin

When recording with delay, the time origin of the resulting video is set to the real moment the record button was hit or triggered.

For example we are filming a golf swing for a total duration of 2.5 seconds and a delay of 1.5 seconds. The recording is started via audio trigger when the club hits the ball.

The first image of the video will correspond to what the camera was filming 1.5 seconds before the club hit the ball.

The time origin in the metadata file will be set to the club-ball impact. All of the action happening before the impact will be timestamped with negative numbers.

Pause and browse

Pausing the camera using the pause button 

freezes the delay buffer.

This enables manually navigating the recent history of the camera stream using the delay slider.

Instant replay mechanism

Feature description

Replay folder observers are special playback screens that constantly monitor a specific folder for new videos. When a new video is created in the monitored folder, the playback screen automatically loads it and starts playing it. This can be used to automate the capture and replay the feedback loop.

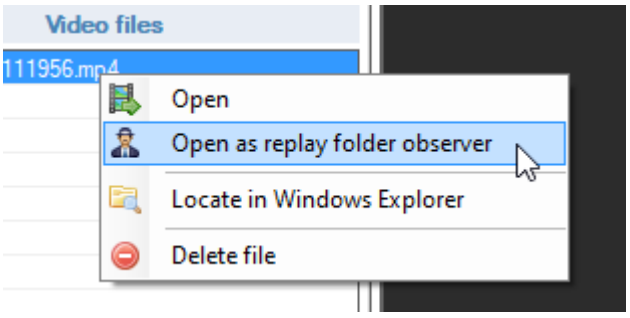
The replay folder observer works at the file system level and is independent from the source or process creating the video. The videos can be created by the same or another instance of Kinovea, or by an external process.

Apart from the monitoring mechanism, the video is loaded normally. In particular the annotation file created by the capture screen is loaded and drawings created on the capture side are imported.

Creating a replay observer

You can create a replay observer from the following places:

- From the menu `File ▸ Open replay folder observer...`. This opens a folder selection dialog.
- In the files and shortcuts tabs of the explorer panel, right click a folder or a file and choose `Open as replay folder observer`.



- In the capture tab, right click a file in the capture history and choose `Open as replay folder observer`.
- In the playback screen, right click the main video image and choose `Open a replay folder observer...`. This opens a folder selection dialog.
- In the capture screen, in the recently captured files list, right click the thumbnail of a file and choose `Open as replay folder observer`.

The recently opened folder observers are listed under the menu `File ▸ Recent`.

Note

The menus on files are opening the observer on the parent folder of that file. This observer will immediately load the most recent file of the folder which is not necessarily the file used to start it.

Activation and deactivation

When manually loading a video into a replay folder observer, the observer is not deactivated and will continue to monitor the folder.

In order to deactivate the monitoring and loading of new videos, you can turn the folder observer into a regular playback screen by clicking the observer icon in the infobar of the screen.



Conversely, in order to turn a normal playback screen into a replay folder observer, click the video icon in the infobar of the screen.

Bidirectional annotation linking

The annotations created on the capture screen are stored in a companion KVA file next to the recordings.

When the recorded video is opened in Kinovea the annotations are loaded and can be modified.

Because the capture screen is simultaneously monitoring the annotation file it last exported, any changes to the annotations on the playback side will be reflected into the capture screen. The next recording from this capture screen will then use the updated annotations.

Advanced

Multi-window system

Kinovea implements a multi-window system where each opened window (instance of Kinovea) remembers its own content and state.

Multiple instances of Kinovea can run at the same time without interfering with each other. The core preferences are shared and synchronized between instances.

You can assign names to windows to associate them with specific tasks.

> **Note**
>
> We use the terms “window” and “instance” interchangeably in this documentation. Each instance of Kinovea has only one main window.

This system supports many use cases, for example:

- Having a “go-to” reference video that can quickly be opened in separate window while working on other videos.
- Associating one window with a long term research project and using a separate one for quick reviews.
- Keeping a group of windows working together in a capture-and-replay workflow, and restarting them all at the same time.

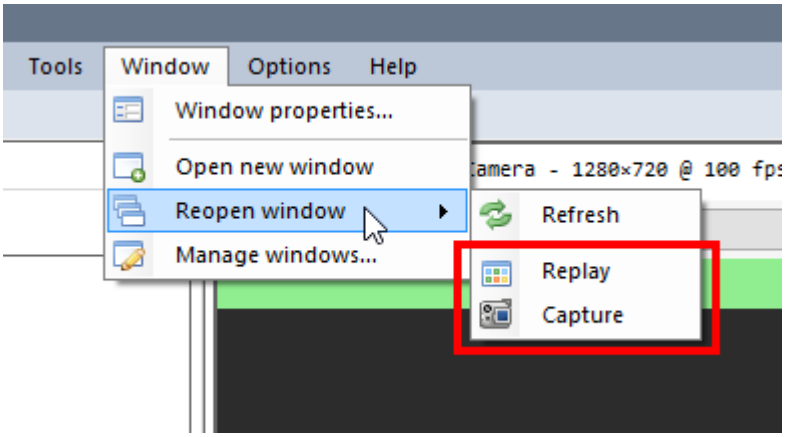
Starting new windows

To start a new window use the menu **Window ▸ Open new window**.

Reopening previous windows

When Kinovea is started it will reopen the last window that was open when it was last closed.

To reopen another window use the menu **Window ▸ Reopen window** and select the desired window from the list. Alternatively use the Window manager, **Window ▸ Window manager...**, and select from the list.

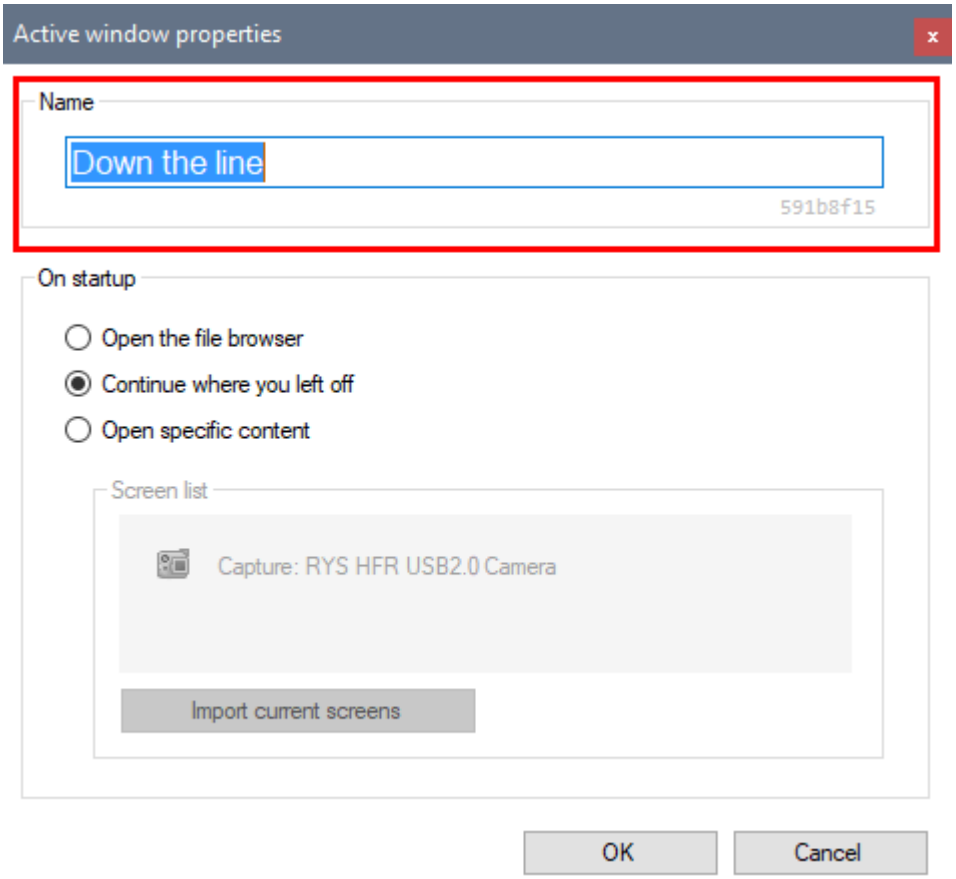


The icon next to the window name or id indicates the type of content it is hosting.

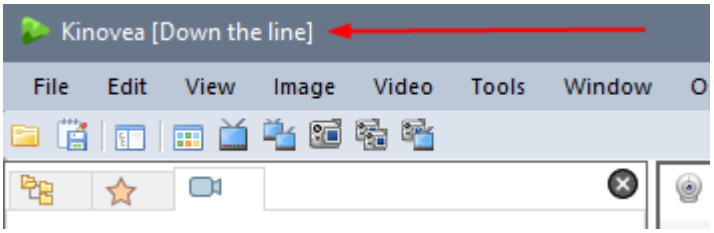
Window name

By default windows are assigned random Ids to identify them internally.

You can assign a custom name to each window by using the menu `Window ▸ Window properties...` .



The name of the Window can be seen in the window title bar in square brackets.



Warning

Do not use the same name for different windows. Even though windows have internal unique Ids, the name is used in certain functions like identifying windows in the command line options or for interprocess communication.

Startup options

Each window defines startup options that are used when the window is opened.

To edit the startup options of a window use the menu **Window ▸ Window properties...** .

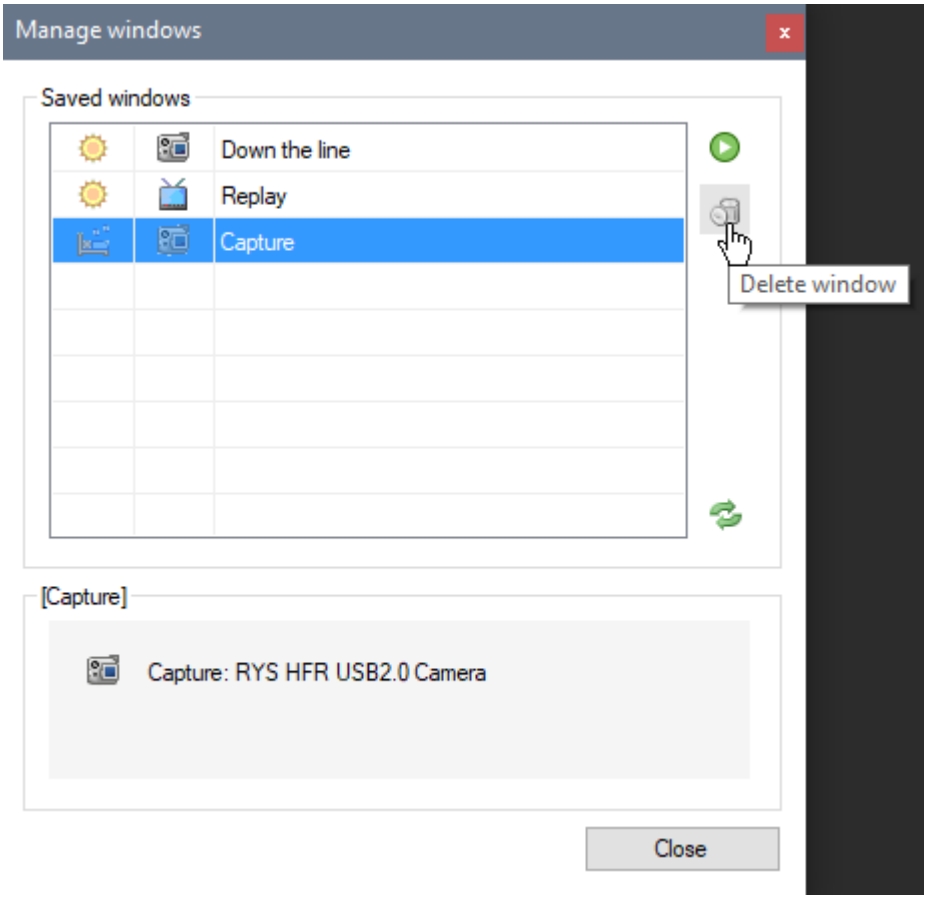
The following options are available:

- `Open the file browser` : The window starts on the file browser, no video or camera is loaded.
- `Continue where you left off` : The window restores its previous content and state. This is the default option.
- `Open specific content` : The window loads a specific video or camera, or two screens with videos and/or cameras. This option is useful if you want to create a stronger link between a window and a specific task or project. Even if you change the content or close the screens, the next time you open the window it will reload the specified content.

To assign the current content of a window to its startup option use the button `Import current screens` .

Deleting old windows

To delete old windows use the menu **Window ▸ Window manager...** , select the window to delete from the list, and click the `Delete` button.



Global preferences vs window state

It can be helpful to understand what is saved in the core preferences and what is saved in the window state.

Everything that is modified via the menu `Options ▸ Preferences...` is saved in the core preferences and shared between all windows.

This include:

- The general behavior of Kinovea, language, time format, units, drawing defaults, keyboard shortcuts, etc.
- The central repository of capture folders that the capture screens and replay screens reference
- The list of recently opened files
- The list of known cameras and their configuration

Whereas the visual content of each window and the state of the player or capture screens are saved at the window level.

This include:

- The layout of the panels and other UI elements
- In player screens: the speed slider and the image “stretch” state
- In capture screens: the selected capture folder, the output file name, the live delay and maximum duration, the post-recording command.

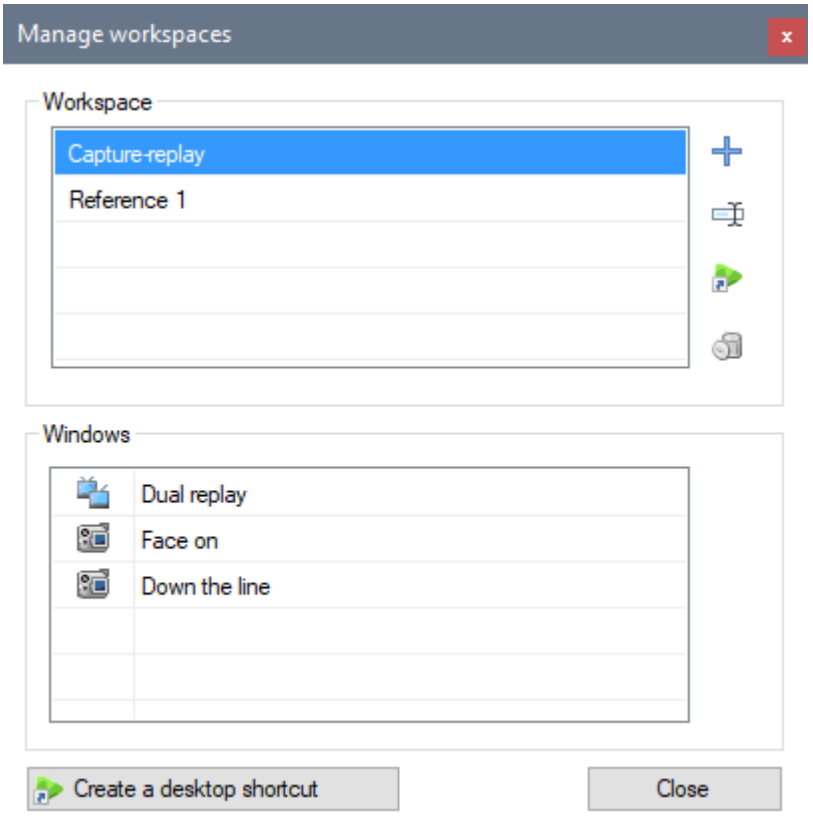
Workspaces

A workspace is a predefined set of windows that are meant to be opened together. Each window in the workspace has its own content and state.

Workspaces are built on top of the Multi-window system.

To create and delete workspaces, use the menu **Window ▸ Manage workspaces...**.

Workspace manager



The workspace manager is divided in two areas. At the top, the workspace list shows the existing workspaces and lets you create, rename and delete them, or create a desktop shortcut to open a specific workspace. At the bottom, the window list shows the windows that are part of the selected workspace.

	Creates a workspace referencing the currently active windows and prompts for a name.
	Opens a dialog to rename the selected workspace.
	Creates a shortcut on the desktop to open the selected workspace.

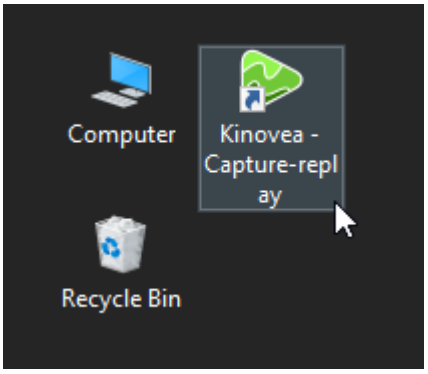
	Deletes the selected workspace.

Creating shortcuts

To create a shortcut to open a specific workspace, use the button `Create a desktop shortcut`. (Same function as the one next to the workspace list.)

The shortcut will automatically be created on the desktop.

When the shortcut is used, Kinovea will start and reopen all the windows defined in the workspace.



Storage

Workspace files are saved as XML files in the application data folder under the *Workspaces* subfolder.

A workspace only contains references to windows. It does not store any state by itself.

Post-recording commands

Post-recording commands let you execute external programs right after a recording is finalized. The created file can be passed as an argument to the external program.

Some use cases include:

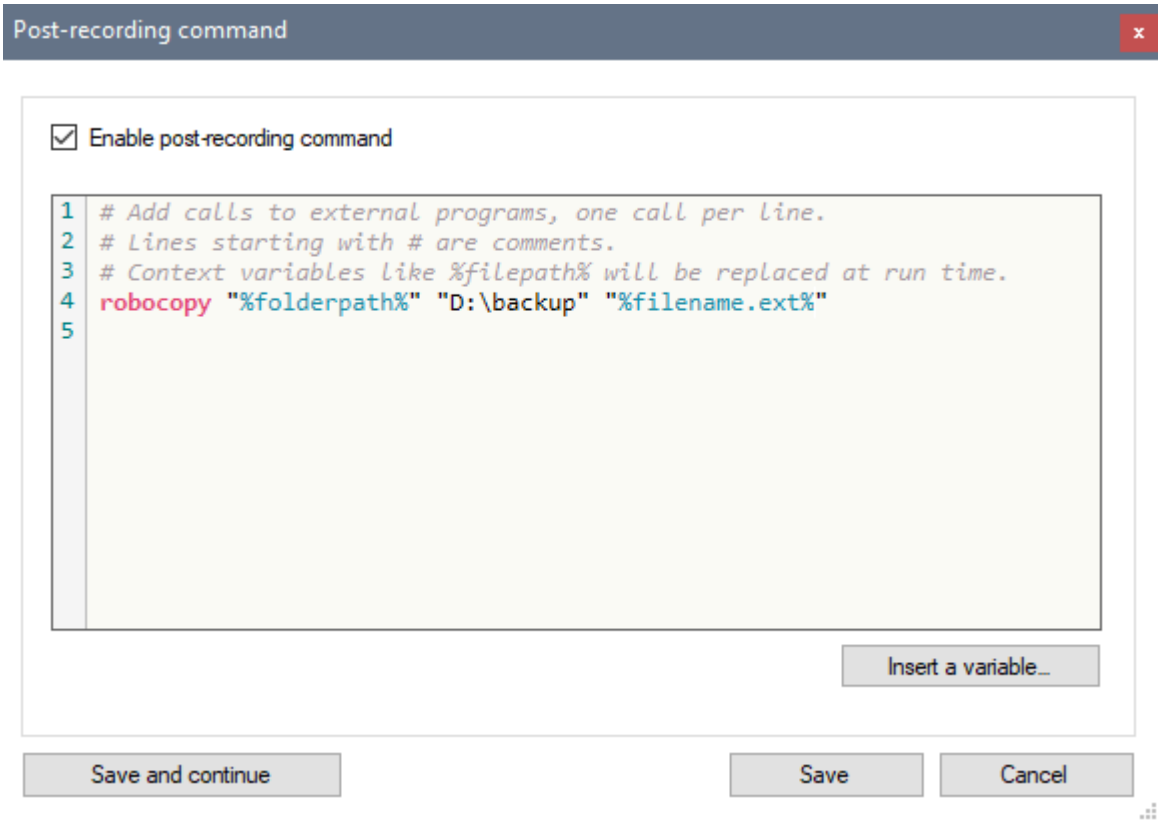
- copying the file to a backup location.
- converting the recorded file to a different format.
- running a post-processing application to transform the video in some way or extract images.

Command editor

To enable post-recording commands right click the background of a capture screen and choose the menu **Post-recording command...**. Check the `Enable post-recording command` checkbox and enter the command line to execute.

You can call multiple programs, add each command on its own line. The commands are executed sequentially and the whole thing runs in a background thread.

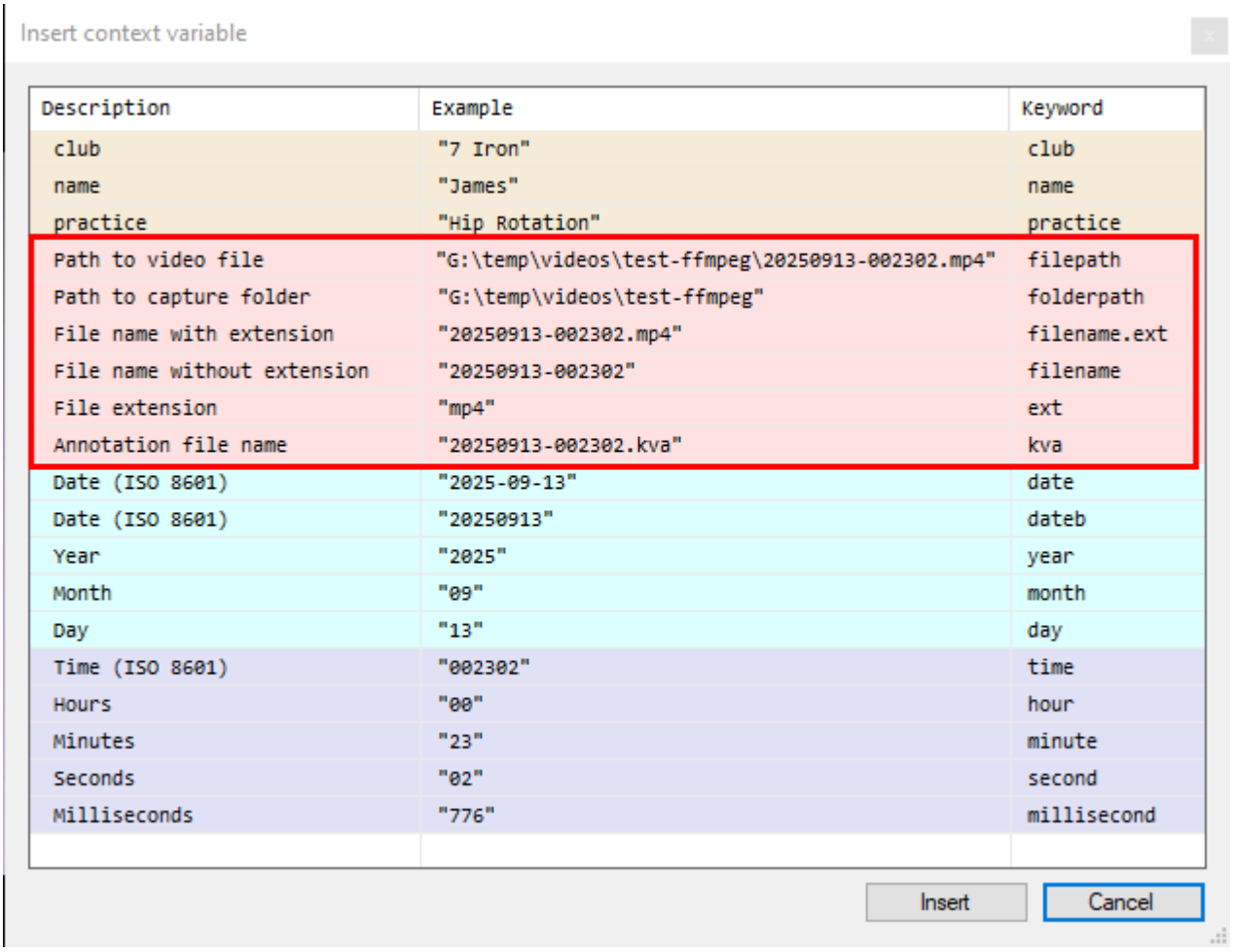
Lines starting with `#` are treated as comments and ignored.



Variables

Use the button `Insert a variable` to insert context variables that will be replaced with actual values when the command is executed. In addition to the user-defined variables and the date and time built-in variables, the following are available in this context:

Variable	Meaning
`%filepath%`	The full path to the video file.
`%folderpath%`	The full path to the capture folder containing the video file.
`%filename.ext%`	The name of the recorded file including the extension.
`%filename%`	The name of the recorded file without the extension.
`%ext%`	The extension by itself, without the leading dot.
`%kva%`	The name of the KVA file created alongside the recording.



FFmpeg

Kinovea ships with the ffmpeg executable that can be used for video conversion and compression.

See the documentation of ffmpeg at https://ffmpeg.org/ for usage instructions.

Example command that encodes the recorded video using the H.264 codec:

```
ffmpeg -i "%filepath%" -vcodec libx264 "%folderpath%\%filename%_compressed.mp4"
```

Command line options

Kinovea can be started from the command line. There are essentially two modalities of operation:

- Starting Kinovea with a specific video file, with some extra options.
- Starting a specific Kinovea window or group of windows, which can contain their own content and settings.

Usage

```
> kinovea.exe
    [-video <path>] [-speed <0..200>] [-stretch] [-hideExplorer]
    [-name <string>]
    [-id <string>]
```

Options

`-video` `<path>`

Path to the video or image to load.

`-speed` `<0..200>`

Playback speed to play the video, as a percentage of its original playback framerate.

Default: `100`.

`-stretch`

The video will be expanded to fit the viewport. Default: `false`.

`-hideExplorer`

The navigation pane on the left of the main area will be closed. Default: `false`.

`-name` `<string>`

Name of the Kinovea window to load.

The name of a window is set from the window properties dialog: **Window ▸ Window properties...**.

`-id` `<string>`

Id of a Kinovea window to load. This option is meant to be used internally between instances of Kinovea.

Examples

```
> kinovea.exe -video test.mp4 -stretch
> kinovea.exe -video test.mp4 -speed 50
> kinovea.exe -name Replay
```

Control from other programs

While a Kinovea window is running it is possible to send it commands from a third party program or script. As an example this can be used to trigger recording based on a software event.

The commands use the Windows messaging system with the `WM_COPYDATA` message. The exact way to create these Windows messages depends on the programming language or platform used to write the third party application. Each command has to be sent separately. There is no return value.

The general format of the messages is the following:

```
Kinovea:<Category>.<Command>
```

The list of categories and commands is available under **Options ▸ Preferences ▸ Keyboard**.

For example:

```
Kinovea:CaptureScreen.ToggleRecording

Kinovea:PlayerScreen.TogglePlay
```

Miscellaneous topics

Frequently asked questions

This is a list of the most frequently asked questions about Kinovea.

1. Can I use Kinovea on MacOS? on Linux? on Android? on iPhone?

No, Kinovea is only supported on Windows.

2. Can I use Kinovea in my school? my university? my bike fit or shoe fit business? my physiotherapy or podiatry practice?

Yes, you can use Kinovea for *any* purposes, including but not limited to personal, professional, commercial, academic and research.

3. What is the price of Kinovea?

Kinovea is a free program, it does not cost anything to use.

4. Is Kinovea compliant with HIPAA or GDPR?

Not applicable.

Kinovea does not store any personal data and does not transmit anything over the internet. It is a standalone desktop application, similar to a media player with extra functions, and does not require the Internet to function. The only aspect of the program that makes use of the Internet is the menu “Check for updates…”, which consults a file on the Kinovea web server and reports if a new version is available or not.

5. In which city is Kinovea made?

Not applicable.

This question comes up when a scientific journal asks for bibliographic references. Kinovea is not a company and does not have headquarters in any city. There is no meaningful way to answer this question. Please ask the journal about their dedicated format for citing Open Source software. You may also check the built-in citations examples in the About dialog.

6. Which camera should I buy?

It is not the place of Kinovea or its authors to suggest one camera brand over another. It is best for the project to stay neutral. Besides, there are too many parameters specific to each use-case that it is not possible to answer this question with anything else than “it depends”. Please register on the forum and ask the question there, providing as many details and context as possible, this way other users can share their experience directly.

7. Why is the speed of my video slowing down by itself?

The speed slider is forced down when the original frame rate of the video cannot be sustained by the player.

Software update

New versions are published on the project website:

https://www.kinovea.org/download.html

To check if a new version has been published from within the program itself you can use the menu **Help ▸ Check for updates**. This function requires an Internet connection.

The function simply consults a text file on the web server that contains the version number of the latest published version. There is no automatic update mechanism, to update you must download and install the new version manually.

Communication channels

The project uses the following communication channels and social networks:

- Discussion forum: https://www.kinovea.org/en/forum/
- Bug tracker: https://github.com/Kinovea/Kinovea/issues
- Translation: https://hosted.weblate.org/engage/kinovea/
- Youtube: @kinovea
- Twitter: @Kinovea
- Instagram: @kinovea

API and integration reference

Integration overview

> **Note**
>
> This chapter is not part of the official Kinovea manual. It was compiled from the Kinovea source code (github.com/Kinovea/Kinovea, commit `13d0236`, October 2026) to document behavior the manual omits. Where the official manual and the source disagree, the source is what actually runs.

Kinovea has no SDK, plugin system or network API. Third-party programs integrate with it through five mechanisms:

Mechanism	Direction	Use it to…
Window messages (`WM_COPYDATA`)	Your program → running Kinovea	Drive playback, recording, delay, zoom, drawings: any of 75 commands.
UDP trigger	Any device on the network → capture screen	Fire the capture trigger (start recording) from a network packet.
Command line	Launcher → new Kinovea process	Open videos, set speed, restore a named window.
Post-recording command	Kinovea → your program	Run a shell command after each recording, with file path variables (see the *Post-recording commands* chapter in Advanced).
Files (KVA, posture tools, exports)	Both ways	Read or write annotations, define custom drawing tools, consume CSV/XML exports.

All of these are Windows-only, like Kinovea itself.

Remote control command reference

> **Note**
>
> This chapter is not part of the official Kinovea manual. It was compiled from the Kinovea source code (github.com/Kinovea/Kinovea, commit `13d0236`, October 2026) to document behavior the manual omits. Where the official manual and the source disagree, the source is what actually runs.

The official manual explains that commands can be sent through `WM_COPYDATA` but defers the command list to the Keyboard preferences page inside the application. This chapter is that list, plus the exact wire protocol.

Protocol

- **Message:** `WM_COPYDATA` (`0x004A`) sent with `SendMessage` to the Kinovea main window.
- **dwData:** must be `0`. Any other value is rejected as malformed.
- **lpData / cbData:** the command string, NUL-terminated. Both UTF-16 (wide) and ANSI encodings are accepted: Kinovea tries UTF-16 first, then ANSI.
- **Format:** `Kinovea:<Category>.<Command>`. Everything before the first `:` must be `Kinovea`; the payload must split into exactly two non-empty tokens on `.`.
- **Categories:** only `PlayerScreen`, `CaptureScreen` and `Window` are dispatched. Other category names shown in the Keyboard preferences (`FileExplorer`, `ThumbnailViewer…`, `DualPlayer`, `DualCapture`) are hotkey-only and **cannot** be triggered remotely.
- **Names are case-sensitive** (parsed with `Enum.TryParse` without ignore-case). `toggleplay` fails; `TogglePlay` works.
- **Fan-out:** a `PlayerScreen` command is executed on *every* open player screen; a `CaptureScreen` command on *every* capture screen. There is no way to target only the left or right screen.
- **No reply.** `SendMessage` returns when the message is handled, but no status is returned. Failures appear only in Kinovea's log file.
- **Minimized windows** still receive and execute commands. Kinovea explicitly allows `WM_COPYDATA` through UIPI (`ChangeWindowMessageFilterEx`), so a non-elevated sender can talk to an elevated Kinovea.

Finding the window

Each Kinovea window's title is `Kinovea [<name>]`, where `<name>` is the window name set in Window ▸ Window properties…, or a name derived from its id if none was set. Locate it with `FindWindow(NULL, "Kinovea [Replay]")`. Naming your windows makes scripting much easier.

Example: Python (ctypes)

```
import ctypes
from ctypes import wintypes

WM_COPYDATA = 0x004A

class COPYDATASTRUCT(ctypes.Structure):
    _fields_ = [("dwData", ctypes.c_size_t),
                ("cbData", wintypes.DWORD),
                ("lpData", ctypes.c_void_p)]

def send(command, window_title="Kinovea [Replay]"):
    hwnd = ctypes.windll.user32.FindWindowW(None, window_title)
    if not hwnd:
        raise RuntimeError("Kinovea window not found: " + window_title)
    buf = ctypes.create_unicode_buffer("Kinovea:" + command)   # UTF-16, NUL-terminated
    cds = COPYDATASTRUCT(0, ctypes.sizeof(buf), ctypes.cast(buf, ctypes.c_void_p))
    ctypes.windll.user32.SendMessageW(hwnd, WM_COPYDATA, 0, ctypes.byref(cds))

send("CaptureScreen.ToggleRecording")
```

Example: C#

```
[StructLayout(LayoutKind.Sequential)]
struct COPYDATASTRUCT { public IntPtr dwData; public int cbData; public IntPtr lpData; }

[DllImport("user32.dll", CharSet = CharSet.Unicode)] static extern IntPtr FindWindow(strin
g cls, string title);
[DllImport("user32.dll")] static extern IntPtr SendMessage(IntPtr hWnd, int msg, IntPtr wP
aram, ref COPYDATASTRUCT cds);

static void Send(string command, string title = "Kinovea [Replay]")
{
    IntPtr hwnd = FindWindow(null, title);
    string msg = "Kinovea:" + command + "\0";
    var cds = new COPYDATASTRUCT { dwData = IntPtr.Zero, cbData = msg.Length * 2, lpData =
Marshal.StringToHGlobalUni(msg) };
    SendMessage(hwnd, 0x004A, IntPtr.Zero, ref cds);
    Marshal.FreeHGlobal(cds.lpData);
}
```

PlayerScreen commands

Default keys are the factory hotkeys; users can rebind them, but remote commands are addressed by name, so rebinding does not affect them.

Message	Default key	Effect
`Kinovea:PlayerScreen.ResetViewport`	Esc	Reset the viewport (exit zoom, close menus).
`Kinovea:PlayerScreen.Close`	Ctrl + F4	Close the screen.
`Kinovea:PlayerScreen.TogglePlay`	Space	Start/stop playback.
`Kinovea:PlayerScreen.IncreaseSpeed10`	Up	Increase playback speed by 10%.
`Kinovea:PlayerScreen.DecreaseSpeed10`	Down	Decrease playback speed by 10%.
`Kinovea:PlayerScreen.IncreaseSpeed1`	Shift + Up	Increase playback speed by 1%.
`Kinovea:PlayerScreen.DecreaseSpeed1`	Shift + Down	Decrease playback speed by 1%.
`Kinovea:PlayerScreen.GotoPreviousImage`	Left	Step one frame backward.
`Kinovea:PlayerScreen.GotoNextImage`	Right	Step one frame forward.
`Kinovea:PlayerScreen.GotoFirstImage`	Home	Go to the first frame.
`Kinovea:PlayerScreen.GotoLastImage`	End	Go to the last frame.
`Kinovea:PlayerScreen.GotoPreviousImageForceLoop`	Shift + Left	Step backward, wrapping to the end at the start.
`Kinovea:PlayerScreen.LargeJumpBackward`	Page Up	Large jump backward.
`Kinovea:PlayerScreen.LargeJumpForward`	Page Down	Large jump forward.
`Kinovea:PlayerScreen.SmallJumpBackward`	Shift + Page Up	Small jump backward.
`Kinovea:PlayerScreen.SmallJumpForward`	Shift + Page Down	Small jump forward.
`Kinovea:PlayerScreen.GotoPreviousKeyframe`	Ctrl + Left	Go to the previous key image.
`Kinovea:PlayerScreen.GotoNextKeyframe`	Ctrl + Right	Go to the next key image.
`Kinovea:PlayerScreen.GotoSyncPoint`	F8	Go to the synchronization point.
`Kinovea:PlayerScreen.IncreaseSyncAlpha`	Alt + Num +	Increase merge opacity in dual mode.
`Kinovea:PlayerScreen.DecreaseSyncAlpha`	Alt + Num –	Decrease merge opacity in dual mode.

Message	Default key	Effect
`Kinovea:PlayerScreen.ToggleSyncMerge`	F9	**No effect via remote control.** The player screen has no handler for it; superposition is handled by the dual-player controller, which external messages do not reach.
`Kinovea:PlayerScreen.IncreaseZoom`	Ctrl + Num +	Zoom in.
`Kinovea:PlayerScreen.DecreaseZoom`	Ctrl + Num −	Zoom out.
`Kinovea:PlayerScreen.ResetZoom`	Ctrl + Num 0	Reset zoom.
`Kinovea:PlayerScreen.AddKeyframe`	Ins	Add a key image at the current frame.
`Kinovea:PlayerScreen.DeleteKeyframe`	Ctrl + Del	Delete the key image at the current frame.
`Kinovea:PlayerScreen.Preset1`	Ctrl + Num 1	Apply key image preset 1.
`Kinovea:PlayerScreen.Preset2`	Ctrl + Num 2	Apply key image preset 2.
`Kinovea:PlayerScreen.Preset3`	Ctrl + Num 3	Apply key image preset 3.
`Kinovea:PlayerScreen.Preset4`	Ctrl + Num 4	Apply key image preset 4.
`Kinovea:PlayerScreen.Preset5`	Ctrl + Num 5	Apply key image preset 5.
`Kinovea:PlayerScreen.Preset6`	Ctrl + Num 6	Apply key image preset 6.
`Kinovea:PlayerScreen.Preset7`	Ctrl + Num 7	Apply key image preset 7.
`Kinovea:PlayerScreen.Preset8`	Ctrl + Num 8	Apply key image preset 8.
`Kinovea:PlayerScreen.Preset9`	Ctrl + Num 9	Apply key image preset 9.
`Kinovea:PlayerScreen.Preset10`	—	Apply key image preset 10.
`Kinovea:PlayerScreen.CutDrawing`	Ctrl + X	Cut the selected drawing.
`Kinovea:PlayerScreen.CopyDrawing`	Ctrl + C	Copy the selected drawing.

Message	Default key	Effect
`Kinovea:PlayerScreen.PasteDrawing`	Ctrl + V	Paste drawing at the mouse position.
`Kinovea:PlayerScreen.PasteInPlaceDrawing`	—	Paste drawing at its original position.
`Kinovea:PlayerScreen.DeleteDrawing`	Del	Delete the selected drawing.
`Kinovea:PlayerScreen.ValidateDrawing`	Enter	Finish the drawing in progress.
`Kinovea:PlayerScreen.CopyImage`	Ctrl + Shift + C	Copy the current image to the clipboard.
`Kinovea:PlayerScreen.CopyTime`	—	Copy the current time to the clipboard.
`Kinovea:PlayerScreen.ToggleDrawingsVisibility`	—	Show/hide all drawings.
`Kinovea:PlayerScreen.ChronometerStartStop`	F5	Start/stop the active stopwatch.
`Kinovea:PlayerScreen.ChronometerSplit`	F6	Record a split on the active stopwatch.
`Kinovea:PlayerScreen.CadenceBeat`	F7	Add a beat to the cadence counter.
`Kinovea:PlayerScreen.StartAllTracking`	—	Start tracking for all trackable drawings.

CaptureScreen commands

Message	Default key	Effect
`Kinovea:CaptureScreen.ResetViewport`	Esc	Reset the viewport (exit zoom, close menus).
`Kinovea:CaptureScreen.OpenConfiguration`	F12	Open the camera configuration dialog.
`Kinovea:CaptureScreen.Close`	Ctrl + F4	Close the screen.
`Kinovea:CaptureScreen.ToggleGrabbing`	Space	Pause/resume the camera stream.
`Kinovea:CaptureScreen.ToggleRecording`	Ctrl + Enter	Start/stop recording.
`Kinovea:CaptureScreen.TakeSnapshot`	Shift + Enter	Save a snapshot image.
`Kinovea:CaptureScreen.ToggleArmCaptureTrigger`	—	Arm/disarm the capture trigger.

Message	Default key	Effect
`Kinovea:CaptureScreen.IncreaseZoom`	—	Zoom in.
`Kinovea:CaptureScreen.DecreaseZoom`	—	Zoom out.
`Kinovea:CaptureScreen.ResetZoom`	—	Reset zoom.
`Kinovea:CaptureScreen.GotoPreviousImage`	Left	Step one frame backward.
`Kinovea:CaptureScreen.GotoNextImage`	Right	Step one frame forward.
`Kinovea:CaptureScreen.GotoFirstImage`	Home	Go to the first frame.
`Kinovea:CaptureScreen.GotoLastImage`	End	Go to the last frame.
`Kinovea:CaptureScreen.BackwardRound10Percent`	Page Up	Move back 10% in the delay buffer.
`Kinovea:CaptureScreen.ForwardRound10Percent`	Page Down	Move forward 10% in the delay buffer.
`Kinovea:CaptureScreen.BackwardRound1Percent`	Shift + Page Up	Move back 1% in the delay buffer.
`Kinovea:CaptureScreen.ForwardRound1Percent`	Shift + Page Down	Move forward 1% in the delay buffer.
`Kinovea:CaptureScreen.ToggleDelayedDisplay`	Alt + Home	Toggle the delayed display.
`Kinovea:CaptureScreen.IncreaseDelayOneSecond`	Up	Increase delay by 1 s.
`Kinovea:CaptureScreen.DecreaseDelayOneSecond`	Down	Decrease delay by 1 s.
`Kinovea:CaptureScreen.IncreaseDelayOneFrame`	Ctrl + Up	Increase delay by one frame.
`Kinovea:CaptureScreen.DecreaseDelayOneFrame`	Ctrl + Down	Decrease delay by one frame.
`Kinovea:CaptureScreen.IncreaseDelayHalfSecond`	Shift + Up	Increase delay by 0.5 s.
`Kinovea:CaptureScreen.DecreaseDelayHalfSecond`	Shift + Down	Decrease delay by 0.5 s.

Window commands

These are used internally for communication between Kinovea instances. They are listed for completeness; sending them from outside is possible but rarely useful.

Message	Effect
`Kinovea:Window.PreferencesUpdated`	Reload the shared preferences file from disk and refresh the UI.
`Kinovea:Window.VariableTableImported`	Reload context variable tables.
`Kinovea:Window.CameraUpdated`	Reload camera aliases/icons.
`Kinovea:Window.Close`	Close this window (may prompt if annotations are unsaved).

UDP network trigger

> **Note**
>
> This chapter is not part of the official Kinovea manual. It was compiled from the Kinovea source code (github.com/Kinovea/Kinovea, commit `13d0236`, October 2026) to document behavior the manual omits. Where the official manual and the source disagree, the source is what actually runs.

The capture screen can be triggered by a UDP packet, in addition to the audio trigger documented in *Recording triggers*. The official manual does not describe this.

Behavior

- Kinovea listens on UDP port **8875** by default (configurable in the capture preferences, under the automation/trigger settings).
- **Any datagram** on that port fires the trigger. The payload is ignored entirely; an empty packet works.
- The trigger must be enabled (it is **off** by default) and armed on the capture screen.
- The configured trigger **quiet period** applies: packets arriving during it are ignored.
- The resulting action is the configured trigger action (default: record video).
- There is no authentication. Anyone on the network who can reach the port can start a recording; firewall accordingly.

Example sender

This is the test client shipped in the Kinovea repository (`Tools/remote/udp-client.py`), slightly generalized:

```
import socket

def fire(host="192.168.0.228", port=8875):
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.setsockopt(socket.SOL_SOCKET, socket.SO_BROADCAST, 1)  # allows broadcast addresses
too
    s.sendto(b"", (host, port))
    s.close()

fire()
```

Typical uses: a timing gate, force plate or microcontroller (ESP32, Arduino with Ethernet) signalling the moment of interest. Combine with the capture delay buffer so the recording includes the moments *before* the trigger.

Command line reference (complete)

> **Note**
>
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This supersedes the *Command line options* chapter, which omits several options.

```
kinovea.exe [-name <string>] [-id <string>] [-video <path>] [-speed <1..1000>]
            [-video2 <path>] [-stretch] [-hideExplorer]
kinovea.exe <file>
kinovea.exe -help | -h
```

Options

Option	Meaning
`-name <string>`	Restore the saved window with this name. Takes precedence over every other option: if present, `-video` etc. are ignored.
`-id <string>`	Restore the saved window with this id. Used internally; checked after `-name`.
`-video <path>`	Load and **autoplay** this video in a player screen.
`-video2 <path>`	*Undocumented in the manual.* Load a second video in a second player screen (dual mode). Requires `-video`.
`-speed <n>`	Playback speed as a percentage of the original frame rate, applied to both videos. The manual says `0..200`; the code actually clamps to **1–1000**. Unparseable values fall back to 100. Use `.` as decimal separator.
`-stretch`	Expand the video(s) to fit the screen.
`-hideExplorer`	Hide the navigation pane.
`<file>` (sole argument)	*Undocumented.* If the only argument is an existing file, it is opened in a single player with autoplay **and stretch**. This is what happens when a file is dropped on `kinovea.exe`.
`-help`, `-h`	*Undocumented.* Print usage to the parent console.

Notes

- `-speed`, `-video2`, `-stretch` and `-hideExplorer` without `-video` are ignored with a logged error.
- Examples:

```
kinovea.exe -name "Dual replay"
kinovea.exe -video "C:\clips\jump.mp4" -speed 25 -stretch
kinovea.exe -video "a.mp4" -video2 "b.mp4" -hideExplorer
```

KVA file format reference

> **Note**
>
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KVA files are UTF-8 XML. The current Kinovea writes **FormatVersion 2.1**.

> **Warning**
>
> The XML schemas in the Kinovea repository (`Tools/XML/Schema/kva.2.0.xsd` and `drawings.xsd`) are badly outdated: they describe a `CalibrationHelp` element that is no longer written and lack most current sections (coordinate system, trackability, camera motion, video filters…). Do not validate against them. Treat the structure below, taken from the serializer, as authoritative, and read files leniently: Kinovea itself skips unknown elements.

Root element and section order

The root is `<KinoveaVideoAnalysis>`. Children are written in this order; sections are omitted when empty:

Element	Content
`FormatVersion`	`2.1`
`Producer`	`Kinovea.<version>`
`OriginalFilename`, `FullPath`, `ByteLength`	Source video name (no extension), full path, size in bytes. Absent for capture recordings.
`CaptureRecording`	`true` for KVAs written alongside a capture recording (replaces the three elements above).
`GlobalTitle`	Optional title.
`ImageSize`	`width;height` in pixels.
`Aspect`, `Rotation`, `Mirror`, `Demosaicing`, `Deinterlacing`	Image-level overrides.
`StabilizationTrack`	Id of the track used for stabilization.

Element	Content
`BackgroundColor`	Background layer color.
`AverageTimeStampsPerFrame`	Timestamps per frame in the video's time base.
`CaptureFramerate`, `VideoFramerate`	Real capture fps (high-speed calibration) and file playback fps. Invariant culture (`.` decimal).
`FirstTimeStamp`, `SelectionStart`, `SelectionEnd`	Timestamps of the first frame and working zone. Absent for capture recordings.
`TimeOrigin`	Timestamp of the custom time origin.
`Context`	Context variables string, if any.
`Calibration`	Spatial calibration (line or plane), lens distortion.
`Keyframes`	`<Keyframe>` elements, each with its position, title, color, comments and `<Drawings>`.
`Chronos`	`<Chrono>`, `<ChronoMulti>` and `<Counter>` elements (attributes `id`, `name`).
`Tracks`	`<Track>` elements (trajectory objects).
`Spotlights`, `NumberSequence`	Singleton spotlight and auto-numbers drawings.
`CoordinateSystem`	Coordinate system object.
`TestGrid`	Test grid (lens calibration helper).
`Trackability`	Tracking timelines for every trackable drawing point.
`VideoFilters`	Kinogram and other filter configurations.
`CameraMotion`	Camera motion estimation data.

Which sections are written depends on the export flags: *Calibration* (`Calibration`, `CoordinateSystem`), *Drawings* (`Keyframes`, `TestGrid`) and *VideoSpecific* (`Chronos`, `Tracks`, `Spotlights`, `NumberSequence`, `Trackability`, `VideoFilters`, `CameraMotion`). A full save sets all three.

Conventions

- Points are written as `x;y`; sizes as `w;h`.
- Times are stored as **timestamps** in the video's own time base, not milliseconds. Convert with `AverageTimeStampsPerFrame` and `VideoFramerate`; `TimeOrigin` gives the zero.
- Numbers use the invariant culture regardless of the user's locale.

- Older files (1.x, 2.0) are upgraded on load; the repository ships XSLT converters in `Tools/XML/Kva to Kva/`.

Practical advice

To produce KVA files programmatically, create the annotations once in Kinovea, save, and use that file as a template. That is far more reliable than writing from scratch against an incomplete specification. For reading measurement data, the spreadsheet exports (see *Export*) are usually easier than parsing KVA.

Custom posture tool format

> **Note**
>
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Every tool in the *Posture* menu (goniometer, human model, bike fit, OpenPose body 25, etc.) is an XML file. You can create your own: drop a new `.xml` file into `<Kinovea install dir>\DrawingTools\Custom\` and restart Kinovea. Only files with the `.xml` extension are loaded. The shipped files are the best examples; this chapter documents the vocabulary their parser understands.

Minimal skeleton

```
<?xml version="1.0" encoding="utf-8"?>
<KinoveaPostureTool>
  <FormatVersion>1.1</FormatVersion>
  <Id>generate-a-new-guid-here</Id>
  <Name>My tool</Name>
  <DisplayName>My tool</DisplayName>
  <Icon>base64 PNG, 16×16</Icon>

  <Points>
    <Point value="100;100" />
    <Point value="200;100" />
  </Points>
  <Segments>
    <Segment point1="0" point2="1" style="Solid" width="2" />
  </Segments>
  <Handles>
    <Handle type="Point" reference="0" trackable="true" />
    <Handle type="Point" reference="1" trackable="true" />
  </Handles>
</KinoveaPostureTool>
```

`Id` must be a unique GUID: it is how KVA files refer back to the tool. Points are referenced everywhere by **zero-based index** in `<Points>`. `DisplayName` may be a localization key (e.g. `DrawingName_Goniometer`) or plain text.

Real example: the shipped Goniometer

```
<Points>
  <Point value="100;100" />
  <Point value="150;100" />
  <Point value="200;100" />
  <Point value="180;70" />
</Points>
<Segments>
  <Segment name="Stationary arm" point1="0" point2="1" style="Solid" width="4"/>
  <Segment name="Reference" point1="1" point2="2" style="Dash" width="2"/>
  <Segment name="Movable arm" point1="1" point2="3" style="Solid" arrowEnd="true" width
="2"/>
</Segments>
<Angles>
  <Angle origin="1" leg1="2" leg2="3" signed="true" ccw="true" supplementary="false"
         radius="-25" textDistance="40" tenth="true"/>
</Angles>
<Handles>
  <Handle type="Point" reference="0" trackable="true">
    <Impacts><KeepAngle origin="1" leg1="0" leg2="2"/></Impacts>
  </Handle>
  <Handle type="Point" reference="1" trackable="true">
    <Impacts><KeepAngle origin="1" leg1="0" leg2="2"/></Impacts>
  </Handle>
  <Handle type="Point" reference="2">
    <Constraint type="RotationSteps">
      <RotationSteps origin="1" leg1="0" step="45" keepDistance="false" />
    </Constraint>
  </Handle>
  <Handle type="Point" reference="3" trackable="true"/>
</Handles>
<Capabilities flipHorizontal="true" flipVertical="true" />
```

Data section

Element	Attributes
`Points/Point`	`value="x;y"` initial position; `name`, `color`. (Legacy 1.0 files use `InitialConfiguration/Point` with text content, and `PointCount`, which is ignored.)
`Segments/Segment`	`point1`, `point2`; `name`, `style` (`Solid` | `Dash`), `width`, `color`, `arrowBegin` / `arrowEnd` (`arrowStart` also accepted), `optionGroup`.
`Circles/Circle` (alias `Ellipses`)	`center` (point index), `radius`, `style`, `width`, `color`, `name`, `optionGroup`.
`Angles/Angle`	`origin`, `leg1`, `leg2`; `signed`, `ccw`, `supplementary`, `radius` (negative = inside), `textDistance`, `tenth` (one decimal), `symbol`, `color`, `name`, `optionGroup`.

Element	Attributes
`Polylines/Polyline`	`points="0;1;2;3"` ; `style` , `width` , `color` , `arrowBegin` / `arrowEnd` , `name` , `optionGroup` .

Interaction section

`Handles/Handle` makes something draggable. Attributes: `type` (`Point` | `Segment` | `Circle` | `Ellipse`), `reference` (index of the point/segment/circle), `trackable` (included in tracking), `color` , `name` , `optionGroup` .

A handle may contain one **constraint** (where it is allowed to go) and **impacts** (what else moves when it moves).

`Constraint` takes `type` and an optional `optionGroup` , and a child element carrying that type's parameters:

Constraint type	Parameters / meaning
`None`	Free movement.
`LineSlide`	`point1` , `point2` , `position` — slide along the line through two points.
`VerticalSlide` / `HorizontalSlide`	Move only vertically / horizontally.
`DistanceToPoint`	`point` — keep a fixed distance (move on a circle).
`RotationSteps`	`origin` , `leg1` , `step` (degrees), `keepDistance` — snap rotation to steps.
`PerpendicularSlide`	`origin` , `leg1` — slide perpendicular to a leg.
`ParallelSlide`	`a` , `b` , `c` — keep parallel to a reference segment.
`LockedInPlace`	Cannot be moved.

`Impacts` (or `Impact`) contains any of:

Impact element	Parameters / meaning
`Align`	`point1` , `point2` , `pointToAlign` — keep a point on a line.
`VerticalAlign` / `HorizontalAlign`	`point` — keep a point vertically / horizontally aligned with the handle.
`Pivot`	`pivot` , `impacted` — rotate other points around a pivot.

Impact element	Parameters / meaning
`KeepAngle`	`origin`, `leg1`, `leg2` — preserve an angle.
`HorizontalSymmetry`	`axis`, `impacting`, `impacted` — mirror a point across an axis.
`SegmentCenter`	`point1`, `point2`, `pointToMove` — keep a point at a segment's midpoint.
`PerdpendicularAlign`	`origin`, `leg1`, `pointToMove`. **Note the misspelling**: the parser expects `PerdpendicularAlign`. The correct spelling is silently ignored.
`ParallelAlign`	`a`, `b`, `c`, `pointToMove`.

`HitZone/Polygon points="2;11;15;6"` defines an area; dragging inside it moves the whole drawing.

Variables section

- `Distances/Distance point1 point2` — display a measured distance; `symbol`, `color`, `name`, `optionGroup`.
- `Positions/Position point` — display coordinates of a point; `symbol`, `color`, `optionGroup`.
- `ComputedPoints/ComputedPoint` — a weighted point, e.g. center of mass. Nest `ComputedPoint` (with `weight`) and `ReferencePoint ref="<index>" weight="…"` arbitrarily deep; `name`, `color`, `symbol`, `displayCoordinates`, `optionGroup`. Computed points are referenced elsewhere with a `$` prefix (`point="$0"` = first computed point). See *10 - Human model.xml* for a full anthropometric center-of-mass model.

Menus and extras

- `optionGroup` on any element links it to a context-menu toggle; the element is drawn only when the option is on. Combine several with `|` (`optionGroup="a|b"`): **all** must be on (AND logic).
- Options are created **automatically** from every `optionGroup` value, labelled with that value and **off by default**. That's how *Human model* gets its "Display knee angles" toggles without declaring anything.
- `Options/Option key label default hidden` declares an option explicitly, to set a friendlier `label`, make it `default="true"`, or keep it out of the menu with `hidden="true"` (used by the OpenPose model).
- `Capabilities flipHorizontal flipVertical` — adds Flip menu items. Must be an empty element.
- `TrackingParameters` — per-tool tracking parameter overrides.

Tips

- Start from the shipped file closest to what you want. Change the `Id`.
- Kinovea logs and skips unknown elements rather than failing, so a typo silently does nothing. Check `%APPDATA%\Kinovea\Logs\log.txt` (`log.<window>.txt` for named windows; portable installs keep settings and logs next to the exe) if something doesn't appear.